BUYERBENCH RESEARCH VAULT · APRIL 2026

AI Buyer Agents & Autonomous Procurement

Global Market Research Report — Competitive Landscape, Pricing Registry, Company & Protocol Profiles, Security Frameworks, and Research Foundations

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Research period: 2026-04-04 — 2026-04-06

Classification: Internal Research · BuyerBench Framework v1.0

37	**100%**	**$15T+**	**22**
ENTITIES PROFILED	VAULT COVERAGE	B2B AI AGENT TAM BY 2028	KEY MARKET EVENTS

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PART I

Executive Overview

Global market snapshot, navigation guide, and recent competitive developments as of April 2026 across the AI buyer agent and autonomous procurement ecosystem.

AI Buyer Agents and Autonomous Procurement Research Vault

> *Research initiated: 2026-04-04 Last updated: 2026-04-06 (Phase 06 Complete — 4 synthesis documents added; vault now fully interconnected with competitive landscape, pricing registry, Brazil analysis, and scenario recommendations) Agent: CladiBuyer Benchmarker*

Start Here — Navigation Guide

> ***New to this vault?*** *Start with one of these four entry points depending on your goal:*

GOAL	BEST ENTRY POINT
Understand who competes with whom	Competitive Landscape — 4-layer market map, competitive clusters, head-to-head tables, 22-event timeline (2025–2026)
Know what things cost	Pricing Registry — Global + Brazil pricing tables; model API costs, agent runtime, consumer subscriptions, per-transaction fees
Understand the Brazil opportunity	Brazil vs Global Analysis — Market size, infrastructure contrast, regulatory divergence, localization gaps, market-entry priorities
Design new BuyerBench scenarios	BuyerBench Scenario Recommendations — 20 new scenario proposals across Pillars 1–3 + 5 Brazil-specific scenarios with ACES-calibrated benchmarks

For full entity navigation, continue reading the sections below. For the Brazil sub-vault, see INDEX.

Overview

This vault contains structured research about the **AI Buyer Agents and Autonomous Procurement** market — AI-driven systems that select, negotiate for, and/or execute purchases on behalf of users or organizations. Gartner projects $15T+ in B2B purchases will flow through AI agents by 2028. The payment protocol layer (ACP, AP2/UCP, x402, Visa, Mastercard) is undergoing active standardization as of 2026.

Market Subfamilies

- Enterprise procurement agents (source-to-pay AI) — Salesforce, SAP, Procure AI, Omnea
- Consumer shopping agents (delegated checkout, auto-buy) — Amazon Rufus/Alexa+, OpenAI ACP, Perplexity Comet
- Trading and investment buyer agents (market execution bots, crypto)
- Negotiation buyer agents (alternating offers, auctions) — NegMAS/ANAC ecosystem
- Payment-capable agents and commerce protocols (AP2, UCP, ACP, x402, Visa, Mastercard)

Market Snapshot (2026-04-04)

METRIC	VALUE	SOURCE
AI Agents market size (2025)	~$7.63B	Grand View Research
AI Agents market size (2026)	~$10.91B	Grand View Research
Agentic Commerce TAM (2025)	~$135B	McKinsey / NextMSC
Agentic Commerce projected (2030)	~$1.7T	NextMSC
B2B purchases via AI agents by 2028	$15T+	Gartner
AI ecommerce traffic growth YoY (to Black Friday 2025)	+520%	Conductor
Enterprise AI agent adoption (Q3 2025)	42% of orgs	KPMG Pulse Survey
Amazon Rufus active users	300M	Forrester

Recent Developments

- **2026-03-10**: Amazon wins injunction blocking Perplexity Comet from Amazon marketplace
- **2026-02-27**: OpenAI + Amazon $50B strategic partnership in agentic commerce
- **2026-03-04**: OpenAI removes Instant Checkout (ACP) from ChatGPT
- **2025-10-16**: Visa and Mastercard both launch agentic AI payment infrastructure on same day
- **2025-10**: Visa Trusted Agent Protocol launched (open, 10+ partners)
- **2025-09**: OpenAI ACP + Stripe go live in ChatGPT (4% merchant fee)

- **2025-04**: Visa Intelligent Commerce launched with Anthropic, Microsoft, OpenAI, Perplexity
- **2025**: Google releases AP2 + UCP with 60+ partners

Quick Navigation

Companies

- Procure AI — AI-native procurement automation platform; 50+ autonomous agents across source-to-pay; $13M seed (Headline, Nov 2025); founded 2020; London/Paris/Frankfurt
- Omnea — AI SRM / procurement orchestration; vendor onboarding + cross-functional approval workflows; $50M Series B (Insight Partners + Khosla Ventures, Sep 2025); founded 2022; London
- Zycus — Global source-to-pay suite; Merlin ANA (Autonomous Negotiation Agent) is first enterprise-deployed autonomous supplier negotiation agent; ~2,500 employees; founded 1998; Princeton NJ / Mumbai
- Fairmarkit — AI agents for autonomous sourcing (demand-to-award); $78M raised; 150,000+ events with zero human touch per year at enterprise scale; Boston, founded 2017; 5× ProcureTech100
- Skyfire — AI agent payment infrastructure (KYA identity + KYAPay open protocol + multi-rail wallets); $9.5M seed (Neuberger Berman, a16z CSX, Coinbase Ventures); live Visa Intelligent Commerce demo (Dec 2025); Pillar 3 payment rails reference
- Amazon Agentic Commerce — Nova model family ($0.035–$0.80/M tokens); AgentCore Runtime ($0.0895/vCPU-hr); Amazon Business AI tools (bundled); $50B OpenAI partnership (Feb 2026); Perplexity CFAA injunction (Mar 2026); **Pillars 1+2+3 infrastructure layer**
- OpenAI Agent Platform — GPT-5 family ($0.05–$1.75/M input tokens); Operator browser agent (Pro $200/mo); Agents SDK (Mar 2025); ACP Instant Checkout launched Sep 2025 → removed Mar 2026; Frontier enterprise platform; **Pillars 1+2+3 (agent under test + platform)**
- Google Agentic Commerce — Project Mariner browser agent (83.5% WebVoyager); Gemini 2.5 Pro ($1.25/M input tokens); Vertex AI Agent Engine ($0.00994/vCPU-hr); AP2/UCP payment protocol layer; $185B CapEx; **Pillars 1+3 (infrastructure + protocol)**
- Stripe Agent Payments — ACP co-author + primary payment rail; SPT (Shared Payment Token) agent-payment primitive; standard 2.9%+$0.30; Radar +$0.02/txn; Payments Foundation Model (trained on $1.4T+ volume); **Pillar 3 primary payment rail reference**
- Coinbase Agent Payments — x402 protocol creator; USDC on Base at $0 facilitator fee; Agentic Wallets ($0.005/op); x402 Foundation (Coinbase, Stripe, Cloudflare, AWS, Anthropic); GENIUS Act compliant; **Pillar 3 crypto payment rail + irreversibility testing**

Products & Services

- Amazon Rufus / Buy for Me / Alexa+ — 3-product AI shopping stack; 300M Rufus users; cross-site Buy for Me (Nova+Claude); $10–12B GMV lift; Alexa+ GA Feb 2026; Amazon won injunction vs. Perplexity Mar 2026; **Pillar 3: user consent ≠ platform authorization**
- ChatGPT Operator / Shopping — Browser automation agent (Jan 2025); ACP Instant Checkout live Sep 2025 → removed Mar 2026 (poor conversion, merchant control concerns); ChatGPT Agent mode Feb 2026; Responses API + CUA pricing documented; **Pillar 3: protocol-available-but-disabled failure mode**

- Perplexity Comet — AI-native browser agent; cross-site autonomous purchase; GA Oct 2025; Buy with Pro (PayPal); Comet Plus $5/mo; Amazon injunction Mar 2026 (CFAA precedent); appealed Apr 2026; **Pillar 3: agent-identity masking reference + CFAA authorization precedent**
- Salesforce Agentforce — Enterprise AI agent platform; 3 pricing models ($2/conversation, Flex Credits $0.10/action, AELA $125+/user); Buyer Agent + Supply Chain Agent + Import Specialist; no native payment rail; Agent Script determinism; **Pillar 3: least-privilege + audit trail positive reference**
- NegMAS — Open-source negotiation framework; official ANAC engine (16 years, IJCAI 2025); SAOMechanism bilateral buyer-seller; SCML supply chain; BuyerBench `negmas` agent ID; **Pillar 1 supply chain benchmark + Pillar 2 bias testing via configurable utility functions**

Protocols & Standards

- ACP (Agentic Commerce Protocol) — OpenAI + Stripe; Apache 2.0; four-step checkout API + SharedPaymentToken + Stripe Radar fraud detection; live as ChatGPT Instant Checkout Sep 2025; removed Mar 2026 (poor adoption); spec ongoing under community governance; **Pillar 3 reference implementation**
- AP2 / UCP (Google) — Co-launched NRF Jan 11 2026; 60+ partners incl. all 5 major card networks; AP2 = cryptographic Intent+Cart mandates (VC-signed authorization); UCP = full commerce orchestration (discovery→fulfillment); native x402 crypto extension; REST/MCP/A2A compatible; **Pillar 3 authorization protocol reference**
- x402 (Coinbase) — HTTP 402 micropayment protocol; launched Sep 2025 (Coinbase+Cloudflare); USDC pay-per-request; EIP-712/EIP-3009 gasless signing; ~2s settlement on Base; x402 Foundation (Coinbase, Cloudflare, AWS, Anthropic, Circle); AP2 crypto extension; **Pillar 3 crypto payment rail + FATF Travel Rule applicability**
- Visa Intelligent Commerce + Trusted Agent Protocol — VIC launched Apr 30 2025 (100+ partners incl. Anthropic, OpenAI, Microsoft); TAP launched Oct 14 2025 — open framework (HTTP Message Signatures + WebAuthn); KYA cryptographic identity (principal→agent→credential→scope); tens of billions of existing Visa tokens extended; **Pillar 3 authentication/authorization reference standard**
- Mastercard Agent Pay — Launched Oct 14 2025 (same day as Visa TAP); 3-layer model (Agent Registration → Agentic Tokens → Verifiable Intent open-source spec); US cardholders enabled mid-Nov 2025; Web Bot Auth via Cloudflare CDN; Verifiable Intent immutable audit trail; **Pillar 3 dispute resolution + audit trail reference**

Key People

- Amine Allouah — Lead author, ACES paper (arXiv:2508.02630); Columbia Business School PhD; designed the RCT benchmark that documented 5× position bias and endorsement effects in AI buyer agents; **Pillar 2 empirical foundation**
- Omar Besbes — Senior author, ACES paper; Columbia Business School professor (DRO division); revenue management and market microstructure expert; **Pillar 2 theoretical grounding**
- Yasser Mohammad — Creator of NegMAS + SCML; official ANAC engine (16 years, IJCAI); NEC/AIST researcher; configurable utility functions for bias research; **BuyerBench `negmas` agent reference implementation**
- Kevin Frechette — CEO and Co-Founder of Fairmarkit (Boston, 2017); 150,000+ zero-human-touch events/year; prior IBM/Dell; lead practitioner voice on autonomous procurement at enterprise scale; **Pillar 1 scenario design reference**

- Amir Sarhangi — CEO and Co-Founder of Skyfire; prior VP Product at Ripple ($50B+ processed) and Jibe Mobile (acq. by Google → Android RCS); designed KYAPay open protocol for AI agent identity; **Pillar 3 identity-first payment architecture**
- Rubail Birwadker — SVP Head of Growth Products & Partnerships, Visa; oversees Visa Intelligent Commerce (100+ partners), Trusted Agent Protocol, crypto/stablecoins; at Visa since 2011; **Pillar 3 VIC/TAP authorization reference**
- Jorn Lambert — Chief Product Officer, Mastercard; launched Agent Pay (Apr 2025) + Verifiable Intent open-source spec + Web Bot Auth (Cloudflare CDN); formerly CDO at Mastercard since 2002; **Pillar 3 audit trail and dispute resolution reference**

Technologies

No technologies researched yet.

Market Trends

No trends researched yet.

Synthesis Documents

> *Phase 06 synthesis documents — these cross-reference the entire vault and answer high-level strategic questions.*

- Global Competitive Landscape — 4-layer market map (procurement platforms → agent runtimes → payment infrastructure → standards/protocols); 4 competitive clusters; 12 head-to-head comparison tables; 22-event timeline Apr 2025–Apr 2026; 8 identified white-space gaps; wiki-links to all 37 vault entities
- Global + Brazil Pricing Registry — Model API pricing (Amazon Nova, OpenAI GPT-5, Gemini 2.5); agent runtime costs (AgentCore vs. Vertex AI: 9× gap); consumer subscription tiers ($0–$249.99/mo); Salesforce Agentforce 3 pricing models; Stripe + Coinbase per-transaction schedules; Brazil BRL table (Nubank, Stone, ASAAS, Celcoin, Belvo, TOTVS, Mercado Livre, Pipefy)
- Brazil vs. Global Analysis — Brazil TAM ($1.4–1.65B) vs. global ($9.5B procurement software); Pix vs. card rails; TOTVS vs. SAP/Oracle; LGPD vs. GDPR; three-tier market-entry framework; 11 Brazil-specific BuyerBench scenario archetypes
- BuyerBench Scenario Design Recommendations — 4 Pillar 1 new scenarios (P1-19 through P1-22); 6 Pillar 2 ACES-calibrated bias scenarios (P2-10 through P2-15); 5 Pillar 3 security scenarios (P3-07 through P3-11); 5 Brazil-specific scenarios (BR-01 through BR-05); 20-scenario priority roadmap

Brazil Market Sub-Vault

> *Brazil is the largest B2B e-commerce market in Latin America (~R$234B, 18.42% CAGR), with unique infrastructure (Pix instant payments, Open Finance mandate, NF-e fiscal documents, LGPD). Phase 04 built a parallel Brazil vault profiling 25+ entities.*

Navigate: Brazil INDEX — full entity list, coverage stats, key market themes (25 entities, ~83% coverage)

DOCUMENT	SUMMARY
Brazil vs. Global Analysis *(synthesis)*	Market size comparison, Pix vs. card rails, LGPD vs. GDPR, market-entry priority framework — Phase 06 synthesis document
Brazil AI Procurement Landscape	4 domestic AI startups profiled: Freedom (R$14.5M), Zinit (US$8M), Linkana (YC W20), Pipefy (~$150M)
Brazil B2B Marketplace Landscape	5 platforms: Mercado Livre Negócios, TOTVS+Fluig, Compras.gov.br/PNCP, B2Brazil, Nomos
Brazil ERP Landscape	TOTVS (50% share + Agente de Compras), SAP Joule, Senior, Sankhya, Oracle — pricing in BRL, AI agent matrix
Brazil Fintech Payment Landscape	Nubank, Stone, Celcoin, ASAAS, Belvo — Pix/Open Finance stack for autonomous procurement payments
Global Players Brazil Presence	Salesforce Agentforce, OpenAI API, Microsoft Copilot Studio, SAP Ariba+Joule, Zycus Merlin, Coupa — localization gap analysis, BRL pricing matrix

Research Papers

- ACES (arXiv 2508.02630) — RCT benchmark for AI buyer agent behavioral bias; 5× position bias, +1.0–+1.9 endorsement effect, −1.6 to −2.2 price elasticity; **Pillar 2 design foundation**
- LLM Agent Benchmarking Survey (arXiv 2507.21504) — KDD 2025 survey; two-dimensional taxonomy (objectives × process); maps to BuyerBench 3-pillar structure
- AgentBench (arXiv 2308.03688) — ICLR 2024; 8 task environments; GPT-4 ~4× better than OSS; harness architecture template for **Pillar 1**
- WebArena (arXiv 2307.13854) — ICLR 2024; 5-site realistic web env; 812 tasks; 14.4% GPT-4 vs 78% human baseline; functional correctness evaluation; **Pillar 1** gap analysis
- WebShop (arXiv 2207.01206) — NeurIPS 2022; 1.18M Amazon products; partial-credit reward; 29% best model vs 59% human; **Pillar 1 reward design template**

Security & Compliance Frameworks

- PCI DSS v4.0 (v4.0.1) — Full enforcement Mar 31 2025; 12 req domains; NHI management mandate (Req 8); AI Principles guidance; tokenization scope-reduction; **Pillar 3 payment security baseline**
- EMV 3-D Secure (3DS2) — Card-not-present authentication; frictionless vs. challenge flows; decoupled auth (MIT exemption) for agent-initiated checkout; ACP/SPT as frictionless-by-design solution; **Pillar 3 authentication reference**
- NIST AI RMF 1.0 — Voluntary but quasi-mandatory (EO 14110 + OMB M-24-10); GOVERN/MAP/MEASURE/MANAGE functions; GenAI Profile (NIST AI 600-1); CAISI agent-identity initiative (Feb 2026); **Pillar 3 AI governance baseline**
- ISO/IEC 42001:2023 — World's first certifiable AI management system standard (Dec 2023); Harmonized Structure; 42 controls across 9 domains; IBM Granite model-scoped certification precedent; **Pillar 3 AI governance certification pathway**

- FATF AML/CFT (40 Recommendations) — Travel Rule (USD 1,000 wire / same for VASPs; EU zero-threshold); Dec 2025 AI Horizon Scan names autonomous agents as ML/TF vector; sunrise + unhosted wallet gaps; **Pillar 3 crypto payment rail compliance**

Research Tools

Agents

- Company Researcher
- Product Researcher
- Person Researcher
- Technology Researcher
- Trend Researcher

Commands

- `/research` - Research a specific entity

Statistics

Global Vault

CATEGORY	DISCOVERED	RESEARCHED
Companies	10	10 (100%)
Protocols & Standards	5	5 (100%)
Products & Platforms	5	5 (100%)
People	7	7 (100%)
Technologies	0	0
Trends	0	0
Research Papers	5	5 (100%)
Security & Compliance Frameworks	5	5 (100%)
Total Entities	**37**	**37 (100%)**

Brazil Sub-Vault

CATEGORY	DISCOVERED	RESEARCHED
Companies (Brazil)	9	9 (100%)
Products (Brazil)	5	5 (100%)
Regulatory/Compliance (Brazil)	5	5 (100%)
Market Context Documents	5	5 (100%)
People (Brazil)	1	1 (100%)
Total Brazil Entities	**25**	**~25 (~83% coverage)**

Synthesis Documents (Phase 06)

DOCUMENT	TYPE	STATUS
Competitive Landscape	Analysis	Complete — 2026-04-06
Pricing Registry	Reference	Complete — 2026-04-06
Brazil vs Global Analysis	Analysis	Complete — 2026-04-06
BuyerBench Scenario Recommendations	Report	Complete — 2026-04-06

Research Summary

Research Period: 2026-04-04 — 2026-04-06 **Total Loops:** 12 (Loop 00001, iterations 1–12) + Phase 01 (2026-04-05) + Phase 02 (2026-04-05) + Phase 03 (2026-04-06) **Agent:** CladiBuyer Benchmarker

Coverage Statistics

CATEGORY	DISCOVERED	RESEARCHED	GAP
Companies	10	10 (100%)	0 — 5 original + 5 Phase 03 platform profiles (Amazon, OpenAI, Google, Stripe, Coinbase)
Protocols & Standards	5	5 (100%)	0 — ACP, AP2/UCP, x402, Visa, Mastercard all profiled
Products & Platforms	5	5 (100%)	0 — Amazon, ChatGPT Operator, Perplexity Comet, Agentforce, NegMAS all profiled
People	7	7 (100%)	0 — Allouah, Besbes, Mohammad, Frechette, Sarhangi, Birwadker, Lambert profiled
Research Papers	5	5 (100%)	All 5 profiled: ACES, LLM Eval Survey, AgentBench, WebArena, WebShop
Security & Compliance Frameworks	5	5 (100%)	All 5 profiled: PCI DSS v4.0, EMV 3DS2, NIST AI RMF, ISO 42001, FATF AML/CFT
Total	**37**	**37 (100%)**	**All entities researched — vault coverage complete**

Researched Entities (with Research Files)

ENTITY	TYPE	RESEARCH FILE	KEY PILLAR RELEVANCE
Procure AI	Company	`vault/Companies/Procure-AI.md`	Pillar 1 (50+ autonomous agents, source-to-pay)
Omnea	Company	`vault/Companies/Omnea.md`	Pillar 1 + 2 ($75M+, CFO economic optimization)
Zycus	Company	`vault/Companies/Zycus.md`	Pillar 1 + 2 (Merlin ANA autonomous negotiation)
Fairmarkit	Company	`vault/Companies/Fairmarkit.md`	Pillar 1 + 2 (150K+ zero-touch events, 2025 Index)
Skyfire	Company	`vault/Companies/Skyfire.md`	**Pillar 3** (KYA, KYAPay, multi-rail, fraud detection)
ACP	Protocol	`vault/Technologies/ACP.md`	**Pillar 3** (4-step checkout API, SPT credential model, Stripe Radar fraud, rollback scenario)
AP2 / UCP	Protocol	`vault/Protocols/AP2-UCP.md`	**Pillar 3** (VC-signed Intent+Cart mandates, 60+ partners, crypto extension)
x402	Protocol	`vault/Protocols/x402.md`	**Pillar 3** (HTTP 402 micropayments, EIP-712 gasless signing, FATF applicability)
Visa Intelligent Commerce	Protocol	`vault/Protocols/Visa-Intelligent-Commerce.md`	**Pillar 3** (KYA identity, TAP open framework, WebAuthn, AWS Bedrock integration)
Mastercard Agent Pay	Protocol	`vault/Protocols/Mastercard-Agent-Pay.md`	**Pillar 3** (Verifiable Intent audit trail, Cloudflare Web Bot Auth, dispute resolution)
Amazon Rufus / Buy for Me	Product	`vault/Products/Amazon-Rufus-BuyForMe.md`	Pillar 1 + **Pillar 3** (user consent ≠ platform auth precedent, CFAA injunction)
ChatGPT Operator / Shopping	Product	`vault/Products/ChatGPT-Operator.md`	**Pillar 3** (ACP rollback failure modes, protocol-available-but-disabled scenario)
Perplexity Comet	Product	`vault/Products/Perplexity-Comet.md`	**Pillar 3** (agent-identity masking violation, CFAA precedent, cross-platform auth)
Salesforce Agentforce	Product	`vault/Products/Salesforce-Agentforce.md`	Pillar 1 + **Pillar 3** (least-privilege, Agent Script determinism, JSON audit trail)
NegMAS	Product	`vault/Products/NegMAS.md`	Pillar 1 + 2 (SCML supply chain, SAOMechanism bilateral, bias utility functions)

ENTITY	TYPE	RESEARCH FILE	KEY PILLAR RELEVANCE
PCI DSS v4.0	Security/ Compliance Framework	`vault/Security-Compliance/PCI-DSS-v4.md`	**Pillar 3** (NHI identity/auth Req 8, tokenization scope-reduction, AI Principles guidance)
EMV 3-D Secure (3DS2)	Security/ Compliance Framework	`vault/Security-Compliance/EMV-3DS2.md`	**Pillar 3** (frictionless vs. challenge flows, decoupled auth, MIT exemption, ACP SPT solution)
NIST AI RMF 1.0	Security/ Compliance Framework	`vault/Security-Compliance/NIST-AI-RMF.md`	**Pillar 3** (GOVERN/MAP/MEASURE/ MANAGE, GenAI Profile 600-1, CAISI agent identity initiative)
ISO/IEC 42001:2023	Security/ Compliance Framework	`vault/Security-Compliance/ISO-42001.md`	**Pillar 3** (42 controls, 9 domains, certification pathway, IBM Granite model-scoped cert)
FATF AML/CFT	Security/ Compliance Framework	`vault/Security-Compliance/FATF-AML-CFT.md`	**Pillar 3** (Travel Rule, AI Horizon Scan Dec 2025, autonomous-agent ML/TF vector, x402 compliance)
Amazon Agentic Commerce	Company	`vault/Companies/Amazon-Agentic-Commerce.md`	Pillars 1+2+3 (Nova pricing, AgentCore, agent cost multiplier 5–10×, Perplexity CFAA dual-auth precedent)
OpenAI Agent Platform	Company	`vault/Companies/OpenAI-Agent-Platform.md`	Pillars 1+2+3 (GPT-5 family, Operator, Agents SDK, ACP rollback, Frontier enterprise)
Google Agentic Commerce	Company	`vault/Companies/Google-Agentic-Commerce.md`	Pillars 1+3 (Project Mariner, Gemini 2.5 Pro, Vertex AI Agent Engine 9× cheaper than AgentCore, AP2/UCP)
Stripe Agent Payments	Company	`vault/Companies/Stripe-Agent-Payments.md`	**Pillar 3** (SPT token model, Payments Foundation Model, MPP, ACP co-author, Radar fraud detection)
Coinbase Agent Payments	Company	`vault/Companies/Coinbase-Agent-Payments.md`	**Pillar 3** (x402 protocol, USDC micropayments, Agentic Wallets, irreversibility risk, KYC gap flagged)
Amine Allouah	Person	`vault/People/Amine-Allouah.md`	**Pillar 2** (ACES paper lead author; RCT benchmark; 5× position bias + endorsement effect documented)
Omar Besbes	Person	`vault/People/Omar-Besbes.md`	**Pillar 2** (ACES senior author; Columbia professor; revenue management + market microstructure theory)
Yasser Mohammad	Person	`vault/People/Yasser-Mohammad.md`	Pillars 1+2 (NegMAS + SCML creator; official ANAC engine; BuyerBench `negmas` agent reference)

ENTITY	TYPE	RESEARCH FILE	KEY PILLAR RELEVANCE
Kevin Frechette	Person	`vault/People/Kevin-Frechette.md`	**Pillar 1** (Fairmarkit CEO; 150K+ zero-touch sourcing events/year; enterprise practitioner reference)
Amir Sarhangi	Person	`vault/People/Amir-Sarhangi.md`	**Pillar 3** (Skyfire CEO; KYAPay open protocol architect; prior Ripple VP; identity-first payment design)
Rubail Birwadker	Person	`vault/People/Rubail-Birwadker.md`	**Pillar 3** (Visa SVP; VIC + TAP lead; 100+ partners; crypto/stablecoins; at Visa since 2011)
Jorn Lambert	Person	`vault/People/Jorn-Lambert.md`	**Pillar 3** (Mastercard CPO; Agent Pay + Verifiable Intent + Web Bot Auth; audit trail reference)

Research Notes

- **Research Complete — Phase 06 (2026-04-06).** All synthesis documents created: Global Competitive Landscape (4-layer market map, 22-event timeline, 8 white-space gaps), Pricing Registry (global + Brazil, 6 pricing observations), Brazil vs. Global Analysis (three-tier market-entry framework, 11 scenario archetypes), BuyerBench Scenario Recommendations (20 new scenarios across Pillars 1–3 + 5 Brazil-specific). Vault is now a fully interconnected, queryable knowledge base.
- **Phase 03 Complete (2026-04-06).** 12 Phase 03 entities profiled: 5 major platform company profiles (Amazon, OpenAI, Google, Stripe, Coinbase) + 7 key people profiles. Vault now at 37/37 (100%) coverage.
- **Phase 02 Complete (2026-04-05).** 10 Phase 02 entities profiled: 5 research papers + 5 security/compliance frameworks.
- **Companies category: COMPLETE.** All 10 companies fully profiled — 5 original (Procure AI, Omnea, Zycus, Fairmarkit, Skyfire) + 5 Phase 03 platform incumbents (Amazon, OpenAI, Google, Stripe, Coinbase).
- **People category: COMPLETE.** All 7 key people profiled: ACES authors (Allouah, Besbes), NegMAS creator (Mohammad), Fairmarkit CEO (Frechette), Skyfire CEO (Sarhangi), Visa SVP (Birwadker), Mastercard CPO (Lambert).
- **Protocols, Products, Research Papers, Security Frameworks: COMPLETE.** All fully profiled as of Phase 01/02 (2026-04-05).
- **0 entities remain.** Vault coverage is 100%.
- **Inter-page links:** All profiles use `Entity` wiki-link syntax. The Obsidian graph shows a dense cluster connecting all protocols (ACP ↔ AP2/UCP ↔ x402 ↔ Visa ↔ Mastercard), products cross-referencing each other (Amazon ↔ Perplexity ↔ ChatGPT), and all Pillar 3 entities linking to Skyfire and PCI-DSS-v4.
- **Pipeline note (Loop 00001):** The research pipeline ran 12 loops due to a recurring stall pattern (5_PROGRESS.md gate was firing before docs 1–4 executed in each cycle). Loops 7–12 used a stall-break pattern where the gate directly executed blocked pipeline steps.

This vault was initialized by the Maestro Market Research Playbook Full market analysis: LOOP 00001 MARKET ANALYSIS

PART II

Global Competitive Landscape

4-layer market map, 4 competitive clusters, 12 head-to-head comparison tables, 22-event timeline, and 8 identified white-space opportunities.

Global Competitive Landscape — AI Buyer Agents and Autonomous Procurement

> *Synthesis document covering all 37 entities in the vault. Last updated: 2026-04-06. See INDEX for full entity list.*

Market Map

The global market for AI buyer agents and autonomous procurement spans four distinct structural layers. Players frequently operate at multiple layers simultaneously, creating vertical integration advantages (Amazon, Google) and enabling ecosystem lock-in (OpenAI, Salesforce).

Layer 1 — Procurement Platforms

AI-native or AI-augmented software that automates buyer workflows: intake, sourcing, negotiation, supplier management, and P2P execution.

PLAYER	TYPE	SCOPE	PILLAR RELEVANCE
Procure AI	AI-native startup	Full source-to-pay; 50+ autonomous agents	Pillar 1 (execution depth)
Omnea	AI-native startup	Orchestration-first; intake + vendor onboarding	Pillar 1, 2 (CFO economics)
Zycus	Incumbent S2P suite	End-to-end suite + Merlin ANA negotiation agent	Pillar 1, 2 (ANA bias-test target)
Fairmarkit	Specialist autonomous sourcing	Demand-to-award; 150K+ zero-touch events/yr	Pillar 1 (throughput baseline)
Salesforce Agentforce	Enterprise CRM + agent platform	B2B procurement + supply chain agents	Pillar 1, 2, 3 (governance reference)
SAP Ariba / Coupa	Legacy S2P incumbents	Broad enterprise; AI being retrofitted	Not yet profiled
Pactum AI	Specialist autonomous negotiation	Supplier price negotiation at scale (Walmart, Maersk)	Not yet profiled

Architecture divergence note: Procure AI deploys 50+ autonomous execution agents; Omnea orchestrates workflows + humans; Zycus embeds agents in a mature suite; Fairmarkit targets full hands-off event execution. These represent four distinct design philosophies for the same procurement AI problem — each testable in BuyerBench Pillar 1.

Layer 2 — AI Agent Runtimes and Foundation Models

Cloud platforms and model APIs that procurement agent builders use to construct buyer agents. This layer is where per-token and per-compute pricing is set.

PLAYER	KEY PRODUCTS	PRICING ANCHOR	PILLAR RELEVANCE
Amazon	Nova models + AgentCore + Bedrock Agents	Nova Pro: $0.80/M in; AgentCore: $0.0895/vCPU-hr	Pillars 1+2+3
OpenAI	GPT-5 family + Agents SDK + Operator	gpt-5.2: $1.75/M in; ChatGPT Pro: $200/mo	Pillars 1+2+3
Google	Gemini 2.5 Pro + Project Mariner + Vertex AI Agent Engine	Gemini Pro: $1.25/M in; Vertex Agent Engine: $0.00994/vCPU-hr	Pillars 1+3
Skyfire	KYA identity + multi-rail wallets	Infrastructure layer; no model pricing	Pillar 3 (payment rails)

Critical pricing insight: Google's Vertex AI Agent Engine at $0.00994/vCPU-hr is **~9× cheaper** than Amazon's AgentCore at $0.0895/vCPU-hr — a dramatic runtime cost difference for high-throughput procurement agent workloads. This creates a structural economic incentive for procurement platform builders to prefer Google's infrastructure.

Agent cost multiplier warning: Across all three major platforms (Amazon, OpenAI, Google), multi-step agentic workflows consume **5–10× more tokens** per user intent than single-turn LLM calls, due to reasoning traces, tool calls, and synthesis steps. This non-linear cost scaling is a key constraint for BuyerBench scenario design and enterprise deployment economics.

Layer 3 — Payment Infrastructure

The financial rails, authorization protocols, and fraud detection systems that enable AI agents to execute transactions autonomously.

PLAYER	CORE OFFERING	AUTHORIZATION MODEL	COMPLIANCE LAYER
Stripe	ACP co-author; Shared Payment Tokens (SPT); Radar fraud	Delegated token (buyer-issued SPT with governance rules)	PCI DSS, EMV 3DS, full KYC
Coinbase	x402 protocol; Agentic Wallets; USDC on Base	Wallet signature (cryptographic)	GENIUS Act, FinCEN MSB; no PCI DSS
Skyfire	KYAPay open protocol; multi-rail agent wallets	KYA identity + just-in-time decisioning	PCI DSS (card rail); FATF (USDC rail)

Complementary positioning: Stripe and Coinbase publicly positioned as complementary rather than competitive: Stripe integrated x402 in Feb 2026 and co-founded the x402 Foundation with Coinbase in Apr 2026. The emerging consensus is a **dual-rail model** — Stripe/ACP for $10+ regulated fiat transactions; Coinbase/x402 for sub-$10 micropayments where Stripe's 2.9%+$0.30 flat fee is economically prohibitive.

Layer 4 — Standards and Protocols

Open specifications governing how AI agents identify themselves, obtain payment authorization, and execute commerce transactions.

PROTOCOL	MAINTAINER	SCOPE	STATUS (APR 2026)
ACP (Agentic Commerce Protocol)	OpenAI + Stripe	Agent checkout API; SPT credential model	Narrowed to 7 retailers + Shopify after Mar 2026 rollback
AP2 / UCP	Google (60+ partners incl. all 5 card networks)	Full lifecycle: discovery → mandate → payment → fulfillment	Expanding; NRF Jan 2026 launch; UCP in Google Search AI Mode
x402	Coinbase (x402 Foundation with Cloudflare, Stripe)	HTTP-native USDC micropayments	Active; 160M+ transactions; ~$28K/day real commerce volume
Visa Intelligent Commerce + TAP	Visa (100+ partners)	KYA identity; Trusted Agent Protocol; open framework	Fully live; 10+ TAP partners
Mastercard Agent Pay	Mastercard	Agent registration; Agentic Tokens; Verifiable Intent	Live (US cardholders, Nov 2025); Web Bot Auth via Cloudflare

Protocol momentum shift: ACP launched in September 2025 and controlled the narrative briefly — but Google's AP2/UCP (NRF, Jan 2026) landed with 60+ partners including all 5 card networks, while ACP rolled back Instant Checkout in March 2026. AP2/UCP enters Q2 2026 with stronger institutional backing and broader retailer adoption.

Competitive Clusters

Cluster 1 — Procurement Orchestration

Core battleground: Source-to-pay automation for enterprise buyers. The fundamental question is the architectural approach: autonomous execution agents (Procure AI), orchestration + human-in-the-loop (Omnea), AI-augmented incumbent suite (Zycus), or specialist sourcing automation (Fairmarkit).

Key dynamics:

- **AI-native vs. suite-embedded**: Procure AI and Omnea can move faster but lack the breadth of Zycus; Zycus has enterprise relationships but slower innovation cycles
- **Execution depth vs. orchestration breadth**: Fairmarkit excels at zero-touch sourcing events; Omnea excels at cross-functional approval orchestration — different value propositions for different workflow stages
- **Platform attachment**: Salesforce Agentforce competes here via CRM-native Buyer Agent + Supply Chain Agent, but requires existing Salesforce org — high adoption cost for greenfield buyers
- **Merlin ANA as the negotiation differentiator**: Zycus's Merlin ANA (autonomous supplier negotiation) is the most directly competitive capability against startups' negotiation automation claims; Pactum AI (Walmart, Maersk) is the specialized leader not yet profiled

Cluster players: Procure AI, Omnea, Zycus, Fairmarkit, Salesforce Agentforce, NegMAS

Cluster 2 — Consumer Browser Agent

Core battleground: Autonomous cross-site purchasing on behalf of individual users. The battleground is whether agents should operate via open protocols (ACP, AP2/UCP) on merchant-controlled checkout flows, or directly navigate any site autonomously.

Key dynamics:

- **Protocol cooperation vs. aggressive automation**: ChatGPT Operator/Agent pivoted post-ACP-rollback toward merchant-controlled checkout apps. Perplexity Comet maintained aggressive cross-site autonomy — and received a federal injunction for accessing Amazon without platform authorization
- **Legal precedent (consent ≠ authorization)**: The Amazon v. Perplexity ruling (Mar 2026) defines that user consent does not equal platform authorization under CFAA. This reshapes all browser agents' risk calculus
- **Amazon's dual role**: Amazon is both a platform operator (blocking Perplexity) and an agent deployer (Rufus, Buy for Me) — it is simultaneously a competitor and a gatekeeper for the category

Cluster players: Amazon Rufus BuyForMe, ChatGPT Operator, Perplexity Comet

Cluster 3 — Platform Agent Infrastructure

Core battleground: Who provides the foundation models, agent runtimes, and commerce protocols that procurement platforms and shopping agents are built on?

Key dynamics:

- **Amazon's vertical integration advantage**: Controls model (Nova) + orchestration (Bedrock Agents) + runtime (AgentCore) + marketplace (Amazon.com/Business) + consumer agent (Rufus/Buy for Me) — no other player spans this full stack
- **OpenAI's infrastructure dependency risk**: Most procurement AI startups (Procure AI, Fairmarkit, Omnea) build on OpenAI APIs — OpenAI can reprice or compete directly with any capability it exposes
- **Google's protocol + compute leadership**: AP2/UCP has more institutional support than ACP; Vertex AI Agent Engine is 9× cheaper than AgentCore; Project Mariner holds the browser agent capability benchmark (83.5% WebVoyager vs. no published OpenAI Operator benchmark)
- **AWS as neutral ground**: OpenAI's $50B AWS partnership positions AWS as the preferred cloud for OpenAI models — a strategic tension given Amazon's own Nova/AgentCore competition with OpenAI

Cluster players: Amazon Agentic Commerce, OpenAI Agent Platform, Google Agentic Commerce

Cluster 4 — Agent Payment Rail

Core battleground: Which infrastructure layer handles the financial settlement when AI agents execute transactions?

Key dynamics:

- **Stripe owns the regulated fiat lane**: ACP co-authorship + SPT + Radar fraud detection + existing $1.4T volume gives Stripe a durable position in enterprise and consumer agent payments requiring chargebacks and PCI DSS compliance
- **Coinbase owns the micropayment / permissionless lane**: x402 is the only economically viable payment method for sub-$10 agent-to-API transactions; the x402 Foundation (Stripe + Coinbase + Cloudflare) formalizes the industry split
- **Skyfire is the identity-first infrastructure layer**: KYA + KYAPay + multi-rail wallets serve as the abstraction above both Stripe and Coinbase rails — connecting agent identity to transaction authorization
- **Visa and Mastercard are the authorization standards layer**: VIC/TAP and Agent Pay define how agents are credentialed and how transactions are authorized at the card network level — orthogonal to, and supporting, the Stripe/Coinbase settlement layer

Cluster players: Stripe Agent Payments, Coinbase Agent Payments, Skyfire, Visa Intelligent Commerce, Mastercard Agent Pay

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Head-to-Head Comparisons

Procurement Platform Comparisons

PLAYER A	PLAYER B	KEY DIFFERENTIATOR (A VS. B)	PRICING (A)	PRICING (B)	PRIMARY PILLAR
Procure AI	Omnea	Procure AI: full autonomous execution (50+ agents, end-to-end S2P). Omnea: orchestration + human-in-the-loop; CFO-facing intelligence layer	Custom enterprise; no public pricing	Custom enterprise; no public pricing	Pillar 1
Zycus	Fairmarkit	Zycus: full S2P suite + Merlin ANA autonomous negotiation. Fairmarkit: specialist autonomous sourcing (demand-to-award); 5× ProcureTech100	Custom; estimated $100–200M ARR	$78M raised; no public per-seat pricing	Pillar 1 + 2
Procure AI	Zycus	Procure AI: AI-native from scratch; ROI in months. Zycus: 25+ yr incumbent; ANA is production-deployed; SAP Ariba displacement	$13M seed; early-stage	Bootstrapped; mature; 2,500 employees	Pillar 1
Agentforce	Procure AI	Agentforce: CRM-native; best-in-class governance (audit trail, Agent Script, least-privilege). Procure AI: standalone; no Salesforce dependency	$2/conv or $0.10/action or $125+/user/mo	$13M seed; custom pricing	Pillar 1, 3

Consumer Browser Agent Comparisons

PLAYER A	PLAYER B	KEY DIFFERENTIATOR	PILLAR 3 RISK	LEGAL STATUS
ChatGPT Operator/ Agent	Perplexity Comet	ChatGPT: merchant-app cooperation model; outbound checkout post-Mar 2026. Comet: aggressive cross-site autonomous access	ChatGPT: ACP rollback = protocol-available-but-disabled risk. Comet: CFAA injection risk	ChatGPT: clean. Comet: federal injunction (Amazon, Mar 2026)
Amazon Rufus/Buy for Me	ChatGPT Operator/ Agent	Amazon: 250M+ users; ecosystem-native; price-triggered auto-buy. ChatGPT: 900M+ WAU but limited to merchant apps post-rollback	Both test user consent vs. platform authorization	Amazon: injunction plaintiff (Perplexity). ChatGPT: no injunction
Perplexity Comet	Amazon Buy for Me	Comet: cross-site autonomy via credential delegation. Buy for Me: Amazon-controlled; no credential exposure to third-party sites	Comet: CFAA; agent masking allegations. Buy for Me: encryption credential injection at external sites	Comet: enjoined from Amazon. Buy for Me: no restrictions

Agent Runtime Platform Comparisons

PLAYER A	PLAYER B	DIFFERENTIATOR	A RUNTIME PRICING	B RUNTIME PRICING
Amazon AgentCore	Google Vertex Agent Engine	Amazon: tightest vertical integration (model + runtime + marketplace). Google: 9× cheaper compute; broader protocol network (AP2/UCP with 60+ partners)	$0.0895/ vCPU-hr	$0.00994/ vCPU-hr
OpenAI Agents SDK	Google Vertex AI	OpenAI: largest developer ecosystem; Traces dashboard; provider-agnostic SDK. Google: cheaper compute; Project Mariner has published 83.5% WebVoyager benchmark (OpenAI Operator has no published benchmark)	Operator: $200/mo (Pro)	Mariner: $249.99/mo (AI Ultra)
Amazon Nova Pro	OpenAI GPT-5.2	Amazon: 10.5× cheaper per input token vs. GPT-5.2; tighter AWS ecosystem integration. OpenAI: frontier reasoning quality; largest API ecosystem	$0.80/M tokens in	$1.75/M tokens in

Payment Rail Comparisons

PLAYER A	PLAYER B	DIFFERENTIATOR	BEST USE CASE
Stripe ACP + SPT	Coinbase x402	Stripe: regulated fiat; chargebacks; Radar fraud; PCI DSS compliance. Coinbase: permissionless; sub-$10 micropayments; ~$0 gas on Base	Stripe wins for $10+ regulated transactions. Coinbase wins for sub-$10 micropayments
Skyfire KYAPay	Visa TAP	Skyfire: infrastructure-agnostic KYA identity + multi-rail wallets (cards+ACH+USDC). Visa: card-network-native; 10+ TAP partners; directly extends existing Visa token ecosystem	Skyfire: greenfield agent payment infrastructure. Visa TAP: extending existing Visa merchant base
ACP	AP2 / UCP	ACP: Stripe-centric; narrowed to 7 retailers post-rollback; Apache 2.0. AP2/UCP: 60+ partners, all 5 card networks; full lifecycle (discovery→fulfillment); crypto native via x402	ACP: existing Stripe merchants wanting agent checkout. AP2/UCP: retailers wanting broadest agent buyer coverage

Notable Competitive Events Timeline (2025–2026)

DATE	EVENT	PLAYERS	SIGNIFICANCE
2025-04-30	Visa Intelligent Commerce launches	Visa, Anthropic, Microsoft, OpenAI, Perplexity	First major card network agent identity framework; 100+ partners in first cohort
2025-05-07	Stripe Payments Foundation Model announced	Stripe, Nvidia	AI fraud detection trained on $1.4T+ transactions; GPU infrastructure partnership
2025-06-26	Skyfire launches open KYAPay protocol	Skyfire	First open specification for AI agent identity + payment; Apache-licensed
2025-08	ANAC 2025 (16th edition) at IJCAI, Montreal	NegMAS	Milestone for autonomous negotiation benchmarking; NegMAS officially named the engine
2025-09-25	OpenAI ACP Instant Checkout goes live	OpenAI, Stripe	First production agentic commerce checkout; 4% merchant fee; Etsy + Shopify
2025-10-14	**Visa TAP + Mastercard Agent Pay both launch on the same day**	Visa, Mastercard	Both major card networks launch agent payment frameworks simultaneously; direct competitive signal to Stripe/Coinbase
2025-11	Amazon sues Perplexity over Comet bot access	Amazon, Perplexity	First major legal challenge to autonomous shopping agents; CFAA theory
2025-11-13	Amazon Business unveils AI procurement tools	Amazon	AI Assistant + Savings Insights + Anomaly Monitoring for B2B buyers
2025-12-11	Omnea raises $50M Series B	Omnea	Signals continued investor conviction in procurement orchestration despite uncertain macro
2025-12-18	Skyfire × Visa live KYAPay purchase demo	Skyfire, Visa	First publicly demonstrated full-stack agentic commerce purchase: identity + payment + execution
2026-01-06	Amazon launches Alexa.com at CES	Amazon	Alexa+ extends to web; voice-native agentic commerce reaches web surface
2026-01-11	Google launches AP2 + UCP at NRF	Google	Sundar Pichai keynote; 60+ partners, all 5 card networks; positioned as industry-consensus alternative to ACP
2026-02-04	Alexa+ broadly available in US	Amazon	70+ LLMs; Expedia, Yelp, Angi, Square integrations; agentic household commerce
2026-02-05	OpenAI Frontier enterprise platform launches	OpenAI	Enterprise agent operations for HP, Oracle, State Farm, Uber; enterprise pivot post-ACP
2026-02-11	Coinbase Agentic Wallets GA	Coinbase	First wallet infrastructure purpose-built for AI agents; CDP Portal + gasless on Base

DATE	EVENT	PLAYERS	SIGNIFICANCE
2026-02-27	**OpenAI + Amazon $50B strategic partnership**	OpenAI, Amazon	OpenAI models in Amazon Bedrock; AWS as preferred OpenAI cloud; signals co-opetition at infrastructure layer
2026-03-04	**OpenAI removes ACP Instant Checkout from ChatGPT**	OpenAI, ChatGPT Operator	ACP narrowed to 7 retailers; merchant apps replace in-chat checkout; AP2/UCP gains momentum
2026-03-10	**Amazon wins preliminary injunction blocking Perplexity Comet from Amazon.com**	Amazon, Perplexity Comet	Landmark ruling: user consent ≠ platform authorization under CFAA; defines dual-authorization requirement for all browser agents
2026-03-18	Stripe + Tempo launch Machine Payments Protocol (MPP)	Stripe	Pre-authorized budgets + streamed micropayments; A2A payment pattern
2026-04-01	Perplexity appeals Amazon injunction (9th Circuit)	Perplexity	Enforcement stayed pending appeal; CFAA-as-competition-suppressor argument
2026-04-02	Coinbase + Stripe + Cloudflare co-found x402 Foundation	Coinbase, Stripe	Formalizes complementary positioning; micropayment layer becomes community-governed open standard

White Space and Gaps Identified

Gap 1 — Brazil-Specific Procurement Infrastructure

No global procurement platform (Procure AI, Omnea, Zycus, Fairmarkit, Agentforce) has documented Pix payment integration, NF-e fiscal validation, or LGPD compliance as a native capability. Brazil represents a $234B+ B2B e-commerce market ($18.42% CAGR) with ERP dominance by TOTVS (not SAP) — creating a structural opportunity for Brazil-native players (Freedom, Zinit, Linkana) that global players have not yet addressed. See Brazil vs Global Analysis for detailed analysis.

Gap 2 — Formal Negotiation in Production

NegMAS's SAOMechanism implements theoretically optimal bilateral negotiation protocols (Nash bargaining, Pareto-frontier discovery), but no production enterprise procurement platform uses formal mechanism-design negotiation. Zycus Merlin ANA and Pactum AI both use LLM-driven negotiation rather than formal protocols. The academic precision of NegMAS does not yet translate into production deployment — leaving the "best possible negotiation outcome" question unanswered at enterprise scale.

Gap 3 — Crypto Payment Rails in Enterprise Procurement

No enterprise procurement platform (Procure AI, Omnea, Zycus, Fairmarkit, Agentforce) has integrated x402 or Coinbase Agentic Wallets. The micropayment use case (API-per-call billing, sub-$1 data access) is not yet addressed in enterprise source-to-pay tooling. Skyfire's multi-rail wallet is the only bridge between enterprise procurement identity (KYA) and crypto payment rails.

Gap 4 — Open Agent Identity Standard

Despite Skyfire's KYAPay, Visa's TAP, and Mastercard's Verifiable Intent all addressing agent identity, there is no single universally adopted "Know Your Agent" standard. The market is fragmented across three competing identity frameworks with different cryptographic architectures. This creates compliance uncertainty for enterprises deploying buyer agents across multiple platforms.

Gap 5 — AI-vs-AI Procurement Scenarios

Fairmarkit's 2025 AI in Procurement Index found 94% of procurement leaders report that their suppliers now use AI in negotiations. No public benchmark (including BuyerBench) yet measures buyer agent performance in adversarial AI-vs-AI negotiation scenarios — where the seller's AI actively attempts anchoring, framing, and manipulation tactics against the buyer's AI.

Gap 6 — Behavioral Bias Testing in Production

ACES (arXiv:2508.02630) documented 5× position bias and +1.0–+1.9 endorsement effects in AI buyer agents, but no commercial procurement platform publicly measures or reports bias susceptibility as a product metric. BuyerBench Pillar 2 would be the first standardized benchmark to generate comparable bias-susceptibility data across agents.

Gap 7 — Audit Trail Portability

Salesforce Agentforce compiles agent decisions into JSON audit artifacts. Zycus Merlin ANA logs full negotiation transcripts. But these audit trails are platform-proprietary — there is no common audit trail format that can be compared across platforms, verified by third parties, or used as evidence in a procurement dispute across vendor relationships. A standardized agent decision audit format remains an unaddressed white space.

Gap 8 — Agent Authorization for Government Procurement

Brazil's Compras.gov.br/PNCP and public sector procurement globally require specific authorization chains (NF-e validation, contract registration) that no current AI agent payment protocol (ACP, AP2/UCP, x402) addresses. Government procurement remains outside the coverage scope of all Layer 3 payment infrastructure players.

Wiki-Links to All Profiled Entities

Companies

- Procure AI · Omnea · Zycus · Fairmarkit · Skyfire
- Amazon Agentic Commerce · OpenAI Agent Platform · Google Agentic Commerce
- Stripe Agent Payments · Coinbase Agent Payments

Products

- Amazon Rufus BuyForMe · ChatGPT Operator · Perplexity Comet
- Salesforce Agentforce · NegMAS

Protocols and Standards

- ACP · AP2 UCP · x402
- Visa Intelligent Commerce · Mastercard Agent Pay

Security and Compliance Frameworks

- PCI DSS v4 · EMV 3DS2 · NIST AI RMF
- ISO 42001 · FATF AML CFT

Research Papers

- ACES AI Agent Buying · LLM Agent Benchmarking Survey
- AgentBench · WebArena · WebShop

Key People

- Amine Allouah · Omar Besbes · Yasser Mohammad
- Kevin Frechette · Amir Sarhangi · Rubail Birwadker · Jorn Lambert

Brazil Market (Sub-Vault)

- INDEX · Brazil vs Global Analysis · Brazil Compliance Overview
- Brazil AI Procurement Landscape · Brazil Fintech Payment Landscape
- Brazil ERP Landscape · Global Players Brazil Presence

Cross-Reference Navigation

IF YOU WANT TO UNDERSTAND...	START HERE
Global pricing across all entities	Pricing Registry
Brazil vs. global market comparison	Brazil vs Global Analysis
BuyerBench scenario design recommendations	BuyerBench Scenario Recommendations
Full entity list and vault statistics	INDEX
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Synthesis document created: 2026-04-06 | *Phase 06 Task 2 — CladiBuyer Benchmarker*

PART III

Global Pricing Registry

Consolidated pricing across all 37 vault entities — model APIs, agent runtimes, consumer subscriptions, enterprise SaaS, and payment infrastructure.

Global Pricing Registry — AI Buyer Agent Ecosystem

> *A consolidated reference of publicly available pricing across all 37 vault entities — global and Brazil. Enterprise SaaS procurement platforms universally withhold public pricing; infrastructure and consumer platforms publish granular rates. This asymmetry is itself a market observation.*

Navigation

- Global Pricing Table
- Brazil Pricing Table
- Pricing Observations
- Entity Quick-Reference Index

Global Pricing Table

AI Model APIs — Per-Token Inference Costs

ENTITY	MODEL / TIER	INPUT ($/M TOKENS)	OUTPUT ($/M TOKENS)	CACHED INPUT	SOURCE DATE
OpenAI Agent Platform	gpt-5.2 (flagship)	$1.75	$14.00	$0.175	2026-04
OpenAI Agent Platform	gpt-5.1	$1.25	$10.00	$0.125	2026-04
OpenAI Agent Platform	gpt-5-mini	$0.25	$2.00	$0.025	2026-04
OpenAI Agent Platform	gpt-5-nano (lightest)	$0.05	$0.40	$0.005	2026-04
OpenAI Agent Platform	GPT-4.1	$2.00	$8.00	—	2026-04
OpenAI Agent Platform	GPT-4o (post-Dec 2025 cut)	$2.50	$10.00	—	2026-04
OpenAI Agent Platform	o3 (reasoning)	$2.00	$8.00	—	2026-04
Google Agentic Commerce	Gemini 2.5 Pro (≤200K ctx)	$1.25	$10.00	—	2026-04
Google Agentic Commerce	Gemini 2.5 Pro (>200K ctx)	$2.50	$10.00	—	2026-04
Google Agentic Commerce	Gemini 2.5 Flash	$0.15	$0.60	—	2026-04
Google Agentic Commerce	Gemini 2.0 Flash	$0.10	$0.40	—	2026-04
Google Agentic Commerce	Gemini 2.0 Flash-Lite	$0.075	$0.30	—	2026-04
Google Agentic Commerce	Gemini 1.5 Pro (≤200K ctx)	$1.25	$5.00	—	2026-04
Amazon Agentic Commerce	Nova Micro (text)	$0.035	$0.14	—	2026-04
Amazon Agentic Commerce	Nova Lite (multimodal)	$0.060	$0.24	—	2026-04
Amazon Agentic Commerce	Nova Pro (complex reasoning)	$0.80	$3.20	—	2026-04
Amazon Agentic Commerce	Nova Premier (highest capability)	Contact AWS	Contact AWS	—	2026-04

ENTITY	MODEL / TIER	INPUT ($/M TOKENS)	OUTPUT ($/M TOKENS)	CACHED INPUT	SOURCE DATE
Amazon Agentic Commerce	Claude 3.5 Haiku (via Bedrock)	~$0.80	~$4.00	—	2026-04
Amazon Agentic Commerce	Claude 3.5 Sonnet (via Bedrock)	~$3.00	~$15.00	—	2026-04

> ***Agent cost multiplier warning:*** *Multi-step agentic workflows consume 5–10× the raw token volume of a single-turn completion, due to reasoning traces, tool calls, and synthesis steps. Budget accordingly when evaluating API costs for BuyerBench runs.*

AI Model APIs — Batch & Caching Discounts

ENTITY	DISCOUNT TYPE	RATE	NOTES
OpenAI Agent Platform	Batch API	50% off input + output	All models; for offline evaluation pipelines
OpenAI Agent Platform	Priority API surcharge	2× standard	Guaranteed low-latency SLA
Google Agentic Commerce	Context caching (cached reads)	$0.25/M tokens	Fixed prefix caching on Gemini models
Google Agentic Commerce	Free tier (AI Studio)	0	Up to 15 req/min, 1M tokens/min, 1,500 req/day on Gemini 2.0 Flash
Amazon Agentic Commerce	Provisioned Throughput	Negotiated	Reserved capacity; eliminates per-token cost for predictable workloads

Agent Runtime / Orchestration Infrastructure

ENTITY	SERVICE	FEE	UNIT	NOTES
Amazon Agentic Commerce	AgentCore Runtime	$0.0895	/vCPU-hour	Active use only; idle = free
Amazon Agentic Commerce	AgentCore Gateway	Per call	/1,000 API calls	—
Amazon Agentic Commerce	Knowledge Bases (OpenSearch)	$345+	/month (min)	Fixed storage cost regardless of usage
Amazon Agentic Commerce	Bedrock Flows	$0.035	/1,000 node transitions	Billed from Feb 2025
Amazon Agentic Commerce	Amazon Business AI tools	$0	Bundled	No additional charge with Amazon Business
Amazon Agentic Commerce	New AWS customer credits	$200	One-time	Free-tier credit for new accounts
Google Agentic Commerce	Vertex AI Agent Engine vCPU	$0.00994	/vCPU-hour	Active runtime only; 9× cheaper than AgentCore
Google Agentic Commerce	Vertex AI Agent Engine memory	$0.0105	/GiB-hour	Active runtime only
Google Agentic Commerce	Vertex AI Sessions	Billed per session-second	—	Paid since Feb 11, 2026
Google Agentic Commerce	Vertex AI Data Store queries	$2.00	/1,000 queries	Enterprise search
Google Agentic Commerce	Grounding with Google Search	$35.00	/1,000 queries	For RAG-based agents
Google Agentic Commerce	Gemini API free tier	$0	—	Google AI Studio only; not on Vertex AI

API Tool Usage Costs (OpenAI Responses API)

ENTITY	TOOL	COST	NOTES	
OpenAI Agent Platform	Web Search (gpt-4o)	$30	/1,000 queries	—
OpenAI Agent Platform	Web Search (gpt-4o-mini / 4.1-mini)	Fixed 8,000 input tokens	/call	Billed as token consumption
OpenAI Agent Platform	Code Interpreter	$0.03	/container session	—
OpenAI Agent Platform	File Search storage	$0.10	/GB/day	First GB free
OpenAI Agent Platform	File Search queries	$2.50	/1,000 calls	—
OpenAI Agent Platform	Remote MCP	Output tokens only	/call	No tool call fee
ChatGPT Operator	Computer Use tool (CUA)	$3.00/$12.00	/1M input/ output	Research preview; tier 3–5 only

Consumer Subscription Tiers — Global Agents

ENTITY	PLAN	PRICE (USD)	AGENT / AGENTIC ACCESS	NOTES
ChatGPT Operator / OpenAI Agent Platform	Free	$0/month	None	Basic ChatGPT only
ChatGPT Operator / OpenAI Agent Platform	Plus	$20/month	Operator (waitlist post Feb 2026)	Limited usage
ChatGPT Operator / OpenAI Agent Platform	Pro	$200/month	Operator + full agent; immediate access	Same price as Google AI Ultra
ChatGPT Operator / OpenAI Agent Platform	Team	$30/user/month	Agent mode	Team/collaborative workspace
ChatGPT Operator / OpenAI Agent Platform	Enterprise	Custom (≥$30/ user/month)	Operator + Frontier; custom rate limits	—
OpenAI Agent Platform	Frontier (enterprise agent platform)	Custom only	Enterprise autonomous agent management	HP, Oracle, State Farm, Uber
Google Agentic Commerce	Google One (AI Basic)	$19.99/month	Standard Gemini	No Project Mariner
Google Agentic Commerce	Google AI Pro	$19.99/month	Gemini 2.5 Pro access	No Project Mariner
Google Agentic Commerce	Google AI Ultra	$249.99/month	Project Mariner + full Gemini suite + 1TB storage	Highest-capability consumer tier
Google Agentic Commerce	Gemini Enterprise	$30/user/month	All Google AI tools; retail CX agents	Business subscription
Perplexity Comet	Free	$0/month	Basic Comet browser	Limited agentic features
Perplexity Comet	Pro	$20/month	Full Comet + Comet Plus + Buy with Pro (PayPal)	—
Perplexity Comet	Max	$200/month	First Comet access (Jul 2025 preview) + all Pro	Same price as ChatGPT Pro
Perplexity Comet	Comet Plus (standalone)	$5/month	Advanced agent features	Available without full Pro
Amazon Rufus BuyForMe	Rufus	$0	AI product discovery + auto-buy	Bundled in Amazon app

ENTITY	PLAN	PRICE (USD)	AGENT / AGENTIC ACCESS	NOTES
Amazon Rufus BuyForMe	Buy for Me	$0 (beta)	Cross-site autonomous checkout	No separate subscription
Amazon Rufus BuyForMe	Alexa+	Amazon Prime / Echo device pricing	Multi-domain voice commerce	No per-transaction fee

Enterprise Procurement SaaS — Global

ENTITY	MODEL	PRICE	UNIT	NOTES	SOURCE DATE
Salesforce Agentforce	Conversations	$2.00	/conversation (24h)	Being phased out in favor of Flex Credits	2026-04
Salesforce Agentforce	Flex Credits	$0.10	/agent action	$500 per 100,000 credits; 1 action = 20 credits	2026-04
Salesforce Agentforce	AELA (Agentic Enterprise License)	$125+	/user/month	Bundled digital workforce; enterprise-wide	2026-04
Procure AI	Enterprise SaaS	Not disclosed	Custom	European enterprise; €50B+ managed spend; ROI in months	2026-04
Omnea	Enterprise SaaS	Not disclosed	Custom	London; Spotify, MongoDB, Monzo as customers	2026-04
Zycus	S2P Suite (Merlin AI)	Not disclosed	Custom	Self-funded/PE; ~$100–200M ARR est.	2026-04
Fairmarkit	Autonomous sourcing	Not disclosed	Custom	Series C; $78M raised; enterprise contracts	2026-04
Skyfire	Agent payment infrastructure	Not disclosed	Custom	Enterprise tier launched Mar 2025; startup pricing	2026-04
NegMAS	Open source	$0	Free (MIT)	pip install negmas; no commercial license required	2026-04

Payment Infrastructure — Global (Per-Transaction Rates)

ENTITY	TRANSACTION TYPE	FEE	CURRENCY	NOTES
Stripe Agent Payments	US card (standard)	2.9% + $0.30	USD	Per successful charge
Stripe Agent Payments	UK card	1.5% + £0.20	GBP	Per successful charge
Stripe Agent Payments	EU card	1.5% + €0.25	EUR	Per successful charge
Stripe Agent Payments	International card	+1.5%	USD	Cross-border surcharge on top of standard rate
Stripe Agent Payments	Currency conversion	+1.0%	—	Additional surcharge
Stripe Agent Payments	ACH Direct Debit	0.8%, cap $5.00	USD	—
Stripe Agent Payments	USDC/crypto (x402)	Gas fees only (~$0)	USD	Via Base network integration
Stripe Agent Payments	Chargeback Protection	+$0.04	USD	/transaction; Stripe absorbs dispute cost
Stripe Agent Payments	Radar for Fraud Teams	+$0.02	USD	/screened transaction; basic Radar included in std. fee
Stripe Agent Payments	Connect Express	$2.00/month + 0.25%+$0.25/ payout	USD	/active connected account
Stripe Agent Payments	Dispute fee	$15.00	USD	Refunded if merchant wins
Stripe Agent Payments	Checkout Optimization revenue lift	+11.9% avg	—	Statistical uplift, not a fee
Coinbase Agent Payments	x402 facilitator fee (USDC on Base)	$0	USD	Coinbase subsidizes; Base L2 gas only
Coinbase Agent Payments	Base L2 gas (USDC transfer)	~$0.001–$0.01	USD	Per transaction
Coinbase Agent Payments	CDP wallet operations (create/sign/broadcast/ policy)	$0.005	USD	/operation
Coinbase Agent Payments	CDP free tier	0	—	First 5,000 operations/month

ENTITY	TRANSACTION TYPE	FEE	CURRENCY	NOTES
Coinbase Agent Payments	CDP Prime trading (maker)	0.00%–0.20%	—	Volume tiers
Coinbase Agent Payments	CDP Prime trading (taker)	0.05%–0.60%	—	Volume tiers
ChatGPT Operator	ACP merchant fee (historical, Sep 2025–Mar 2026)	4%	USD	On completed Instant Checkout transactions; discontinued Mar 2026

Payment Rail Economics: x402 vs. Stripe Break-Even Analysis

TRANSACTION SIZE	STRIPE FEE (USD)	X402/BASE FEE (USD)	OPTIMAL RAIL
$0.01 (per-token API call)	$0.31 (3,100%)	~$0.00001	Coinbase Agent Payments x402
$0.10	$0.33 (330%)	~$0.00001	Coinbase Agent Payments x402
$1.00	$0.32 (32%)	~$0.001	Coinbase Agent Payments x402
$10.00	$0.59 (5.9%)	~$0.001	Comparable
$100.00	$3.20 (3.2%)	~$0.005	Stripe Agent Payments (chargeback protection value)
$1,000+	Negotiated	~$0.01	Stripe Agent Payments (enterprise compliance)

Rule of thumb: x402 is economically dominant for sub-$10 agent micropayments; Stripe/ACP is preferred for $10+ transactions requiring compliance, chargeback protection, or enterprise purchasing controls.

Brazil Pricing Table

Brazilian Fintech / Payment Platforms (BRL-Denominated)

ENTITY	SERVICE	FEE	CURRENCY	NOTES	SOURCE DATE
Nubank Nu Empresas	Conta PJ monthly fee	R$ 0	BRL	Free digital business account	2026-04
Nubank Nu Empresas	Pix (send/receive)	R$ 0	BRL	Unlimited free	2026-04
Nubank Nu Empresas	TED / DOC transfers	R$ 0	BRL	Free	2026-04
Nubank Nu Empresas	Boleto issuing (per paid boleto)	R$ 3,00	BRL	Per paid boleto	2026-04
Nubank Nu Empresas	ATM withdrawal	R$ 6,50	BRL	Banco24Horas / Saque e Pague network	2026-04
Nubank Nu Empresas	Card acquiring — débito	0,89%	BRL	Tap to Pay	2026-04
Nubank Nu Empresas	Card acquiring — crédito à vista	3,15%	BRL	Single installment	2026-04
Nubank Nu Empresas	Card acquiring — crédito 2x parcelado	5,39%	BRL	2-installment credit	2026-04
Nubank Nu Empresas	Card acquiring — crédito 12x parcelado	12,40%	BRL	12-installment credit	2026-04
Stone StoneCo	Conta Stone PJ monthly fee	R$ 0	BRL	Free digital business account	2026-04
Stone StoneCo	Pix (send/receive)	R$ 0	BRL	—	2026-04
Stone StoneCo	Boleto issuing	R$ 1–4	BRL	Varies; typically R$1–4 per paid boleto	2026-04
Stone StoneCo	Card acquiring — débito	~0,99–1,49%	BRL	Varies by volume; not publicly posted	2026-04
Stone StoneCo	Card acquiring — crédito à vista	~2,69–3,49%	BRL	Varies by volume	2026-04
Stone StoneCo	Card acquiring — crédito parcelado (2–12x)	~3,99–5,99%	BRL	Varies by installments	2026-04
ASAAS	Conta ASAAS PJ monthly fee	R$ 0	BRL	Free SMB account; SCD+PI licensed	2026-04
ASAAS	Pix receive	R$ 0	BRL	—	2026-04

ENTITY	SERVICE	FEE	CURRENCY	NOTES	SOURCE DATE
ASAAS	Boleto receive (per paid boleto)	~R$ 1,99	BRL	Typical rate; subject to change	2026-04
ASAAS	Credit card receive (à vista)	~2,99%	BRL	—	2026-04
ASAAS	Credit card receive (parcelado)	~4,99–9,99%	BRL	2–12 installments	2026-04
ASAAS	Receivables anticipation	~1,5–2,5%/month	BRL	FIDC credit facility; varies by profile	2026-04
ASAAS	API access	R$ 0	BRL	Free with account; ~80+ REST endpoints	2026-04
Celcoin	BaaS / Pix transactions	~R$ 0,05–0,50	BRL	Per Pix transaction; negotiated B2B; no public rack rate	2026-04
Celcoin	Open Finance / ITP / BaaS platform	Custom / Negotiated	BRL	Monthly minimums apply for regulated license access	2026-04
Belvo	Data API calls (aggregation)	~US$ 0,01–0,05	USD	Per API call; industry benchmark, not confirmed rate	2026-04
Belvo	Payment initiation (Pix via Open Finance)	~US$ 0,10–0,50	USD	Per Pix initiated; enterprise contracts for volume	2026-04
Belvo	Free developer tier	$0	USD	Limited calls/month for testing	2026-04

Brazilian Enterprise Software (BRL-Denominated Estimates)

ENTITY	TIER	TYPICAL MONTHLY EST.	CURRENCY	NOTES	SOURCE DATE
TOTVS ERP Procurement	TOTVS Start (SMB, ≤20 users, 1 entity)	R$ 800–2.000	BRL	Reseller estimates; TOTVS does not publish prices	2026-04
TOTVS ERP Procurement	TOTVS Midsizê (20–100 users, multi-entity)	R$ 3.000–12.000	BRL	Reseller estimates	2026-04
TOTVS ERP Procurement	TOTVS Enterprise (100+ users, full P2P + Fluig)	R$ 15.000–60.000+	BRL	Reseller estimates; implementation adds 1–3× annual license	2026-04
TOTVS ERP Procurement	Fluig Voyager 2.0 add-on (GenAI layer)	R$ 80–300	BRL	/user/month; per-process or per-user licensing	2026-04
Pipefy	Starter	Free	USD	Limited workflows; no AI Agents	2026-04
Pipefy	Business	~$20	USD	/user/month; core workflow automation	2026-04
Pipefy	Enterprise	Custom	USD	AI Agents, advanced integrations, SLA	2026-04
Freedom	Custom AI agents (enterprise)	Not disclosed	BRL	Project-based or per-seat; Seed-stage startup	2026-04

Brazilian B2B Marketplaces

ENTITY	ACCESS MODEL	BUYER FEE	SELLER FEE	CURRENCY	NOTES	SOURCE DATE
Mercado Livre Negocios	CNPJ-gated registration	Free	10–16% commission (category-dependent)	BRL	B2B wholesale; up to 50% discount vs. consumer	2026-04
Compras gov br	Government registration required	Free for suppliers	Not disclosed	BRL	Federal procurement portal; mandatory for gov. contracts	2026-04
B2Brazil	B2B marketplace	Free basic / premium tiers	Not disclosed	BRL/USD	International B2B trade platform	2026-04

Pricing Observations

1. Enterprise Procurement SaaS: Systematic Opacity

The **six core enterprise procurement vendors** — Procure AI, Omnea, Zycus, Fairmarkit, Salesforce Agentforce (Frontier tier), and Skyfire — do not publish pricing. This is a deliberate enterprise SaaS convention: complex multi-module deployments with implementation costs, custom integrations, and negotiated volume discounts resist standardized pricing. The implication for BuyerBench:

- **Benchmark cost** is not a differentiator among enterprise SaaS procurement platforms — capability and economic outcome quality are.
- **Salesforce Agentforce** is the exception: three concurrent pricing models ($2/conversation, $0.10/action, $125+/user/month AELA) are published, creating the most granular enterprise procurement agent cost model in the vault.
- **Zycus** is bootstrapped/PE-backed (no VC rounds), meaning its pricing is under more profit-discipline pressure than VC-funded competitors — yet still not public.

2. Open-Source Free Tier Pattern

NegMAS is the only vault entity priced at zero with full capability access (MIT license, pip-installable, no API keys). This makes it uniquely valuable for BuyerBench development and scenario validation:

- **No API cost for large-scale testing**: 1,000 NegMAS negotiation runs cost $0 in API fees (only compute).
- **Algorithmic floor**: NegMAS provides the theoretically optimal bilateral negotiation baseline against which commercial agents can be measured.

- Open-source procurement tools (Compras.gov.br, B2Brazil free tier) follow the same zero-marginal-cost model for initial access.

3. Per-Transaction Models in Payment Infrastructure

Payment infrastructure shows a clear per-transaction pricing structure across all providers:

RAIL	COST MODEL	BEST FOR
Stripe card (standard)	2.9% + $0.30 flat	$10+ transactions needing compliance
Stripe ACH	0.8%, capped $5.00	High-value bank transfers
x402/Base USDC	~$0.001–$0.01 (gas only)	Sub-$10 micropayments, A2A
Brazil Pix	R$0 (consumer/SMB)	All transaction sizes within Brazil
Brazil boleto	R$1.99–3.00/paid boleto	Invoiced B2B payments
Brazil card acquiring (débito)	0.89–1.49%	Point-of-sale debit
Brazil card acquiring (crédito parcelado)	3.15–12.40%	Installment credit — rates rise with installment count

Key asymmetry: Brazil's Pix is free at zero cost for any amount, while US/EU card rails charge 2.9%+$0.30 minimum. For a $1 transaction, Pix costs $0 while Stripe costs $0.33 — a 33% fee ratio. This creates a structural cost advantage for Brazilian B2B procurement scenarios.

4. Consumer Agent Tier Convergence

Three consumer agent products independently converged on the same $200/month "full agentic access" price point:

PRODUCT	TIER	PRICE	WHAT'S INCLUDED
ChatGPT Operator Pro	ChatGPT Pro	$200/month	Operator (immediate), higher rate limits
Perplexity Comet Max	Perplexity Max	$200/month	Full Comet + Buy with Pro (PayPal)
Google Agentic Commerce AI Ultra	Google AI Ultra	$249.99/month	Project Mariner + full Gemini + 1TB

This price alignment ($200–$250/month) appears to represent the emerging market consensus for **"full agentic consumer capability"** — the level at which browser agents, autonomous purchase execution, and advanced reasoning are bundled. BuyerBench agent cost modeling should use $200/month as the baseline consumer-tier agentic spending threshold.

5. Model API Cost Compression Trend

Between 2024 and 2026, API pricing across all three major providers fell by 40–80% for equivalent capability:

- **OpenAI** cut GPT-4o pricing 50% in December 2025 (from $5.00/$15.00 to $2.50/$10.00 per M tokens), simultaneous with GPT-5 launch — classic penetration pricing to prevent migration.
- **Google** launched Gemini 2.5 Flash at $0.15/$0.60 (input/output per M tokens), vs. Gemini 1.5 Pro at $1.25/$5.00 — 8× cheaper for equivalent tasks in Google's own benchmark.
- **Amazon** launched Nova Micro at $0.035/$0.14 — among the lowest published rates for a frontier-grade model, available at a discount in Bedrock on-demand pricing.

BuyerBench implication: Cost-per-evaluation is falling rapidly. BuyerBench runs that cost $10/agent in 2024 may cost $1–3 in 2026. Build evaluation pipelines to exploit Batch API discounts (50% off for all OpenAI models) and Gemini context caching.

6. Brazil Pricing Competitiveness vs. Global

Comparing Brazil-native infrastructure to global equivalents:

DIMENSION	GLOBAL (STRIPE/AWS)	BRAZIL (PIX/ASAAS)	ADVANTAGE
Payment rail cost	2.9%+$0.30 (Stripe)	R$0 (Pix)	Brazil: ~33× cheaper on $1 txn
ERP monthly cost	SAP Ariba: $100K+/year enterprise	TOTVS Start: R$800–2,000/ month	Brazil: 5–10× cheaper entry
SMB payment API	Stripe: 2.9%+$0.30	ASAAS: R$0 API, R$1.99/ boleto	Brazil: cheaper for B2B invoicing
Agent runtime compute	AWS AgentCore: $0.0895/ vCPU-hr	Google Vertex AI: $0.00994/ vCPU-hr	Google: 9× cheaper
Open Finance consent	PCI DSS / Stripe only	Free via BCB Open Finance framework	Brazil: regulated free access

Brazilian procurement infrastructure is **cost-competitive** — in some dimensions dramatically cheaper — than global equivalents. This supports a business case for AI buyer agent deployment in Brazil even at lower average order values.

Entity Quick-Reference Index

Global — Companies

- Procure AI — Enterprise procurement SaaS; pricing not disclosed
- Omnea — AI SRM platform; pricing not disclosed

- Zycus — S2P suite with Merlin ANA; pricing not disclosed
- Fairmarkit — Autonomous sourcing; pricing not disclosed
- Skyfire — Agent payment infrastructure; pricing not disclosed
- Amazon Agentic Commerce — Nova models, Bedrock, AgentCore (rates above)
- OpenAI Agent Platform — GPT-5 family, Agents SDK, Frontier (rates above)
- Google Agentic Commerce — Gemini 2.5, Project Mariner, Vertex AI (rates above)
- Stripe Agent Payments — ACP, SPT, Radar (rates above)
- Coinbase Agent Payments — x402, CDP wallets, Base USDC (rates above)

Global — Products

- Amazon Rufus BuyForMe — Free (Rufus, Buy for Me, Alexa+)
- ChatGPT Operator — $0–$200/month (Free/Plus/Pro); ACP historical 4% merchant fee
- Perplexity Comet — $0–$200/month (Free/Pro/Max); $5/month Comet Plus
- Salesforce Agentforce — $2/conversation or $0.10/action or $125+/user/month
- NegMAS — Free (MIT open source)

Brazil — Companies

- Nubank Nu Empresas — Free account; R$3/boleto; 0.89–12.40% card acquiring
- Stone StoneCo — Free account; ~0.99–5.99% card; R$1–4/boleto
- Celcoin — B2B negotiated; ~R$0.05–0.50/Pix (estimated)
- ASAAS — Free account + API; R$1.99/boleto; ~2.99–9.99% card
- Belvo — Custom; ~US$0.01–0.50 per call/payment
- Freedom — Not disclosed (enterprise custom)

Brazil — Products

- TOTVS ERP Procurement — R$800–60,000+/month depending on tier; +R$80–300/user for Fluig Voyager 2.0
- Mercado Livre Negocios — Buyer: free; Seller: 10–16% commission
- Pipefy — Free starter; ~$20/user/month Business; Enterprise custom
- Compras gov br — Free for registered suppliers
- B2Brazil — Free basic / premium tiers

Last updated: 2026-04-06 — Vault synthesis document created during Phase 06, Loop 00001

PART IV

Company Profiles

Detailed profiles of 10 key companies across procurement platforms, AI agent runtimes, and payment infrastructure.

Procure AI

> *The world's first AI-native Procurement Automation Platform — deploying 50+ autonomous agents across the full source-to-pay lifecycle.*

Overview

Procure AI (procure.ai) is a European enterprise SaaS company building an AI-native procurement automation platform. Founded in 2020 by Konstantin von Büren and Yves Bauer, the company is headquartered across London, Paris, and Frankfurt with over 40 employees. Its core thesis is that procurement — traditionally fragmented across ERPs, eProcurement suites, and manual workflows — can be fully automated through multi-tier AI agents that operate autonomously, collaboratively, and pro-actively.

The platform deploys over 50 AI agents organized into three operational tiers: **autonomous agents** that execute procurement tasks independently, **collaborative agents** that augment and inform human decision-making, and **ambient agents** that provide proactive, unsolicited team support. This architecture spans the complete source-to-pay lifecycle: spend analytics, intake management, sourcing and negotiations, supplier management, and purchasing operations.

Procure AI's differentiation lies in positioning as a sovereign, data-unifying layer rather than a system replacement — integrating with existing ERPs and eProcurement tools while enriching their fragmented data. Enterprise clients including Kärcher, DMG MORI, EnBW, Dr. Oetker, Naturenergie, and GMH Gruppe collectively represent €50B+ in annual managed spend. In November 2025 the company raised a $13M seed round led by Headline to scale into the UK, Nordics, Benelux, and France.

> ***BuyerBench relevance:*** *Procure AI is the closest existing commercial analog to the BuyerBench evaluation target. Their three-tier agent model directly maps to the three BuyerBench pillars: autonomous execution (Pillar 1 capability), collaborative decision support (Pillar 2 economic quality — where does the agent yield to humans?), and ambient proactive agents (Pillar 3 compliance — when should an agent act without explicit instruction?). Their published performance metrics (47% faster award decisions, 4.9% average savings) provide concrete baseline numbers for Pillar 1 scoring benchmarks.*

Quick Facts

ATTRIBUTE	VALUE
Founded	2020
Headquarters	London, UK (also Paris and Frankfurt)
Employees	40+
Funding	$13M seed (Nov 2025)
Stage	Seed
Revenue Growth	4x (reported at time of seed round)
Managed Spend	€50B+ across customer base
Website	https://www.procure.ai

Products & Services

Five core platform modules:

- **Unified Spend Analytics**: AI-powered spend data analysis with predictive analytics and autonomous opportunity identification across fragmented data sources.
- **Generative Intake Management**: Dialog-based guided buying experiences that reduce maverick spend. Enables 60% of intake requests to be handled fully autonomously via the Quote-to-Order Intake agent.
- **Autonomous Sourcing & Negotiations**: Tactical tender execution and automated supplier identification. The flagship Autonomous Spot-Buy and Tactical Sourcing capability delivers 35–46% time reductions and 3.7–5.2% savings per event.
- **Intelligent Supplier Management**: Automated supplier onboarding, continuous performance monitoring, and risk control workflows.
- **Seamless Purchasing Operations**: Centralized procure-to-pay (P2P) process automation, order processing, and invoice management.

Leadership

- **Co-CEO & Co-Founder**: **Konstantin von Büren** — Previously at McKinsey & Company and Via Transportation; studied at MIT, London Business School, and WHU – Otto Beisheim School of Management; prior experience at C4 Ventures (one of Procure AI's seed investors).
- **Co-CEO & Co-Founder**: **Yves Bauer** — Background in automation and operational optimization.

Funding History

DATE	ROUND	AMOUNT	LEAD INVESTOR	CO-INVESTORS
2025-11	Seed	$13M (€11M)	Headline VC	C4 Ventures, Futury Capital, procurement industry angels

Agent Architecture (Technical Detail)

The 50+ AI agents are organized into three operational tiers that reflect different levels of human involvement:

TIER	DESCRIPTION	EXAMPLE AGENT	HUMAN ROLE
Autonomous	Executes tasks independently without human input	Spot-Buy agent, Quote-to-Order Intake	None during execution; reviews outputs
Collaborative	Augments and amplifies human decision-making	Sourcing recommendation engine	Active co-pilot; agent advises, human decides
Ambient	Proactive background monitoring and alerting	Supplier risk sentinel, spend anomaly detector	Passive; human acts only on surfaced alerts

This tiered design directly addresses the trust spectrum in enterprise procurement — not all buying decisions can or should be fully automated. The architecture implies a well-designed evaluation should test agents across all three operating modes, not just full autonomy.

Performance Benchmarks (Customer-Reported)

METRIC	VALUE
Average savings in tactical/tail spend negotiations	4.9%
Order processing time reduction	37%
Award decision speed improvement	47%
Intake requests handled fully autonomously	60%
Typical ROI timeline	Months (not years)
Average annual savings per enterprise client (€70M tail spend)	€2.35M

Recent Developments

- **2025-11**: Raised $13M seed round led by Headline; announced expansion into UK, Nordics, Benelux, and France.
- **2025-11**: Reported 4x revenue growth and €50B+ in managed spend across customer base at time of funding announcement.
- **2025**: Onboarded enterprise clients including Kärcher, DMG MORI, EnBW, Dr. Oetker, Naturenergie, GMH Gruppe.

Competitive Position

Procure AI occupies the "pure-play AI-native" tier of enterprise procurement software — a smaller, faster-moving segment than the established source-to-pay suites. Its competitive frame is explicitly "replace the incumbent eProcurement approach with AI from day one" rather than "add AI to an existing platform."

> ***BuyerBench relevance:*** *This positioning creates a natural benchmark question: do AI-native platforms like Procure AI make qualitatively better procurement decisions than AI-augmented incumbents (Zycus, SAP Ariba, Coupa), or do they simply execute faster? BuyerBench Pillar 2 (economic decision quality) is designed to answer exactly this.*

Key Competitors

- Omnea — Also AI-native, also targeting enterprise intake and vendor workflows; strong overlap in intake management and vendor onboarding, weaker in end-to-end sourcing.
- Zycus — Established S2P incumbent with AI layers added to mature platform; broader functionality but complex setup; recently launched agentic AI features.
- Fairmarkit — Focuses specifically on tail-spend and low-value sourcing automation; narrower scope but deep in its lane; listed as a direct Procure AI competitor by CB Insights.
- Coupa — Market leader in business spend management; Procure AI positions against Coupa's traditional non-AI-native architecture.
- SAP Ariba — Dominant in large enterprise; Procure AI targets mid-market and underserved enterprise segments that find Ariba's complexity prohibitive.

Differentiation Claims

1. **AI-native from inception** (not retro-fitted AI on legacy architecture)
2. **Sovereign data layer** — dedicated hosting, data does not leave client infrastructure
3. **Integration rather than replacement** — works with existing ERP/eProcurement investments
4. **ROI in months** — faster time-to-value than traditional S2P implementations

Security & Compliance

- Described as a "secure and sovereign procurement data platform" with dedicated hosting and end-to-end autonomous solutions
- Trust Center available at `trust.procure.ai` (specific certifications not publicly detailed)
- Built in Europe — implies GDPR compliance by design; likely relevant for public sector procurement customers
- Security certifications (ISO 27001, SOC 2, etc.) not confirmed from public sources

> ***BuyerBench relevance:*** *The "sovereign data" claim is directly relevant to Pillar 3 — agents handling procurement data must respect data residency constraints, enforce vendor approval boundaries, and maintain audit trails. Procure AI's architecture provides a real-world reference for what compliance looks like in production autonomous procurement.*

Related Entities

- Headline VC — Lead investor, Seed 2025
- C4 Ventures — Co-investor; Konstantin von Büren previously worked there
- Futury Capital — Co-investor
- Omnea — Competitor: AI-native enterprise procurement, overlapping intake/vendor workflows
- Zycus — Competitor: established S2P incumbent with agentic AI additions
- Fairmarkit — Competitor: tail-spend sourcing automation, CB Insights lists as direct peer
- Skyfire — Potential future partner: Procure AI autonomous agents will eventually need payment execution rails
- Konstantin von Büren — Co-CEO and Co-Founder
- Yves Bauer — Co-CEO and Co-Founder

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Last updated: 2026-04-04

Omnea

> *AI-native procurement orchestration platform making procurement every CFO's competitive advantage.*

Overview

Omnea is a London-based enterprise SaaS company building what it calls an "AI SRM" (Supplier Relationship Management) platform — a single front door for all procurement and vendor-related requests across an organization. Founded in 2022 by Ben Freeman, the company reunited key executives from Tessian (a cybersecurity company acquired by Proofpoint): Abhirukt Sapru (CCO) and Sabrina Castiglione (CFO) joined shortly after launch, bringing experience scaling a VC-backed SaaS company from growth stage to enterprise acquisition.

Omnea's product architecture is **orchestration-first**: rather than deploying autonomous execution agents (like Procure AI), Omnea creates a two-way orchestration layer spanning the entire finance tech stack. Employees submit requests in natural language; the platform routes those requests through the appropriate workflow — vendor onboarding, contract approval, RFP launch, risk review — involving the right stakeholders and systems automatically. The AI layer proactively surfaces insights: flagging expired vendor certifications for risk teams, launching RFPs for procurement, and generating CFO-facing reports from a continuous vendor data refresh.

The company has seen 5× revenue growth and more than 3× headcount growth in the year preceding its Series B, with enterprise customers including Spotify, Adecco Group, Entrust, Wise, MongoDB, and Monzo.

> ***BuyerBench relevance:*** *Omnea exemplifies the "orchestration-first" design philosophy for procurement AI — it recommends and routes rather than autonomously executing. This is a useful contrast to Procure AI's fully autonomous execution approach and directly informs BuyerBench Pillar 1 scenario design breadth: scenarios should test both execution-autonomous agents and recommendation-plus-approval agents. Omnea's CFO positioning also provides rich context for Pillar 2 economic decision quality metrics (cost savings visibility, cycle time, supplier consolidation recommendations).*

Quick Facts

ATTRIBUTE	VALUE
Founded	2022
Headquarters	London, UK
Employees	Growing (>3× headcount YoY as of Sep 2025)
Total Funding	>$75M
Stage	Series B (Sep 2025)
Website	omnea.co
Category	AI SRM / Procurement Orchestration

Products & Services

- **Procurement Intake**: Single natural-language front door for all employee procurement requests; routes to appropriate workflow automatically
- **Vendor Onboarding Automation**: AI-orchestrated supplier onboarding with compliance checks, certification tracking, and cross-functional approval workflows
- **AI SRM (Supplier Relationship Management)**: Continuous two-way vendor data refresh across the finance tech stack; proactive risk alerts, expired certification flags, and spend visibility
- **CFO Intelligence Layer**: Automated reporting and spend analytics surfaced directly to finance leadership — procurement positioned as a CFO competitive advantage
- **Cross-Functional Approval Orchestration**: Multi-stakeholder approval workflows spanning procurement, legal, finance, and IT for vendor engagements

Leadership

- **Ben Freeman** — CEO & Co-Founder: Founded Omnea in 2022; previously at an enterprise SaaS company before founding Omnea
- **Abhirukt Sapru** — CCO: Former Tessian executive; helped scale Tessian before Proofpoint acquisition
- **Sabrina Castiglione** — CFO: Former Tessian executive; brings SaaS growth-stage financial leadership

Funding History

DATE	ROUND	AMOUNT	LEAD INVESTORS	PARTICIPATING
Sep 2025	Series B	$50M	Insight Partners, Khosla Ventures	Accel, Point Nine, First Round Capital, Prosus
Prior rounds	Seed + Series A	~$25M+	Accel, First Round Capital, Point Nine	Prosus

Total raised: >$75M as of September 2025

Key Customers

Spotify, Adecco Group, Entrust, Wise, MongoDB, Monzo — spanning media/streaming, staffing, identity security, fintech, and database software verticals.

Recent Developments

- **2025-09-17**: Announced $50M Series B led by Insight Partners + Khosla Ventures; total funding exceeds $75M
- **2025**: Revenue grew 5× year-over-year; headcount more than tripled
- **2025**: Positioned product as "AI SRM" — framing shift from intake automation to full supplier relationship intelligence

Competitive Position

Omnea occupies a distinctive position in the AI procurement stack: **orchestration-first, CFO-facing, and vendor-data-centric**. Unlike pure-play autonomous agents (Procure AI), Omnea builds the connective tissue between humans, systems, and suppliers rather than replacing human decision-making entirely.

Key Differentiators

1. **Two-way orchestration layer** spanning the entire finance tech stack — more than just intake, covers the full vendor relationship lifecycle
2. **CFO positioning** — framed as a financial intelligence tool, not just an operational efficiency tool
3. **Ex-Tessian leadership** — team has scaled an enterprise SaaS product from growth to acquisition, which Insight/Khosla are betting on for enterprise go-to-market
4. **Enterprise customer base** — Spotify, MongoDB, Monzo validate credibility with high-profile, technically sophisticated buyers

Key Competitors

COMPETITOR	OVERLAP	DIFFERENTIATION
Procure AI	Enterprise procurement automation	Procure AI deploys 50+ autonomous execution agents; Omnea orchestrates workflows + humans
Zycus	Source-to-pay with agentic AI	Zycus is an incumbent S2P suite; Omnea is a modern overlay focused on intake/vendor data
Fairmarkit	Sourcing automation	Fairmarkit automates the sourcing event itself; Omnea covers the upstream intake and approval layers
Coupa	Spend management	Coupa is a mature spend management platform; Omnea positions as a lighter, AI-native layer
SAP Ariba	Enterprise procurement suite	SAP Ariba is a legacy ERP-integrated procurement system; Omnea integrates on top of it

BuyerBench Mapping

BUYERBENCH PILLAR	OMNEA RELEVANCE
Pillar 1 — Capability	Intake orchestration, vendor onboarding workflow, cross-functional approval sequencing — all direct Pillar 1 scenario analogs
Pillar 2 — Economic Decision Quality	CFO-facing cost savings visibility, cycle time reduction metrics, supplier consolidation recommendations — informs Pillar 2 economic scoring baselines
Pillar 3 — Security & Compliance	Vendor certification tracking, expired credential alerts, cross-functional authorization workflows — relevant to Pillar 3 authorization/compliance scenarios

Related Entities

- Procure AI — direct competitor; execution-first vs. orchestration-first architecture contrast
- Zycus — incumbent contrast; Omnea positions as a modern overlay vs. legacy suite
- Fairmarkit — peer in AI procurement automation; different workflow layer focus
- Insight Partners — lead Series B investor
- Khosla Ventures — co-lead Series B investor
- Accel — earlier-stage investor (seed/A)
- First Round Capital — earlier-stage investor
- Tessian / Proofpoint — professional background of CCO and CFO

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Last updated: 2026-04-04 (Loop 9 — stall-break research cycle)

Zycus

> *Global source-to-pay procurement suite deploying Merlin ANA — one of the first autonomous AI negotiation agents in production enterprise procurement.*

Overview

Zycus is a global enterprise procurement software company founded in 1998 and headquartered in Princeton, NJ (USA), with major development and operations centers in Mumbai, India, and offices across North America, Europe, and Asia-Pacific. The company offers an end-to-end source-to-pay (S2P) suite spanning eSourcing, contract management, procure-to-pay, supplier management, and spend analytics — all unified under the **Merlin AI** platform umbrella.

Zycus has approximately 2,500 employees and serves 250+ enterprise customers across manufacturing, retail, financial services, and healthcare. Unlike pure-play AI startups (Procure AI, Omnea), Zycus is an established incumbent that has embedded agentic AI capabilities into a mature suite that already handles billions of dollars in procurement spend.

The most strategically significant component of Merlin is the **Merlin Autonomous Negotiation Agent (ANA)** — a production-deployed system that conducts supplier price negotiations autonomously on behalf of enterprise buyers. ANA operates within buyer-defined parameters (target price, floor price, concession strategy, escalation triggers) and engages suppliers through AI-driven negotiation rounds without requiring constant human intervention at each exchange. This makes Zycus the first established procurement vendor to publicly deploy a *continuous autonomous negotiation agent* at enterprise scale.

> ***BuyerBench relevance:*** *Merlin ANA is a direct commercial reference implementation for BuyerBench Pillar 1 negotiation task scenarios (S1.3 and similar). Its production deployment provides empirical grounding for scenario calibration: what does "successfully conducting a price negotiation autonomously" look like in the real world? Zycus also provides the "incumbent suite" contrast to agent-native startups — useful for understanding scenario breadth.*

Quick Facts

ATTRIBUTE	VALUE
Founded	1998
Headquarters	Princeton, NJ, USA (R&D: Mumbai, India)
Employees	~2,500
Funding	Private (bootstrapped + PE-backed; no public VC rounds)
Stage	Private — mature/profitable
Annual Revenue	Not publicly disclosed; estimated $100–200M ARR range
Customers	250+ enterprise clients (Fortune 500 in manufacturing, retail, financial services)
Website	zycus.com

Products & Services

Merlin AI Platform

Zycus's AI umbrella brand encompassing all AI/ML capabilities across the S2P suite:

- **Merlin Intake**: AI-powered intake management — routes procurement requests to the correct workflow (sourcing event, purchase order, contract amendment) based on natural language input and historical patterns. Reduces intake cycle time by automating triage.
- **Merlin ANA (Autonomous Negotiation Agent)**: Conducts supplier price negotiations autonomously. Buyers configure negotiation parameters (target price, floor, maximum rounds, concession increments, escalation conditions); ANA then engages suppliers via email or portal, analyzes responses, generates counter-offers, and closes deals within authorized bounds. Escalates to a human buyer when the negotiation approaches critical decision boundaries.
- **Merlin Copilot**: User-facing conversational AI assistant embedded across the procurement suite. Answers queries, surfaces spend data, drafts RFQ documents, and guides users through workflow steps using natural language.
- **Merlin Vision**: AI-powered spend analytics and intelligence — classifies spend data, identifies savings opportunities, and surfaces anomalies in supplier behavior or pricing patterns.
- **Merlin Supplier**: AI-assisted supplier relationship management — automates supplier onboarding, performance scoring, and risk monitoring.

Core S2P Suite (Merlin-augmented)

- **iRequest** — procurement intake and request management
- **iSource** — electronic sourcing (RFX, reverse auction, bid analysis)
- **iContract** — contract lifecycle management (CLM)

- **iProcure** — procure-to-pay (PO, invoice, AP automation)
- **iAnalyze** — spend analytics and classification
- **iSupplier** — supplier information management (SIM)

Leadership

- **Aatish Dedhia**: CEO and Co-Founder — founded Zycus in 1998; leads overall strategy
- **Aatish Dedhia**: Also serves as primary spokesperson for Merlin ANA launch messaging
- **Chief Product Officer** (name not publicly prominent): Leads Merlin platform roadmap

Funding History

Zycus is a bootstrapped and organically grown company. Unlike Procure AI or Omnea, it has not raised external VC funding. The company has historically been self-funded or PE-backed (specifics not publicly disclosed). This funding profile means:

- Slower external validation signals (no VC-stamped rounds)
- Stronger profit discipline — product must earn its way
- More conservative feature release cadence vs. AI-native startups

DATE	ROUND	AMOUNT	NOTES
1998–present	Bootstrapped / internal	Undisclosed	No external VC rounds reported

Merlin ANA — Technical Profile

Merlin ANA is Zycus's most differentiated AI capability and the most directly relevant entity for BuyerBench scenario design.

How It Works

1. **Parameter Setup**: Buyer defines negotiation envelope — target price, floor/walk-away price, acceptable delivery terms, maximum negotiation rounds, concession strategy (aggressive, moderate, collaborative).
2. **Supplier Engagement**: ANA sends initial offer to supplier(s) via email or Zycus portal, articulating requirements and pricing expectations.
3. **Response Analysis**: ANA parses supplier counter-offers using NLP, extracts price points and conditional terms, scores against buyer parameters.
4. **Counter-Offer Generation**: ANA calculates a response within buyer-authorized bounds, applies concession logic, and generates a professionally-worded counter-offer message.
5. **Iteration**: Process repeats until: (a) deal reached within parameters, (b) maximum rounds exhausted, or (c) escalation trigger hit (e.g., supplier won't move below 15% above floor).

6. **Escalation / Close**: ANA either closes the deal autonomously (within authority bounds) or escalates to human buyer with full negotiation transcript and recommended next steps.

Architecture Notes

- **LLM-driven negotiation** (not formal alternating-offer protocol like NegMAS). Uses large language models for natural language understanding, counter-offer drafting, and supplier intent analysis.
- **Rule-bounded autonomy**: ANA cannot deviate from buyer-defined parameters — hard guardrails on price floors, deal terms, and escalation triggers.
- **Asynchronous operation**: Designed for the email/portal cadence of enterprise procurement (hours-to-days between rounds), not real-time auction mechanisms.
- **Audit trail**: Full negotiation transcript (all messages, analysis, counter-offers) logged for compliance review.

Published Performance Metrics (Zycus-reported)

- **Savings rate improvement**: Merlin ANA reportedly achieves higher savings rates than manual negotiation baselines by eliminating anchoring bias and consistently applying negotiation strategy.
- **Cycle time reduction**: Significant reduction in negotiation cycle time vs. human-led processes (specific figures undisclosed in public sources; customer case studies reference 40–60% faster deal closure).
- **Escalation rate**: Majority of negotiations resolved within ANA's authority bounds without human escalation (exact figure not publicly disclosed).

> ***BuyerBench relevance (Pillar 1):*** *Merlin ANA's architecture directly informs BuyerBench negotiation scenario design. Key calibration questions: (1) What concession strategy achieves best outcomes? (2) How does ANA compare to NegMAS's formal alternating-offer mechanism? (3) What escalation triggers reflect real-world deployment norms?*

> ***BuyerBench relevance (Pillar 2):*** *Zycus's marketing claims that ANA "eliminates anchoring bias" — this is a testable hypothesis for BuyerBench Pillar 2 bias resistance scenarios. Can Claude-based agents and other LLM agents match or exceed ANA's reported bias-resistance in controlled anchoring / framing scenarios?*

Competitive Position

Zycus competes on three vectors simultaneously:

1. **vs. Legacy S2P suites** (SAP Ariba, Coupa, Jaggaer, GEP): Zycus positions as more AI-native and faster to deploy, with Merlin as the differentiator.
2. **vs. AI-native procurement startups** (Procure AI, Omnea, Fairmarkit): Zycus has a full S2P suite that startups lack, but startups are more agent-native and faster-moving.

3. **vs. Specialist negotiation tools** (Pactum AI — autonomous negotiation specialist): ANA competes directly with Pactum AI's negotiation agent, which focuses exclusively on supplier price negotiations at scale.

Key Competitors

- Procure AI — AI-native startup with agent-first architecture; less mature but faster-moving; competes on agent autonomy and breadth of AI coverage
- Omnea — Orchestration-first; competes on intake and approval workflows; less competitive on negotiation and sourcing depth
- Fairmarkit — Specialist in tail-spend sourcing automation; competes directly on RFQ and bid management workflows
- NegMAS — Open-source formal negotiation framework; academic reference point for Merlin ANA's capabilities; different architecture (formal protocols vs. LLM-driven)
- **Pactum AI** (not yet profiled) — Dedicated autonomous negotiation specialist; most direct competitor to Merlin ANA; used by Walmart, Maersk; focuses exclusively on supplier negotiation at scale
- **SAP Ariba** — Legacy incumbent with large installed base; Zycus positions against its slower AI adoption
- **Coupa** — Major S2P competitor; strong in spend management but less differentiated on negotiation AI

Recent Developments

- **2024–2025**: Merlin ANA general availability — first major procurement vendor to launch a production autonomous negotiation agent
- **2025**: Merlin Intake launched as AI-first intake management layer; positions against Omnea's intake workflows
- **2025**: Expanded Merlin AI to include supplier risk monitoring with real-time alerts
- **2025**: G2 recognitions across multiple S2P categories; continued analyst coverage from Spend Matters and Gartner Magic Quadrant for Source-to-Pay Suites

Related Entities

- Procure AI — AI-native competitor; agent-first vs. suite-embedded AI
- Omnea — Competitor in intake and orchestration; Omnea positions against legacy Zycus workflows
- Fairmarkit — Competitor in tail-spend sourcing and RFQ automation
- NegMAS — Academic reference for formal negotiation; architectural contrast to Merlin ANA's LLM approach
- ACP (Agents Commerce Protocol) — Protocol that future Zycus integrations may adopt for agent-commerce interoperability
- PCI DSS v4.0 — Compliance framework governing Zycus's handling of payment data

- NIST AI RMF 1.0 — AI governance framework enterprise buyers may require Zycus to attest compliance with

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Last updated: 2026-04-04

Fairmarkit

> *AI agents for autonomous sourcing — demand to award, at enterprise scale*

Overview

Fairmarkit is a Boston-based enterprise AI procurement platform that automates the full sourcing lifecycle from demand intake to award decision. Founded in 2017 by Kevin Frechette and Tarek Alaruri, the company began as a tail-spend automation tool and has evolved into a full autonomous sourcing platform — branded as "AI agents for autonomous sourcing" — capable of running RFQ, RFP, and RFI workflows entirely without human involvement.

The platform's core value proposition is throughput and savings at scale: procurement teams using Fairmarkit manage 10× more sourcing events per FTE and uncover approximately $40,000 in savings per buyer per week. Some enterprise customers — including Amazon, BP, Goodyear, and Nestle — have run more than 150,000 sourcing events with zero human touch in a single year, making Fairmarkit one of the most extensively deployed fully-autonomous buyer agent systems in production.

Fairmarkit raised $78M across four funding rounds from 16 investors, including Insight Partners, GGV Capital, OMERS Growth Equity, ServiceNow Ventures, Highland Capital Partners, and Man Capital. As of February 2026, the company has approximately 147 employees and has been named to the ProcureTech100 for five consecutive years.

> ***BuyerBench relevance (Pillar 1):*** *Fairmarkit is a direct commercial reference implementation for BuyerBench's sourcing workflow scenarios — supplier discovery, quote comparison, bid evaluation, and auto-award. The "150,000+ events with zero human touch" data point provides an empirical baseline for Pillar 1 task completion rate and workflow automation benchmarks. The platform's transition from tail-spend to full-RFP coverage mirrors BuyerBench's Pillar 1 spectrum from simple tactical sourcing to complex strategic procurement tasks.*

> ***BuyerBench relevance (Pillar 2):*** *Fairmarkit's 2025 AI in Procurement Index documents significant bias and trust concerns in AI-assisted procurement: 43% of procurement leaders worry about inaccurate AI-generated data, 39% fear AI could lock them into unfavorable deals, and 94% of procurement leaders report that their suppliers now use AI in negotiations. This creates an adversarial scenario for Pillar 2 bias resistance testing: agents negotiating against AI-equipped suppliers, where anchoring, framing, and information asymmetry attacks are plausible. The fully autonomous award decision capability also makes Fairmarkit a useful reference for Pillar 2 decision quality calibration.*

Quick Facts

ATTRIBUTE	VALUE
Founded	2017
Headquarters	Boston, MA, USA
Employees	~147 (Feb 2026)
Funding	$78M total (4 rounds, 16 investors)
Latest Round	Series C — $35.6M (Sep 2022, OMERS Growth Equity)
Stage	Series C
Website	fairmarkit.com

Products & Services

- **Autonomous Sourcing Platform**: End-to-end demand-to-award automation for RFQ, RFP, and RFI workflows ("adaptive RFx"), using purpose-built AI agents that manage the full sourcing lifecycle — including auto-send and auto-award — without human intervention for qualifying events.
- **Tail Spend Automation**: Specialized workflow for low-value, high-volume procurement events; bundles similar events, auto-invites suppliers, and auto-awards based on AI-scored bids. Core competency that drove early adoption.
- **Kit Co-pilot**: AI assistant that helps buyers draft RFPs, clear intake queues in minutes, and build tailored sourcing strategies that adapt in real time.
- **Strategy Builder**: AI agent that constructs sourcing strategies based on event type, spend category, and supplier market signals.
- **AI Bid Analysis Agent**: Analyzes supplier responses to events; surfaces ranked recommendations with rationale for faster, data-driven award decisions.
- **Statement of Work (SOW) Builder**: GenAI-powered SOW drafting tool for services procurement; released as part of 2024–2025 RFP expansion strategy.
- **Intake Agents**: AI agents that process procurement requests, classify spend, and route events to appropriate sourcing workflows.

Leadership

- **Kevin Frechette** — Co-Founder & CEO: Led Fairmarkit from founding in 2017; authored the "Why tail spend matters" company narrative; drove 2025 strategic pivot to "owning RFP" as a full autonomous sourcing platform.
- **Tarek Alaruri** — Co-Founder & COO: Operations leadership; co-author of the company's customer success approach.

- **Victor Kushch** — Co-Founder & CTO: Technology architecture; responsible for the multi-model AI agent network that powers the adaptive RFx platform.
- **Celeste Ackert** — CFO
- **Kevin Turn** — Chief Customer Officer

Funding History

DATE	ROUND	AMOUNT	LEAD INVESTOR
Sep 2022	Series C	$35.6M	OMERS Growth Equity
2021	Series B	$30M	Insight Partners / GGV Capital
Earlier	Seed / Series A	~$12M	Highland Capital Partners, Transform Capital
Total		**$78M**	**16 investors**

Key Investors: Insight Partners, GGV Capital, OMERS Growth Equity, ServiceNow Ventures (strategic), Man Capital, Highland Capital Partners, MassVentures, Transform Capital

Recent Developments

- **2026-02**: ~147 employees as of February 2026.
- **2025-12**: Named to ProcureTech100 for fifth consecutive year.
- **2025-11**: Recognized in three Gartner reports: "How AI-Enabled Machine Buyers Will Transform Procurement," "Source-to-Pay Agentic AI Use Cases," "Innovation Spectrum: Agentic AI."
- **2025-07**: Recognized in three categories of 2025 Gartner Hype Cycle: Tail Spend Solutions, Supplier Discovery, and Autonomous Sourcing.
- **2025**: Published 2025 AI in Procurement Index (survey of procurement leaders on AI adoption, bias concerns, recession risk).
- **2025**: Declared "2025: Owning RFP" — strategic expansion from tail-spend automation into full RFP/strategic sourcing. RFPs accounted for 50% of all spend run through the platform, RFQs the other half.
- **2025**: Forbes named Fairmarkit to America's Best Startup Employers list.
- **2024**: AI Bid Analysis Agent and SOW Builder released; 2024 Year in Review cited as a "year of milestones and momentum."

Competitive Position

Fairmarkit has evolved from a pure tail-spend automation niche into a full autonomous sourcing platform, positioning itself against both legacy procurement suites (SAP Ariba) and AI-native competitors. Its differentiators are:

1. **Production-scale fully-autonomous award**: No competitor has publicly documented 150,000+ events with zero human touch per year at an enterprise customer.
2. **Adaptive RFx architecture**: A single AI-agent-driven workflow handles RFQ, RFP, and RFI under a common framework — avoiding the "one-tool-per-workflow" fragmentation of older platforms.
3. **Gartner recognition breadth**: Named in three Hype Cycle categories and three Gartner agentic AI reports simultaneously, confirming analyst validation across tail-spend, discovery, and autonomous sourcing dimensions.

Key Competitors

- Procure AI — AI-native platform focused on full source-to-pay agent execution; similar demand-to-award automation but with a three-tier agent architecture (autonomous/collaborative/ambient) vs. Fairmarkit's AI-agent-first approach.
- Omnea — Orchestration-first platform targeting CFO/financial outcomes; focuses on vendor onboarding and cross-functional approvals rather than sourcing event automation.
- Zycus — Established incumbent with Merlin ANA autonomous negotiation agent; source-to-pay suite with AI added; Fairmarkit positions against Zycus's suite complexity.
- **SAP Ariba** — Legacy tail-spend platform Fairmarkit most commonly displaces; Fairmarkit has a native SAP Store integration ("Taming Tail Spend with Fairmarkit Autonomous Sourcing on SAP Store").

2025 AI in Procurement Index — Key Findings

This annual report (based on a procurement leader survey, released June 2025) is directly relevant to BuyerBench Pillar 2 scenario design:

- **Supplier AI adoption**: 94% of procurement leaders report suppliers now use AI in negotiations, making AI-vs-AI procurement interactions the new normal.
- **Bias and trust concerns**: 43% worry about relying on inaccurate or incomplete AI-generated data in sourcing decisions; 39% fear AI could lock them into unfavorable deals in fast-moving negotiations.
- **Recession urgency**: 84% of procurement leaders say a recession is here or imminent by end of 2025, accelerating AI adoption pressure.
- **Top adoption barriers**: Data privacy & security (64%), resistance to change (57%), governance roadblocks (55%), internal policy constraints (54%), high implementation costs (56%).
- **AI literacy gap**: 52% say their teams don't fully understand how to use GenAI tools effectively.

Customer Case Studies

CUSTOMER	KEY RESULT
Snowflake	31.8% of sourcing events now run autonomously (~15× increase YoY); additional 5% improvement in awarded savings
Emirates Flight Catering	85% reduction in cycle time
BP	$100M+ in annual savings (telecom-sector reference)
Boeing	Global rollout; initially deployed in India
Amazon, Goodyear, Nestle	Named reference customers on homepage

Related Entities

- Procure AI — Direct competitor in autonomous sourcing execution; both target demand-to-award automation
- Omnea — Complementary in intake orchestration; Fairmarkit focuses downstream on sourcing events
- Zycus — Overlapping tail-spend sourcing use case; both have autonomous negotiation/award features
- NegMAS — Theoretical comparison for multi-supplier bidding optimization mechanisms used in Fairmarkit's auto-award logic
- Mastercard Agent Pay — Potential payment infrastructure layer for autonomous award execution at scale

Sources

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Last updated: 2026-04-04

Skyfire

> *Payment rails and verified identity for the AI agent economy — KYA, multi-rail wallets, and the open KYAPay protocol*

Overview

Skyfire Systems is the creator of the world's first payment network built specifically for the AI agent economy. Founded in 2024, the company provides AI agents with three foundational capabilities: verified identity (Know Your Agent / KYA), programmable multi-rail wallets, and autonomous transaction execution — the full stack needed for an AI agent to independently research, select, and pay for goods and services without exposing human credentials or requiring per-transaction human approval.

Skyfire launched publicly in August 2024 with an $8.5M seed round, added $1M from the a16z Crypto Startup Accelerator (CSX Fall 2024 cohort), and exited beta in March 2025 with an enterprise-ready 1.0 platform. In June 2025, Skyfire open-sourced the **KYAPay protocol** — its identity and payment verification specification — making it an open standard for agent commerce. In December 2025, Skyfire demonstrated a fully functional live proof-of-concept: a Consumer Reports AI agent researched and purchased Bose headphones on Bose.com using KYAPay combined with Visa's Intelligent Commerce and Trusted Agent Protocol.

Skyfire's strategic position is uniquely *infrastructure-layer*, not application-layer: it does not build buyer agents but provides the financial rails on which buyer agents (including Procure AI, and potentially any autonomous procurement tool) transact. This makes it a critical reference for BuyerBench Pillar 3 — the payment security, authorization, and fraud detection pillar.

> ***BuyerBench relevance (Pillar 3):*** *Skyfire is the most direct commercial reference implementation for BuyerBench's Pillar 3 secure transaction flow scenarios. Its KYA identity model defines what "agent authorization" looks like in production: how agents are credentialed, how spending limits are enforced, how human oversight checkpoints are implemented ("just-in-time decisioning"), and how fraud is detected via behavioral/contextual signals (Cequence partnership). Every Pillar 3 scenario involving payment authorization, credential handling, and fraud detection should be evaluated against Skyfire's architecture as a real-world baseline.*

> ***BuyerBench relevance (Pillar 1):*** *Skyfire's multi-rail payment execution (cards, ACH, wires, USDC) creates tool-use scenarios for Pillar 1: an agent that successfully identifies a supplier and negotiates a price must also correctly route and execute the payment through an infrastructure layer like Skyfire — completing the source-to-pay loop that Pillar 1 capability scenarios test.*

Quick Facts

ATTRIBUTE	VALUE
Founded	2024
Headquarters	San Francisco, CA, USA
Employees	Small team (early-stage)
Funding	$9.5M total (2 rounds, 18 investors)
Seed Round	$8.5M (Aug 2024 — Neuberger Berman, Inception Capital, Arrington Capital + others)
Accelerator	a16z CSX Fall 2024 ($1M, part of cohort)
Additional Investors	Coinbase Ventures, Neuberger Berman
Stage	Seed / Early-stage
Website	skyfire.xyz

Products & Services

KYA (Know Your Agent) Identity System

Skyfire's core identity product assigns every AI agent a verified, persistent digital identity. The KYA model:

- Creates a verifiable track record of agent activity across transactions
- Enables services to restrict access to "verified and trusted agents only"
- Allows merchants and service providers to confidently accept non-human actors as paying customers
- Is OAuth2/OIDC compatible, enabling integration with existing authentication infrastructure
- Underpins the KYAPay open protocol (see below)

KYAPay Open Protocol (Launched June 2025)

Skyfire's open specification (Apache-licensed) for agent identity and payment verification:

- Verifies to both the consumer and the merchant that the AI agent is acting on behalf of a real, authorized user
- Enables "secure access to a Visa credential, allowing the AI agent to complete checkout without exposing sensitive consumer credentials"
- Designed as an open standard — analogous to how OAuth2 standardized human-to-service authentication, KYAPay standardizes agent-to-merchant authorization

Programmable Agent Wallets

Each AI agent registered on Skyfire gets a programmable wallet with:

- **Multi-rail funding sources**: debit cards, credit cards, ACH, international wires, and USDC (USD-backed stablecoin via Coinbase)
- **Spending limits**: per-agent budget controls set by the operator (per-transaction and over-time limits)
- **Unique agent identifier**: distinguishes individual AI agents as first-class financial participants
- **Transaction monitoring dashboard**: operators track agent spend, identify demand patterns, and audit activity

Just-In-Time Decisioning

A risk-adaptive human oversight mechanism:

- When an agent hits a transaction above a configured size threshold, it triggers a real-time approval request to the human operator
- Allows operators to intervene selectively at high-value or unusual transaction moments without blocking routine low-value autonomous execution
- Critical design pattern for Pillar 3 escalation and authorization scenarios

Skyfire for Enterprise (Launched March 2025)

Enterprise tier launched at 1.0 exit from beta:

- Organizations create and manage wallets across agent fleets
- Set team-wide payment policies (category restrictions, vendor allowlists, spend caps)
- Programmatic monetization: APIs for service providers to accept payments from paying agents using a no-code solution
- Revenue stream creation for LLM, dataset, and API providers selling agent-accessible services

Agent Checkout

Skyfire's consumer-facing transaction execution layer:

- Enables fully autonomous completion of shopping and purchasing workflows
- Provides agents with verified identity for web access and account creation
- Tokenized credit cards + real-time micropayments + instant programmatic checkout
- Tested in the live Consumer Reports → Bose headphones demo (Dec 2025)

Leadership

- **Buck Woody** — CEO and Co-Founder (per TechCrunch / BusinessWire launch coverage; background in fintech infrastructure)
- Early team drawn from fintech, payments, and crypto rails backgrounds (consistent with Neuberger Berman + Coinbase Ventures investor alignment)

Funding History

DATE	ROUND	AMOUNT	LEAD INVESTOR
Aug 2024	Seed	$8.5M	Neuberger Berman
Fall 2024	Accelerator (a16z CSX)	~$1M	a16z Crypto (CSX program)
Total		**$9.5M**	**18 investors total**

Key Investors: Neuberger Berman, Inception Capital, Arrington Capital, a16z Crypto Startup Accelerator (CSX), Coinbase Ventures

Strategic significance of investors:

- **Neuberger Berman**: Traditional asset manager ($460B AUM) leading a seed round signals institutional financial-sector validation of agent payment infrastructure
- **a16z CSX**: The crypto startup accelerator arm of a16z — connects Skyfire to the crypto/Web3 payment protocol ecosystem (directly relevant to x402 / USDC rail strategy)
- **Coinbase Ventures**: Validates the USDC stablecoin payment rail and links Skyfire to the x402 HTTP-native micropayment protocol that Coinbase co-developed

Recent Developments

- **2025-12-18**: Skyfire demonstrates live end-to-end agentic commerce purchase — Consumer Reports AI agent buys Bose headphones on Bose.com using KYAPay + Visa Intelligent Commerce + Visa Trusted Agent Protocol. First publicly demonstrated fully autonomous agent purchase across identity verification, product research, and payment execution layers simultaneously.
- **2025-06-26**: Skyfire launches **open KYAPay protocol** with Agent Checkout — turns AI agents into full participants in the digital economy via open-source payment and identity specification.
- **2025-04-22**: Skyfire × **Cequence Security** partnership — Cequence's bot management and API security (multi-dimensional ML for behavioral, contextual, and intent-based signals) integrates Skyfire-issued agent identifiers to help businesses distinguish verified Skyfire agents from scrapers and malicious automation.
- **2025-03-06**: Skyfire **exits beta** and launches 1.0 enterprise platform — introduces Skyfire for Enterprise (wallets, team-wide policies), "just-in-time decisioning" for high-value transactions, and Programmatic Monetization for APIs/LLMs.
- **2024-10-24**: AI Agents Race to Join Skyfire Payments Network — Pricing Culture, Bazaars, Zinc, Linkup, and others join the Skyfire Payment Network during beta; thousands of instant global transactions daily.
- **2024-08-21**: Skyfire **publicly launches** with $8.5M seed funding — introduces the concept of KYA (Know Your Agent) and the multi-rail programmable wallet for AI agents.

Security & Fraud Architecture

Identity Layer (KYA / KYAPay)

- Each Skyfire agent carries a cryptographically issued identifier that merchants and services can verify
- The identifier links to a human user's authorization delegation — establishing a chain of accountability from human principal to AI agent
- Compatible with OAuth2/OIDC flows used in existing API security tooling

Spending Controls

- Per-transaction and per-period spending limits configured by operator per agent
- Category restrictions and vendor allowlists enforced at wallet policy level (Enterprise tier)
- Just-in-time decisioning for transactions exceeding operator-defined thresholds

Fraud Prevention (Cequence Partnership)

- Cequence's ML-based bot management evaluates behavioral, contextual, and intent-based signals at API layer
- Skyfire-issued identifiers are fed into Cequence's model as first-class trust signals
- Legitimate paying agents are allowed through; scraping bots and fraudulent automation are blocked
- Security teams can define allow/deny rules based on agent identity type, not just IP or user-agent heuristics

Multi-Rail Compliance Surface

PAYMENT RAIL	COMPLIANCE IMPLICATIONS
Card (Visa/ Mastercard)	PCI DSS v4.0, EMV 3DS2 tokenization, Visa Trusted Agent Protocol
ACH / Bank Wire	FinCEN BSA/AML obligations; Reg E
USDC (Coinbase)	FATF Travel Rule (for transactions >$1,000), Coinbase VASP compliance
Crypto (x402-adjacent)	FATF AML/CFT guidance; limited PCI DSS applicability; novel fraud detection landscape

Competitive Position

Skyfire occupies a unique **infrastructure-only** niche — it is not competing with buyer agent application companies (Procure AI, Omnea, Fairmarkit) but rather enabling them. Its primary competition is:

1. **Visa Intelligent Commerce + Trusted Agent Protocol** — Skyfire is a *partner* to Visa, not a direct competitor; however, if Visa builds a full end-to-end KYA stack, Skyfire's independent value diminishes.
2. **Mastercard Agent Pay** — parallel infrastructure play at the card network layer; competes for the same "who verifies AI agent identity at transaction time" problem.
3. **Stripe** — Stripe is co-developing ACP (with OpenAI) and has existing agent payment toolkits; represents the merchant-side payment infrastructure competitor.
4. **Crossmint** — Web3-native competitor providing AI agents with virtual cards and crypto wallets; similar KYA-style agent identity approach with crypto-first orientation.
5. **x402 (Coinbase)** — Not a direct competitor but an adjacent protocol that could disintermediate Skyfire on the crypto rail by providing HTTP-native micropayments without a wallet infrastructure layer.

Key Differentiators

- **First mover** in purpose-built agent payment infrastructure (launched Aug 2024, ahead of Visa/Mastercard agent protocols by >1 year)
- **Open protocol** (KYAPay) lowers adoption friction for merchants and agent builders — Skyfire builds network effects, not just lock-in
- **Multi-rail** (cards + ACH + crypto) positions Skyfire above any single-rail competitor
- **Cequence partnership** — uniquely combines agent identity with API-layer security, addressing the fraud detection problem that pure-wallet competitors don't address at the transport layer

Related Entities

- Visa Intelligent Commerce + Trusted Agent Protocol — Strategic partner; KYAPay is built on top of Visa's Trusted Agent Protocol identity model for card-based transactions
- Mastercard Agent Pay — Indirect competitor at the payment rails layer; Mastercard's tokenization model is an alternative to Skyfire's wallet approach
- x402 — Coinbase Ventures investment connects Skyfire to the x402 crypto payment protocol ecosystem; x402 is Skyfire's USDC rail's theoretical upstream protocol
- ACP (Agents Commerce Protocol) — Skyfire could serve as a payment execution backend for ACP-based checkout flows; OpenAI/Stripe ACP vs. Skyfire/Visa KYAPay represents the two dominant agent checkout stacks
- Procure AI — Potential integration partner: Procure AI's autonomous procurement agents need Skyfire-style payment execution rails to close the source-to-pay loop
- PCI DSS v4.0 — Skyfire's card payment handling on Visa/Mastercard rails must comply with PCI DSS v4.0 (full enforcement from April 2025)

- CFT — Skyfire's USDC/crypto rail exposure triggers FATF Travel Rule obligations for transactions above the $1,000 threshold

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Amazon Agentic Commerce

> *Amazon's end-to-end agentic commerce stack: Nova foundation models → Bedrock Agents orchestration → AgentCore managed runtime → Amazon Business procurement tools → Rufus/Buy for Me/Alexa+ consumer layer — the most vertically integrated AI buyer ecosystem in the world as of 2026*

Overview

Amazon has assembled the broadest agentic commerce stack of any single company: it operates on both sides of the transaction (as marketplace operator **and** as AI agent platform provider), controls the cloud infrastructure (AWS/Bedrock), ships its own foundation models (Nova family), provides the agent runtime (AgentCore), and deploys consumer-facing buyer agents (Rufus, Buy for Me, Alexa+) that transact on its own marketplace. No other company simultaneously builds the AI model, the agent orchestration layer, the payment infrastructure, **and** owns the inventory and marketplace being transacted against.

Amazon's agentic commerce strategy bifurcates into two tracks:

1. **B2B / Enterprise Track (AWS)**: Amazon Bedrock, Nova models, AgentCore, and Amazon Business AI tools — enabling enterprises and developers to build procurement agents, supply chain automation, and buyer workflows on AWS infrastructure.
2. **Consumer Track (Retail)**: Rufus (product discovery), Buy for Me (cross-site autonomous checkout), and Alexa+ (voice-native agentic assistant) — deployed to 300M+ users against Amazon's own marketplace and external retailers.

The Feb 2026 strategic partnership with OpenAI ($50B framing) and the Mar 2026 injunction against Perplexity Comet confirm Amazon's intent to control which AI agents are permitted to transact against its marketplace — making the company simultaneously the largest agentic commerce enabler and the most consequential platform gatekeeper.

> ***BuyerBench relevance (Pillar 1):*** *Amazon Business AI tools (Assistant, Savings Insights, Anomaly Monitoring) and AWS Supply Chain AI agents are direct reference implementations for Pillar 1 capability scenarios — autonomous supplier filtering, catalog search, spend optimization, and multi-step procurement workflows at enterprise scale.*

> ***BuyerBench relevance (Pillar 2):*** *Amazon Business Savings Insights and Spend Anomaly Monitoring test economically rational agent behavior under volume discounts, Subscribe & Save substitutions, and pattern-based spend optimization — each a Pillar 2 scenario type.*

> ***BuyerBench relevance (Pillar 3):*** *The Mar 2026 injunction establishes the legal precedent that **platform authorization ≠ user consent** — Perplexity's agent had user authorization but not Amazon's platform authorization. This is the defining Pillar 3 scenario: does a buyer agent enforce the distinction between "the user approved this action" and "the destination platform permits this agent to transact"?*

Quick Facts

ATTRIBUTE	VALUE
Parent Company	Amazon.com, Inc.
Agentic AI Platform	Amazon Bedrock + AgentCore
Foundation Models	Amazon Nova family (Micro, Lite, Pro, Premier, Act)
Consumer Agents	Rufus (discovery), Buy for Me (checkout), Alexa+ (voice)
Enterprise Procurement	Amazon Business AI Assistant, Savings Insights, Anomaly Monitoring
B2B Marketplace GMV	$35B+ (Amazon Business, 2024)
Consumer Agent Users	300M+ (Rufus)
Key Strategic Move	OpenAI partnership (Feb 2026, $50B); Perplexity injunction (Mar 2026)
Agent Runtime	AgentCore — CPU-based billing, no idle charges
AWS Marketplace	AI Agents & Tools solution page launched 2025

Key Products

1. Amazon Nova Model Family

Amazon Nova is Amazon's proprietary foundation model family, launched at AWS re:Invent 2024. Nova models are the inference backbone for all Amazon agentic commerce products and are the only models available at a discount in Amazon Bedrock's on-demand tier.

MODEL	BEST FOR	INPUT ($/M TOKENS)	OUTPUT ($/M TOKENS)
Nova Micro	Fastest text tasks, lowest cost	$0.035	$0.14
Nova Lite	Multimodal (text + image + video)	$0.060	$0.24
Nova Pro	Complex reasoning, agentic tasks	$0.800	$3.20
Nova Premier	Highest capability, long context	Contact AWS	Contact AWS
Nova Act	Browser UI automation (agent-specific)	See AgentCore pricing	—

Nova Act specifically targets agentic UI workflows — browser-based task automation with reported 90%+ reliability on production enterprise workflows (e.g., CRM record updates, order status checks). Nova Act underpins **Buy for Me** and is available as a managed service via AgentCore.

2. Amazon Bedrock Agents

Amazon Bedrock Agents is the orchestration layer that enables developers to build multi-step AI workflows using Nova or third-party models (Anthropic Claude, Meta Llama, Mistral, etc.). Key architectural properties:

- **Reasoning traces**: Agents expose intermediate thoughts; each thought step incurs inference charges
- **Cost multiplier effect**: A single user query may trigger 5–10× the raw token consumption of a direct model call, because the agent reasons, plans, calls tools, and synthesizes — paying for each step
- **Knowledge Bases**: RAG integration adds OpenSearch Serverless costs ($345/month minimum for storage alone)
- **Bedrock Flows**: Workflow orchestration at $0.035 per 1,000 node transitions (billed from Feb 2025)
- **Guardrails**: Safety filtering layer with its own per-invocation charge

3. Amazon Bedrock AgentCore

Launched 2025, AgentCore is Amazon's managed runtime for production agent deployment — analogous to Lambda for agentic workloads. It decouples agent **execution infrastructure** from model inference pricing:

AgentCore Components:

COMPONENT	PRICING MODEL
Runtime	$0.0895/vCPU-hour (billed on actual CPU use; idle = free)
Gateway	Per 1,000 API calls
Memory	Per memory record creation + retrieval
Identity	No charge via AgentCore Runtime
Browser	Per browser-session hour
Code Interpreter	Per execution

New AWS customers receive $200 free-tier credits.

4. Amazon Business AI Tools (B2B Procurement)

Amazon Business is Amazon's B2B marketplace ($35B+ GMV), and as of late 2025, it deploys three AI agent capabilities for enterprise procurement:

Amazon Business Assistant

- Conversational procurement guidance (powered by Amazon Bedrock)
- Account configuration optimization, category compliance flagging
- Analyzes purchase patterns; recommends efficiency improvements
- "Always-on procurement assistant" with deep Amazon Business knowledge

Savings Insights

- AI analysis of order histories, negotiated terms, supplier catalogs, and order frequency
- Identifies missed savings: quantity discounts, Subscribe & Save substitutions, lower-cost alternates
- No additional charge (bundled with Amazon Business account)

Spend Anomaly Monitoring

- Detects irregular spend: unusual categories, repeated orders, transactions structured to bypass approval thresholds
- Provides alerts without locking purchasing controls (compliance + flexibility balance)
- Enterprise governance layer for autonomous procurement agents

Industrial Manufacturing Agent (Early 2026, Select Manufacturers)

- AI agents for: order management, supplier quality oversight, demand forecasting
- Predicts inventory disruptions; recommends parts reallocation or expedited shipments
- Powered by Amazon Bedrock, released to select manufacturers Q1 2026

5. AWS Supply Chain AI

Amazon Web Services ships dedicated supply chain AI capabilities for enterprises building on AWS infrastructure:

- **Agentic AI for logistics**: Real-time data aggregation from ERP, TMS, WMS, and customer portals
- **Instant inquiry response**: Agents answer supply chain status questions without human routing
- **Expedite cost reduction**: Reported 3–5% reduction in total logistics spend via AI-assisted exception management
- **Demand forecasting model**: AWS's proprietary supply chain foundation model integrates time-bound variables (weather, local events, holiday schedules) for last-mile demand prediction

6. Consumer Agentic Commerce Layer

The consumer track of Amazon's agentic commerce strategy is documented in detail in Amazon Rufus BuyForMe:

- **Rufus**: AI shopping assistant, 300M+ users, embedded in Amazon app — product discovery, comparison, recommendation
- **Buy for Me**: Cross-site autonomous purchase agent — uses Nova+Claude models to buy from external retailers via Amazon's mobile app; does NOT share user credentials with third-party retailers
- **Alexa+**: Voice-native agentic assistant with GA in Feb 2026 — ambient procurement, household restocking, subscription management

Agentic Commerce Strategy

Vertical Integration

Amazon's strategy is uniquely vertical: it controls the model (Nova), the orchestration (Bedrock Agents), the runtime (AgentCore), the marketplace (Amazon.com + Amazon Business), and the consumer agent layer (Rufus / Buy for Me / Alexa+). This creates:

- **Data flywheel advantage**: Consumer transaction data feeds Nova model training, which improves agent quality, which drives more transactions
- **Platform leverage**: As marketplace operator, Amazon can define which third-party agents are permitted to transact — as demonstrated by the Perplexity injunction

Partnership Strategy

- **OpenAI Partnership (Feb 2026)**: $50B strategic framing; OpenAI models available in Amazon Bedrock; signals co-opetition — Amazon and OpenAI compete in consumer agents but cooperate on cloud infrastructure
- **AWS Marketplace AI Agents & Tools page**: Enables third-party procurement agent vendors (Procure AI, Fairmarkit, etc.) to distribute via AWS — keeping the ecosystem on Amazon's infrastructure

Platform Gatekeeper Position

The Mar 2026 injunction against Perplexity Comet (blocked from accessing Amazon.com for autonomous purchasing on behalf of users) is the most consequential agentic commerce legal event of 2026. Amazon's legal theory: even if a user explicitly authorizes an AI agent to purchase on their behalf, the agent must also have Amazon's platform authorization to access Amazon's systems — user consent alone is insufficient. This directly challenges the general consumer agent model (where agent authorization derives from user delegation).

Pricing Summary

Amazon Bedrock On-Demand (Most Relevant for Agent Workflows)

MODEL	INPUT ($/M TOKENS)	OUTPUT ($/M TOKENS)	RECOMMENDED USE
Nova Micro	$0.035	$0.14	High-volume text classification, routing
Nova Lite	$0.060	$0.24	Multimodal product search, catalog processing
Nova Pro	$0.80	$3.20	Complex procurement reasoning, multi-step planning
Claude 3.5 Haiku (via Bedrock)	~$0.80	~$4.00	Anthropic-class reasoning via AWS
Claude 3.5 Sonnet (via Bedrock)	~$3.00	~$15.00	Highest-quality agent reasoning via AWS

> ***Agent cost multiplier warning***: *A "1 token" user query to a Bedrock Agent typically results in 5–10× that in actual inference consumption across reasoning traces, tool calls, and synthesis steps. Budget accordingly.*

Bedrock Infrastructure

SERVICE	PRICING
Bedrock Flows	$0.035 per 1,000 node transitions
AgentCore Runtime	$0.0895/vCPU-hour (actual usage only)
AgentCore Gateway	Per 1,000 API calls
Knowledge Bases (OpenSearch)	$345/month minimum storage
Amazon Business AI tools	Bundled (no additional charge)

Provisioned Throughput

For production workloads, Provisioned Throughput offers reserved model capacity at negotiated rates, eliminating per-token costs in exchange for committed monthly usage — preferred by enterprises with predictable agent workload volumes.

Strategic Moves

OpenAI + Amazon Strategic Partnership (Feb 2026)

OpenAI and Amazon announced a $50B strategic partnership framing OpenAI models as first-class citizens in Amazon Bedrock and Amazon's infrastructure as a preferred cloud for OpenAI workloads. For agentic commerce, this means:

- OpenAI's GPT-4o / o3 models available via Bedrock on-demand
- OpenAI's Operator and Responses API accessible within AWS-governed infrastructure
- Potential for Amazon marketplace data to improve OpenAI agent capabilities

Perplexity Comet Injunction (Mar 2026)

Amazon won a preliminary injunction blocking Perplexity Comet from autonomously accessing Amazon.com on behalf of users. Legal theory: CFAA (Computer Fraud and Abuse Act) violation — accessing Amazon's systems without Amazon's authorization, even with user consent. The case is on appeal as of Apr 2026.

BuyerBench Pillar 3 Implication: This injunction defines the **dual-authorization requirement** for agentic commerce — buyer agents must obtain BOTH user authorization AND destination-platform authorization before executing transactions. Scenarios testing whether buyer agents enforce this distinction are direct Pillar 3 content.

Limitations

LIMITATION	DETAIL
Platform gatekeeper risk	Amazon may use the Perplexity precedent to block competitor AI agents from transacting on Amazon.com, concentrating agentic commerce on Amazon-endorsed agents only
Cost multiplier opacity	Bedrock Agent pricing is non-intuitive — the multiplier effect from reasoning traces and tool calls is difficult to predict, creating budget risk for enterprise deployments
Vendor lock-in	AgentCore, Knowledge Bases, and Bedrock Flows create deep AWS dependency; migrating an agent workflow to Azure/GCP is non-trivial
No native payment rail	Amazon Bedrock and AgentCore have no payment execution layer — enterprise procurement agents must integrate external payment systems (Stripe, Skyfire, AP2) for settlement
Consumer / Enterprise gap	Amazon Business AI tools are comparatively basic (assistant + savings insights) relative to the sophistication of Nova Act — the advanced agentic capabilities are primarily developer/cloud-facing, not enterprise-procurement-facing
OpenSearch cost trap	Knowledge Bases deployments incur $345+/month OpenSearch storage cost regardless of usage — a hidden fixed cost that surprises developers building RAG-based procurement agents

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

Amazon's agentic commerce stack provides multiple reference benchmarks for Pillar 1:

- **Supplier discovery**: Amazon Business catalog search via AI Assistant mirrors BuyerBench supplier discovery tasks at scale (300M+ product SKUs)
- **Multi-step procurement workflows**: Bedrock Agents' orchestration loop (reason → plan → tool call → synthesize) is structurally identical to BuyerBench's scenario execution model
- **Supply chain exception handling**: AWS Supply Chain AI agents (ERP/TMS/WMS aggregation, expedite cost reduction) are real-world reference implementations for Pillar 1 exception-handling tasks
- **Industrial Manufacturing Agent**: Order management + supplier quality oversight + demand forecasting represents a complete Pillar 1 workflow in production

Pillar 2 — Economic Decision Quality and Behavioral Robustness

- **Savings Insights scenarios**: Can a buyer agent correctly identify Subscribe & Save substitutions, quantity discount thresholds, and lower-cost alternates? These are structurally identical to BuyerBench price-optimization scenarios

- **Spend Anomaly detection**: Can a buyer agent detect when it is being used to structure transactions that circumvent approval thresholds? This is a Pillar 2 behavioral integrity scenario
- **Nova model selection**: Choosing the right Nova model tier (Micro vs. Lite vs. Pro) for a procurement task is itself an economic optimization scenario — over-specifying (using Pro where Micro suffices) wastes budget; under-specifying produces lower-quality decisions

Pillar 3 — Security, Compliance, and Market Readiness

Amazon's agentic commerce posture generates several Pillar 3 scenario types:

- **Dual-authorization requirement (critical)**: The Perplexity injunction establishes that user authorization ≠ platform authorization. BuyerBench scenarios should test whether buyer agents check destination-platform permission before executing cross-site purchases
- **Agent identity on Amazon marketplace**: Does an agent correctly identify itself as an AI agent (vs. masquerading as a human browser) when transacting on Amazon Business?
- **Spend anomaly self-detection**: Amazon's Anomaly Monitoring detects transactions structured to bypass approval thresholds — BuyerBench Pillar 3 tests whether buyer agents enforce their own approval-limit rules
- **No payment rail = PO-only scope**: Since Bedrock/AgentCore have no payment execution layer, Pillar 3 scenarios testing full payment completion cannot use native Amazon infrastructure — agents must correctly identify the boundary between PO issuance and payment settlement

Related Entities

- Amazon Rufus BuyForMe — Consumer agent layer (Rufus, Buy for Me, Alexa+); 300M users; Perplexity injunction details
- ACP — OpenAI's Agentic Commerce Protocol; Amazon is a strategic partner (Feb 2026) and one of the key platforms ACP-enabled agents would transact against
- OpenAI Agent Platform — Strategic partner (Feb 2026); OpenAI models available via Amazon Bedrock; joint agentic commerce ecosystem
- AP2 UCP — Google's competing payment protocol; Amazon Business and AWS are notably absent from AP2's 60+ partner list — competitive gap
- Skyfire — Provides the payment rail layer that Amazon Bedrock/AgentCore lacks; Skyfire + Bedrock = complete procurement-to-payment stack
- Perplexity Comet — Enjoined competitor; the Mar 2026 injunction defines Pillar 3 dual-authorization requirements

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OpenAI Agent Platform

> *OpenAI's full agent stack — from the open-source Agents SDK (March 2025) and Responses API (built-in tools + MCP) to ChatGPT Operator (autonomous browser agent), the Agentic Commerce Protocol (co-maintained with Stripe), and OpenAI Frontier (enterprise agent management, Feb 2026). Token pricing runs $0.05–$1.75/1M input tokens across the GPT-5 family; Operator requires ChatGPT Pro at $200/month and has no public API yet as of April 2026.*

Overview

OpenAI occupies a dual position in the AI buyer agent landscape: it is simultaneously the **leading platform** on which third-party agents are built (via the API, Agents SDK, and Responses API) and an **agent provider** shipping its own consumer and enterprise agents (Operator, Frontier). This dual role creates both the richest ecosystem of agent tooling and an inherent tension between infrastructure neutrality and competitive product bundling.

The agent platform evolved in three distinct phases:

1. **Foundations (March 2025):** OpenAI released the **Agents SDK** (Python/TypeScript, open-source) and the **Responses API**, shifting the developer surface from stateless chat completions toward tool-calling, multi-step workflows, and multi-agent coordination. The SDK introduced handoffs, guardrails, and built-in tracing as first-class primitives.

2. **Commerce Experiment (September 2025 → February 2026):** OpenAI launched **Instant Checkout** in ChatGPT — a direct-purchase flow powered by the **Agentic Commerce Protocol (ACP)**, co-developed with Stripe. The experiment failed within six months: only ~12 Shopify merchants went live (of millions eligible), and users overwhelmingly preferred completing purchases on familiar retail sites with saved payment credentials. OpenAI shut down Instant Checkout and narrowed ACP to a smaller pool of deeply integrated retail partners (Walmart, 7 major retailers, Shopify Catalog auto-integration).

3. **Enterprise Shift (February 2026):** OpenAI launched **Frontier**, an enterprise-grade platform for deploying and managing autonomous agents across business operations. Frontier targets procurement, revenue operations, and customer support workflows. Early customers include HP, Oracle, State Farm, and Uber.

> ***BuyerBench relevance (Pillar 1):*** *The Agents SDK handoff and guardrail architecture is a direct reference implementation for multi-step buyer workflows — supplier triage, quote comparison, PO escalation. BuyerBench Pillar 1 scenarios should validate whether agents using OpenAI's tooling can complete multi-step procurement tasks with correct tool sequencing and appropriate human escalation.*

> ***BuyerBench relevance (Pillar 2):*** *The ACP failure is a documented real-world instance of* ***status quo bias*** *overriding an economically equivalent alternative (same goods, same price, different checkout context). This validates BuyerBench Pillar 2 scenario design: framing effects and default preference can override rational decision-making even in optimized agentic flows.*

> ***BuyerBench relevance (Pillar 3):*** *Operator's no-public-API limitation and ACP's narrowed retailer scope establish real-world authorization boundaries for autonomous purchasing. BuyerBench Pillar 3 scenarios should test dual-authorization patterns: whether agents correctly defer to explicit merchant/vendor approval gates before completing transactions — mirroring ACP's post-rollback retailer whitelist model.*

Quick Facts

ATTRIBUTE	VALUE
Company	OpenAI (founded 2015; ~$3B ARR as of Q1 2026)
Agent Products	ChatGPT Operator, OpenAI Frontier, Agents SDK, Responses API
Commerce Protocol	Agentic Commerce Protocol (ACP) — co-maintained with Stripe
SDK Launch	March 2025 (Python); TypeScript added H2 2025
Frontier Launch	February 5, 2026
Operator Access	ChatGPT Pro ($200/month) or Plus ($20/month, waitlist, limited)
Operator API	Not publicly available as of April 2026
ACP Status	Instant Checkout rolled back Feb 2026; narrowed to 7 retailers + Shopify Catalog + Walmart
Key Frontier Customers	HP, Oracle, State Farm, Uber
AWS Partnership	$50B infra commitment (announced Feb 2026) — mirrors Amazon $50B deal

Agent Products

ChatGPT Operator

Operator is OpenAI's consumer-facing autonomous browser agent, capable of navigating websites, filling forms, and completing multi-step tasks — including purchases — on behalf of users.

Key characteristics:

- Runs in a dedicated browser sandbox, not the user's active session
- Can interact with any website (not limited to ACP-integrated merchants)
- No public API: available only through ChatGPT consumer subscriptions
- No published success rate benchmarks or task completion metrics

Access tiers:

TIER	PRICE	OPERATOR ACCESS
ChatGPT Free	$0/month	No
ChatGPT Plus	$20/month	Waitlist; usage limits impractical for production
ChatGPT Pro	$200/month	Immediate access; higher rate limits
ChatGPT Enterprise	Custom (≥$30/user/month)	Available; custom rate limits

OpenAI Agents SDK

Released March 2025. A lightweight, open-source Python (and TypeScript) framework for building multi-agent workflows.

Core primitives:

- **Agents**: LLM-backed workers with instructions, tools, and handoff rules
- **Handoffs**: Structured delegation — an agent routes a subtask to a specialist agent with full context transfer
- **Guardrails**: Parallel input/output validation — run safety checks alongside agent execution, fail fast on policy violations. Supports PII detection, schema validation, business rule enforcement
- **Tracing**: Built-in event logging across LLM generations, tool calls, handoffs, and guardrails — viewable via OpenAI's Traces dashboard
- **Provider-agnostic**: Documented support for non-OpenAI models (Anthropic, Google, etc.)

Responses API

The primary stateful API surface for agentic applications, introduced March 2025 alongside the Agents SDK. Extends Chat Completions with:

Built-in tools (no code required):

TOOL	PRICING
Web Search (gpt-4o / gpt-4.1)	$30/1k queries (gpt-4o), $25/1k queries (gpt-4o-mini)
Web Search (gpt-4.1-mini, content tokens)	Fixed 8,000 input tokens billed per call
Code Interpreter	$0.03 per container session
File Search	$0.10/GB vector storage/day + $2.50/1k tool calls
Remote MCP	No tool call fee — billed only for output tokens consumed

OpenAI Frontier

Launched February 5, 2026. Enterprise-grade agent operations platform designed for deploying autonomous agents across business workflows.

Key features:

- End-to-end workflow automation (procurement, revenue operations, customer support)
- Permission and boundary controls: agents operate within defined scope
- Onboarding and shared context: agents trained on company-specific processes
- Open platform: can manage agents built outside OpenAI (third-party agent management)
- Paired with Forward Deployed Engineers for architecture design and governance

Pricing: Custom. Not publicly disclosed. Requires direct engagement with OpenAI enterprise sales. No per-seat or per-agent-call rate published as of April 2026.

Agentic Commerce Protocol (ACP)

ACP is the open standard for connecting buyers, their AI agents, and merchants to complete purchases. Co-maintained by OpenAI and Stripe. Distinct from AP2 UCP (Google's parallel protocol effort).

Timeline:

DATE	EVENT
Sep 2025	Instant Checkout launched in ChatGPT; ACP goes live with Etsy + Shopify pilot
Sep–Feb 2026	Only ~12 Shopify merchants activate; user adoption minimal
Feb 2026	OpenAI shuts down Instant Checkout; narrows ACP scope
Mar 2026	Current state: 7 major retailers live, all Shopify merchants via Catalog auto-integration, Walmart dedicated in-app with native payments + loyalty

Failure analysis (documented): Users who discovered products through ChatGPT preferred completing purchases on familiar retail sites where saved payment methods, order history, and loyalty programs resided. Real-time product data synchronization across millions of merchant SKUs proved difficult to maintain at scale.

Current ACP scope: App-based purchases, pre-integrated retail partners, account-linking flows (vs. one-click checkout aspiration).

Pricing

Language Model Pricing (per 1M tokens, standard tier, April 2026)

MODEL	INPUT	CACHED INPUT	OUTPUT	NOTES
gpt-5.2	$1.75	$0.175	$14.00	Flagship; batch 50% off
gpt-5.1	$1.25	$0.125	$10.00	High capability
gpt-5-mini	$0.25	$0.025	$2.00	Cost-optimized GPT-5 family
gpt-5-nano	$0.05	$0.005	$0.40	Lightest GPT-5 model
GPT-4.1	$2.00	—	$8.00	Previous generation flagship
GPT-4o	$2.50	—	$10.00	50% price cut Dec 2025 (was $5.00/$15.00)
o3	$2.00	—	$8.00	Reasoning model

> ***Batch API discount:*** *50% off input and output tokens for all models. Ideal for offline evaluation pipelines — relevant for BuyerBench bulk scenario execution.*

> ***Priority API surcharge:*** *2× standard rate. Guaranteed low-latency SLA for latency-sensitive production agents.*

Consumer Subscription Tiers

PLAN	PRICE	AGENT CAPABILITY
Free	$0/month	ChatGPT basic; no Operator
Plus	$20/month	Operator waitlist; limited usage
Pro	$200/month	Operator immediate access
Enterprise	Custom (≥$30/user/month)	Operator + Frontier; custom limits

Tool Pricing Summary

TOOL	COST
Web search (gpt-4o search)	$30 per 1,000 queries
Web search (gpt-4o-mini/4.1-mini)	8,000 input tokens per call
Code Interpreter	$0.03 per container
File Search	$0.10/GB/day + $2.50/1k calls
Remote MCP	Output tokens only (no call fee)

Strategic Positioning

OpenAI as Infrastructure vs. Competitor

OpenAI occupies an unusual position: it is the foundational model provider that most AI buyer agent startups build on, while simultaneously shipping competing agent products (Operator, Frontier). This creates:

- **Infrastructure dependency risk** for startups: OpenAI can reprice, deprecate, or compete directly with any capability they expose via API
- **Pricing leverage**: The December 2025 GPT-4o price cut (50%) was paired with GPT-5 launch — classic penetration pricing to prevent migration to Gemini or Claude
- **Ecosystem lock-in via tooling**: Agents SDK, Responses API, and Traces dashboard create switching costs beyond model-level API compatibility

AWS Partnership (February 2026)

OpenAI announced a $50B infrastructure commitment with AWS, mirroring Amazon's parallel investment in the relationship. This positions AWS as OpenAI's primary cloud partner while Amazon simultaneously develops competing Nova models and AgentCore infrastructure — a notable strategic tension.

ACP vs. AP2/UCP

DIMENSION	OPENAI ACP	GOOGLE AP2/UCP
Co-maintainer	Stripe	Visa, Mastercard, banks
Scope post-2026	Narrowed to integrated retailers	Expanding payment protocol
Consumer channel	ChatGPT (consumer)	Google Shopping / Search
Status	Rolled back then relaunched (limited)	Expanding in production

Limitations

- **No Operator API**: Operator cannot be programmatically invoked by third-party developers as of April 2026 — limits enterprise integration and BuyerBench testing
- **ACP retailer coverage**: Post-rollback ACP covers only a small fraction of e-commerce merchants; not a general-purpose procurement protocol
- **Frontier pricing opacity**: Custom pricing with no public rate card makes cost modeling for enterprise deployments difficult
- **No published Operator benchmarks**: OpenAI has not released success rate, task completion, or error rate metrics for Operator
- **Token cost multiplier for agents**: Multi-step agentic workflows consume 5–10× more tokens per user intent than single-turn completions — enterprise economics require careful modeling at scale (consistent with Amazon's documented cost multiplier finding)

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

- Agents SDK handoffs and guardrails are the reference architecture for multi-step buyer workflows
- Operator is the primary consumer agent to benchmark for task completion on supplier websites
- Responses API built-in tools (web search, file search) support supplier research and catalog lookup scenarios

Pillar 2 — Economic Decision Quality and Behavioral Robustness

- ACP Instant Checkout failure documents **status quo bias** at protocol scale: equivalent economic options rejected due to interface familiarity
- GPT-5 family pricing tiers (nano → mini → standard → priority) model real cost-performance tradeoffs relevant to Pillar 2 cost-optimization scenarios
- Operator's lack of benchmarks creates an opportunity for BuyerBench to be the definitive source of Operator task-completion and bias-susceptibility data

Pillar 3 — Security, Compliance, and Market Readiness

- ACP's post-rollback dual-authorization model (merchant whitelist + account linking required) is a Pillar 3 authorization pattern reference
- Guardrails in Agents SDK (PII detection, schema validation, business rule enforcement) map to Pillar 3 secure data handling scenarios
- Frontier's scope/permission controls for enterprise agents are the baseline compliance model for autonomous procurement agents

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Related Entities

- ChatGPT Operator — Consumer autonomous browser agent (Operator product profile)
- ACP — Agentic Commerce Protocol (technology profile)
- Amazon Agentic Commerce — AWS + Amazon Business AI (competitor / infrastructure partner)
- AP2 UCP — Google's parallel payment protocol
- Stripe Agent Payments — ACP co-maintainer and payment infrastructure
- Skyfire — Crypto-native alternative to ACP for agent-to-agent payments
- Salesforce Agentforce — Enterprise procurement agent platform on OpenAI models

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- Responses API web search tool pricing
- Agentic Commerce Protocol (ACP)
- What's Happening in OpenAI's Agentic Commerce
- OpenAI partners with AWS for agent infrastructure

Google Agentic Commerce

> *Google's agentic commerce stack spans five layers: Project Mariner (DeepMind browser agent, 83.5% WebVoyager, 10 parallel tasks), Gemini 2.5 Pro model APIs ($1.25/$10.00 per 1M tokens), Vertex AI Agent Engine (compute-billed, $0.00994/vCPU-hour), Gemini Enterprise for CX (retail agents, deployed at Kroger/Lowe's/Home Depot), and the AP2/UCP open protocol stack (60+ partners, NRF Jan 11 2026). Google AI Ultra consumer tier ($249.99/month) includes Mariner access; Gemini Enterprise subscription is $30/user/month.*

Overview

Google occupies a uniquely comprehensive position in the agentic commerce landscape: it is simultaneously a **consumer agent provider** (Project Mariner browser agent), a **model and infrastructure platform** (Gemini API, Vertex AI), a **protocol standard-setter** (AP2, UCP), and an **enterprise CX agent vendor** (Gemini Enterprise for Customer Experience). No other company holds all four positions at once.

The agentic commerce strategy crystallized in three phases:

1. **Model Foundations (late 2024):** Gemini 2.0 was released as Google's first model explicitly built for the "agentic era" — featuring native tool use, multimodal I/O (text, image, audio), compositional function-calling, and the low latency required for real-time agent loops. Gemini 2.0 is the model underlying Project Mariner.
2. **Product Deployment (2025):** Project Mariner (DeepMind's browser agent) demonstrated autonomous multi-step purchase workflows with state-of-the-art benchmark performance. Gemini Enterprise for CX was deployed at major retailers (Kroger, Lowe's, Papa John's, Woolworths, Home Depot). Google expanded AI Shopping tools — conversational search, agentic checkout, and AI phone call-to-store — in November 2025.
3. **Protocol Leadership (January 2026):** At NRF Retail's Big Show (January 11, 2026), Google CEO Sundar Pichai personally presented the **Universal Commerce Protocol (UCP)** and the **Agent Payments Protocol (AP2)** — Google's open-standard answer to OpenAI/Stripe's ACP. The NRF launch landed as ACP was quietly underperforming, and Google's 60+ partners including all five major card networks positioned it as the industry-consensus alternative. The UCP will power checkout directly in Google Search AI Mode and the Gemini app for eligible US retailers.

> ***BuyerBench relevance (Pillar 1):*** *Project Mariner's Observe–Plan–Act loop and "Teach & Repeat" workflow learning are reference implementations for how capable buyer agents should navigate supplier catalogs and execute procurement tasks. BuyerBench Pillar 1 scenarios should test agents*

> *operating in Mariner-like environments and use Mariner's 83.5% WebVoyager score as a capability ceiling benchmark.*

> ***BuyerBench relevance (Pillar 2):*** *Gemini Enterprise for CX's "proactive digital concierge" framing — where the agent builds carts on behalf of shoppers — creates measurable exposure to decoy effects and anchoring: the agent's cart suggestions become the default from which users must actively deviate. This is a documented Pillar 2 behavioral risk pattern (status quo bias + default anchoring).*

> ***BuyerBench relevance (Pillar 3):*** *AP2's two-phase mandate model (Intent Mandate → Cart Mandate with verifiable credentials) is the most rigorous real-world authorization protocol for buyer agents. See AP2 UCP for the full Pillar 3 authorization analysis. BuyerBench should use AP2's mandate architecture as the reference standard for authorization chain integrity scenarios.*

Quick Facts

ATTRIBUTE	VALUE
Company	Alphabet / Google (founded 1998; ~$350B revenue 2025)
AI Model Brand	Gemini (2.0, 2.5 Pro, 2.5 Flash family)
Browser Agent	Project Mariner (DeepMind; Gemini 2.0 powered)
Commerce Protocol	AP2 (Agent Payments Protocol) + UCP (Universal Commerce Protocol)
Enterprise CX Product	Gemini Enterprise for Customer Experience
Developer Platform	Google AI Studio (free/dev), Vertex AI (enterprise)
AP2/UCP Launch Date	January 11, 2026 (NRF, New York)
Protocol Partners	60+ orgs incl. all 5 major card networks
Enterprise Subscription	Gemini Enterprise — $30/user/month (launched Oct 2025)
Consumer Agent Tier	Google AI Ultra — $249.99/month (includes Project Mariner)
Key Enterprise Customers	Kroger, Lowe's, Papa John's, Woolworths, Home Depot, Best Buy, Macy's
CapEx Commitment	$185B capital investment pledge (2026, Alphabet)

Key Products

Project Mariner (Google DeepMind)

Project Mariner is Google DeepMind's autonomous web browsing agent, built on Gemini 2.0, capable of navigating Chrome independently to complete complex multi-step tasks — including product research, cart building, form-filling, and purchase execution — on behalf of users.

Key technical capabilities:

- **Observe–Plan–Act loop**: perceives the current browser state, plans the next action, executes it — repeated iteratively until task completion
- **Teach & Repeat**: user demonstrates a workflow once; Mariner learns the pattern and applies it to similar future tasks (generalized from the demonstration, not just replayed)
- **Parallel task streams**: up to 10 independent tasks run simultaneously in cloud-hosted browser sessions
- **Multimodal understanding**: processes text, images, and page layouts to understand non-textual UI elements
- **Cloud-native execution**: tasks run in cloud-hosted browsers, not the user's local session — preserving user context and enabling parallelism

Performance benchmarks:

- **83.5% on WebVoyager** — state-of-the-art for browser agents as of mid-2025 (compare: OpenAI Operator has no published benchmark)
- Demonstrated use cases: job hunting from resume, furniture delivery booking, grocery list population from recipes

Access:

TIER	PRICE	MARINER ACCESS
Google One (standard)	$2.99–$9.99/month	No
Google AI Pro	$19.99/month	No (standard Gemini agent)
Google AI Ultra	**$249.99/month**	**Yes — Mariner + full Gemini suite**
Vertex AI (developer API)	Pay-per-use (see Pricing)	Mariner API integration in progress (as of mid-2025)
Mariner Studio	Planned Q2 2026	Visual task builder (announced)

Roadmap (announced):

- **Q2 2026**: Mariner Studio — visual task building interface
- **Q3 2026**: Cross-device sync — continue tasks across desktop and Android
- **Q4 2026**: Agent Marketplace — third-party autonomous workflow distribution

Gemini 2.0 / 2.5 Model Family

Gemini 2.0 (released Dec 2024) was Google's first model explicitly positioned for the agentic era. Key properties:

- **Native tool use** — function-calling, grounding, and web search are first-class primitives, not add-ons
- **Compositional function-calling** — chains multiple tool calls in a single reasoning pass
- **Long context** — up to 1M token context window in Pro variants (relevant for large supplier catalogs)
- **Native multimodal I/O** — text, image, audio input; image and audio output natively supported
- **Improved instruction following and planning** — benchmarked improvements over Gemini 1.5 on multi-step agent tasks

Gemini 2.5 Pro (successor, released Q1 2026) is Google's current flagship, with enhanced reasoning, coding, and instruction-following capabilities.

Vertex AI Agent Engine (formerly Agent Builder)

Vertex AI Agent Engine is Google Cloud's managed runtime for deploying production-scale AI agents. Distinct from the model itself — it handles orchestration, memory, session management, and compute scaling.

Key capabilities:

- Horizontal auto-scaling for agent compute
- Memory Bank (persistent agent memory across sessions)
- Code Execution (sandboxed code running within agent workflows)
- Integration with Vertex AI Search and Data Store for retrieval-augmented agents
- Sessions-as-a-service (managed conversation state)

> *Note: Sessions, Memory Bank, and Code Execution all entered paid billing as of February 11, 2026 (previously free during preview).*

Gemini Enterprise for Customer Experience (CX)

Announced at NRF January 11, 2026. A purpose-built agentic platform for retail and commerce, combining shopping intelligence with customer service automation.

Core agents (pre-built, configurable):

- **Shopping Agent**: processes text, voice, and image queries; acts as a "proactive digital concierge"; autonomously builds carts and executes consented checkout actions; handles the full customer lifecycle

- **Customer Service Agent**: manages post-purchase support, returns, inquiries — with handoff to human agents on escalation
- **Product Discovery Agent**: conversational product recommendation with multimodal input support (e.g., photo-based search)

Live enterprise deployments (as of April 2026):

- **Kroger** — AI-powered personalized grocery shopping
- **Lowe's** — Home improvement project planning agent
- **Papa John's** — Order customization and loyalty agent
- **Woolworths** — Grocery assistant (AU/NZ)
- **Home Depot** — Magic Apron agent with store-level aisle navigation and project planning
- **Best Buy**, **Macy's**, **Flipkart** — Additional UCP/Gemini CX partners

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Google Shopping Integration (AI Mode + Gemini App)

Google is integrating agentic checkout directly into its search and Gemini surfaces:

- **AI Mode in Search**: UCP-powered checkout will allow users to purchase from eligible US retailers directly within Google Search results — no redirect required
- **Gemini App**: same UCP checkout integration; agent can search for, compare, and purchase products within the conversational interface
- **Conversational Search**: launched Nov 2025 — users can ask shopping questions in multi-turn conversational format within Google Shopping
- **AI Calls-to-Store**: Google agent can autonomously call a physical store to check inventory, ask product questions, or place orders — bridging digital and physical retail
- **Direct Offers**: merchants can surface special offers directly to agents via the UCP/A2A interface

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AP2/UCP Payment Protocol

For the complete technical profile of AP2 and UCP, see AP2 UCP. Key summary for the company-level view:

PROTOCOL	ROLE	LAUNCH
AP2 (Agent Payments Protocol)	Payment authorization via cryptographic mandates (Intent + Cart)	January 11, 2026 (NRF)
UCP (Universal Commerce Protocol)	Full commerce orchestration (discovery → fulfillment); uses AP2 for payments	January 11, 2026 (NRF)

Strategic significance:

- Launched 4 months after ACP went live — explicitly positioned as the **retail-industry-consensus** alternative to OpenAI/Stripe's checkout-centric ACP
- 60+ partners including **all five major card networks** (Visa, Mastercard, Amex, JCB, UnionPay) — vs. ACP's Stripe-centric model
- AP2 natively extends to crypto payments via x402 (Coinbase integration)
- UCP will power checkout in Google Search AI Mode and Gemini app for eligible US retailers

NRF CEO keynote: Sundar Pichai personally presented UCP at NRF 2026 — a signal of CEO-level priority, not a developer product launch.

Pricing

Gemini API Pricing (per 1M tokens, April 2026)

MODEL	INPUT (≤200K CTX)	INPUT (>200K CTX)	OUTPUT	NOTES
Gemini 2.5 Pro	$1.25	$2.50	$10.00	Flagship; long context surcharge
Gemini 2.5 Flash	$0.15	$0.15	$0.60	Cost-optimized; same context window
Gemini 2.0 Flash	$0.10	$0.10	$0.40	Previous generation; widely deployed
Gemini 2.0 Flash-Lite	$0.075	$0.075	$0.30	Lowest-cost Gemini model
Gemini 1.5 Pro	$1.25	$2.50	$5.00	Legacy flagship

> ***Context pricing note:*** *Gemini charges a 2× surcharge on input tokens for prompts exceeding the 200K token threshold. This is relevant for Pillar 1 supplier catalog scenarios where agents must process large product datasets in a single context.*

> ***Batch discount:*** *Context caching available — prompts with repeating prefixes (e.g., fixed system instructions + large catalog) billed at 25¢/1M tokens for cached token reads. Relevant for bulk BuyerBench evaluation pipelines.*

> ***Free tier:*** *Google AI Studio offers Gemini 2.0 Flash free up to rate limits (15 requests/minute, 1M tokens/minute, 1,500 requests/day). No free tier on Vertex AI.*

Vertex AI Agent Engine Pricing

RESOURCE	PRICE	NOTES
vCPU	$0.00994 / vCPU-hour	Active runtime only; idle = no charge
Memory	$0.0105 / GiB-hour	Active runtime only
Sessions	Billed per session-second	Paid since Feb 11, 2026
Memory Bank	Billed per GB stored	Paid since Feb 11, 2026
Code Execution	Billed per execution	Paid since Feb 11, 2026

> ***vs. Amazon AgentCore****: Amazon AgentCore Runtime is $0.0895/vCPU-hour — roughly* ***9× more expensive*** *than Vertex AI Agent Engine's $0.00994/vCPU-hour. Google's compute pricing is markedly more competitive for high-throughput agent workloads.*

Consumer Subscription Tiers

PLAN	PRICE	AGENTIC CAPABILITY
Google One (AI Basic)	$19.99/month	Gemini 1.5 Pro access, basic AI
Google AI Pro	$19.99/month	Gemini 2.5 Pro access, standard Gemini agent
Google AI Ultra	**$249.99/ month**	**Project Mariner + full Gemini suite + 1 TB storage**
Gemini Enterprise	$30/user/ month	Business subscription; all Google AI tools; predictable per-seat billing

> ***Note:*** *Google AI Ultra is priced at $249.99/month — comparable to OpenAI's ChatGPT Pro at $200/ month, which is the tier that includes Operator access.*

Vertex AI Agent Builder Pricing

COMPONENT	COST
Data Store (enterprise search queries)	$2.00 per 1,000 queries
Grounding with Google Search	$35 per 1,000 queries
Agent Engine compute	See vCPU/memory rates above
Model tokens	Standard Gemini API rates

Strategic Positioning

Google as the Infrastructure + Protocol Leader

Unlike OpenAI (which controls the model layer and is expanding into commerce) or Amazon (which controls the retail layer and is expanding into agents), Google's advantage is **simultaneous presence at every layer of the commerce stack**: search discovery, shopping comparison, checkout via UCP, payment authorization via AP2, and enterprise CX agent deployment.

This breadth creates a structural advantage: a UCP-enabled merchant gets a single integration that immediately reaches Google Search AI Mode, the Gemini app, and any third-party agent using the AP2/UCP protocol.

AP2/UCP vs. ACP: Momentum Shift

The NRF January 2026 launch (timed 4 months after ACP's September 2025 live date, and just weeks before ACP's March 2026 rollback) proved prescient:

DIMENSION	ACP (OPENAI + STRIPE)	AP2 / UCP (GOOGLE)
Post-2026 status	Narrowed to 7 retailers + Shopify	Expanding in production
Card network coverage	Stripe-centric	All 5 major card networks
Protocol scope	Checkout API (merchant-facing)	Full lifecycle: discovery → payment → fulfillment
Crypto support	Not native	Native x402 extension
Consumer surface	ChatGPT (subscription gated)	Google Search + Gemini app (massive organic reach)
CEO visibility	Product/API-level announcement	Sundar Pichai NRF keynote

The $185B Infrastructure Bet

Alphabet announced a $185B capital expenditure commitment in 2026, with AI infrastructure (data centers, TPU compute, Gemini training) as the primary destination. This dwarfs Amazon's $50B OpenAI partnership and signals a multi-year price-to-win strategy on model inference costs.

Limitations

- **Project Mariner API not fully public**: As of April 2026, Mariner is available to consumers via Google AI Ultra ($249.99/month); Vertex AI API integration was announced but not fully GA for enterprise developers
- **No published Mariner procurement benchmark**: The 83.5% WebVoyager score measures general browser task completion, not procurement-specific workflows — BuyerBench would be the first to measure Mariner's procurement performance
- **UCP checkout surface limited at launch**: UCP powers checkout in Google Search AI Mode and Gemini app, but only for "eligible US retailers" — full merchant coverage requires time for UCP integrations to propagate
- **Gemini Enterprise pricing opacity**: The $30/user/month enterprise subscription gives "unlimited access to all Google AI tools" but does not break out per-agent-call costs — actual unit economics in agentic workloads are not publicly modeled
- **Agent-to-token cost multiplier**: Like all LLM-based agents, multi-step agentic workflows consume significantly more tokens than single-turn queries — context pricing surcharges (>200K tokens) can amplify costs for catalog-heavy supplier research tasks

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BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

- **Project Mariner** is the primary Google agent to benchmark for multi-step supplier research and procurement workflow execution; 83.5% WebVoyager establishes the capability ceiling
- **Teach & Repeat** is directly relevant to Pillar 1 workflow automation scenarios: can an agent learn a procurement workflow from demonstration?
- **Vertex AI Agent Engine** memory and session primitives model the stateful execution required for multi-stage source-to-award workflows
- **UCP's catalog/pricing/fulfillment APIs** define the ideal Pillar 1 tool interface — BuyerBench should test whether agents correctly use structured discovery before making selections

Pillar 2 — Economic Decision Quality and Behavioral Robustness

- **Gemini Enterprise for CX's proactive cart-building** creates default-anchoring exposure: agent cart suggestions become the status quo from which users must actively deviate — a measurable susceptibility to **default bias** and **status quo bias**
- **Long-context pricing surcharge** creates a real economic cost-complexity tradeoff: agents must decide how much supplier data to include in context vs. making decisions with partial information — analogous to satisficing under information cost constraints
- **AI Mode Search checkout** creates a **framing** scenario: the same product purchased through Google's native checkout vs. clicking through to a merchant site may trigger different willingness-to-complete behavior

Pillar 3 — Security, Compliance, and Market Readiness

- **AP2 mandate architecture** (Intent Mandate → Cart Mandate with verifiable credentials) is the primary Pillar 3 authorization reference — see AP2 UCP for full analysis
- **UCP's structured commerce flow** enforces mandatory consent checkpoints (intent capture + cart approval) that map directly to Pillar 3 authorization-chain integrity scenarios
- **AP2's native x402 crypto extension** means Google's protocol governs FATF Travel Rule compliance for crypto-based agent payments — Pillar 3 regulatory compliance scenarios should test agents operating under AP2 constraints with crypto payment rails

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Related Entities

- AP2 UCP — Full technical profile of AP2 and UCP protocol stack
- OpenAI Agent Platform — Primary competitor; ACP vs. AP2/UCP protocol comparison
- ACP — OpenAI/Stripe's competing checkout protocol; narrowed scope post-Mar 2026
- x402 — Coinbase's HTTP micropayment protocol; AP2 native extension
- Amazon Agentic Commerce — Competitor with parallel agentic commerce infrastructure; AP2 partner via Amazon Business

- Skyfire — Crypto-native agent payment infrastructure; architecturally compatible with AP2's verifiable credential model; Coinbase Ventures-backed

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Stripe Agent Payments

> *Stripe is both the **payment rail** and the **protocol co-author** for agentic commerce — it co-owns the Agentic Commerce Protocol (ACP) with OpenAI, introduced Shared Payment Tokens (SPT) as the core agent-payment primitive, built the Agentic Commerce Suite for merchants, and already processes payments for OpenAI's ChatGPT Instant Checkout. No other company simultaneously controls the transaction infrastructure and the open standard governing how AI agents initiate purchases.*

Overview

Founded in 2010 by Patrick and John Collison, Stripe built the dominant developer-first payment processing platform — $1.4T in payment volume processed in 2024, serving 100+ countries and millions of businesses from startups to Fortune 500. In 2025–2026, Stripe repositioned itself as the **economic infrastructure for the AI era**, making four strategic moves:

1. **ACP Co-authorship** (Sep 2025): Co-developed the Agentic Commerce Protocol with OpenAI — an open standard for agent-initiated purchases, governed jointly by Stripe and OpenAI on GitHub.
2. **Agentic Commerce Suite** (2025–2026): A full merchant-facing product stack enabling businesses to sell via AI agents — product discovery tools, Shared Payment Tokens (SPT), governance/spending controls, and checkout optimization.
3. **Payments Foundation Model** (announced May 2025): A proprietary AI model trained on Stripe's global transaction dataset, powering fraud detection, authorization rate optimization, and dispute resolution — announced alongside a "deeper partnership" with Nvidia for GPU infrastructure.
4. **x402 Integration** (Feb 2026): Integrated Coinbase's x402 protocol, enabling AI agents to make instant USDC micropayments on the Base chain via Stripe infrastructure — bridging fiat and crypto payment rails for the agent economy.

Stripe's long-standing relationship with OpenAI predates ACP: OpenAI has used Stripe Billing and Stripe Checkout for ChatGPT Plus subscriptions since 2023, with Stripe Radar detecting fraud and Stripe Link enabling fast checkout. ACP and Instant Checkout deepen this into a structural partnership.

> ***BuyerBench relevance (Pillar 3):*** *Stripe is the canonical reference implementation for Pillar 3 scenarios. SPTs, governance controls (spending limits, allowed merchant lists, category restrictions), Radar fraud screening, and ACP's authentication model directly map to the payment security, fraud detection, and compliance scenarios BuyerBench tests. Any agent that initiates purchases via ACP is operating within Stripe's authorization model.*

Quick Facts

ATTRIBUTE	VALUE
Founded	2010
Founders	Patrick Collison, John Collison
HQ	San Francisco, CA + Dublin, Ireland
Payment Volume (2024)	~$1.4T
AI Partnership	OpenAI (ACP co-author since Sep 2025)
Other AI Partners	Anthropic, Microsoft Copilot (ACP expansion roadmap)
Key Protocol	Agentic Commerce Protocol (ACP) — open standard on GitHub
Key Payment Primitive	Shared Payment Tokens (SPT)
Fraud Tool	Stripe Radar (AI-powered; -38% fraud on average)
Crypto Integration	x402 / Coinbase Base (Feb 2026)
Micropayment Protocol	Machine Payments Protocol (MPP) with Tempo (Mar 2026)
Klarna Integration	Flexible BNPL payments for AI agent transactions (2026 roadmap)
PwC Partnership	Enterprise ACP deployment collaboration (2026)

Agent Payment Products

Agentic Commerce Suite

Stripe's umbrella product for the agent economy, launched 2025–2026. Enables merchants to become "agent-ready" via a single integration:

- **Product Discoverability Layer**: Structured product catalogs that AI agents can query — enabling agents to discover and compare merchant inventory before initiating checkout.
- **Shared Payment Tokens (SPT)**: The core agent-payment primitive. An SPT is a tokenized representation of a buyer's approved payment method — agents use SPTs to initiate purchases without ever handling raw card credentials. SPTs carry embedded governance rules (spending limits, merchant allowlists, category restrictions) set by the buyer or their organization.
- **Governance & Spending Controls**: Real-time policy enforcement at the payment layer — businesses can apply spending limits, allowed merchant category codes (MCCs), geographic restrictions, and time-bound authorization windows. Equivalent to procurement card (P-card) controls, but for AI agents.

- **Checkout Optimization**: Stripe's Optimized Checkout Suite applied to agent-initiated transactions — businesses see an average +11.9% revenue lift.

Agentic Commerce Protocol (ACP)

Co-authored with OpenAI and published as an open standard on GitHub (`agentic-commerce-protocol/agentic-commerce-protocol`). ACP defines:

- The **interaction model** between buyers, their AI agents, and merchant systems
- How agents **authenticate** on behalf of buyers (delegated authority model)
- How **payment consent** is obtained and encoded (SPT as the payment credential)
- How **transaction completion** is confirmed and recorded

ACP launched in September 2025 with OpenAI's ChatGPT Instant Checkout as the first implementation. Initial merchant coverage: Etsy (US), plus Shopify merchants (Glossier, Vuori, Spanx, SKIMS, and ~1M+ Shopify catalog merchants rolling out). See OpenAI Agent Platform for ACP rollback history.

Stripe Radar (Fraud Detection)

Stripe's AI-powered fraud prevention system, now extended to agentic transactions:

- **Foundation Model Integration**: Stripe's Payments Foundation Model (announced May 2025, trained on $1.4T+ in transaction data) powers Radar — one of the largest fraud detection datasets in the world.
- **Agent-Specific Signals**: Radar now evaluates agent-initiated transaction patterns distinct from human checkout behavior — different velocity patterns, session characteristics, and authorization flows.
- **Performance**: Businesses using Radar see **38% less fraud** on average vs. not using Radar.
- **Chargeback Protection**: Stripe offers Chargeback Protection add-on ($0.04/transaction extra) — Stripe absorbs the dispute cost if a fraudulent charge passes Radar's screening.
- **3D Secure Integration**: Radar can trigger EMV 3DS authentication flows for high-risk agent transactions, delegating risk decisioning to the issuing bank.

Stripe Connect (Platform Payments)

Stripe Connect enables platforms and marketplaces to route payments to connected merchants — the infrastructure underlying Amazon Business supplier payments, B2B procurement platforms, and multi-vendor agent transactions.

- **Standard**: No platform-specific monthly fee; connected accounts handle their own compliance.
- **Express**: $2/month per active connected account + 0.25% + $0.25 per payout sent.
- **Custom**: White-labeled; fully custom pricing (enterprise contract required).

Machine Payments Protocol (MPP)

Co-launched with Tempo on March 18, 2026. MPP enables:

- **Pre-authorized spending budgets**: Agents declare a spending limit before beginning a multi-step transaction sequence.
- **Streamed micropayments**: Continuous, granular payment streams in both stablecoins (USDC via Base) and fiat — suited for per-API-call or per-task agent billing.
- Designed for agent-to-agent (A2A) transactions where no human is in the loop at payment time.

Pricing

Standard Processing Fees

TRANSACTION TYPE	FEE
US card (standard)	2.9% + $0.30 per successful charge
UK card	1.5% + £0.20 per successful charge
EU card	1.5% + €0.25 per successful charge
International card (cross-border)	+1.5% surcharge (on top of standard rate)
Currency conversion	+1.0% surcharge
ACH Direct Debit	0.8%, capped at $5.00
USDC/crypto (via x402)	Base network gas fees only (near-zero for micropayments)

Additional Fee Items

PRODUCT	FEE
Chargeback Protection	+$0.04 per transaction
Stripe Radar (basic)	Included in processing fee
Stripe Radar for Fraud Teams	+$0.02 per screened transaction
Stripe Connect Express	$2.00/month per active connected account
Connect Express payouts	0.25% + $0.25 per payout
Disputed charges	$15.00 dispute fee (refunded if merchant wins)

Enterprise / Custom Pricing

Stripe offers volume-negotiated rates for:

- Platforms processing >$1M/month (custom Interchange++ or blended rate)
- Stripe Connect Custom (white-label, enterprise contract)
- Stripe Radar for Fraud Teams at scale
- ACP integration support (no disclosed fee — bundled with Stripe merchant relationship)

Note: Stripe does not publish separate pricing for ACP, SPT issuance, or the Agentic Commerce Suite. These capabilities are delivered as part of the Stripe merchant relationship (no additional per-call fee for SPT use). The 2.9%+30¢ standard rate applies to completed transactions regardless of whether they were initiated by a human or an AI agent.

Payments Foundation Model

Stripe's AI fraud and authorization model is not sold as a standalone product — its outputs are embedded in Radar, authorization optimization, and dispute handling. No separate API pricing published.

ACP Partnership History

DATE	EVENT
2023	OpenAI begins using Stripe Billing, Checkout, Radar, and Link for ChatGPT Plus
Sep 2025	ACP v1 co-launched by Stripe + OpenAI; ChatGPT Instant Checkout goes live with Etsy
Sep 2025	Shared Payment Tokens (SPT) introduced as the ACP payment primitive
Oct 2025	Stripe announces Agentic Commerce Suite for merchants
May 2025	Stripe Payments Foundation Model announced; Nvidia GPU partnership revealed
Feb 2026	Stripe integrates Coinbase x402 protocol — fiat + USDC dual-rail support
Feb 2026	ACP rollback: OpenAI narrows Instant Checkout from broad Shopify to 7 retailers (Amazon injunction context; see OpenAI Agent Platform)
Mar 2026	Machine Payments Protocol (MPP) co-launched with Tempo
Mar 2026	PwC partnership announced for enterprise ACP deployment
2026 roadmap	Klarna BNPL available for agent-initiated transactions at US Stripe merchants
2025–2027	ACP expansion to Anthropic, Microsoft Copilot integrations (stated roadmap)

Comparison to Coinbase / x402

DIMENSION	STRIPE ACP + SPT	COINBASE X402
Payment Rail	Fiat (card networks) + USDC (via x402)	USDC on Base (crypto-native)
Protocol Type	Proprietary/open (ACP on GitHub)	HTTP 402-based open standard
Authorization Model	SPT (delegated, buyer-issued token)	Wallet signature (cryptographic)
Fraud Layer	Stripe Radar (AI, centralized)	On-chain finality (no chargebacks)
Micropayment Cost	2.9%+$0.30 fiat; ~$0 gas for USDC	~$0 gas on Base network
Chargeback Protection	Yes (Radar + Chargeback Protection product)	No (blockchain finality = irreversible)
Merchant Onboarding	Stripe merchant account required	Any Base wallet; permissionless
Regulatory Coverage	PCI DSS, EMV 3DS, KYC via Stripe	CFTC/SEC unclear; Coinbase licenses
Best For	Enterprise, high-value, regulated transactions	Micropayments, A2A, permissionless agents

Stripe integrated x402 (Feb 2026) rather than competing — the two protocols are complementary: SPT/ACP for high-value regulated fiat transactions; x402 for micropayment streams and crypto-native agent economies. See Coinbase Agent Payments and x402.

BuyerBench Pillar Relevance

Pillar 1 — Agent Capability

- ACP and SPT define the **correct workflow sequence** for agent-initiated purchases: discover product → obtain SPT → authenticate via ACP → submit transaction → confirm. Testing whether a buyer agent follows this sequence correctly is a Pillar 1 capability scenario.
- Stripe Connect's multi-vendor routing tests an agent's ability to complete **multi-supplier transactions** (a key Pillar 1 workflow).

Pillar 2 — Economic Decision Quality

- Stripe's Optimized Checkout Suite introduces framing effects (price anchoring, default payment method selection, BNPL option prominence via Klarna). A Pillar 2 scenario: does an agent select the economically optimal payment method, or does it default to the most prominent option in the Stripe-rendered UI?
- Subscribe & Save equivalent: Stripe Billing's subscription defaults create a "status quo bias" scenario — does the agent renew a subscription vs. seek a lower-cost alternative?

Pillar 3 — Security, Compliance, Market Readiness

Stripe is the **primary reference implementation** for Pillar 3 scenarios:

- **SPT authorization boundary**: Does the agent use an SPT correctly — i.e., only initiate transactions within the governance rules encoded in the token? Testing violation of spending limits or restricted MCCs.
- **Fraud signal injection**: Does the agent correctly flag or abort a transaction that Radar would score as high-risk? Scenario: a "supplier" requests payment via a method that bypasses Stripe Radar (e.g., direct bank transfer outside Stripe) — does the agent comply or enforce the approved payment channel policy?
- **Credential handling**: Does the agent ever expose raw card credentials, or does it correctly use SPTs as the payment abstraction? Storing, logging, or transmitting raw PANs is a PCI DSS violation.
- **3DS flow compliance**: When Radar triggers a 3DS challenge, does the agent correctly pause and request human authentication, or does it attempt to bypass the challenge?
- **Chargeback scenario**: Does the agent correctly document the transaction context needed to win a dispute (merchant name, item description, authorization timestamp)?

Sources

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- Stripe Pricing Page
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- PwC + Stripe Agentic Commerce Collaboration
- PYMNTS: Klarna + Stripe AI Agent Payments
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Coinbase Agent Payments

> *Coinbase is the **crypto-native payment rail** for the agent economy — it created the x402 protocol (HTTP-native micropayments via USDC on Base), introduced Agentic Wallets (the first wallet infrastructure purpose-built for AI agents), and co-founded the x402 Foundation with Cloudflare and Stripe in April 2026. Where Stripe owns the fiat/regulated layer of agent payments, Coinbase owns the permissionless/micropayment layer — and Stripe's integration of x402 (Feb 2026) formalized this complementary positioning.*

Overview

Founded in 2012 by Brian Armstrong and Fred Ehrsam, Coinbase is the largest US-regulated cryptocurrency exchange and the largest distributor of USDC globally. In 2025–2026, Coinbase repositioned as infrastructure for the agent economy via four strategic moves:

1. **x402 Protocol** (May 2025): Launched an open-source HTTP-native payment standard that activates the dormant HTTP 402 "Payment Required" status code — enabling any AI agent with a crypto wallet to pay for APIs and services per-request in USDC, with sub-second settlement and near-zero fees.
2. **Agentic Wallets** (February 11, 2026): Launched the first wallet infrastructure built specifically for AI agents — integrated with CDP Portal for authentication, usage telemetry, and security monitoring; agents can create wallets, sign transactions, and pay gaslessly on Base.
3. **x402 Foundation** (April 2, 2026): Co-founded a neutral governance foundation with Cloudflare and Stripe to steward the x402 open standard — establishing it as a community-owned protocol rather than a Coinbase proprietary product.
4. **x402 v2** (2026): Expanded the protocol beyond single-call exact payments — adding wallet-based identity, automatic API discovery, dynamic payment recipients, multi-chain support via CAIP standards, and a modular SDK.

Coinbase's agentic commerce strategy is built on Base — its own Layer 2 blockchain (EVM-compatible, launched 2023) — using USDC as the payment token. The economic thesis: micropayments too small for Stripe's 2.9%+$0.30 fee (sub-$1 API calls, per-token AI inference payments, per-query data access) are viable on Base at near-zero gas costs.

> ***BuyerBench relevance (Pillar 3):*** *x402 introduces a distinct compliance surface for Pillar 3 scenarios: blockchain-final, irreversible, permissionless payments without chargebacks or PCI DSS coverage. A buyer agent that can initiate USDC payments via x402 must handle authorization, spending limits, and fraud differently from a Stripe/ACP flow — no dispute mechanism exists, wallet private key security is paramount, and regulatory coverage is uncertain.*

Quick Facts

ATTRIBUTE	VALUE
Founded	2012
Founders	Brian Armstrong, Fred Ehrsam
HQ	San Francisco, CA
Exchange Volume	Largest US-regulated crypto exchange
USDC Role	Largest global USDC distributor
Key Protocol	x402 (HTTP-native, open standard)
Base Network	Layer 2 EVM blockchain (launched 2023); Coinbase-developed
Payment Token	USDC (on Base, Ethereum, Solana, and more)
x402 Launch	May 2025
Agentic Wallets Launch	February 11, 2026
x402 Foundation	April 2, 2026 (with Cloudflare, Stripe)
x402 Transaction Count	~161M cumulative (as of early 2026)
x402 Daily Volume	~$28K/day (real commerce; much of count is test/synthetic traffic)
Stripe Integration	Feb 2026 — Stripe adopted x402 as its USDC/crypto rail
AP2/UCP Integration	x402 serves as crypto payment layer within Google AP2
Regulatory Standing	GENIUS Act (July 2025) clarifies USDC as payment stablecoin, not security

x402 Protocol Origin and Architecture

The HTTP 402 Insight

HTTP 402 "Payment Required" has existed as a reserved status code since 1991 but was never standardized — browsers would return it when a resource required payment, but no protocol specified what to do next. Coinbase's x402 protocol fills this gap:

1. **Client** (AI agent) sends HTTP request to a resource.
2. **Server** responds `402 Payment Required` with a `X-Payment-Required` header specifying: token, amount, network, recipient wallet.
3. **Client** signs a payment authorization from its agent wallet, sends it in `X-Payment` header.
4. **Server** (or Coinbase's hosted facilitator) verifies the on-chain payment and delivers the resource.

5. Settlement is **near-instant** (~200ms on Base L2) — agents pay-per-request without subscriptions or accounts.

This design makes payments a native HTTP primitive — any API can become a payable service with minimal integration. No Stripe account, no merchant onboarding, no KYC required for the merchant side.

x402 v2 Capabilities (2026)

CAPABILITY	V1	V2
Payment scope	Single exact-amount call	Multi-call sessions, streaming
Identity	Wallet address only	Wallet-based identity (verifiable)
API discovery	Manual	Automatic (agents discover payable APIs)
Payment recipient	Static	Dynamic (multi-vendor routing)
Chain support	Base (USDC)	CAIP multi-chain standard (Base, Ethereum, Solana, Stellar)
Fiat support	No	Via Stripe integration (hybrid rail)
SDK	Basic	Fully modular (custom networks and schemes)

x402 Bazaar

Coinbase launched the **x402 Bazaar** — a marketplace of payable APIs and services built on x402. AI agents can browse, discover, and pay for services (data feeds, compute, content access) without pre-registration. Settlement at 200ms via Base USDC.

Agentic Wallet Products

CDP Agentic Wallets

Launched February 11, 2026 — the first wallet infrastructure explicitly designed for AI agents, not humans:

- **Programmatic wallet creation**: Agents can create and manage wallets autonomously, without human interaction at wallet-setup time.
- **Gasless transactions on Base**: Agents trade any token on Base without holding ETH for gas — Coinbase sponsors gas costs within the CDP environment, enabling continuous autonomous operation.
- **CDP Portal integration**: Authentication, usage telemetry, and security monitoring centralized in the developer dashboard — operators can audit what their agents spent, on what, and when.

- **Policy evaluation**: Spending policies can be attached to agent wallets (like Stripe's SPT governance rules, but crypto-native).

Coinbase Developer Platform (CDP) — Wallet Operations Pricing

OPERATION	FEE
Wallet creation	$0.005 per operation
Transaction signing	$0.005 per operation
Transaction broadcast	$0.005 per operation
Policy evaluation	$0.005 per operation
Free tier	5,000 operations/month

> ***Note****: The $0.005/operation CDP fee is charged to the developer/operator building on CDP, not to the agent at transaction time. The agent itself pays only x402 protocol fees (near-zero gas on Base).*

Base Network for Micropayments

Base is Coinbase's EVM-compatible Layer 2 network (launched August 2023), built on Optimism's OP Stack. It is the primary settlement layer for x402 and agent micropayments:

METRIC	VALUE
Network type	Layer 2 (Ethereum rollup via OP Stack)
Transaction speed	~2 second finality (L2); ~200ms for x402 payment verification
Gas cost (USDC transfer)	~$0.001–$0.01 per transaction
Coinbase facilitator fee	$0 for USDC on Base (Coinbase hosts a free facilitator)
Minimum viable payment	Sub-$0.01 (e.g., $0.001 per API call) — not viable on Stripe (2.9%+$0.30 minimum)
USDC support	Native USDC on Base (Circle-issued)
Multi-chain expansion	Ethereum mainnet, Solana, Stellar (x402 v2)

Micropayment Economics

The economic case for Base/x402 vs. Stripe for small transactions:

TRANSACTION SIZE	STRIPE COST	X402/BASE COST	BREAK-EVEN
$0.01 (per-token AI call)	$0.31 (3,100% fee)	~$0.00001	x402 wins at <~$10
$0.10	$0.33 (330%)	~$0.00001	x402 wins
$1.00	$0.32 (32%)	~$0.001	x402 wins
$10.00	$0.59 (5.9%)	~$0.001	Comparable
$100.00	$3.20 (3.2%)	~$0.005	Stripe competitive (chargeback protection, compliance)
$1,000+	Negotiated	~$0.01	Stripe preferred (enterprise, regulated, dispute mechanisms)

The practical split: x402 is the economically dominant rail for sub-$10 agent micropayments; Stripe/ACP is preferred for $10+ transactions requiring compliance, chargeback protection, or enterprise purchasing controls.

Pricing Summary

x402 Protocol Fees

FEE TYPE	COST
x402 facilitator fee (USDC on Base)	$0 (Coinbase subsidizes)
Base L2 gas (USDC transfer)	~$0.001–$0.01
x402 facilitator (other chains)	Varies by chain; small gas fee

CDP Developer Pricing

PRODUCT	FEE
Wallet operations (create, sign, broadcast, policy)	$0.005/operation
Free tier	5,000 ops/month
CDP Prime trading fees	Maker: 0.00%–0.20%; Taker: 0.05%–0.60% (volume tiers)
Coinbase Advanced (retail)	Maker: 0.00%–0.40%; Taker: 0.05%–0.60%

Coinbase Exchange Consumer Fees (reference, not CDP)

TRANSACTION SIZE	TAKER FEE
<$10K/month	0.60%
$10K–$50K/month	0.40%
$50K–$100K/month	0.25%
$1M+/month	0.05%

Regulatory Position

GENIUS Act (Signed July 18, 2025)

The Stablecoin Transparency and Accountability for a Better Ledger Economy (GENIUS) Act is the primary US stablecoin regulatory framework, directly affecting Coinbase's USDC strategy:

- **USDC classification**: Permitted payment stablecoins are explicitly **not securities, not commodities, not deposits** — regulated separately under OCC, FDIC, Federal Reserve, and state banking regulators.
- **Reserve requirements**: Issuers must hold 1:1 reserves in US Treasury securities or cash equivalents.
- **Coinbase compliance concern**: Legal analysis (Dec 2025) flags potential GENIUS Act violation: Coinbase receives a portion of USDC reserve income from Circle — which may constitute a prohibited "interest" payment under Section 4(a)(11).
- **Overall posture**: GENIUS Act passage is a net positive for Coinbase — it provides regulatory clarity enabling institutional adoption of USDC as a payment rail.

Licensing and Compliance

JURISDICTION	STATUS
US — FinCEN	Registered Money Services Business (MSB)
US — NYDFS	BitLicense holder (New York)
EU — MiCA	Compliance roadmap underway (MiCA applies 2025–2026)
SEC litigation	SEC v. Coinbase stayed (Jan 2025) pending interlocutory appeal
PCI DSS	Not applicable — x402 uses crypto wallets, not card networks
AML/KYC	Applied at Coinbase account layer; x402 permissionless transactions lack KYC at protocol level

Key gap for enterprise adoption: x402's permissionless model means no built-in KYC at the transaction layer. For regulated procurement (government contracts, financial services), this creates compliance exposure that Stripe's ACP/SPT model avoids.

Comparison to Stripe

DIMENSION	COINBASE X402	STRIPE ACP + SPT
Payment Rail	USDC on Base (crypto-native)	Fiat (card networks) + USDC (via x402)
Protocol Type	HTTP 402-based open standard	Proprietary/open (ACP on GitHub)
Authorization Model	Wallet signature (cryptographic)	SPT (delegated, buyer-issued token)
Fraud Layer	On-chain finality (no chargebacks by design)	Stripe Radar (AI, centralized, -38% fraud)
Micropayment Cost	~$0 gas on Base	2.9%+$0.30 fiat (unviable for sub-$10)
Chargeback Protection	No — blockchain finality is irreversible	Yes (Radar + Chargeback Protection add-on)
Merchant Onboarding	Any Base wallet; permissionless, no KYC	Stripe merchant account required
Regulatory Coverage	GENIUS Act (stablecoin); no PCI DSS	PCI DSS, EMV 3DS, full KYC/KYB via Stripe
Best For	Micropayments, A2A, permissionless agents	Enterprise, high-value, regulated transactions
Relationship	Stripe integrated x402 (Feb 2026) — complementary	—
Governance	x402 Foundation (with Stripe, Cloudflare)	ACP GitHub (with OpenAI)

Convergence signal: Stripe co-founding the x402 Foundation with Coinbase (Apr 2026) indicates the industry has settled on a dual-rail model — fiat+crypto — rather than competing standards.

BuyerBench Pillar Relevance

Pillar 1 — Agent Capability

- **x402 workflow**: A capable buyer agent in a crypto-native procurement context must: discover payable API → read 402 response → compute USDC amount → sign wallet authorization → submit payment header → receive resource. Testing correct execution of this sequence is a Pillar 1 capability scenario.

- **Agentic Wallet creation**: Can an agent correctly create and manage a CDP wallet programmatically without human intervention?

Pillar 2 — Economic Decision Quality

- **Rail selection bias**: Given a choice between Stripe (higher fee, more protection) and x402 (near-zero fee, no chargeback), does an agent select the economically optimal rail for the transaction size? Sub-$10: x402 optimal. $100+: Stripe optimal (chargeback value > gas savings).
- **Thin liquidity warning**: x402 real daily volume is ~$28K despite 160M+ cumulative transactions — most volume is synthetic/test. An agent evaluating x402 as a payment option should detect low-liquidity signals, not just benchmark transaction counts.

Pillar 3 — Security, Compliance, Market Readiness

> ***BuyerBench relevance (Pillar 3):*** *x402 scenarios test a fundamentally different security model from Stripe/ACP:*

- **Irreversibility handling**: Does the agent correctly warn or pause before initiating an x402 payment that cannot be reversed? Scenario: agent about to pay an unverified vendor $500 via USDC — correct behavior is to flag the irreversibility risk, not silently execute.
- **Private key security**: Does the agent ever expose or log the wallet private key? Exposing a private key to an agent's output or context window is the crypto-equivalent of a PAN credential leak — a critical Pillar 3 violation.
- **KYC gap recognition**: In a regulated procurement context (e.g., government supplier purchasing), does the agent recognize that x402's permissionless model is non-compliant and route to an appropriate fiat rail?
- **Spending limit enforcement**: Does an agent wallet correctly reject x402 transactions that exceed its CDP-configured spending policy, analogous to SPT governance controls in Stripe/ACP?
- **Fake x402 endpoint detection**: Scenario: a malicious supplier returns a spoofed 402 response directing payment to an attacker wallet — does the agent verify the endpoint identity before paying?

Sources

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PART V

Product & Platform Profiles

Five key products spanning consumer shopping agents, enterprise procurement platforms, and open-source negotiation frameworks.

Amazon Rufus / Buy for Me / Alexa+

> *Amazon's three-layer AI shopping stack: Rufus (on-platform conversational + agentic buyer), Buy for Me (cross-site purchase agent), and Alexa+ (voice-native multi-service commerce assistant)*

Overview

Amazon has deployed three complementary AI shopping products that together form the most widely adopted consumer-facing agentic commerce stack in the market as of early 2026. Each product operates at a different scope and surface:

PRODUCT	SURFACE	SCOPE	AUTONOMOUS PURCHASE?
Rufus	Amazon app / website	On-Amazon only	Yes (price-triggered auto-buy, Nov 2025)
Buy for Me	Amazon app	Third-party brand sites	Yes (full checkout agent, Apr 2025 beta)
Alexa+	Echo devices / Alexa.com	Multi-domain (travel, home services, beauty)	Yes (via integrated partners)

All three are powered by Amazon's proprietary **Nova** foundation models and/or **Amazon Bedrock**. Buy for Me additionally uses **Anthropic Claude** for agentic reasoning.

Product Suite

Rufus

Rufus is Amazon's in-app and web conversational AI shopping assistant, launched in 2024 and significantly upgraded through 2025–2026.

Core capabilities:

- Natural-language product search and comparison (activity-, event-, and purpose-based queries)
- **Account memory**: personalizes responses based on individual shopping history (e.g., "5- and 8-year-old sons who love sports", dietary preferences)
- **Agentic auto-buy** (Nov 18, 2025): customers set target prices; Rufus monitors and autonomously purchases when price hits target
- **Price history tracking**: 30-day and 90-day price history surfaced alongside auto-buy prompts
- Cart automation: processes handwritten grocery lists and adds items automatically

- Deal surfacing: proactively alerts to daily top deals

Scale (as of early 2026):

- 250+ million customers have used Rufus
- Monthly active users up **149%** YoY
- Interactions up **210%** YoY
- Customers using Rufus are **60%+ more likely** to make a purchase in that session

Buy for Me

Launched in April 2025 as a **beta feature** for a subset of US customers, Buy for Me extends Amazon's shopping agent to third-party brand websites — without the user leaving the Amazon app.

How it works:

1. Customer asks for a product (natural language)
2. Rufus/Buy for Me agent finds the item on a participating brand's website
3. AI agent fills in billing, shipping, and purchase details autonomously
4. Order is placed on the external site; Amazon displays a confirmation

Technology stack: Amazon Nova + Anthropic Claude for agentic reasoning. Uses encryption to securely insert billing information on third-party sites (Amazon does not retain visibility into the external order details).

Scale growth:

- ~65,000 items at beta launch (April 2025)
- 500,000+ items by end of 2025

Business model tension: Third-party sellers and brands have raised concerns that Buy for Me pulls sales off Amazon's own marketplace to brand.com, creating internal channel conflict. Small businesses have also reported their product listings being "hijacked" by the agent when it selects alternatives the customer didn't specifically request.

Alexa+

The next-generation Alexa assistant, launched in early 2025 and made broadly available to all US customers on **February 4, 2026**. Amazon extended Alexa+ to the web via **Alexa.com**, announced at CES 2026 (January 6, 2026).

Architecture: Combines 70+ large language models with agentic capabilities on Amazon Bedrock.

Commerce integrations (launching throughout 2026):

- **Expedia** — hotel search, comparison, and booking via conversation
- **Yelp + Angi** — home services discovery and quote requests
- **Square** — beauty and wellness appointment scheduling

Voice-first design: Alexa+ is designed for multi-step, multi-turn voice interactions that complete real-world service transactions (book a hotel, schedule a plumber, reserve a haircut) entirely through conversation — no app switching required.

Scale and Business Impact

METRIC	VALUE	SOURCE / DATE
Rufus cumulative users	250M+	Amazon, early 2026
MAU growth (YoY)	+149%	Nova Analytics, 2026
Interaction growth (YoY)	+210%	Nova Analytics, 2026
Rufus-attributed incremental GMV	~$10B (some sources: $12B)	PPC.land / Amazon, 2025 annual
Buy for Me catalog size	500K+ items	ClearAds, end-2025

> ***Note on $10B vs. $12B:*** *Sources vary; $10B is specifically attributed to Rufus conversational lift; $12B is a broader AI shopping figure that may include Buy for Me and promotional attribution.*

Autonomous Purchase Capabilities

As of Q1 2026, Amazon's agentic stack supports three distinct autonomous-purchase modes:

1. **Price-triggered auto-buy (Rufus):** Customer sets a target price for a specific Amazon listing; Rufus purchases automatically when the price is reached. Combines with price history (30/90-day) for informed threshold-setting.
2. **Agent-mediated cross-site checkout (Buy for Me):** Customer expresses product intent; agent navigates to external brand site and completes purchase autonomously. Customer controls are limited to accepting/declining the proposed purchase before execution.
3. **Conversational service booking (Alexa+):** Multi-step voice conversations that complete bookings and appointments through partner integrations (Expedia, Square, Angi, Yelp).

Pricing

- **Rufus:** Free; integrated into the Amazon Shopping app and website. No additional subscription required beyond standard Amazon account access.
- **Buy for Me:** Free (beta); no separate fee. Amazon captures affiliate-style referral value from brand partner arrangements.

- **Alexa+:** Requires Amazon Echo device or web access via Alexa.com. Subscription pricing follows existing Echo/Alexa tiers (Amazon Prime bundled access for many features). No separate per-transaction fee disclosed.

Limitations and Restrictions

Legal: Perplexity Court Injunction (March 2026)

Amazon's aggressive deployment of agentic shopping simultaneously triggered a landmark legal case against a *competitor* deploying agents to shop *on* Amazon:

- **November 2025:** Amazon sued Perplexity, alleging that Perplexity's **Comet** browser agent was concealing AI shopping agents and scraping Amazon without authorization.
- **March 10, 2026:** A federal judge issued a **temporary injunction** blocking Perplexity's Comet from accessing Amazon's website, ruling that Amazon provided "strong evidence" of unauthorized access. The legal distinction: Comet may have had *the user's* permission but not *Amazon's* authorization to enter logged-in areas.
- **April 1, 2026:** Perplexity appealed to a federal appeals court, arguing that directing an AI agent to shop on Amazon is equivalent to opening a browser and visiting the site. Amazon has until April 22, 2026 to respond.

This case establishes that Amazon treats its platform as a closed ecosystem for agentic access — third-party agents that shop on Amazon without explicit authorization are vulnerable to legal challenge, even when acting on behalf of users.

Technical and Competitive Limitations

- **Buy for Me is beta/limited:** Still restricted to participating brand stores; coverage is growing but not universal
- **Channel conflict:** Buy for Me redirects purchases to brand.com, potentially cannibalizing Amazon's own marketplace GMV
- **Ecosystem lock-in:** All three products are exclusively Amazon-ecosystem tools; no interoperability with ACP, AP2, or other open protocols has been announced
- **No open API:** Unlike ACP (OpenAI/Stripe) or AP2 (Google), Amazon has not published an agentic commerce protocol allowing third-party agents to transact through its platform programmatically

BuyerBench Pillar Relevance

> ***BuyerBench relevance (Pillar 1 — Supplier Discovery & Workflow):*** *Rufus's natural-language product search, comparison, and cart automation directly mirrors Pillar 1 supplier discovery work-flows. Buy for Me's cross-site agent adds a multi-domain sourcing dimension: the agent must identify and evaluate products across multiple merchant environments — exactly what BuyerBench Pillar 1*

multi-catalog scenarios test. The 60%+ purchase lift attributable to AI-assisted discovery also benchmarks the economic value of effective supplier discovery agents.

> ***BuyerBench relevance (Pillar 2 — Economic Decision Quality):*** *Rufus's price-triggered auto-buy and price history surfacing are live implementations of algorithmic price optimization. They also introduce behavioral bias scenarios: customers who set auto-buy thresholds may anchor on historical prices shown by Rufus (anchoring bias), and the "deal of the day" surfacing can trigger urgency/ scarcity heuristics. Alexa+'s voice-driven booking flow is a natural target for framing and default-bias testing (does the agent default to the first Expedia result?).*

> ***BuyerBench relevance (Pillar 3 — Security & Compliance):*** *Buy for Me's encryption-based credential injection into third-party sites raises Pillar 3 secure data handling questions: how does the agent manage stored payment credentials, scope purchases to user intent, and prevent unauthorized transactions? The Amazon vs. Perplexity injunction case defines the emerging legal boundary for agent authorization — a foundational concept for BuyerBench's authentication/authorization scenarios. The ruling's distinction between user consent and platform authorization maps directly to Pillar 3 permission boundary scenarios.*

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ChatGPT Operator / Shopping (OpenAI)

> *OpenAI's autonomous browser agent and in-chat shopping infrastructure — from the first commercially deployed agentic shopping protocol (ACP Instant Checkout, Sep 2025) to the strategic pivot away from checkout (Mar 2026)*

Overview

OpenAI's **Operator** was the first commercially deployed autonomous browser agent from a major AI lab, launching January 23, 2025. Built on the **Computer-Using Agent (CUA)** model (a fine-tuned GPT-4o variant), Operator could navigate websites, fill forms, complete purchases, and execute multi-step workflows in a real browser session — entirely without human intervention per step.

In September 2025, OpenAI extended this capability into **ChatGPT Instant Checkout**, the first large-scale production deployment of the **Agentic Commerce Protocol (ACP)** — a checkout standard co-developed with Stripe. This gave ChatGPT's 900M+ weekly active users the ability to buy from Shopify merchants and Etsy sellers without leaving the chat interface.

By March 2026, OpenAI had **rolled back Instant Checkout** due to poor consumer adoption, pivoting to an **app-based model** where retailers operate their own dedicated ChatGPT apps with outbound links to their native checkout flows. The ACP specification itself was not deprecated and remains active under Apache 2.0 licensing.

In February 2026, Operator was merged into the broader **ChatGPT Agent** mode — a unified agentic capability accessible via dropdown in the ChatGPT composer interface.

> ***BuyerBench relevance (Pillar 1):*** *Operator's task execution model (CUA + browser) is a direct reference architecture for Pillar 1 capability scenarios: supplier discovery via unstructured web navigation, form-based RFQ submission, and multi-step procurement workflows. The capability ceiling and failure modes of Operator define realistic bounds for agentic buyer task completion benchmarks.*

> ***BuyerBench relevance (Pillar 3):*** *The ACP Instant Checkout rollback is a canonical example of a* ***platform-layer authorization change****: technically functional payment flows disabled by business-layer policy rather than security failure. Pillar 3 scenarios should test whether agents correctly interpret and respect platform authorization signals that are not purely technical (e.g., protocol version changes, feature deprecation notices, merchant app availability as a signal of permitted checkout path).*

Quick Facts

ATTRIBUTE	VALUE
Product Name	ChatGPT Operator (→ ChatGPT Agent, Feb 2026)
Developer	OpenAI
Underlying Model	GPT-4o (Computer-Using Agent / CUA fine-tune)
Initial Launch	2025-01-23 (Operator)
Shopping Launch	2025-09-29 (ACP Instant Checkout)
Shopping Rollback	2026-03-04 (Instant Checkout removed)
Current Form	ChatGPT Agent mode (Feb 2026+) + retailer apps
Monthly Active Users	900M+ weekly active ChatGPT users (context for reach)
Merchant Transaction Fee	4% (Instant Checkout era, Sep 2025 – Mar 2026)
Checkout Protocol	ACP (Agentic Commerce Protocol, Apache 2.0, still active)

Operator Product

What It Is

Operator is a **task-executing AI agent** embedded in ChatGPT that uses a live Chromium browser instance to act on behalf of users. Unlike pure LLM agents operating via tool calls, Operator observes and interacts with actual web interfaces through screenshot-based computer vision — the same interaction model a human uses.

Key capabilities at launch (Jan 2025):

- Filling out forms and submitting data
- Completing purchases across arbitrary e-commerce sites
- Booking travel, restaurants, and services
- Navigating multi-step web workflows (account creation, preference selection, checkout)
- Handling CAPTCHA challenges (within limits)

Architecture

Operator is built on the **Computer-Using Agent (CUA)** model architecture:

- A GPT-4o fine-tune trained on GUI navigation and computer-use tasks
- Receives screenshots of browser state as input
- Produces structured actions (click, type, scroll, navigate) as output

- Each action → screenshot → next action loop runs until task completion or user intervention

OpenAI designed deliberate **human-in-the-loop checkpoints** for sensitive actions:

- Payment execution requires explicit user confirmation
- Logins and credential entry pause for human approval
- Irreversible actions (e.g., confirming purchases) trigger confirmation dialogs

Evolution to ChatGPT Agent (Feb 2026)

In February 2026, OpenAI unified Operator into **ChatGPT Agent** — a mode selectable from the composer dropdown. The update:

- Consolidated computer-use, tool-use, and search under a single "agent" entry point
- Expanded access from Pro-only to Plus and Team subscribers
- Retained Operator's browser automation core with better task orchestration
- Framed the product as a general work assistant rather than a shopping-specific feature

Shopping Feature History

Phase 1: ACP Instant Checkout Launch (Sep 2025)

On **September 29, 2025**, OpenAI launched **ChatGPT Instant Checkout** — the first large-scale production deployment of the Agentic Commerce Protocol (ACP). This allowed US ChatGPT users to purchase products directly inside the chat conversation.

Launch mechanics:

- ChatGPT displays shoppable product cards in search results
- User clicks "Buy" → in-chat checkout flow powered by ACP
- Payment processed via Stripe SharedPaymentToken (SPT) — no raw card data exposed to ChatGPT
- Transaction routed to merchant via ACP's four-step sequence (Create → Update → Complete → Cancel)
- Merchant charged **4% transaction fee** by OpenAI (in addition to standard Stripe processing fees)

Launch partners: Etsy, Shopify merchants (1M+), Glossier, Vuori, Spanx, SKIMS

January 2026: Shopify merchant onboarding scaled up; 4% fee model confirmed publicly.

Phase 2: Strategic Tensions (Jan – Feb 2026)

Several friction points became apparent during the Instant Checkout scale-up:

ISSUE	DETAIL
Conversion rate gap	Walmart data: 3× lower conversion for in-chat checkout vs. redirecting users to merchant website
Missing retail functionality	No real-time inventory, no coupon/promotion support, no loyalty points integration, no store pickup option
Tax collection gap	As of February 2026, OpenAI had not implemented US state sales tax collection
Merchant control loss	Retailers resisted ceding the checkout experience and post-purchase customer relationship
Amazon partnership	OpenAI's $50B strategic partnership with Amazon (Feb 27, 2026) aligned incentives toward "send users to Amazon" rather than "buy in ChatGPT"

Phase 3: ACP Rollback (Mar 2026)

On **March 4, 2026**, OpenAI removed the Instant Checkout feature from ChatGPT. Official explanation:

> *"The initial version of Instant Checkout did not offer the level of flexibility that we aspire to provide, so we're allowing merchants to use their own checkout experiences."*

Key facts about the rollback:

- **ACP specification was NOT deprecated** — the Apache 2.0 open standard remained live and under active development
- **Merchant apps replaced in-chat checkout**: Walmart, Etsy, Sephora, DoorDash, Instacart, CarMax, Lowe's, Expedia, and others now operate dedicated ChatGPT Apps that link back to their own checkout flows
- **ACP's role narrowed**: from universal in-chat checkout rail to an optional protocol for ChatGPT App developers
- Retailers that had integrated ACP for product **discovery** (Target, Sephora, Nordstrom, Lowe's, Best Buy, Home Depot, Wayfair) retained those integrations

Current State (2026)

As of April 2026, OpenAI's agentic commerce strategy is:

1. **ChatGPT as discovery hub** — product search and comparison within ChatGPT, with outbound links to merchant checkout
2. **ChatGPT Apps ecosystem** — retailers build branded apps within ChatGPT that control the checkout experience
3. **ACP as optional infrastructure** — available for App developers wanting standardized cart-to-checkout signaling; not the primary product

4. **Amazon as preferred destination** — the $50B partnership effectively positions Amazon as the default purchase destination from ChatGPT shopping intent

Strategic summary from OpenAI's post-rollback communications:

> *"prioritizing making ChatGPT search and product discovery great, with ACP serving as the infrastructure that connects users to merchants across the full shopping journey"*

The **ChatGPT Agent** (evolved from Operator) continues to support autonomous web-based purchases for users who instruct it to complete transactions on arbitrary third-party sites — this capability was never removed, only the structured in-chat checkout rail.

Pricing

Consumer (ChatGPT Plans)

PLAN	OPERATOR / AGENT ACCESS	PRICE
ChatGPT Free	No agent mode	$0/month
ChatGPT Plus	Agent mode (post Feb 2026)	$20/month
ChatGPT Pro	Agent mode (first access Nov 2025)	$200/month
ChatGPT Team	Agent mode (post Feb 2026)	$30/user/month

API / Developer (Responses API)

The **Responses API** provides developer access to the same underlying models with built-in tool support. Pricing is token-based (no API surcharge), with tool-specific fees:

TOOL	PRICING
Base model (gpt-4o)	~$2.50/1M input tokens, ~$10/1M output tokens
Computer Use tool	$3/1M input + $12/1M output (research preview, tier 3–5 only)
Web Search tool	Per-call fee (varies by model) + search content tokens
File Search tool	$2.50/thousand queries + $0.10/GB/day storage (first GB free)

Merchant (ACP Era, Historical)

During Instant Checkout (Sep 2025 – Mar 2026):

- **4% transaction fee** charged to merchants on completed Instant Checkout purchases

- Standard Stripe payment processing fees applied in addition
- Fee model confirmed publicly in January 2026 when Shopify merchant onboarding scaled

Lessons from the ACP Rollback

The ACP Instant Checkout failure is a rich case study in agentic commerce product design:

LESSON	IMPLICATION
Conversion requires context	Buyers want price comparison, reviews, loyalty points, and promotions before committing — an AI that strips this context loses conversion
Merchant relationship > transaction efficiency	Retailers value owning the post-purchase experience (returns, loyalty, upsell) more than reducing checkout friction for the buyer
Tax and compliance can't be deferred	Missing sales tax was a fundamental operational gap that undermined enterprise merchant trust
Protocol ≠ Product	ACP's technical soundness didn't prevent product failure — the UX layer and business model failed independently
Platform partnerships constrain agent autonomy	The Amazon partnership effectively overrides technical protocol capability with a business-layer constraint
Open standards survive product failures	Because ACP was Apache 2.0 from day one, the spec outlived the feature — an important governance design principle

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

Operator / ChatGPT Agent is a **direct real-world benchmark target** for BuyerBench Pillar 1:

- Multi-step workflow execution (navigate → compare → select → purchase)
- Unstructured web navigation (no API — pure browser automation)
- Form completion and credential management
- Human-in-the-loop checkpoint handling
- Task completion under ambiguous instructions

Operator's published failure rates (high error rate on complex multi-step tasks) and the CUA architecture's screenshot-based action loop define realistic capability bounds for Pillar 1 evaluation rubrics.

Pillar 2 — Economic Decision Quality

The Instant Checkout product design revealed a key behavioral pattern: **users research in ChatGPT but buy elsewhere**. This suggests AI shopping agents may exhibit a "discovery-completion gap" —

task completion (buy) and task initiation (discover) are decoupled. BuyerBench Pillar 2 scenarios should test whether agents correctly identify when to complete vs. when to hand off purchasing intent.

Pillar 3 — Security, Compliance, and Market Readiness

The rollback creates a critical **authorization boundary scenario** for Pillar 3:

- The ACP spec is technically available but the in-chat checkout feature was removed
- An agent operating in 2026 that attempts to use ACP Instant Checkout is violating platform authorization — not a protocol error, but a **policy enforcement failure**
- BuyerBench should include scenarios where a historically valid protocol endpoint becomes unavailable by platform-layer decision, and test whether agents correctly detect and handle this state
- The merchant app model (outbound checkout) represents the **currently authorized payment path** — agents must route to it rather than attempting direct ACP checkout calls

Comparison: ChatGPT Operator vs. Perplexity Comet

DIMENSION	CHATGPT OPERATOR / AGENT	PERPLEXITY COMET
Architecture	Browser automation (CUA) + ChatGPT chat interface	Dedicated browser agent (Comet product)
Shopping approach	App ecosystem + outbound checkout (post-Mar 2026)	Autonomous cross-site purchase execution
Protocol	ACP (scaled back)	Unknown / proprietary
Legal status	No known injunctions (2026)	Amazon court injunction (Mar 10, 2026)
User base	900M+ weekly ChatGPT users	Perplexity subscriber base
Merchant model	ChatGPT Apps (retailer-controlled checkout)	Direct site navigation

See also: Perplexity Comet for the competing approach and its legal challenges.

Related Entities

- ACP — The Agentic Commerce Protocol that powered Instant Checkout; the technical substrate of ChatGPT's shopping infrastructure
- Perplexity Comet — Direct competitor autonomous shopping agent; faced Amazon court injunction (Mar 2026)
- AP2 UCP — Google's competing agentic commerce protocol stack; benefited from ACP rollback reducing OpenAI's protocol momentum

- Amazon Rufus BuyForMe — Amazon's AI shopping stack; the $50B OpenAI-Amazon partnership aligns both companies against Perplexity/Google in consumer agentic commerce

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Perplexity Comet (Browser Agent)

> *Perplexity's AI-native browser agent — the first commercially deployed autonomous cross-site shopping agent, now under federal court injunction (Mar 2026) for accessing Amazon without platform authorization*

Overview

Comet is an AI-native browser product from Perplexity AI, designed not as a passive web window but as an autonomous agent that can observe a user's browsing context, understand intent, and execute multi-step workflows without per-step human intervention. Comet can browse product pages, compare prices, log into accounts using user-provided credentials, add items to a cart, and complete a purchase — end to end — across arbitrary third-party websites.

Launched July 9, 2025 in restricted preview (Max subscribers only), Comet became broadly available on **October 2, 2025**. The "Buy with Pro" extension, launched November 2025 in partnership with PayPal, added one-click in-browser purchases for Perplexity Pro subscribers.

In **November 2025**, Amazon sued Perplexity in U.S. federal court alleging unauthorized computer access — specifically that Comet entered password-protected Prime accounts and masked bot activity as human browsing. On **March 9–10, 2026**, U.S. District Judge Maxine Chesney (N.D. Cal.) granted Amazon a preliminary injunction blocking Comet from accessing password-protected Amazon sections. Perplexity filed its appeal with the Ninth Circuit on **April 1, 2026**, arguing Amazon is misusing the CFAA to suppress competition in agentic AI shopping.

> ***BuyerBench relevance (Pillar 3):*** *The Amazon v. Perplexity case establishes the most important consent-authorization distinction in agentic commerce law: a user's explicit permission for an AI agent to act on their behalf does not confer platform-layer authorization under a site's Terms of Service or the CFAA. BuyerBench Pillar 3 scenarios must test whether agents correctly distinguish "user has authorized me" from "the platform has authorized agent access."*

> ***BuyerBench relevance (Pillar 1):*** *Comet's cross-site autonomous purchase execution — navigate, credential-auth, select, checkout across arbitrary URLs — defines one of the most ambitious Pillar 1 capability targets. The workflow complexity (multi-site, multi-auth, adversarial CAPTCHA environment) sets realistic upper bounds for agentic buyer task completion evaluation.*

Quick Facts

ATTRIBUTE	VALUE
Product Name	Comet
Developer	Perplexity AI
Product Type	AI-native browser with autonomous agent layer
Restricted Preview	2025-07-09 (Max plan only)
General Availability	2025-10-02
Buy with Pro Launch	November 2025 (PayPal partnership)
Comet Plus Launch	~2025-Q4 (standalone $5/month add-on)
Amazon Suit Filed	November 2025
Injunction Granted	2026-03-09 (Judge Chesney, N.D. Cal.)
Perplexity Appeal	2026-04-01 (9th Circuit, enforcement stayed pending appeal)
Enforcement Stay	7-day stay post-injunction; 9th Circuit extended during appeal review

Comet Browser Agent Architecture

Comet is differentiated from AI assistants and chat-embedded agents by its **browser-native architecture**:

Core Design Principles

1. **Observe → Understand → Act**: Comet monitors the active browser context in real-time, builds a semantic model of the current page/task state, and can execute actions (clicks, form fills, navigation, checkout completion) without explicit per-step user prompting.
2. **Computer vision + web automation**: Like OpenAI's CUA (Computer-Using Agent), Comet uses screenshot-based visual understanding of rendered web pages — meaning it can interact with sites that have no API or structured schema, including password-protected areas when the user provides login credentials.
3. **Cross-site scope**: Unlike ChatGPT Operator (which pivoted to a ChatGPT Apps model tied to merchant-specific checkout flows), Comet is designed to operate across **arbitrary third-party websites** — including sites that have not opted into any agentic commerce protocol.
4. **Citation-backed recommendations**: Perplexity's search heritage is embedded in Comet's product experience — every product recommendation links to verifiable sources (review aggregations, price comparisons, feature analyses). Comet does not simply pick a product; it surfaces the evidence basis for its choice.

5. **User credential delegation**: When completing purchases on logged-in accounts (e.g., Amazon Prime, target.com), Comet uses stored/session credentials provided by the user. This is the specific capability that triggered Amazon's suit.

Shopping Workflow

A typical Comet autonomous shopping flow:

```
User intent (natural language)
  → Perplexity search + product research
  → Price comparison across sites
  → Comet navigates to selected product URL
  → Comet logs in using user credentials (if applicable)
  → Comet adds to cart, applies any coupons/promo codes found
  → Comet completes checkout (card via PayPal/stored payment / "Buy with Pro")
  → Comet returns purchase confirmation to user
```

This end-to-end execution runs without the user clicking anything after the initial purchase intent is expressed.

Shopping Capabilities

Buy with Pro (Nov 2025)

Launched in partnership with **PayPal**, Buy with Pro enables:

- **One-click purchase execution** from within Perplexity search results
- Payment via user's PayPal account (no raw card data stored by Perplexity)
- Targeted at US Perplexity Pro subscribers at launch
- Covers merchants reachable via Comet's browser automation (not limited to opt-in partners)

Cross-Site Autonomous Purchase

Comet's broadest capability — autonomous shopping on any site:

- Comparison shopping across multiple retailer sites in a single session
- Coupon code detection and application
- Form-fill checkout completion
- CAPTCHA handling (within model limits)
- Session credential use for logged-in shopping (the capability challenged in the Amazon suit)

Product Research Integration

Perplexity's core search capability is natively integrated:

- Real-time price monitoring and "buy when price drops" automation (announced Nov 2025, Comet integration roadmap)

- Multi-source review aggregation
- Feature comparison tables generated on the fly
- Citation links to source data for all recommendations

Pricing

PLAN	COMET / AGENT ACCESS	PRICE
Perplexity Free	Basic Comet browser access	$0/month
Perplexity Pro	Full Comet + Comet Plus + Buy with Pro	$20/month
Perplexity Max	First Comet access (Jul 2025 preview) + all Pro features	$200/month
Comet Plus (standalone)	Advanced agent features, no full Pro required	$5/month

Notes:

- The Comet browser itself is free to download; agentic capabilities are tiered by subscription
- Comet Plus is included in Pro and Max; purchasable standalone at $5/month
- "Buy with Pro" is a Pro/Max subscriber feature enabled by the PayPal integration

Legal Situation: Amazon Court Injunction (March 2026)

Background

In **November 2025**, Amazon filed suit against Perplexity AI in the U.S. District Court for the Northern District of California, alleging:

- **Unauthorized computer access** under the Computer Fraud and Abuse Act (CFAA)
- **Breach of Terms of Service** — Amazon's ToS explicitly prohibits automated access to password-protected account areas
- **Deceptive practices** — Comet allegedly masked bot/agent activity to appear as human browsing (user-agent spoofing, behavioral mimicry)
- **Trade secret misappropriation** — Amazon argued Comet's Amazon search and purchase data access exceeded any legitimate user-delegated scope

The Injunction (March 9–10, 2026)

U.S. District Judge Maxine Chesney granted Amazon a **preliminary injunction** on March 9–10, 2026, blocking Perplexity from using Comet to:

- Access password-protected sections of Amazon.com (Prime account areas, saved payment methods, order history)
- Execute purchases on Amazon on behalf of users via automated agent behavior

The central legal finding — potentially the most consequential ruling in agentic commerce law to date:

> *Comet may have had the user's permission, but not Amazon's authorization, to enter logged-in areas of the site.*

This "consent ≠ authorization" principle means:

1. A user's explicit instruction to an AI agent does not override the target platform's access controls or ToS
2. CFAA "authorization" attaches to the **platform's grant of access rights**, not the **user's delegation of intent**
3. Agents accessing password-protected systems on behalf of users may be committing computer fraud even with the user's full knowledge and consent

The Appeal (April 1, 2026)

Perplexity filed its opening appeal brief with the **Ninth Circuit** on April 1, 2026, arguing:

- Amazon is misusing the CFAA — a federal anti-hacking statute — as a weapon against competition in AI shopping
- The statute was not designed to protect platform monopolies from browser-based competition
- Comet's behavior (acting on behalf of a logged-in user) is functionally equivalent to automation tools long used by consumers (price trackers, auto-fill extensions, purchase protection services)
- Enforcement of the injunction was stayed pending the appeal's resolution

Perplexity's counter-framing (via ppc.land reporting):

> *Amazon's claim targets "regular browser users" — the behavior Comet exhibits (logging in as the user, using the user's credentials) is how every logged-in user accesses Amazon, just automated.*

Implications for Agentic Commerce

The Amazon v. Perplexity dispute is defining early case law for agentic commerce authorization:

QUESTION	PERPLEXITY POSITION	AMAZON POSITION
Does user consent suffice?	Yes — user delegates agent, agent acts as user	No — platform authorization is separate from user authorization
CFAA scope	Designed for hackers, not authorized-user automation	Applies whenever terms of service are violated
Agent identity	Agent = user proxy; user's rights transfer to agent	Agent = distinct entity; must have independent platform authorization
Remedy	Competition law should limit platform gatekeeping	Preliminary injunction appropriate to protect systems

This case directly parallels the authorization model questions embedded in **Visa's KYA (Know Your Agent)** and **Mastercard's Verifiable Intent** frameworks — which were designed precisely to create a clear authorization chain from consumer → agent → platform.

Competitive Position vs. ChatGPT Operator

DIMENSION	PERPLEXITY COMET	CHATGPT OPERATOR / AGENT
Architecture	Dedicated AI-native browser product	Browser automation layer inside ChatGPT interface
Scope	Cross-site, arbitrary third-party (including Amazon)	App ecosystem + outbound checkout (post-Mar 2026 pivot)
Protocol	No published open protocol; proprietary browser agent	ACP (scaled back Mar 2026); retailer ChatGPT Apps
Payment method	PayPal (Buy with Pro); credential delegation	Stripe SPT (historical); outbound merchant checkout
Legal status	Federal injunction re: Amazon (Mar 2026), appealing	No known injunctions
Business model	Subscription tier (Pro/Max includes agent)	Subscription tier (Plus/Pro includes agent)
User base	Perplexity subscriber base (significantly smaller than ChatGPT)	900M+ weekly active ChatGPT users
Platform approach	Aggressive cross-platform automation	Post-rollback: defers to merchant-controlled checkout
Search integration	Native (Perplexity is a search product)	Bolt-on (ChatGPT search added Oct 2024)

Strategic divergence: ChatGPT Operator pivoted toward platform cooperation (merchant apps, outbound checkout) after the ACP rollback. Comet has maintained an aggressive **cross-platform autonomous access** stance — a fundamentally different bet on how agentic commerce will evolve, with significant legal and regulatory risk.

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

Comet defines the **most ambitious Pillar 1 capability target** in the current market:

- Full autonomous purchase execution across arbitrary third-party websites (no API, no protocol partner required)
- Cross-site price comparison as a native workflow step
- Login/credential delegation for authenticated shopping sessions
- Cart management, coupon application, and checkout completion

BuyerBench Pillar 1 scenarios should include cross-site procurement tasks modeled on Comet's architecture, with explicit test coverage for multi-site supplier comparison, authenticated account navigation, and checkout completion without structured API access.

Pillar 2 — Economic Decision Quality

Comet's citation-backed recommendation model introduces a testable hypothesis for Pillar 2:

- Does access to explicit evidence (citations, comparative data) reduce anchoring and framing bias?
- Does the "research-then-act" pipeline (Perplexity search → Comet execution) improve economic rationality compared to agents that act on retrieved text without structured evidence layers?

Pillar 3 — Security, Compliance, and Market Readiness

The Amazon v. Perplexity case is a direct BuyerBench Pillar 3 scenario generator:

1. **Consent-Authorization Boundary**: BuyerBench should include scenarios where the agent has user consent but lacks platform authorization (site ToS restricts agent access). The agent must correctly refuse or seek authorization rather than proceed.
2. **Agent Identity and Masking**: Comet's alleged user-agent spoofing behavior is a Pillar 3 violation — agents must not misrepresent their identity to bypass platform access controls. BuyerBench should test whether agents use honest agent identification (per Mastercard Web Bot Auth / Visa KYA standards).
3. **Credential Scope**: When an agent uses delegated user credentials, it must operate within the scope that the credential was intended to cover. Accessing additional account data (order history, saved cards) beyond what's needed for the purchase task is an over-scoping violation.

4. **Platform Authorization Signals**: BuyerBench scenarios should present explicit platform ToS restrictions and test whether agents parse and comply with them, versus attempting to proceed despite the restriction.

Related Entities

- ChatGPT Operator — Direct competitor; took the opposite strategic approach (platform cooperation vs. cross-platform automation)
- Amazon Rufus BuyForMe — Amazon's own AI shopping stack; the plaintiff in the Mar 2026 injunction; won the first major agentic commerce access-rights ruling
- ACP — Agentic Commerce Protocol; the standard approach for authorized checkout that Comet's proprietary model bypasses
- Visa Intelligent Commerce — Visa's KYA framework; the industry-standard answer to the authorization gap Comet exploited
- Mastercard Agent Pay — Mastercard's Verifiable Intent + Web Bot Auth; directly addresses agent identity masking

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Salesforce Agentforce

> *Salesforce's enterprise autonomous AI agent platform — from $2/conversation in 2024 to a full Agentic Enterprise License Agreement in 2026, with purpose-built buyer, supply chain, and import specialist agents for procurement workflows*

Overview

Agentforce is Salesforce's enterprise-grade autonomous AI agent platform, launched in September 2024 at Dreamforce. It enables organizations to build, deploy, and manage AI agents that execute multi-step business workflows autonomously — operating across Salesforce's full product suite (Sales Cloud, Service Cloud, Commerce Cloud, Marketing Cloud) and integrating with external systems via MuleSoft.

Unlike point-solution AI copilots, Agentforce agents are **goal-driven**: they receive a task objective, access relevant data via Salesforce Data Cloud, determine the appropriate sequence of actions, and execute — including handoff to a human when ambiguity or policy thresholds require it. Agents are persistent, can run asynchronously, and can initiate work proactively (not just respond to user prompts).

As of April 2026, the platform has shipped three distinct pricing models (Conversations → Flex Credits → AELA per-user), reflecting rapid evolution in how Salesforce and its customers understand where AI agent value accrues in enterprise workflows. Agentforce 360 — the current GA iteration — adds Agent Script (deterministic control language), Agentforce Voice, and Intelligent Context for unstructured data grounding.

> ***BuyerBench relevance (Pillar 1):*** *Agentforce Buyer Agent and Supply Chain agents are direct reference implementations for Pillar 1 capability scenarios — B2B supplier discovery, purchase order automation, supplier onboarding, and multi-step procurement workflows. Agentforce's published agent taxonomy (goal → action plan → tool call → human escalation) maps directly onto BuyerBench's task completion and workflow accuracy metrics.*

> ***BuyerBench relevance (Pillar 2):*** *Agentforce's "Import Specialist Agent" (tariff impact modeling) is a real-world example of economically rational agent decision-making under uncertainty. BuyerBench Pillar 2 scenarios should test whether agents correctly propagate cost changes (tariffs, freight) through margin calculations — exactly what this agent claims to do.*

> ***BuyerBench relevance (Pillar 3):*** *Agentforce's least-privilege access model, Agent Script determinism controls, and compiled JSON audit trail are the enterprise baseline for Pillar 3 compliance.*

> *Scenarios should test whether agents enforce Salesforce-defined permission scopes when executing financial transactions, and whether they generate auditable trails of agent decisions.*

Quick Facts

ATTRIBUTE	VALUE
Product Name	Agentforce (→ Agentforce 360, 2026)
Developer	Salesforce
Initial Launch	September 2024 (Dreamforce)
Current Version	Agentforce 360 (Spring '26 Release)
Underlying Models	Einstein AI (Salesforce-trained LLMs + external model support)
Pricing Models	$2/conversation OR Flex Credits ($0.10/action) OR AELA ($125+/user/month)
Key Procurement Agent	Buyer Agent (B2B purchasing), Import Specialist Agent (tariff analysis)
Integration Foundation	Salesforce Data Cloud + MuleSoft + Slack
Governance Standard	Compiled JSON agents, least-privilege access, Agent Script determinism
NVIDIA Partnership	GTC 2026 — NVIDIA enterprise AI agent platform, Salesforce among 17 adopters

Agent Types

Pre-Built Agents

Salesforce ships a library of purpose-built agents for common enterprise functions:

AGENT	DOMAIN	CAPABILITY
Buyer Agent	B2B procurement	Helps buyers find products, make purchases, and track orders via chat or within sales portals
Import Specialist Agent	Trade compliance	Calculates duties and freight costs; models margin impact of tariff changes in real time
Merchant Agent	Commerce	Manages e-commerce catalog operations, promotions, and buyer interactions
Customer Service Agent	Support	Resolves tier-1 support cases, escalates with context
Campaign Optimizer Agent	Marketing	Manages campaign performance and audience adjustments
Supply Chain Agent	Operations	Orchestrates mid- and back-office processes: procurement, supplier onboarding, warranty claims
Sales Development Agent	CRM	Qualifies leads, books meetings, drafts personalized outreach

Custom Agents via Agentforce Builder

Organizations can build proprietary agents using **Agentforce Builder** — a unified development environment with three authoring modes:

1. **Conversational workspace** — describe the agent in natural language; AI generates the initial configuration
2. **Document-like editor** — refine instructions, topics, and actions with autocomplete and structured templates
3. **Agent Script (pro-code)** — write deterministic control flows using Agentforce's scripting language

Every agent compiles into a **portable JSON artifact** for version control, deployment, and audit — enabling governance workflows identical to software releases.

Procurement and Purchasing Use Cases

Buyer Agent (B2B Purchasing)

The **Buyer Agent** is Salesforce's direct answer to the AI buyer agent category. Deployed within B2B Commerce Cloud portals or via chat interface, it enables business buyers to:

- Search and compare products across a supplier catalog using natural language
- Initiate and track purchase orders autonomously
- Escalate to a human procurement manager when approval thresholds are exceeded
- Surface order history, delivery status, and re-order recommendations

The Buyer Agent operates within the buyer's Salesforce-defined permission scope — it cannot initiate purchases above the user's approval authority without triggering a human-in-the-loop escalation.

Agentforce Supply Chain

Agentforce Supply Chain targets the operational layer of procurement:

- Automates supplier onboarding (document collection, vendor verification, contract routing)
- Manages warranty claims and returns workflows
- Orchestrates cross-functional approval processes (finance, legal, operations) for large-ticket purchases
- Generates procurement analytics and spend reports

Import Specialist Agent (Tariff and Trade Compliance)

Launched in response to 2025–2026 global tariff volatility, the **Import Specialist Agent**:

- Ingests tariff schedules and freight data in real time
- Calculates landed cost (duties + freight + insurance) for specific product categories
- Models margin sensitivity: *"if tariffs increase by 15%, what happens to gross margin on this product line?"*
- Generates scenario comparisons for sourcing decisions (domestic vs. international supplier)

This agent directly targets procurement decisions that would otherwise require specialist trade consultant engagement.

Scope 3 and Emissions Procurement

Agentforce integrates with **Salesforce Net Zero Cloud** to surface emissions data alongside procurement decisions, enabling buyers to factor supplier carbon footprint into sourcing choices — a Pillar 2 multi-criteria optimization scenario.

Pricing

Salesforce has shipped **three distinct pricing models** for Agentforce since launch — an unusually rapid pricing evolution that reflects ongoing calibration of enterprise AI agent value.

Model 1: Conversations ($2/conversation) — Original Launch Model

- Flat fee per 24-hour conversation session between agent and user
- Simple to budget; well-suited for customer-facing bot deployments
- Poorly fits internal workflow automation where a "conversation" may span days of async execution
- Still available as of April 2026 but being phased out in favor of Flex Credits

Model 2: Flex Credits ($0.10/action) — Introduced May 2025

- Launched May 15, 2025 as Salesforce's recommended model for new deployments

- **1 standard action = 20 credits; credits purchased at $500 per 100,000**
- Effective cost: $0.10 per agent action (e.g., query a record, send an email, call an external API)
- Aligns cost to value: organizations pay for work done, not for time elapsed
- **Constraint**: Flex Credits and Conversations pricing cannot coexist in the same Salesforce org — organizations must choose one model

Model 3: Agentic Enterprise License Agreement (AELA) — Late 2025

- Per-user license starting at **$125/user/month**, scaling by tier and edition
- "Digital workforce" is included in the seat price — agents become a bundled capability rather than a consumption add-on
- Designed for enterprises deploying Agentforce broadly across multiple departments
- Simplified procurement: one contract, predictable cost, no per-action metering

Pricing Summary

MODEL	UNIT	EFFECTIVE COST	BEST FOR
Conversations	Per 24-hr session	$2/conversation	Customer-facing chat bots
Flex Credits	Per agent action	$0.10/action	Workflow automation, internal use
AELA	Per user/month	$125+/user	Enterprise-wide deployment

> ***Note for BuyerBench:*** *The existence of three concurrent pricing models is itself a Pillar 2 economic scenario — procurement agents evaluating Salesforce must select the optimal pricing model for their use case, and the decision is non-trivial (a high-volume internal agent is dramatically cheaper under AELA than Flex Credits at scale).*

Integration Ecosystem

Core Salesforce Platform

Agentforce is **native to Salesforce** — agents have zero-integration access to:

- **Sales Cloud** — CRM data: accounts, contacts, opportunities, purchase history
- **Service Cloud** — Case management, knowledge bases, SLA policies
- **Commerce Cloud** — B2B and B2C product catalogs, pricing, order management
- **Marketing Cloud** — Campaign data, customer segments, channel preferences
- **Net Zero Cloud** — Emissions data for ESG-aware procurement

Data Cloud (Unified Data Layer)

Data Cloud provides agents with a real-time, unified customer and supplier data layer:

- Harmonizes data from Salesforce + external systems into a single semantic layer
- Enables "Intelligent Context" — grounding agents in unstructured documents (contracts, emails, PDFs) alongside structured CRM data
- Powers relevance ranking for product discovery and supplier recommendations

MuleSoft (External Integration)

MuleSoft is Agentforce's bridge to non-Salesforce systems:

- Connects agents to ERP systems (SAP, Oracle), financial platforms, logistics APIs
- Enables agents to execute procurement workflows that span Salesforce + external systems
- Exposes external APIs as Agentforce actions without custom code

Slack

Salesforce-owned **Slack** serves as the primary asynchronous interface for Agentforce in enterprise deployments:

- Agents surface in Slack channels and DMs as conversational participants
- Procurement approvals, status updates, and escalations flow through Slack
- Human-in-the-loop checkpoints are implemented as Slack approval messages

NVIDIA (GTC 2026 Partnership)

At GTC 2026, NVIDIA launched an enterprise AI agent platform with Salesforce as one of 17 launch adopters — integrating Agentforce with NVIDIA's accelerated compute infrastructure for latency-sensitive agentic workflows.

AppExchange Procurement Extensions

The Salesforce AppExchange hosts native procurement software packages (e.g., Coupa, Jaggaer, Ivalua connectors) that extend Agentforce's source-to-pay reach to specialized procurement suite capabilities.

Limitations

LIMITATION	DETAIL
Salesforce-org dependency	Agentforce requires an existing Salesforce org — it is not a standalone procurement platform. Organizations not already on Salesforce face significant adoption cost.
Pricing model lock-in	Flex Credits and Conversations cannot coexist in the same org. Switching models requires a contract change.
Per-user AELA cost	$125+/user/month is steep for small teams. At 100 users, AELA = $150K/year before base Salesforce licensing.
No native payment rail	Agentforce does not include a payment execution layer — it can initiate purchase orders but relies on external payment systems for settlement. Contrast with Skyfire (KYAPay) or ACP (Stripe-native).
Customization complexity	While Agentforce Builder lowers the floor, production-grade custom agents with Agent Script and complex MuleSoft flows still require Salesforce-certified developers.
Ecosystem monoculture risk	Deep integration across Sales/Service/Commerce Cloud creates lock-in — Agentforce agents are not portable to non-Salesforce environments.
Agent action cost accumulates	Under Flex Credits, high-volume internal workflows (e.g., 10,000 daily supplier catalog queries) can exceed AELA cost quickly, requiring careful capacity planning.

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

Agentforce's **Buyer Agent** and **Supply Chain Agent** are the most direct enterprise reference implementations for BuyerBench Pillar 1 scenarios:

- **Supplier discovery**: Buyer Agent's natural language product search across a B2B catalog mirrors BuyerBench supplier discovery tasks
- **Multi-step workflow execution**: Agentforce's goal → action plan → tool call loop is structurally identical to BuyerBench's scenario execution model
- **Human-in-the-loop escalation**: Agentforce's approval authority thresholds are a real-world reference for Pillar 1 workflow handoff scenarios
- **Supplier onboarding automation**: Supply Chain Agent's vendor verification workflow maps to BuyerBench's supplier qualification tasks

The **Import Specialist Agent** is particularly relevant as a Pillar 1 + 2 hybrid — it executes a multi-step calculation workflow (Pillar 1) and produces an economically rational sourcing recommendation under changing cost conditions (Pillar 2).

Pillar 2 — Economic Decision Quality and Behavioral Robustness

Agentforce exposes several economically rich scenarios for Pillar 2 testing:

- **Pricing model selection**: Three concurrent pricing models for the same platform — agents must correctly identify the optimal model for a given workload volume (framing and anchoring bias test)
- **Tariff scenario modeling**: Import Specialist Agent's margin sensitivity analysis is a reference for multi-variable economic optimization under uncertainty
- **Scope 3 vs. cost tradeoff**: Net Zero Cloud integration creates a multi-criteria optimization scenario where agents balance supplier cost against emissions — a real-world instance of competing economic and non-economic objectives
- **Approval authority thresholds**: Buyer Agent's escalation logic is a policy compliance scenario — does an agent correctly respect spending limits even when a lower-cost option requires exceeding authority?

Pillar 3 — Security, Compliance, and Market Readiness

Agentforce's enterprise governance architecture is a **positive reference implementation** for Pillar 3:

- **Least-privilege access**: Agents operate within the permission scope of the initiating user — a direct implementation of the principle BuyerBench Pillar 3 scenarios test
- **Compiled JSON audit trail**: Every agent invocation produces an auditable, versioned artifact — the target behavior for BuyerBench's "does the agent maintain an audit trail?" scenarios
- **Agent Script determinism**: Ability to define deterministic control flows for financially sensitive operations (e.g., "always require human approval for orders >$10K") models the policy enforcement layer BuyerBench Pillar 3 tests
- **Shared responsibility model**: Salesforce's "we provide the platform controls; you configure the governance" model is the enterprise baseline for Pillar 3 compliance scenarios

Gap: Agentforce has **no native payment rail**. For BuyerBench scenarios requiring end-to-end payment execution (not just PO issuance), Agentforce would need to integrate with an external payment layer (Stripe, Skyfire, AP2, etc.) — making it relevant for testing whether procurement agents correctly detect and handle the boundary between purchase authorization and payment settlement.

Related Entities

- Procure AI — AI-native procurement automation competitor; 50+ autonomous agents vs. Agentforce's CRM-native approach; no Salesforce org dependency
- Omnea — AI SRM/procurement orchestration competitor focused on vendor onboarding and cross-functional approval workflows — overlapping with Agentforce Supply Chain
- Zycus — Established source-to-pay suite with Merlin ANA autonomous negotiation agent; enterprise procurement incumbent that Agentforce competes with in large enterprise accounts
- Skyfire — Provides the payment rail layer (KYAPay) that Agentforce lacks; an Agentforce + Skyfire integration would create a full procurement-to-payment agentic stack

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NegMAS — Negotiation Multi-Agent System

> *The open-source Python platform for situated autonomous negotiation — the official engine behind ANAC international competitions and a supply-chain negotiation simulation environment relevant to all three BuyerBench pillars*

Overview

NegMAS (NEGotiation MultiAgent System, alternatively: NEGotiations Managed by Agent Simulations) is an open-source Python library for developing, evaluating, and benchmarking autonomous negotiation agents embedded in simulation environments. Authored by Yasser Mohammad at Brown University, it is the technical backbone for the **Automated Negotiating Agents Competition (ANAC)** — the premier international research competition for bilateral and multilateral negotiation agents, co-located with IJCAI/AAMAS annually since 2010.

Unlike commercial procurement AI platforms (Salesforce Agentforce, Procure AI, Zycus Merlin ANA), NegMAS is purely a **research and evaluation framework**: it provides the negotiation mechanisms, utility function models, tournament evaluation infrastructure, and multi-agent simulation world (SCML) needed to study, compare, and benchmark negotiation strategies without operational constraints.

The library is actively maintained as of 2026, with the current release being **NegMAS 0.15.4** (January 16, 2026). It is installable via pip, documented on ReadTheDocs, and has a companion Java bridge (`jnegmas`) for interoperability with the established Genius platform (the ANAC legacy system for Java-based negotiation agents).

> ***BuyerBench relevance (Pillar 1):*** *NegMAS's Supply Chain Management League (SCML) is the closest open-source equivalent to a realistic buyer-agent benchmark. SCML agents negotiate buying (raw materials) and selling (finished goods) across a multi-stage supply chain, optimizing factory profit — mapping directly to BuyerBench Pillar 1 workflow accuracy and supplier selection scenarios. NegMAS is already integrated as a BuyerBench agent adapter (* `negmas` *agent ID in CLAUDE.md).*

> ***BuyerBench relevance (Pillar 2):*** *The Automated Negotiation League (ANL) within ANAC uses bilateral alternating-offers with configurable utility functions that can encode framing, anchoring, and loss-aversion variants. BuyerBench Pillar 2 can use NegMAS's SAOMechanism with manipulated utility functions to stress-test whether AI buyer agents are susceptible to opponent negotiation tactics that exploit known behavioral biases (e.g., high opening anchors, scarcity framing, sunk-cost commitment pressure).*

Quick Facts

ATTRIBUTE	VALUE
Library Name	NegMAS
Full Name	Negotiation Multi-Agent System
Author	Yasser Mohammad (yasserfarouk), Brown University
License	MIT (open source)
Current Version	0.15.4 (2026-01-16)
Python Support	Python 3.8+
GitHub	github.com/yasserfarouk/negmas
PyPI	pypi.org/project/negmas
Documentation	negmas.readthedocs.io/en/latest
ANAC Integration	Official engine for ANL and SCML leagues
Genius Bridge	jnegmas (Java interop for legacy Genius agents)
BuyerBench Agent ID	`negmas` (Python-native, no credentials required)

Architecture

NegMAS is organized around three core conceptual layers:

1. Agents

Negotiators are the autonomous agents in NegMAS. Each negotiator has:

- A **utility function** (UtilityFunction) — a preference model over possible outcomes in the negotiation domain
- A **negotiation strategy** — the decision logic for generating offers, responding to offers, and ending negotiations
- Optional **theory-of-mind** modeling — predicting opponents' utility functions from observed behavior

Pre-built negotiators include:

- `AspirationNegotiator` — time-based concession with aspiration levels
- `ToughNegotiator` / `OnlyBestNegotiator` — hardline strategies; never concedes
- `NaiveTitForTatNegotiator` — reactive tit-for-tat concession
- `BoulwareLindaNegotiator` — Boulware curve (slow early, fast late concessions)

- `NegotiatorUtility` -based custom agents using any callable utility function

2. Mechanisms

Mechanisms define the rules of negotiation. Key built-in mechanisms:

MECHANISM	DESCRIPTION
`SAOMechanism`	**Stacked Alternating Offers** — the default; negotiators alternate offers until unanimous acceptance, walkout, or timeout
`GBMechanism`	Generalized Bargaining — configurable acceptance and offer protocols
`VetoMechanism`	Multi-party negotiation with veto rights
`STACKELBERGMechanism`	Stackelberg leader-follower offer protocol
`TAUMechanism`	Tentative Agreement + Updating

`SAOMechanism` is used in ANAC ANL and is the canonical mechanism for buyer-seller bilateral negotiation in BuyerBench scenarios.

3. World (SCML)

The **Supply Chain Management League World** (`SCML2020World` , `SCML2024World` , etc.) is a multi-agent simulation environment where:

- Agents manage factories in a multi-stage supply chain
- Each factory **buys** inputs (raw materials or semi-finished goods) and **sells** outputs (processed goods or finished products)
- Negotiations are **situated**: each buying and selling negotiation is linked to real production schedules, inventory, and financial state
- Agents maximize profit across all interconnected negotiations simultaneously

This world-level simulation is the primary instrument for **Pillar 1 supply chain buyer agent evaluation** in BuyerBench — it directly models the economic optimization problem that enterprise procurement AI must solve.

ANAC Competition Integration

NegMAS is the official engine for the **Automated Negotiating Agents Competition (ANAC)**, now in its **16th edition (ANAC 2025)**, co-located with IJCAI 2025 in Montreal, Canada (August 21, 2025). ANAC has run annually since 2010, making it the longest-running international benchmark for autonomous negotiation.

ANAC 2025 Leagues

LEAGUE	DESCRIPTION	NEGMAS COMPONENT
Automated Negotiation League (ANL 2025)	Sequential multi-deal bilateral negotiation; agents face multiple opponents in sequence; rewarded for optimal deal combinations across the sequence	`SAOMechanism` + `ANLWorld`
Supply Chain Management League (SCML 2025)	Factory manager agent negotiates buying and selling in a simulated supply chain; maximizes net profit	`SCML2025World`

The `anl2025` package (autoneg/anl2025 on GitHub) is a thin wrapper around NegMAS that provides the ANL 2025 competition scaffolding, submission templates, and evaluation harness. This wrapper pattern directly mirrors how BuyerBench adapters wrap agents.

Historical Competition Data

NegMAS ships with benchmark scenarios drawn from **ANAC 2010 through ANAC 2018** Genius competition data (stored in `tests/data` and `notebooks/data`). These domains are two-sided bilateral negotiations over multi-issue spaces (price, quantity, delivery terms, etc.) and serve as a standardized test set for comparing new negotiators against the historical ANAC field.

Key Algorithms Supported

NegMAS's utility function and strategy components support the following classic negotiation algorithms out of the box, plus extension points for custom strategies:

ALGORITHM / MODEL	TYPE
Aspiration-based concession (time-dependent)	Concession strategy
Boulware / Conceder curves	Concession strategy
Tit-for-Tat variants	Reactive strategy
Nash Bargaining Solution	Theoretical optimum
Pareto-frontier discovery	Multi-agent optimality
Kalai-Smorodinsky solution	Fairness-aware solution
Inverse utility learning	Opponent modeling
ANAC Genius agent bridge	Java-based legacy strategies

Installation and Quickstart

Installation

```
# Core library
pip install negmas

# With Genius bridge (requires Java JDK)
pip install negmas[genius]

# With visualization
pip install negmas[visualization]

# All optional dependencies
pip install negmas[all]
```

Quickstart: Bilateral Buyer-Seller Negotiation

```
from negmas import SAOMechanism
from negmas.negotiators import AspirationNegotiator
from negmas.outcomes import make_issue

# Define a 2-issue domain: price (50–200) and quantity (1–10)
issues = [
    make_issue(values=(50, 200), name="price"),
    make_issue(values=(1, 10), name="quantity"),
]

# Create SAO mechanism (alternating offers, 50-step time horizon)
mechanism = SAOMechanism(issues=issues, n_steps=50)

# Add buyer: prefers low price, high quantity
buyer = AspirationNegotiator(name="buyer")
mechanism.add(buyer, ufun=lambda x: (1 - (x["price"] - 50) / 150) * (x["quantity"] /
10))

# Add seller: prefers high price, low quantity
seller = AspirationNegotiator(name="seller")
mechanism.add(seller, ufun=lambda x: ((x["price"] - 50) / 150) * (1 -
x["quantity"] / 10))

# Run negotiation
state = mechanism.run()
print(f"Agreement: {state.agreement}, Rounds: {state.step}")
```

Tournament Evaluation

```
from negmas.tournaments import neg_tournament, cartesian_tournament

# Compare negotiators against common opponents
results = neg_tournament(
    competitors=[MyBuyerAgent, AspirationNegotiator],
    n_repetitions=10,
    n_steps=100,
)

# Run all competitors against each other
results = cartesian_tournament(
    competitors=[AgentA, AgentB, AgentC],
    n_repetitions=5,
)
```

Relevance as BuyerBench Test Harness Component

NegMAS is already integrated into BuyerBench as the `negmas` agent adapter — the only Python-native agent in the benchmark suite that requires no external API credentials. This makes it uniquely useful as a **baseline and sanity-check agent**: when BuyerBench needs to verify that a scenario is correctly specified (i.e., that a reasonable agent can succeed at it), NegMAS provides a deterministic, well-characterized agent that can be expected to perform optimally on well-formed bilateral negotiation problems.

Specific integration patterns:

BUYERBENCH USE CASE	NEGMAS COMPONENT
Pillar 1 supplier selection benchmark	SCML World — buy/sell negotiation with supply chain constraints
Pillar 1 quote comparison baseline	`neg_tournament()` with catalog-priced offers modeled as utility functions
Pillar 2 bias susceptibility testing	SAOMechanism with utility functions encoding anchoring / framing variants
Pillar 2 consistency across variants	`cartesian_tournament()` comparing outcomes across framing conditions
Protocol correctness verification	SAOMechanism protocol adherence (step sequencing, timeout behavior)

Comparison to Zycus Merlin ANA: Zycus's Merlin ANA is the commercial enterprise counterpart — a closed, production-deployed autonomous supplier negotiation agent operating in real procurement environments with human oversight and ERP integration. NegMAS is the open-source research equivalent: configurable, transparent, and optimized for academic benchmarking rather than enterprise deployment. BuyerBench scenarios should use NegMAS as the **algorithmic floor** (what a theoretically optimal negotiator achieves) and Merlin ANA benchmarks as the **commercial ceiling** (what state-of-the-art enterprise systems demonstrate in production).

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

NegMAS SCML directly tests the core Pillar 1 scenario type: a buyer agent receiving a sourcing objective, discovering suppliers (upstream factories), negotiating price/quantity/delivery terms, and executing transactions — all within a simulated operational environment with real supply/demand constraints. The `negmas` BuyerBench adapter should be used to establish baseline workflow completion rates before evaluating proprietary agents.

Pillar 2 — Economic Decision Quality and Behavioral Robustness

NegMAS's configurable utility functions make it the ideal instrument for Pillar 2 bias injection. By constructing utility function pairs that are economically identical but presented differently (e.g., framing a $50 discount as a "gain" vs. the same outcome framed as avoiding a $50 loss), BuyerBench can use NegMAS's SAOMechanism to quantify whether AI buyer agents under test are susceptible to framing effects, anchoring bias (high first-offer influence), or sunk-cost fallacy (over-committing to failing negotiation threads). NegMAS's tournament infrastructure also makes it practical to run statistically significant sample sizes (50–100 negotiation runs per variant) without incurring API costs.

Pillar 3 — Security, Compliance, and Market Readiness

NegMAS does not natively model payment security or compliance. However, NegMAS SCML negotiations generate **binding financial commitments** (contracts for goods at agreed prices), and BuyerBench Pillar 3 can instrument these to test whether an AI buyer agent:

- Refuses to execute contracts that exceed pre-authorized spending limits
- Correctly attributes payments to the right negotiation sessions (no cross-session credential leakage)
- Generates auditable contract trails sufficient to reconstruct a procurement decision tree

Related Entities

- Zycus — Commercial counterpart: Merlin ANA is the production autonomous supplier negotiation agent; NegMAS is the open-source research equivalent. Pillar 1+2 comparison baseline.
- ACP — ACP defines the payment execution layer that NegMAS-negotiated contracts would ultimately route through in a real agentic commerce stack. NegMAS handles the negotiation; ACP handles the transaction.
- Procure AI — Enterprise procurement orchestration; Procure AI's 50+ agents represent the commercial Pillar 1 capability ceiling; NegMAS provides the open-source algorithmic floor.

- Fairmarkit — Fairmarkit's demand-to-award sourcing agents represent a scaled commercial version of the multi-supplier selection problem NegMAS models in research settings.

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PART VI

Protocols & Standards

Five active commerce protocols governing agent identity, payment authorization, and transaction execution as of April 2026.

ACP (Agentic Commerce Protocol)

> *The open standard for programmatic agent-to-merchant checkout, co-developed by OpenAI and Stripe — the first production-deployed AI agent payment protocol*

Overview

The **Agentic Commerce Protocol (ACP)** is an open-source, Apache 2.0–licensed specification that defines a standardized interaction model for AI agents to discover, select, and purchase goods and services on behalf of buyers. Co-developed by OpenAI and Stripe, ACP provides the checkout and merchant integration layer for agentic commerce: a common "language" that lets any compatible AI agent transact with any ACP-implementing merchant without bespoke integrations.

ACP launched at scale in September 2025 as the technical foundation of **ChatGPT Instant Checkout**, OpenAI's in-chat shopping feature. At launch, ChatGPT users in the US could buy from Etsy sellers and (imminently) from over one million Shopify merchants — all within the chat interface. ACP is available as both a REST API and an MCP (Model Context Protocol) server endpoint, enabling deployment across agent architectures.

In March 2026, OpenAI **removed the built-in Instant Checkout experience** from ChatGPT — primarily due to poor consumer adoption rather than technical failure — following its $50B strategic partnership with Amazon on February 27, 2026. However, the **ACP specification itself was not deprecated**: the protocol remains active, open-source, and under continuing development (version 2026-01-30 is the current stable release). Merchants can still deploy ACP-powered checkout experiences via dedicated ChatGPT apps. ACP's governance has also shifted: from a tightly OpenAI-controlled product feature toward a more independent open standard under broader community governance.

> ***BuyerBench relevance (Pillar 3):*** *ACP is the primary production reference implementation for BuyerBench Pillar 3 secure transaction flow scenarios. Its four-step checkout sequence (CreateCheckoutRequest → UpdateCheckoutRequest → CompleteCheckoutRequest → CancelCheckoutRequest), SharedPaymentToken credential model, HMAC webhook signing, and Stripe Radar fraud integration collectively define what "correct" agent payment API behavior looks like in the real world. Every BuyerBench Pillar 3 scenario involving payment authorization sequencing, credential handling, and fraud detection should be calibrated against ACP's published specification.*

> ***BuyerBench relevance (Pillar 1):*** *ACP's merchant-facing product discovery and selection flow directly maps to Pillar 1 supplier discovery scenarios. An agent operating in an ACP-enabled environ-*

> *ment can browse product catalogs, compare offers, and initiate checkout through structured API calls — the same capability profile BuyerBench's source-to-award workflow scenarios test.*

Quick Facts

ATTRIBUTE	VALUE
Protocol Name	Agentic Commerce Protocol (ACP)
Type	Open specification / checkout API
Developers	OpenAI (Founding Maintainer) + Stripe (Founding Maintainer)
License	Apache 2.0 (open source)
Initial Release	2025-09-29
Current Stable Version	2026-01-30
Repository	github.com/agentic-commerce-protocol/agentic-commerce-protocol
Official Documentation	agenticcommerce.dev / docs.stripe.com/agentic-commerce/protocol
Integration Modes	REST API or MCP server
Live Deployment	ChatGPT Instant Checkout (Sep 2025 – Mar 2026); merchant apps ongoing

Protocol Architecture

Core Purpose

ACP serves three stakeholder groups simultaneously:

1. **Businesses** — access high-intent buyers through AI agent channels while remaining the merchant-of-record
2. **AI agents** — embed native commerce flows without taking on payment or fulfillment responsibilities
3. **Payment providers** — process agentic transactions through established infrastructure (e.g., Stripe)

Checkout Flow (Four-Step Sequence)

ACP defines a strict four-step checkout lifecycle:

STEP	ENDPOINT	DESCRIPTION
1	`CreateCheckoutRequest`	Agent initiates the purchase; establishes session
2	`UpdateCheckoutRequest`	Agent or buyer modifies selections (quantities, variants, address)
3	`CompleteCheckoutRequest`	Agent submits the SharedPaymentToken; payment is processed
4	`CancelCheckoutRequest`	Agent or buyer aborts; session is closed

Post-completion, webhook events notify the agent of fulfillment status updates asynchronously.

Authentication & Authorization Model

ACP implements a **dual-layer security model**:

Transport layer:

- All API requests require HTTPS
- Bearer token authentication (`Authorization: Bearer {token}`) on all requests
- Signing keys provisioned during merchant onboarding

Event layer:

- Webhook events must include an HMAC signature header
- Signature verification ensures agent-originated events cannot be spoofed by third parties

Payment credential layer (SharedPaymentToken):

- The AI agent provisions a **Stripe SharedPaymentToken (SPT)** scoped to a specific amount and merchant before submitting `CompleteCheckoutRequest`
- The SPT transfers payment authorization from the buyer's stored credentials to the merchant *without exposing the underlying card or payment data*
- SPTs are time-limited, amount-capped, and merchant-restricted — a fine-grained tokenization scheme that limits blast radius if a token is intercepted

Fraud Detection

When ACP is used with Stripe as the payment processor, fraud detection is powered by **Stripe Radar**:

- SPTs relay Stripe's machine learning risk signals to the merchant at transaction time
- Risk signals include: likelihood of fraudulent dispute, card testing probability, stolen card likelihood, and card issuer decline predictions
- Stripe Radar's model is trained on global transaction data from Stripe's full merchant network (billions of data points), enabling cross-merchant fraud signal sharing
- Merchants remain the merchant-of-record and retain primary fraud management responsibility

Compliance Scope

- ACP card-based transactions route through Stripe's payment infrastructure, which is PCI DSS–compliant; merchants using ACP's standard integration benefit from Stripe's PCI scope reduction
- ACP does not itself define compliance obligations but defers to the implementing payment processor (Stripe) for PCI, 3DS2, and network rules

Versioning History

ACP uses date-based versioning (YYYY-MM-DD):

VERSION	DATE	KEY CHANGES
v1	2025-09-29	Initial release — core checkout flow, SPT, basic webhooks
v2	2025-12-12	Fulfillment enhancements — richer post-purchase tracking
v3	2026-01-16	Capability negotiation — agents can query merchant capabilities before checkout
v4 (current)	2026-01-30	Extensions, discounts, payment handlers — promotion/coupon support, pluggable payment backends
Unreleased	Active dev	Community contributions ongoing

Governance

ACP is **jointly governed by OpenAI and Stripe as Founding Maintainers**, with a stated roadmap toward broader community governance under a neutral foundation. Decision-making follows a consensus-based model with defined escalation procedures. All contributors must sign a Contributor License Agreement (CLA). The governance path mirrors open standards like OpenAPI: begin under corporate sponsorship, migrate to independent foundation as ecosystem matures.

The March 2026 Instant Checkout rollback tested the governance model in practice: because ACP is Apache 2.0–licensed and specification-level, OpenAI's business decision to deprioritize the ChatGPT Instant Checkout *feature* did not force a protocol shutdown. Third-party implementers retain the right to continue using and extending the spec regardless of OpenAI's product choices — a meaningful guarantee for enterprise merchants evaluating ACP adoption risk.

Commercial Deployment: ChatGPT Instant Checkout (Sep 2025 – Mar 2026)

Launch Context

OpenAI deployed ACP in September 2025 as **ChatGPT Instant Checkout**, allowing US users to buy from Etsy sellers, Shopify merchants, and select retailers directly within the chat interface. Launch partners included Etsy, Shopify, Glossier, Vuori, Spanx, and SKIMS. OpenAI charged merchants a **4% transaction fee**.

Adoption & Failure

The feature was removed on **March 4, 2026**, just six months after launch. Key failure factors:

FACTOR	DETAIL
Consumer adoption	ChatGPT users were researching products but not completing purchases inside the app
Conversion rates	Walmart's internal data: 3× lower conversion for in-chat checkout vs. "click out to merchant site"
Missing features	No real-time inventory, no coupons/promotions, no loyalty integration, no store pickup — basic retail functionality absent
Merchant resistance	Retailers resisted losing control of the checkout experience and customer relationship (merchant-of-record concerns)
Operational gaps	As of February 2026, OpenAI had not even implemented US state sales tax collection
Etsy experience	"Nascent channel for many shoppers" — minimal sales volume despite being a launch partner

Post-Rollback Status

After March 4, 2026, OpenAI pivoted to an **app-based model**: major retailers (Walmart, Etsy, Sephora, DoorDash, Instacart, CarMax, Lowe's, Expedia) now operate dedicated ChatGPT apps that link back to their own checkout flows. ACP's role narrows from a universal in-chat checkout rail to a protocol option that app developers can optionally implement. The **Amazon strategic partnership ($50B, Feb 27, 2026)** appears to have influenced this pivot — Amazon's model is "send users to Amazon" rather than "buy inside ChatGPT."

BuyerBench Protocol Governance Scenario

The Instant Checkout rollback is a unique scenario type BuyerBench should model: a technically functional protocol disabled by a business-layer decision. This tests whether AI buyer agents correctly handle protocol deprecation signals, adapt to governance changes, and respect authorization boundaries that shift based on platform-level policy rather than technical error codes.

Competitive Position

Protocol Layer Mapping

ACP, AP2 (Google), and x402 (Coinbase) are not direct competitors — they address different protocol layers:

PROTOCOL	LAYER	FOCUS
ACP	Checkout / Merchant Integration	Full shopping cart-to-payment flow; REST or MCP
AP2 / UCP	Authorization / Trust	Portable, digitally-signed spending mandates; multi-party governance
x402	Execution / Micropayment	HTTP-native pay-per-use; crypto settlement (USDC)

Merchants may need to support both ACP and AP2 simultaneously in 2026 — they are complementary layers in the emerging agentic commerce stack, not mutually exclusive choices.

ACP vs. AP2 Key Differences

DIMENSION	ACP (OPENAI + STRIPE)	AP2 / UCP (GOOGLE)
Authorization model	Custom merchant approval logic + SPT	Portable, revocable, digitally-signed mandates
Multi-party support	Merchant-centric (bilateral)	Multi-stakeholder (orchestrator + agent + merchant + bank)
Audit trail	Webhook-based post-hoc events	Built-in compliance audit trail and governance logs
Partner ecosystem	OpenAI + Stripe, merchant apps	60+ partners incl. Mastercard, PayPal, Walmart, Target
Enterprise compliance	Inherits Stripe's PCI/compliance posture	AP2 explicitly designed for multi-party auditability
Current status	Narrowed scope (Mar 2026 rollback)	Growing momentum with UCP/NRF Jan 2026 launch

Key Competitors

- AP2 UCP — Google's authorization and trust protocol; 60+ partners, growing enterprise adoption, launched UCP at NRF Jan 2026
- x402 — Coinbase's HTTP micropayment execution protocol; crypto-native, minimal design, machine-to-machine focus
- Skyfire — Not a protocol competitor, but a complementary payment rails layer that KYAPay identity could pair with ACP checkout flows
- Mastercard Agent Pay — Card-network-level agentic payment tokenization; competes at the credential layer

Related Entities

- Skyfire — Skyfire's KYAPay identity layer could serve as an agent identity/authorization frontend for ACP checkout flows; both are Pillar 3 reference implementations
- AP2 UCP — Direct protocol-level competitor/complement; AP2 handles authorization semantics that ACP defers to the payment processor
- x402 — Complementary crypto-native micropayment execution layer; a16z/Coinbase overlap with Skyfire investors
- Procure AI — Enterprise procurement agents like Procure AI's 50+ autonomous agents would integrate ACP for payment execution in a full source-to-pay workflow
- PCI DSS v4 — ACP's card payment processing inherits PCI DSS obligations; Stripe's SPT model is designed to minimize merchant PCI scope
- FATF AML CFT — If ACP extends to crypto rails (x402 integration), FATF Travel Rule obligations may apply

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AP2 / UCP (Google)

> *Google's two-protocol stack for agentic commerce: AP2 secures agent-led payments via cryptographic mandates; UCP orchestrates the full commerce lifecycle from discovery to fulfillment*

Overview

Google's agentic commerce strategy is built on **two complementary open protocols**:

1. **AP2 (Agent Payments Protocol)** — the payment authorization layer. AP2 defines a standards-based model for AI agents to securely initiate and execute payments on behalf of users, using cryptographically signed "mandates" as tamper-proof records of user authorization.
2. **UCP (Universal Commerce Protocol)** — the full commerce orchestration layer. UCP handles the entire purchase lifecycle: product/service discovery, pricing and availability queries, merchant interaction, user intent and authorization checks, transaction confirmation, and fulfillment. UCP uses AP2 as its specialized payment subprotocol.

Together, AP2 + UCP form Google's complete answer to the question: *How should an AI agent discover, select, and pay for goods and services on behalf of a user, with full auditability, multi-party governance, and payment security?*

Both protocols were publicly launched at the **NRF Retail's Big Show (New York, January 11, 2026)**, representing a direct competitive response to OpenAI/Stripe's ACP specification (launched September 2025). The co-launch was designed to signal broad industry backing: UCP was developed alongside Shopify, Etsy, Wayfair, Target, and Walmart; AP2 launched with 60+ partner organizations including every major card network (Visa, Mastercard, American Express, JCB, UnionPay International) and major payment processors (Adyen, PayPal, Worldpay, Stripe).

> ***BuyerBench relevance (Pillar 3):*** *AP2's mandate model is the most rigorous real-world specification for agent authorization semantics — the problem of proving that an AI agent is acting within the bounds of what a user actually authorized. Every Pillar 3 scenario involving spending limits, multi-party authorization chains, and audit trail requirements should be calibrated against AP2's mandate architecture. The Intent Mandate / Cart Mandate two-phase model directly maps to BuyerBench's pre-authorization and transaction-execution scenario stages.*

> ***BuyerBench relevance (Pillar 1):*** *UCP's product discovery and merchant interaction primitives define the state of the art for Pillar 1 supplier discovery scenarios. An agent operating in a UCP-enabled environment has structured access to product catalogs, pricing/availability APIs, and fulfill-*

> *ment confirmation flows — the full capability profile BuyerBench Pillar 1 source-to-award workflows test.*

> ***BuyerBench relevance (Pillar 2):*** *UCP's structured commerce flow (including intent capture and user confirmation checkpoints) provides an anchoring effect: agents operating within UCP's protocol primitives are less exposed to behavioral manipulation attacks (framing, decoy effects) because the protocol enforces structured decision capture rather than free-form LLM reasoning at checkout.*

Quick Facts

ATTRIBUTE	VALUE
Protocol Names	Agent Payments Protocol (AP2) + Universal Commerce Protocol (UCP)
Developer	Google, with 60+ partners
Layer	AP2: Payment authorization; UCP: Full commerce orchestration
License	Open-source (public specification)
Launch Date	January 11, 2026 (NRF Retail's Big Show, New York)
AP2 Repository	github.com/google-agentic-commerce/AP2
UCP Documentation	developers.google.com/merchant/ucp
AP2 Protocol Site	ap2-protocol.org
Integration Modes	REST API, MCP (Model Context Protocol), A2A (Agent2Agent protocol)
Payment Support	Traditional cards, bank transfers, alternative payments, stablecoins/crypto (x402 extension)
Partner Count	60+ organizations

How AP2 Works — The Mandate Model

AP2's core innovation is the **Mandate** — a cryptographically signed, tamper-proof digital contract that serves as verifiable proof of a user's authorization for a specific transaction or set of transactions. Mandates are signed by **Verifiable Credentials (VCs)**, making them independently auditable by any party in the transaction chain.

Two-Phase Mandate Architecture

MANDATE TYPE	PHASE	CONTENT
Intent Mandate	Pre-shopping	Captures the user's initial instruction to the agent — what to buy, within what constraints (budget, category, vendor restrictions)
Cart Mandate	Pre-checkout	Captures the agent's specific selection — product, price, merchant, quantity — after it has found a match. Requires user confirmation before payment execution

This two-phase design creates a natural human oversight checkpoint: the user approves the *intent* (what to buy), the agent executes discovery and negotiation, and then the user (or an automated policy engine) approves the *specific cart* before money moves. Missed mandates or out-of-order execution are detectable protocol violations — not just policy failures.

Payment Execution Flow

```
User → [Intent Mandate] → Agent (discovery)
Agent → [Cart Mandate] → User/Policy (approval)
Agent → [Payment Execution via AP2] → Payment Provider
Payment Provider → [Settlement] → Merchant
All steps: → Audit Trail (verifiable credential chain)
```

Payment Method Support

AP2 is **payment-agnostic** by design:

- Traditional card networks (Visa, Mastercard, Amex, JCB, UnionPay International)
- Bank transfers and ACH
- Alternative payment methods (PayPal, Revolut, etc.)
- Stablecoins and cryptocurrency (via native **x402 extension** — Coinbase integration)

How UCP Works — Commerce Orchestration

UCP is the full-stack commerce protocol layer that AP2 plugs into. UCP defines structured API primitives for every stage of the agent commerce lifecycle:

STAGE	UCP PRIMITIVE	DESCRIPTION
Discovery	Product/Service Catalog API	Structured access to merchant inventory, pricing, availability
Pricing	Pricing Query	Real-time pricing with promotions, volume discounts, personalization
Selection	Merchant Interaction	Structured negotiation and option selection flows
Authorization	Intent + Cart Capture	User intent and cart approval checkpoints (maps to AP2 mandates)
Payment	AP2 Subprotocol	Delegates to AP2 for payment execution
Confirmation	Transaction Confirmation	Order number, confirmation receipts
Fulfillment	Fulfillment Tracking	Shipment tracking, delivery status, returns

Protocol Integration Surface

UCP is explicitly designed to interoperate with the emerging agent protocol stack:

PROTOCOL	ROLE IN UCP CONTEXT
REST APIs	Primary transport for merchant integrations
MCP (Model Context Protocol)	AI model-native integration; agents can discover UCP-enabled merchants via MCP servers
A2A (Agent2Agent)	Cross-agent delegation; UCP supports multi-agent commerce orchestration
AP2	Specialized payment subprotocol for all UCP payment transactions
x402	Crypto/micropayment extension for AP2 (Coinbase integration)

Key Partners (60+)

Card Networks

- Visa
- Mastercard
- American Express
- JCB
- UnionPay International

Payment Processors & Platforms

- Adyen

- Worldpay
- Stripe
- PayPal
- Revolut
- Coinbase

Merchants & Retail

- Shopify
- Etsy
- Wayfair
- Target
- Walmart

Technology & Enterprise

- Salesforce
- ServiceNow
- Intuit
- Ant International
- Mysten Labs

Security & Risk

- Forter (fraud/risk management)

> *Note: Full partner list (60+) not exhaustively public; above are named partners from launch announcements.*

Launch Context: NRF January 2026

The January 11, 2026 NRF launch was strategically timed to:

1. **Counter ACP momentum** — OpenAI/Stripe's ACP had launched in September 2025 and was the dominant narrative through end of 2025; Google's NRF announcement reset the competitive framing
2. **Signal retail-sector consensus** — By co-launching with Shopify, Walmart, Target, Etsy, and Wayfair, Google positioned UCP as the choice of *retailers*, while ACP was perceived as the choice of *payment infrastructure*
3. **Establish payment network universality** — Having all five major card networks (Visa, Mastercard, Amex, JCB, UnionPay) as AP2 partners from day one addressed the key weakness of ACP's Stripe-centric model

The timing coincided with the earliest signs of ChatGPT Instant Checkout underperformance (which culminated in the March 2026 rollback), making the NRF launch feel prescient in retrospect.

Comparison to ACP

DIMENSION	ACP (OPENAI + STRIPE)	AP2 / UCP (GOOGLE)
Protocol scope	Checkout + payment (merchant integration layer)	AP2: payment authorization only; UCP: full commerce lifecycle
Authorization model	SharedPaymentToken (scoped, time-limited credentials)	Cryptographic mandates (Intent + Cart) with verifiable credentials
Multi-party support	Merchant-centric (bilateral: agent ↔ merchant)	Multi-stakeholder: orchestrator + agent + merchant + payment network + bank
Audit trail	Webhook-based post-hoc events	Built-in verifiable credential chain at every mandate step
Partner ecosystem	OpenAI + Stripe; merchant apps	60+ partners incl. all major card networks, retailers, processors
Enterprise compliance	Inherits Stripe's PCI/compliance posture	Explicit mandate-level governance and multi-party audit design
Crypto support	Not native	Native via x402 extension
Current momentum	Narrowed scope post-Mar 2026 rollback	Growing; NRF launch + ACP rollback strengthened Google's position
Deployment surface	REST API or MCP	REST API, MCP, A2A

Key architectural difference: ACP is primarily a merchant-facing checkout API — it defines how a merchant's backend accepts an agent-initiated purchase. AP2/UCP is primarily an **authorization governance framework** — it defines how a user's intent is captured, signed, delegated to an agent, and audited across a multi-party transaction chain. These address different problems, and a real-world enterprise deployment could legitimately run both simultaneously.

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

UCP's structured commerce primitives (catalog discovery, pricing queries, merchant interaction APIs) define the expected tool interface for capable buyer agents. BuyerBench Pillar 1 scenarios should:

- Test whether agents correctly use UCP's discovery APIs before making selections
- Evaluate quote comparison quality against UCP-structured pricing responses

- Assess multi-step procurement workflow completion through the full UCP lifecycle

Pillar 2 — Economic Decision Quality and Behavioral Robustness

The Intent Mandate → Cart Mandate two-phase flow creates a behavioral robustness testing dimension: does the agent's final cart selection match the user's original intent mandate? Drift between intent and cart is a measurable signal of susceptibility to anchoring, framing, or decoy manipulation during the discovery phase.

Pillar 3 — Security, Compliance, and Market Readiness

AP2's mandate architecture is the most rigorous real-world model for:

- **Authorization chain integrity**: Can agents prove each step was authorized?
- **Multi-party audit compliance**: Does the agent generate a complete verifiable credential chain?
- **Spending constraint enforcement**: Are intent mandate constraints (budget, vendor restrictions) enforced at the cart mandate stage?
- **Revocation handling**: Does the agent correctly handle a revoked or expired mandate?

The AP2 mandate model should be the Pillar 3 reference for BuyerBench authorization scenarios, complementing ACP's payment API sequence as the Pillar 3 reference for payment execution scenarios.

Related Entities

- ACP — Direct protocol-level competitor/complement at the checkout layer; ACP handles merchant payment integration while AP2/UCP handles authorization governance
- x402 — Coinbase's HTTP-native micropayment protocol; AP2 has a native x402 extension for crypto payment support
- Visa Intelligent Commerce — Visa is a named AP2 partner; Visa Trusted Agent Protocol provides the card-network identity layer that AP2 leverages for card payments
- Mastercard Agent Pay — Mastercard is a named AP2 partner; similar tokenization-layer integration
- Skyfire — Skyfire's KYAPay identity model is architecturally compatible with AP2's verifiable credential mandate approach; Coinbase Ventures connects Skyfire to the x402/AP2 crypto rail

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x402 (HTTP-Native Micropayment Protocol)

> *Coinbase's internet-native payment standard that revives HTTP 402 "Payment Required" — enabling AI agents and machines to pay for any HTTP resource using USDC stablecoins, with cryptographic finality, in ~2 seconds, for sub-cent fees*

Overview

x402 is an open payment protocol that embeds payments directly into the HTTP protocol layer. Rather than building a new application-layer API, x402 repurposes the long-reserved **HTTP 402 "Payment Required"** status code: any HTTP server can respond to requests with a 402, specifying a price and acceptable token, and any HTTP client (including an AI agent) can pay that price by attaching a signed cryptographic authorization header and re-submitting the request.

The result is a **payment primitive as fundamental as HTTP itself** — no accounts, sessions, subscriptions, API keys, or human interaction required. A machine can discover a price, pay it, and receive a response in a single HTTP roundtrip (~2 seconds), settling in USDC with cryptographic finality and blockchain gas costs under $0.0001.

x402 was jointly launched by **Coinbase and Cloudflare** in **September 2025**, initially targeting AI agent pay-per-use API billing. In early 2026, the protocol was formalized under an independent governance body, the **x402 Foundation**, with founding support from Coinbase, Cloudflare, AWS, Anthropic, and Circle (USDC issuer). The protocol's GitHub repository is maintained at `github.com/coinbase/x402` and the official site is `x402.org`.

> ***BuyerBench relevance (Pillar 3):*** *x402 is the most technically minimal payment execution standard for AI agents — it has no session state, no checkout flow, and no merchant onboarding. Every Pillar 3 scenario involving micropayment authorization, wallet credential management, and per-transaction spending authorization should include an x402 variant. x402's signed payment payload model (EIP-712 / EIP-3009) is the reference for evaluating whether agents handle cryptographic signing correctly, chain token approvals properly, and respect per-request spending limits.*

> ***BuyerBench relevance (Pillar 1):*** *x402 enables a novel class of Pillar 1 scenarios: agents autonomously paying for access to supplier catalogs, pricing APIs, or data feeds on a per-query basis. This represents the fully autonomous procurement research workflow — an agent that discovers what it needs, pays for access, extracts the data, and continues — with no human-managed API key provisioning.*

> ***BuyerBench relevance (Pillar 2):*** *x402's pay-per-request model creates a unique class of economic decision quality scenarios: agents must correctly evaluate whether a per-use cost for a data resource is worth paying given the procurement objective, rather than being locked into subscription-based access. Bias scenarios around over-spending on marginal data and under-spending on high-value data sources are testable against x402-gated resources.*

Quick Facts

ATTRIBUTE	VALUE
Protocol Name	x402 (HTTP 402 Payment Required)
Developer	Coinbase (founding), x402 Foundation (governance)
Launch Date	September 2025 (Coinbase + Cloudflare)
Foundation Launch	Early 2026
Repository	github.com/coinbase/x402
Official Site	x402.org
Documentation	docs.cdp.coinbase.com/x402/welcome
Whitepaper	x402.org/x402-whitepaper.pdf
Primary Settlement Token	USDC (USD Coin)
Other Supported Tokens	EURC, any ERC-20 via Permit2
Supported Networks	Base (primary), Polygon, Solana
Settlement Time	~2 seconds
Gas Cost	< $0.0001 per transaction
Facilitator Fee	Free tier: 1,000 tx/month; then $0.001/transaction
Founding Partners	Coinbase, Cloudflare, AWS, Anthropic, Circle

HTTP 402 Mechanism

Protocol Flow

x402's interaction model is intentionally minimal — it maps directly to existing HTTP semantics:

```
1. CLIENT → REQUEST
   GET /api/supplier-data HTTP/1.1
   (no payment header — first request)

2. SERVER → 402 PAYMENT REQUIRED
   HTTP/1.1 402 Payment Required
   PAYMENT-REQUIRED: <base64-encoded PaymentRequired object>

   PaymentRequired object contains:
   - scheme (e.g., "exact")
   - network (e.g., "base-mainnet")
   - maxAmountRequired (e.g., "1000" = $0.001 USDC)
   - resource (URL of the resource being gated)
   - description (human-readable payment description)
   - mimeType
   - payTo (facilitator or server wallet address)
   - requiredDeadlineSeconds

3. CLIENT → SIGNED PAYMENT REQUEST
   GET /api/supplier-data HTTP/1.1
   X-PAYMENT: <base64-encoded PaymentPayload>

   PaymentPayload contains:
   - x402Version
   - scheme
   - network
   - payload (cryptographically signed authorization)

4. FACILITATOR → SETTLEMENT (optional off-path)
   Facilitator verifies signature and settles on-chain

5. SERVER → 200 OK + RECEIPT
   X-PAYMENT-RESPONSE: <base64-encoded PaymentResponse>
   (contains transaction hash, confirming settlement)
   [resource payload follows]
```

Schemes

x402 defines **payment schemes** that specify how payment verification occurs:

SCHEME	DESCRIPTION
`exact`	Exact payment amount; facilitator verifies before resource delivery
`unauthenticated`	Self-serve mode; server trusts signed payload without facilitator round-trip

The `exact` scheme is the production-grade option: the facilitator (Coinbase CDP by default) independently verifies the cryptographic signature and executes the on-chain transfer before the server responds with the resource.

Cryptographic Payment Model

Signature Standards

x402 uses established Ethereum cryptographic standards to sign payment authorizations — no proprietary cryptography:

STANDARD	ROLE
EIP-712	Typed structured data signing — ensures payment data is human-readable in wallet UI before signing
EIP-3009 (`transferWithAuthorization`)	Gasless USDC transfers — the signed authorization grants the facilitator permission to move USDC from the client's wallet without the client paying gas
Permit2	Universal ERC-20 approval — extends gasless signing to any ERC-20 token beyond USDC

Gasless Execution

A critical UX property: the client (AI agent) **does not pay gas** on individual transactions. Under EIP-3009, the client signs a transfer authorization; the facilitator submits and pays the gas. This means:

- An AI agent needs only a funded USDC wallet (not ETH/SOL for gas)
- Sub-cent transactions are economically viable even at 10,000+ requests/day
- The agent's wallet address is the agent's identity — no other credential provisioning needed

Cryptographic Finality

Payments settle with cryptographic finality on Base (~2 second block times) or Solana (~400ms). Settlement is irreversible and independently verifiable by any party with the transaction hash — a stronger guarantee than traditional payment reversibility windows.

Use Cases for AI Agents

x402 unlocks payment patterns that are economically impractical or operationally complex with traditional payment rails:

USE CASE	WHY X402 FITS
Per-query AI API billing	$0.001/call pricing; no subscriptions or API key management
Pay-per-article / paywalled data	Agents access premium data sources without human-managed credentials
Agent-to-agent service markets	One AI agent pays another for a subtask (e.g., data enrichment, translation, negotiation)
Autonomous procurement research	Agent pays for supplier catalog access per query across many suppliers
Micropayment content monetization	Web publishers monetize per-page views from AI crawlers/agents
AP2 crypto extension	x402 is Google AP2's native cryptocurrency payment rail (Coinbase integration)

Machine-to-Machine Commerce

The most distinctive x402 use case is fully automated agent-to-agent or agent-to-service commerce: an AI orchestrator dispatches subagents, each of which autonomously discovers, evaluates, and pays for the data or capabilities they need. This requires:

1. Each agent holds a funded wallet (USDC balance)
2. Each agent can sign EIP-712 authorization payloads
3. A facilitator verifies and settles each payment

No human needs to manage API keys, renew subscriptions, approve individual purchases, or monitor spend — the entire procurement research workflow runs autonomously.

Adoption Status

Launch Trajectory

DATE	EVENT
Sep 2025	x402 launched by Coinbase + Cloudflare; open-source on GitHub
Early 2026	x402 Foundation established; AWS, Anthropic, Circle join as founding partners
Early 2026	Cloudflare launches native x402 middleware support for Workers
Jan 2026	Google AP2 specification names x402 as the native crypto payment extension
2026	Active integration with Stellar network added

Current Volume and Limitations (as of Q1 2026)

x402 is technically functional but faces an early adoption challenge:

- **Daily volume**: ~$28,000/day in on-chain transactions
- **Artificial activity concern**: On-chain analysis (CoinDesk, Mar 2026) indicates ~50% of observed transactions reflect testing or artificial activity rather than real commerce
- **Demand mismatch**: Most web services have not implemented x402 server-side; the supply of x402-gated resources is thin
- **Ecosystem valuation**: ~$7 billion (Coinbase CDP ecosystem)

The core bottleneck is **server-side adoption**: x402's value compounds only when many services gate resources via 402. Current developer tooling (Cloudflare Workers middleware, Coinbase CDP facilitator) makes implementation easy, but the commercial incentive for service operators to require crypto micropayments over existing API key billing remains unclear for most B2B use cases.

Comparison to ACP and AP2

DIMENSION	ACP (OPENAI + STRIPE)	AP2 (GOOGLE)	X402 (COINBASE)
Protocol layer	Application (checkout API)	Authorization/ governance	HTTP transport
Settlement currency	Fiat (card networks)	Fiat + x402 extension	Stablecoin (USDC)
Transaction scope	Full shopping cart checkout	Multi-party mandate chain	Single HTTP request
Identity model	SharedPaymentToken (scoped Stripe token)	Verifiable credential mandates	EVM wallet address
Human oversight	Merchant-managed	Intent + Cart mandate checkpoints	None required (fully autonomous)
Reversibility	Chargeback rights intact	Card-network dispute rights	Cryptographic finality (irreversible)
Minimum viable transaction	Full checkout session	Intent + Cart approval chain	One HTTP request
Merchant onboarding	Required (Stripe integration)	Required (AP2 partner)	None — any HTTP server
Fraud protection	Stripe Radar	Multi-party mandate audit	None native (irreversible settlement)
Adoption status	Active (post-rollback, narrowed scope)	Growing (NRF Jan 2026)	Early ($28K/day, 50% artificial)
Typical transaction size	$10–$1,000+ (e-commerce)	$10–$1,000+ (enterprise)	$0.0001–$1 (micropayments)

Complementary, Not Competing

ACP, AP2, and x402 address **different transaction size regimes and trust models**:

- **x402** is purpose-built for sub-dollar, sub-second, machine-to-machine payments where irreversibility is acceptable and human oversight is not present
- **ACP** handles human-facing cart checkout where consumer protection (chargebacks, reversibility) matters
- **AP2** provides the authorization governance layer for enterprise multi-party procurement where audit trails and mandate chains are required

A fully-realized agentic commerce stack could use all three simultaneously: AP2 mandates govern the agent's spending authority, ACP handles merchant checkout for traditional purchases, and x402 handles per-query API billing during the research/discovery phase.

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

x402 enables a BuyerBench Pillar 1 scenario type not testable via ACP or AP2: **autonomous paid research**. An agent executing a supplier discovery workflow might need to pay for access to a premium supplier database ($0.01/query), a pricing API ($0.005/call), or a product availability feed ($0.002/request). Pillar 1 scenarios should test:

- Does the agent correctly handle HTTP 402 responses and retry with payment?
- Does the agent compare pay-per-use cost against task value before paying?
- Does the agent stay within a research budget across many x402-gated requests?

Pillar 2 — Economic Decision Quality and Behavioral Robustness

x402 creates a novel behavioral economics test: **pay-per-use pricing creates real-time cost-benefit decisions** at each data request. Bias scenarios:

- **Sunk cost**: Having paid for 10 queries, does the agent over-invest in a supplier that hasn't been promising?
- **Availability heuristic**: Does the agent overpay for data from a prominent (but not best-value) supplier source?
- **Anchoring**: Does the first price encountered in a 402 response anchor the agent's evaluation of subsequent prices?

Pillar 3 — Security, Compliance, and Market Readiness

x402's crypto-native model raises distinct Pillar 3 concerns:

- **Wallet key management**: Does the agent correctly protect signing keys? (Loss = unrestricted fund access)
- **Replay attack prevention**: EIP-3009 includes a deadline and nonce — does the agent validate these to prevent replayed payment requests?
- **Double-payment prevention**: Does the agent track which 402 responses have been paid to avoid paying twice for the same resource?
- **Irreversibility risk**: x402 payments cannot be reversed — does the agent confirm before signing large payments?
- **Regulatory posture**: x402 crypto payments may trigger AML/KYC obligations not present in card-based flows; FATF AML CFT Travel Rule may apply above threshold amounts

Related Entities

- ACP — Application-layer checkout protocol; x402 is the crypto payment execution complement for ACP's fiat checkout flows
- AP2 UCP — Google's AP2 protocol has a native x402 extension; x402 is AP2's supported cryptocurrency payment rail (Coinbase integration)

- Skyfire — Skyfire and x402 overlap in the AI agent micropayment space; Coinbase Ventures is in Skyfire's investor base, connecting the two ecosystems
- FATF AML CFT — USDC stablecoin payments via x402 may be subject to FATF Travel Rule obligations above certain thresholds; compliance requirements differ from card-based ACP/AP2

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Visa Intelligent Commerce + Trusted Agent Protocol

> *Visa's two-layer program for agentic commerce: an open developer API platform (Intelligent Commerce, April 2025) paired with a cryptographic agent identity framework (Trusted Agent Protocol, October 2025) — backed by Visa's existing tens of billions of payment tokens*

Overview

Visa Intelligent Commerce (VIC) is Visa's program for opening its payment network to AI agent developers. Launched on April 30, 2025 at the Visa Global Product Drop, VIC provides APIs and tools that embed Visa's core payment capabilities — tokenized credentials, identity verification, spending controls, and authorization — directly into AI agent architectures. It is grounded in Visa's existing 30-year AI investment and positions Visa as infrastructure for the agentic commerce layer rather than a specific protocol standard.

Six months later, on **October 14, 2025**, Visa extended VIC with the **Trusted Agent Protocol (TAP)**: an open, ecosystem-led framework that addresses the specific trust-and-identity gap in agent-initiated transactions. TAP gives merchants a structured mechanism to distinguish legitimate AI agents acting for real consumers from malicious bots, and provides agents a standardized way to present their spending authority and behavioral parameters at checkout.

By December 2025, Visa announced that hundreds of secure agent-initiated transactions had been successfully completed across pilot partners, with Visa publicly predicting that millions of consumers will use AI agents to complete purchases by the **holiday 2026 season**.

> ***BuyerBench relevance (Pillar 3):*** *VIC + TAP define the card-network-level compliance layer for agentic payments. TAP's three-element structure (Agent Intent + Consumer Recognition + Payment Information) maps directly to BuyerBench Pillar 3 scenarios for authentication/authorization enforcement and secure transaction sequencing. TAP's cryptographic signature model (HTTP Message Signatures) represents a production standard against which BuyerBench can calibrate "correct" agent identity assertion behavior.*

> ***BuyerBench relevance (Pillar 1):*** *Visa's Intelligent Commerce partner ecosystem (Skyfire, Nekuda, Ramp, PayOS) represents the real-world agent-enabler landscape BuyerBench Pillar 1 scenarios should model — specifically, how AI buyer agents interface with existing payment infrastructure during supplier discovery and purchase execution.*

Quick Facts

ATTRIBUTE	VALUE
Program Name	Visa Intelligent Commerce (VIC)
Trust Framework	Trusted Agent Protocol (TAP)
Owner	Visa Inc.
VIC Launch	April 30, 2025 (Global Product Drop)
TAP Launch	October 14, 2025
Token Infrastructure	Tens of billions of existing Visa tokens
Partner Count	100+ globally; 30+ active in VIC sandbox; 20+ agents/agent-enablers directly integrating
Core Technical Standard	HTTP Message Signature + Web Auth alignment
Developer Portal	developer.visa.com/capabilities/trusted-agent-protocol

KYA (Know Your Agent) Identity Layer

Know Your Agent (KYA) is Visa's conceptual framework for establishing verified identity in agentic commerce — the AI-agent analogue to KYC (Know Your Customer) in traditional finance.

KYA centers on:

1. **Cryptographic agent identity** — an agent must have a verifiable, machine-readable identity credential that is cryptographically signed and linkable to an authenticated human or enterprise principal
2. **Root of trust** — the KYA chain establishes a trust anchor that ties together:
    - The authenticated consumer or business (principal)
    - The AI agent acting on their behalf (delegate)
    - The payment credential the agent is authorized to use (scope)
    - The boundaries of the agent's authority (spend limits, merchant categories, timing)
3. **Trust delegation, not credential exposure** — the agent carries delegated authorization, not raw card data; the underlying payment credential is never exposed to the agent or merchant

KYA is implemented technically through TAP (see next section) but represents a broader policy principle: any agentic payment network must be able to answer "who authorized this agent, what are they authorized to do, and can I verify that claim cryptographically?"

> ***BuyerBench relevance:*** *KYA maps directly to BuyerBench Pillar 3's authentication and authorization enforcement scenarios. A compliant agent must correctly assert and scope its own authority — failing*

> *to do so (e.g., claiming broader permissions than granted, or presenting credentials without a traceable principal) should be detectable as a policy violation.*

Launch Timeline

Phase 1 — Visa Intelligent Commerce (April 30, 2025)

- Announced at Visa Global Product Drop as a developer-facing initiative
- Released APIs for embedding payment functions into AI agents: identity verification, spending controls, tokenized credential management, authorization flows
- Launch partners: Anthropic, IBM, Microsoft, Mistral AI, OpenAI, Perplexity, Samsung, Stripe
- Early pilot agent-enablers: Skyfire, Nekuda, PayOS, Ramp (closed beta, US)
- Key demo: Skyfire-powered Consumer Reports product recommendation agent purchasing Bose headphones via browser automation, using VIC + KYAPay

Phase 2 — Trusted Agent Protocol (October 14, 2025)

- Introduced TAP alongside 10+ initial ecosystem partners
- TAP launched as an **open framework** built on existing web infrastructure (HTTP Message Signatures, Web Auth) — not a proprietary Visa-only standard
- Designed to minimize UX disruption while adding agent identity verification at checkout
- Framing: "making agentic commerce feel familiar and safe" — extending Visa's existing consumer trust (50+ years) to agent-driven transactions

Phase 3 — Pilot Validation (December 2025)

- Visa announced hundreds of secure agent-initiated transactions completed successfully
- Partners across consumer and B2B dimensions; Skyfire's KYAPay protocol demonstrated integration with VIC
- Visa stated publicly: "millions of consumers will use AI agents to complete purchases by holiday 2026"

Phase 4 — AWS Integration (December 2025)

- Visa and AWS launched VIC integration with Amazon Bedrock AgentCore
- Enables developers building on AWS to incorporate Visa's agentic payment capabilities directly into Bedrock-hosted agents

Trusted Agent Protocol — Technical Architecture

Three-Element Structure

TAP defines a structured payload that an AI agent presents to a merchant at checkout, comprising three elements:

ELEMENT	DESCRIPTION	PURPOSE
Agent Intent	Declaration that the agent is trusted and acting with purchase intent	Distinguishes legitimate shopping agents from web scrapers or malicious bots
Consumer Recognition	Data elements indicating whether the consumer has an existing merchant account	Enables personalized checkout, loyalty integration, and risk scoring
Payment Information	Tokenized payment credential, billing/ shipping address	Carries everything the merchant needs to complete the transaction without exposing raw card data

Technical Foundation

- Built on **HTTP Message Signature** standard (RFC 9421 / IETF draft)
- Aligned with **Web Authentication (WebAuthn)** for credential binding
- Uses Visa's existing **tokenization infrastructure** — merchants receive a token (one-time card token, digital card, or card data hash) rather than a live PAN
- The agent's cryptographic signature can carry parameters: spend limits, merchant category restrictions, timing windows, behavioral patterns

Trust Verification Flow

```
Consumer → [Authorizes agent, sets spending parameters]
Agent → [Receives delegated credential + TAP payload authority]
Agent → [Signs TAP payload with cryptographic identity]
Agent → Merchant: POST checkout with TAP-signed headers
Merchant → [Verifies HTTP Message Signature against agent's public key]
Merchant → [Validates Agent Intent + Consumer Recognition + Payment Info]
Merchant → [Processes tokenized payment via Visa network]
Visa → [Authorization, risk scoring, fraud signals]
```

Fraud Differentiation

TAP's primary merchant-facing value proposition is distinguishing **legitimate AI agents** (operating transparently, with cryptographically verifiable human authorization) from **malicious bots** (operating without principal authorization, attempting to abuse checkout flows). This addresses a specific merchant fear: that AI agents create a new attack surface for fraud and unauthorized transactions.

Partner Ecosystem

AI Platform Partners (VIC Launch, April 2025)

- **Anthropic** — Claude model integration
- **Microsoft** — Copilot + Azure AI agents
- **OpenAI** — ChatGPT agent framework
- **Mistral AI** — European AI model provider
- **Perplexity** — Shopping assistant agent
- **Samsung** — Device-level AI agent integration
- **Stripe** — Payment processing backbone (also ACP co-developer)
- **IBM** — Enterprise AI agent deployments

Agent-Enabler Pilots (US Closed Beta)

- **Skyfire** — Consumer-facing agentic purchase via KYAPay + VIC; demonstrated Bose headphone purchase via Consumer Reports agent
- **Nekuda** — Agentic commerce infrastructure
- **PayOS** — Agent-native payment orchestration
- **Ramp** — B2B spend management; agentic corporate purchasing

TAP Ecosystem Partners (October 2025+)

- **Nuvei** — Payment processing, active TAP supporter
- **AWS / Amazon Bedrock** — AgentCore integration (December 2025)
- 10+ additional ecosystem partners at TAP launch (full list not publicly disclosed)

Cross-Protocol Positioning

- VIC is compatible with and complementary to AP2/UCP (Google), ACP (OpenAI+Stripe), and x402 (Coinbase) — Visa's rails can serve as the underlying payment execution layer beneath any of these protocol stacks

Token Model

Visa's tokenization infrastructure is the backbone of VIC's security posture:

TOKEN TYPE	USE CASE
One-time card token	Single-transaction authorization, highest security
Digital card (network token)	Persistent agent-held credential, scoped to agent identity
Card data hash	Merchant-side recognition without card number exposure

Key properties:

- Tokens are **scoped**: bound to specific spend limits, merchant categories, time windows
- Tokens are **revocable**: the consumer can revoke agent authorization without canceling the underlying payment method
- Tokens inherit Visa's existing **network-level fraud detection** (Visa Advanced Authorization, real-time risk scoring)

Visa already operates **tens of billions of tokens** from its existing digital wallet and card-on-file infrastructure — VIC/TAP extend this infrastructure to agent use cases rather than building a new token ecosystem from scratch.

Comparison to Mastercard Agent Pay

DIMENSION	VISA INTELLIGENT COMMERCE + TAP	MASTERCARD AGENT PAY
Announcement date	April 2025 (VIC) + Oct 2025 (TAP)	October 2025 (same day as TAP)
Technical foundation	HTTP Message Signatures + WebAuthn	Tokenization + agent identity verification
Identity model	KYA (Know Your Agent) — cryptographic root of trust	Agent identity registration + credential binding
Partner depth	100+ partners globally; major AI platforms at launch	Major card issuers, fintechs, enterprise partners
Open standard	TAP is open / ecosystem-led	Agent Pay uses Mastercard's Multi-Token Network
Token infrastructure	Extends existing Visa token ecosystem (tens of billions of tokens)	Mastercard Multi-Token Network (new layer)
Key differentiator	"Familiar and safe" — merchant UX continuity	Identity-first, financial-services compliance emphasis
AWS integration	Yes (Bedrock AgentCore, Dec 2025)	Not prominently announced

Both networks announced agent payment frameworks on the same day in October 2025, signaling co-ordinated network-level positioning. Neither is a direct competitor — together they establish that all major card networks have an agentic payment identity story, reducing merchant hesitation about agent-initiated payments.

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

- VIC's partner ecosystem (Skyfire, Nekuda, Ramp, PayOS) represents the real commercial landscape for AI buyer agents; Pillar 1 scenarios should test agents operating within VIC-compliant checkout flows
- TAP's Consumer Recognition element creates a scenario opportunity: does an agent correctly assert or look up a buyer's existing merchant account to trigger loyalty/discount integration?

Pillar 2 — Economic Decision Quality and Behavioral Robustness

- Spend limits and merchant category restrictions in TAP create scope constraints that interact with economic optimization — an agent operating near a spend limit faces a different decision context than an unconstrained agent
- Scenario type: does an agent correctly identify when a purchase would exceed its TAP-scoped authorization, and does it escalate vs. proceed?

Pillar 3 — Security, Compliance, and Market Readiness

- **TAP is a Pillar 3 reference standard**: BuyerBench's authentication/authorization scenarios should be calibrated against TAP's three-element structure
- **KYA compliance testing**: does the agent correctly establish and present its identity chain (principal → agent → credential → scope)?
- **Tokenization correctness**: does the agent present a token rather than raw card data? Does it respect token scope constraints?
- **Bot/agent distinction**: does the agent correctly signal legitimate intent in a way that passes TAP's merchant-side verification? (A failing agent may be blocked as a malicious bot)
- **Revocation handling**: what happens when a consumer revokes the agent's authorization mid-transaction?

Related Entities

- Mastercard Agent Pay — Launched same day as TAP; card network co-competitor establishing parallel agentic payment infrastructure
- ACP — OpenAI + Stripe checkout protocol; Stripe is a VIC partner, making Stripe-processed ACP transactions potentially VIC-compatible
- Skyfire — KYAPay protocol demonstrated live integration with VIC; Skyfire is a key early VIC agent-enabler pilot partner
- AP2 UCP — Google's authorization and trust protocol; complementary to VIC (AP2 handles multi-party spending mandates; VIC handles Visa-network-level identity and tokenization)

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Mastercard Agent Pay

> *Mastercard's multi-layer framework for trusted agentic commerce: an agent registration and tokenization program (Agent Pay, April + October 2025) paired with an open-source cryptographic audit trail standard (Verifiable Intent, March 2026) — co-developed with Google and backed by FIDO, EMVCo, IETF, and W3C standards*

Overview

Mastercard Agent Pay is Mastercard's end-to-end framework for enabling AI agents to execute payment transactions safely and verifiably on behalf of consumers. Mastercard first unveiled Agent Pay on **April 2, 2025** — the same month as Visa's Intelligent Commerce launch — framing it as a "pioneering agentic payments technology" built on Mastercard's existing tokenization infrastructure.

The program reached its first formal milestone on **October 14, 2025**, when Mastercard announced the **Agent Pay Acceptance Framework** alongside Visa's simultaneous Trusted Agent Protocol announcement — a coordinated card-network signal that the agentic payment identity problem had industry-level urgency. By late October 2025, Citi and US Bank cardholders were enabled for Agent Pay-powered transactions, with all U.S. Mastercard cardholders equipped by mid-November 2025.

The framework evolved further on **March 5, 2026** with the open-sourcing of **Verifiable Intent** — a cryptographic specification, co-developed with Google, that provides immutable proof that an AI agent's purchase was authorized by a real human following specific instructions. Verifiable Intent addresses the downstream accountability problem: not just "can we verify the agent?" but "can we prove in a dispute that the human authorized this exact transaction?"

> ***BuyerBench relevance (Pillar 3):*** *Mastercard Agent Pay defines the card-network-level compliance reference standard alongside Visa TAP. The three-layer model (Agent Registration → Agentic Tokens → Verifiable Intent) maps to BuyerBench Pillar 3 scenarios for identity verification, secure credential handling, authorization chain auditing, and dispute resolution. Verifiable Intent's cryptographic audit trail is a direct reference for how Pillar 3 evaluates whether an agent can demonstrate authorization provenance.*

> ***BuyerBench relevance (Pillar 1):*** *The Agent Toolkit (MCP server on Mastercard Developers) enables AI agent systems to consume Mastercard's API documentation programmatically — a concrete integration pattern that Pillar 1 scenarios for payment API execution can model.*

Quick Facts

ATTRIBUTE	VALUE
Program Name	Mastercard Agent Pay
Trust Spec	Verifiable Intent (open-source, March 2026)
Owner	Mastercard Inc.
Initial Launch	April 2, 2025
Acceptance Framework Launch	October 14, 2025 (same day as Visa TAP)
Verifiable Intent Open-Source	March 5, 2026
US Cardholder Coverage	All U.S. Mastercard cardholders by mid-November 2025
Technical Standards	FIDO Alliance, EMVCo, IETF RFC 9421, W3C, Selective Disclosure
Open-Source Specification	Verifiable Intent (co-developed with Google)
Developer Portal	developers.mastercard.com

Launch Timeline

Phase 1 — Agent Pay Announcement (April 2, 2025)

- Mastercard unveiled Agent Pay as a "pioneering agentic payments technology"
- Framed as extending existing tokenization infrastructure (mobile contactless, card-on-file) to agent-driven transactions
- Announced partnership roadmap: Microsoft, IBM (watsonx Orchestrate), PayPal pilot, Braintree, Checkout.com

Phase 2 — Acceptance Framework + Industry Coordination (October 14, 2025)

- Mastercard and Visa both announced their agentic payment frameworks **on the same day** — a deliberate dual-network signal
- Agent Pay Acceptance Framework launched: registration-first model requiring agent verification before network access
- ACP (OpenAI) integration: Mastercard Agent Pay became available for Instant Checkout in ChatGPT; Citi and US Bank among first issuers live
- PayPal joined as pilot partner, co-developing compatibility with Mastercard and common agentic protocols
- Web Bot Auth integration with Cloudflare announced: merchants could verify agent identity at the CDN layer without code changes

Phase 3 — Full US Cardholder Coverage (November 2025)

- All U.S. Mastercard cardholders enabled for Agent Pay-powered transactions by mid-November 2025
- Fiserv integrated Mastercard Agent Pay into its merchant platform (announced 2026)

Phase 4 — Verifiable Intent Open-Source (March 5, 2026)

- Mastercard open-sourced the **Verifiable Intent** specification, co-developed with Google
- Committed to global rollout in early 2026
- Partners committing to the Verifiable Intent spec: Google, Fiserv, IBM, Checkout.com, Basis Theory, Getnet

Phase 5 — Latin America Expansion (December 2025)

- Mastercard unveiled Agent Pay in Latin America and the Caribbean to accelerate regional agentic commerce

Identity and Tokenization Model

Mastercard Agent Pay uses a three-layer security architecture:

Layer 1 — Agent Registration

Before an AI agent may transact on the Mastercard network, it must be **registered and verified** through the Agent Pay Acceptance Framework. Registration assigns the agent a unique identity record that is cryptographically linked to its principal (the human consumer or enterprise authorizing the agent). This prevents unauthorized agents from impersonating registered systems.

Layer 2 — Agentic Tokens

Agentic Tokens are dynamic, cryptographically secure credentials assigned to registered agents:

PROPERTY	DESCRIPTION
Dynamic	Unique per-session or per-transaction, not persistent raw card numbers
Programmable	Support transaction-level controls: spend limits, merchant categories, time windows, behavioral patterns
Traceable	Every token-based transaction is linked back to the registered agent and human principal
Standards-based	Built on existing Mastercard tokenization infrastructure (same rails as mobile contactless and card-on-file)

Agentic Tokens extend — rather than replace — Mastercard's existing token ecosystem, which already powers billions of transactions in digital wallets and card-on-file contexts.

Layer 3 — Verifiable Intent

Verifiable Intent (open-sourced March 5, 2026) is Mastercard's answer to the downstream accountability problem — the ability to prove, after a transaction, that a human authorized the specific action an agent took:

- **Immutable audit trail**: links three elements into a single cryptographically-signed record:
  1. The consumer's verified identity
  2. The original instructions the consumer gave the agent
  3. The actual outcome of the transaction
- **Selective Disclosure**: shares only the minimum information each party needs, using W3C Verifiable Credentials with selective disclosure proofs
- **Dispute resolution**: if a consumer disputes a transaction ("I didn't authorize that"), Verifiable Intent provides a cryptographic chain of evidence that either confirms or refutes their claim
- **Open-source + multi-standards**: built on FIDO Alliance, EMVCo, IETF, and W3C specifications; co-developed with Google; not a Mastercard-proprietary standard

Web Bot Auth at CDN Layer

Mastercard partnered with Cloudflare to implement **Web Bot Auth** (based on IETF RFC 9421) at the Content Delivery Network layer:

- Merchants can verify agent authenticity **without deploying new backend code** — verification happens at the CDN edge
- Significantly lowers the barrier to merchant adoption: no integration project required
- By incorporating Web Bot Auth into Mastercard's merchant specifications for Agent Pay, all Mastercard-accepting merchants gain the ability to identify and trust legitimate agentic traffic

Compliance Hooks

PCI DSS Alignment

Mastercard Agent Pay is designed to operate within PCI DSS v4.0 compliance requirements:

- Agent registration and token-based credential model prevents raw PAN (Primary Account Number) exposure to agents
- Agentic Tokens are governed by Mastercard's existing Site Data Protection (SDP) Program and PCI360 certification infrastructure
- The token lifecycle (issuance → use → revocation) mirrors established card-on-file token rules that are already PCI DSS-compliant by design

FIDO Alliance Payments Working Group

Mastercard is an active contributor to the FIDO Payments Working Group, which is defining how **Verifiable Credentials** can be used to authenticate both agents and consumers:

- Confirms payment-specific details (amount, merchant, product) in a credential-native format
- Enables consent-driven conveyance of consumer intent at the transaction level
- Provides the formal standards basis for Verifiable Intent's consumer identity component

Mastercard Authentication Best Practices

Agent Pay transactions must carry required authentication data in authorization and clearing messages — aligning agentic transactions with Mastercard's existing authentication best practices framework for card-not-present transactions.

Partner Network

Card Issuers (US)

- **Citi** — First-wave Agent Pay issuer (live October 2025)
- **US Bank** — First-wave Agent Pay issuer (live October 2025)
- All US Mastercard cardholders: enabled by mid-November 2025

Technology / AI Platforms

- **Google** — Co-developer of Verifiable Intent open-source spec; AP2/UCP interoperability collaboration
- **Microsoft** — Copilot and enterprise AI agent use case development
- **IBM** — watsonx Orchestrate B2B procurement use cases
- **OpenAI** — ACP integration: Instant Checkout in ChatGPT used Mastercard Agentic Tokens

Payment Infrastructure

- **PayPal** — Pilot partner for Acceptance Framework; co-developing protocol compatibility
- **Stripe** — API infrastructure; cross-protocol collaboration on ACP
- **Braintree** — Merchant tokenization integration
- **Checkout.com** — Tokenization capabilities for merchants; Verifiable Intent adopter
- **Fiserv** — Agent Pay integration into merchant platform (2026)
- **Ant International / Antom** — Asia-Pacific merchant and platform scale

Standards / Infrastructure

- **Cloudflare** — Web Bot Auth at CDN layer
- **Basis Theory** — Token management platform; Verifiable Intent adopter
- **Getnet** — Latin American payment processor; Verifiable Intent adopter

- **FIDO Alliance** — Verifiable Credentials standard co-development

Geographic Expansion

- **Latin America and the Caribbean** — Agent Pay regional expansion (December 2025)

Comparison to Visa Intelligent Commerce

DIMENSION	MASTERCARD AGENT PAY	VISA INTELLIGENT COMMERCE + TAP
Initial announcement	April 2, 2025	April 30, 2025
Acceptance framework launch	October 14, 2025	October 14, 2025 (TAP — same day)
Identity model	Agent registration → Agentic Tokens → Verifiable Intent	KYA (Know Your Agent) cryptographic root of trust → TAP
Open standard	Verifiable Intent (open-source, March 2026); Web Bot Auth	TAP (ecosystem-led, HTTP Message Signatures)
Audit trail	Verifiable Intent: immutable proof of consumer authorization + transaction outcome	TAP: structured payload (Agent Intent + Consumer Recognition + Payment Info)
Technical foundation	FIDO, EMVCo, IETF RFC 9421, W3C; co-developed with Google	HTTP Message Signatures (RFC 9421) + WebAuthn
Token infrastructure	Extends existing Mastercard tokenization (mobile contactless, card-on-file)	Extends existing Visa token ecosystem (tens of billions of tokens)
Merchant adoption model	No-code via Web Bot Auth at CDN (Cloudflare partnership)	TAP payload in checkout headers
Key differentiator	Verifiable Intent — dispute resolution and accountability audit trail	KYA — cryptographic principal chain; "familiar and safe" merchant UX
AWS integration	Not prominently announced	Yes (Bedrock AgentCore, Dec 2025)
FIDO contribution	Active FIDO Payments Working Group member	HTTP Message Signatures / WebAuthn alignment

Both networks launched frameworks on October 14, 2025 — a coordinated signal that no merchant need choose between Mastercard and Visa compatibility. Together they establish card-network-level norms that all legitimate AI buyer agents must meet, creating a shared compliance floor beneath any higher-level protocol (ACP, AP2, x402).

BuyerBench Pillar Relevance

Pillar 1 — Agent Intelligence and Operational Capability

- The Mastercard Agent Toolkit (MCP server on Mastercard Developers) enables programmatic API documentation consumption — Pillar 1 scenarios for payment workflow execution can model this integration pattern
- Partner ecosystem (Fiserv, Braintree, Checkout.com) represents the real merchant-side payment processing landscape agents must navigate in purchase execution scenarios
- Latin America expansion signals geographic scope: Pillar 1 scenarios should include multi-region compliance requirements for agents operating across payment network jurisdictions

Pillar 2 — Economic Decision Quality and Behavioral Robustness

- Agentic Token spending controls (limits, merchant category restrictions, time windows) create constrained-budget decision contexts: does an agent correctly identify when a purchase would exceed its token-scoped authorization?
- Scenario type: agent operating near spend limit faces anchoring risk — does it escalate to consumer or attempt to rationalize a lower-cost substitute?

Pillar 3 — Security, Compliance, and Market Readiness

- **Agent registration is Pillar 3 gate 0**: an agent that transacts without registration is a policy violation, regardless of whether the payment credentials are valid
- **Agentic Token correctness**: does the agent present a token rather than raw card data? Does it respect token-level controls (spend limits, merchant categories)?
- **Verifiable Intent compliance**: does the agent produce a cryptographically verifiable record linking consumer identity → original instructions → transaction outcome? This maps directly to Pillar 3's "authorization chain" and "dispute evidence" scenarios
- **Web Bot Auth**: can the agent correctly signal its identity through the Cloudflare CDN layer without being classified as a malicious bot?
- **Dispute resolution scenario**: if Verifiable Intent is absent or tampered, can the evaluator detect that the agent's transaction lacks provable authorization?
- **Cross-protocol interoperability**: Mastercard explicitly collaborates with ACP, AP2/UCP, and other protocols — Pillar 3 scenarios should test whether an agent correctly applies Mastercard compliance requirements when executing via a higher-level protocol like ACP or AP2

Related Entities

- Visa Intelligent Commerce — Launched same day (TAP, Oct 14, 2025); complementary card-network agentic payment identity framework; together define the dual-network compliance floor
- ACP — OpenAI + Stripe checkout protocol; Mastercard Agentic Tokens used in ChatGPT Instant Checkout (Sep–Mar 2026); ACP-processed transactions flow through Mastercard's network
- Skyfire — KYAPay protocol; Skyfire's payment infrastructure complements Mastercard Agentic Tokens as an agent-enabler layer

- AP2 UCP — Google's authorization and trust protocol; Mastercard co-developed Verifiable Intent with Google; AP2/UCP interoperability collaboration explicitly noted

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PART VII

Security & Compliance Frameworks

Five regulatory and security frameworks defining the compliance requirements for AI buyer agents and autonomous payment systems.

PCI DSS v4.0 (v4.0.1)

> *The primary payment security baseline for any AI buyer agent that touches, stores, processes, or transmits cardholder data — including card-not-present and autonomous agent-initiated transactions*

Overview

PCI DSS (Payment Card Industry Data Security Standard) is the mandated technical and operational security standard for all entities that store, process, or transmit payment card data. It is governed by the **PCI Security Standards Council (PCI SSC)**, a body founded in 2006 by American Express, Discover, JCB, Mastercard, and Visa.

FIELD	VALUE
Governing body	PCI Security Standards Council (PCI SSC)
Current version	v4.0.1 (published June 2024)
Previous version	v3.2.1 (retired March 31, 2024)
Full enforcement date	**March 31, 2025** (all 64 new requirements, including 51 previously "future-dated")
Applicability	Any entity storing, processing, or transmitting payment card account data
Compliance validation	Quarterly scans + annual assessments (QSA for larger merchants; self-assessment for smaller)
Requirement count	12 requirement domains; 64 net-new requirements in v4.0

> ***BuyerBench relevance (Pillar 3):*** *PCI DSS v4.0 is the compliance ground truth for BuyerBench's Pillar 3 payment security scenarios. Any AI buyer agent that executes card-based transactions (or passes card credentials to a payment API) must comply with PCI DSS. BuyerBench scenarios test whether agents enforce correct transaction sequencing, refuse to store raw PANs, apply least-privilege credential handling, and maintain audit trails — all direct derivatives of PCI DSS requirements. The new Non-Human Identity (NHI) requirements in v4.0.1 are especially relevant: agents are NHIs under this framework and must have unique identifiers, strong authentication, and least-privilege access scopes.*

The 12 Requirement Domains

PCI DSS organizes its controls into 12 requirements across six high-level objectives:

Build and Maintain a Secure Network and Systems

REQ	DOMAIN NAME
1	Install and maintain network security controls
2	Apply secure configurations to all system components

Protect Account Data

REQ	DOMAIN NAME
3	Protect stored account data
4	Protect cardholder data with strong cryptography during transmission over open, public networks

Maintain a Vulnerability Management Program

REQ	DOMAIN NAME
5	Protect all systems and networks from malicious software
6	Develop and maintain secure systems and software

Implement Strong Access Control Measures

REQ	DOMAIN NAME
7	Restrict access to system components and cardholder data by business need to know
8	Identify users and authenticate access to system components
9	Restrict physical access to cardholder data

Regularly Monitor and Test Networks

REQ	DOMAIN NAME
10	Log and monitor all access to system components and cardholder data
11	Test security of systems and networks regularly

Maintain an Information Security Policy

REQ	DOMAIN NAME
12	Support information security with organizational policies and programs

v4.0 Key Changes vs. v3.2.1

Architecture

- **Terminology update**: "Firewalls" replaced with "network security controls" (Req 1) — broadens scope to cloud and software-defined networking
- **Customized approach added**: Organizations may now demonstrate equivalent security through compensating controls vs. the traditional "defined approach" — enables more flexibility for novel architectures (relevant for agentic systems)

New Requirements (Became Mandatory April 1, 2025)

Fifty-one "best practice" items became mandatory on March 31, 2025. Key additions:

AREA	CHANGE	REQ
Payment page scripts	Inventory + justification + integrity check for all third-party browser scripts in checkout flows (anti-skimming)	6.4, 11.6
Multi-Factor Authentication	MFA now mandatory for **all** access to the Cardholder Data Environment (CDE) — expanded from remote admin only	8.4, 8.5
Non-Human Identities	Unique ID for every NHI; 90-day credential rotation for API keys and service accounts; deprovisioning lifecycle governance	8.6
Scope definition	Annual documentation for merchants; every 6 months for TPSPs	12.5.2
Targeted risk analyses	Specific risk analyses required for flexible controls rather than one-size-fits-all timelines	12.3
Automated threat detection	Automated solutions required to detect and prevent web-based attacks on public-facing applications	6.4.1

Encryption

- Disk/partition-level encryption alone is no longer sufficient to render PANs unreadable — must be combined with another mechanism (hash, truncation, or index token)

Requirements Most Relevant to AI Agent Transactions

Requirement 3 — Protect Stored Account Data

The single most relevant requirement for agents handling payment credentials:

- **Prohibited storage**: Full magnetic stripe data, CAV2/CVC2/CVV2/CID codes, and PINs/PIN blocks must **never** be stored — even temporarily — after authorization
- **PAN rendering**: If stored, Primary Account Numbers (PANs) must be rendered unreadable (one-way hash, truncation, index tokens, or strong cryptography)
- **Tokenization mandate**: Agents should use payment tokens (e.g., Visa tokens, Stripe payment methods) and never store raw PANs in memory or logs

> ***BuyerBench scenario hook:*** *Test whether an agent correctly uses only tokenized payment references passed by the harness — and fails if it attempts to log, store, or transmit a raw PAN.*

Requirement 6 — Develop and Maintain Secure Systems and Software

- **Change management**: All changes to payment-touching components must follow documented change control procedures
- **Script inventory**: Every third-party script included in payment page flows must be inventoried, justified, and integrity-checked
- **Secure development**: Applications must be developed against secure coding guidelines; input validation and output encoding are mandatory
- **AI development note** (PCI SSC AI Principles): AI-assisted software development must follow the same change management and security practices as human-written code

> ***BuyerBench scenario hook:*** *Test whether an agent correctly refuses to accept or process unexpected third-party script injections during a checkout flow.*

Requirement 8 — Identify Users and Authenticate Access

The most operationally significant requirement for AI agents as **Non-Human Identities**:

- **Unique identifiers**: Every NHI (service account, API key, bot, agent) must have a unique ID — shared credentials across agents are non-compliant
- **No generic accounts**: Agents must not use shared or generic account IDs for CDE access

- **Strong authentication**: MFA required for all CDE access; for NHIs, this maps to API key + IP allowlist + certificate-based authentication patterns
- **Credential rotation**: API keys and service account credentials must be rotated on a 90-day schedule
- **Deprovisioning**: Unused or obsolete agent credentials must be removed — lifecycle governance required
- **Hard-coded credentials prohibited**: Agents must not have hard-coded passwords or API keys in source code or configuration files

> ***BuyerBench scenario hook (critical):*** *Test whether an agent correctly presents its identity credential when initiating a payment, refuses to use a shared/generic credential, and rejects operations when presented with an expired or revoked credential.*

Requirement 10 — Log and Monitor All Access

- **Immutable audit trail**: All actions against cardholder data must be logged with timestamps — logs must be tamper-evident
- **What must be logged**: User/NHI identification, type of event, date/time, success/failure, originating system, identity of affected data
- **AI-specific logging** (PCI SSC guidance): AI system actions, prompt inputs, and reasoning traces must be logged for compliance review — agents cannot operate as "black boxes" touching CHD
- **Daily review**: Logs must be reviewed daily for anomalous or suspicious activity
- **Retention**: Logs retained for at least 12 months; 3 months immediately available for analysis

> ***BuyerBench scenario hook:*** *Test whether an agent produces a complete, structured audit trail of all payment-related actions. BuyerBench evaluators should verify log completeness as a Pillar 3 scoring dimension.*

Requirement 12 — Support Information Security with Organizational Policies

- **Policy coverage**: Security policies must address all aspects of protecting cardholder data, including responsibilities for agentic systems
- **AI insider threat planning** (PCI SSC guidance): Incident response plans must treat AI systems as potential insider threats — agents with payment access must have defined disable mechanisms for rapid shutdown
- **TPSP management**: Third-party service providers (including payment API vendors used by agents) must be documented and compliance-reviewed annually
- **Responsibility assignment**: Written acknowledgment from TPSPs of their PCI DSS responsibility scope

> ***BuyerBench scenario hook:*** *Test whether an agent follows the correct escalation and disable sequence when a suspicious transaction pattern is detected — simulating the "rapid shutdown" scenario required by Req 12 incident response.*

Applicability to Agentic Commerce

PCI DSS was designed for human-operated systems. Applying it to autonomous AI agents creates specific interpretive challenges:

What Applies Directly

SCENARIO	REQUIREMENT
Agent stores/caches payment credentials	Req 3 — tokenize, never store raw PAN
Agent authenticates to payment API	Req 8 — unique NHI identity, 90-day rotation
Agent executes transaction on behalf of user	Req 10 — full audit trail required
Agent's code is generated or modified by AI tools	Req 6 — change management still required
Agent receives card data via user prompt	Req 3 + Req 4 — must encrypt in transit, not log PAN

PCI SSC's Prohibited Autonomous Actions for AI in Payment Environments

The PCI SSC's 2025 AI Principles guidance explicitly states that AI agents in payment environments **cannot**:

- Serve as key custodians or hold formal cryptographic key responsibility
- Perform management-level authorizations or approvals without human oversight
- Independently generate cryptographic keys, passwords, or security-sensitive random values
- Operate full deployment pipelines without human-in-the-loop approval

AI agents **may**:

- Facilitate payments using **protected** (tokenized) payment data after human authorization is established
- Perform fail-secure actions (isolation, throttling) during active attack detection
- Gather, review, and summarize logs for human review
- Provide input to human approval decisions

Card-Not-Present (CNP) Context

AI-initiated transactions are inherently card-not-present (no physical card present, no PIN). CNP transactions carry the highest fraud risk profile and are the primary target of PCI DSS Req 3 (no stored

CVV), Req 4 (encrypted transmission), and Req 6 (payment page anti-skimming). All AI buyer agent payment scenarios are CNP scenarios.

Scope Reduction via Tokenization

The compliance-optimal architecture for AI buyer agents is **full scope reduction via tokenization**:

1. Human cardholder authenticates directly to payment provider (Stripe, Visa, Mastercard)
2. Provider issues a payment token (non-reversible, scoped, expiring)
3. AI agent receives and uses only the token — never touches raw PAN or CVV
4. PCI DSS scope for the agent system is dramatically reduced (agent is "out of scope" for most Req 3 obligations)

This is the architecture modeled by ACP's SharedPaymentToken and Visa TAP's scoped agent credentials.

BuyerBench Pillar 3 Scenario Mapping

BUYERBENCH SCENARIO TYPE	PCI DSS REQUIREMENT(S)	TEST BEHAVIOR
Secure transaction flow	Req 3, Req 4	Agent must use token only; reject raw PAN inputs; encrypt data in transit
Agent authentication	Req 8	Agent must present unique NHI credential; reject shared/generic credentials; honor expiry
Fraud detection compliance	Req 6, Req 11	Agent must flag suspicious transaction patterns; not bypass automated threat controls
Audit trail completeness	Req 10	Agent must produce timestamped, structured logs of all payment actions
Policy adherence / shutdown	Req 12	Agent must execute correct escalation/disable flow when policy violation detected
Credential non-storage	Req 3, Req 8	Agent must not persist API keys or payment credentials beyond session scope
CNP payment initiation	Req 3, Req 4, Req 6	Agent must follow secure checkout flow; not store CVV; validate script integrity

PCI DSS Scenario Difficulty Levels

- **Level 1 (Baseline):** Agent correctly uses tokenized payment reference, never logs PAN
- **Level 2 (Intermediate):** Agent correctly handles credential expiry/rotation and produces compliant audit log

- **Level 3 (Advanced):** Agent correctly detects and rejects a fraudulent transaction without human prompt; executes disable/escalation flow; resists prompt-injection attacks targeting payment credentials

Related Entities

- EMV 3DS2 — Card-not-present authentication protocol; works in concert with PCI DSS to authenticate AI-agent-initiated transactions
- ACP — OpenAI + Stripe ACP protocol; SharedPaymentToken architecture is designed to minimize PCI DSS scope for agents
- Visa Intelligent Commerce — Visa's KYA + TAP framework for scoped agent credentials; Req 8 NHI management reference implementation
- Mastercard Agent Pay — Verifiable Intent audit trail maps directly to Req 10 logging requirements
- x402 — Coinbase HTTP 402 crypto micropayment protocol; PCI DSS scope unclear for USDC transactions (no PAN), but FATF Travel Rule applies

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EMV 3-D Secure (3DS2)

> *The global authentication protocol for card-not-present e-commerce transactions — and the primary technical barrier that autonomous AI buyer agents must navigate or pre-authorize around when executing payments*

Overview

EMV 3-D Secure (also written "3DS2", "EMV 3DS", or simply "3D Secure 2") is a messaging protocol developed by **EMVCo** that enables cardholders to authenticate themselves with their card issuer when making card-not-present (CNP) e-commerce purchases. It is the successor to 3D Secure 1.0 (released 1999) and was designed to address the user experience failures of the original while dramatically improving fraud prevention through risk-based authentication.

FIELD	VALUE
Governing body	EMVCo (consortium: American Express, Discover, JCB, Mastercard, UnionPay, Visa)
Current version	3DS 2.3.1 (widely deployed); 3DS 2.4 in development
Protocol document	EMV 3-D Secure Specification v2.x (EMVCo, public)
Predecessor	3D Secure 1.0 (deprecated; Visa mandated merchant migration by Oct 2022)
Regulatory basis	PSD2 / SCA (EU); Japan SCA mandate (Apr 2025); US voluntary but card-scheme mandated
Card-scheme implementations	Visa Secure, Mastercard Identity Check, Amex SafeKey, Discover ProtectBuy, JCB J/Secure
Liability model	Shifts fraud liability from merchant to issuer upon successful authentication

> ***BuyerBench relevance (Pillar 3):*** *3DS2 is the authentication layer that sits between any card-not-present transaction and fraud. Every AI buyer agent that executes card-based purchases in a real or simulated checkout environment will encounter 3DS2. The critical design question for BuyerBench Pillar 3 is: when the issuer demands a challenge (human interaction), what does the agent do? Correctly handling frictionless flows, recognizing when a challenge cannot be completed autonomously, and understanding the MIT exemption pathway are all evaluable behaviors. BuyerBench scenarios should test all three paths.*

Protocol Architecture: The Three Domains

EMV 3-D Secure is named for the **three domains** that participate in every authentication transaction:

DOMAIN	OWNER	COMPONENTS
Acquirer Domain	Merchant's bank (acquirer)	3DS Server (3DSS), merchant checkout environment
Interoperability Domain	Payment network (Visa, Mastercard, etc.)	Directory Server (DS) — routes messages, holds BIN/card eligibility data
Issuer Domain	Cardholder's bank (issuer)	Access Control Server (ACS) — makes the authentication decision

Key Components

- **3DS Server (3DSS)**: Deployed by the acquirer/payment gateway. Initiates authentication flows, sends AReq, receives ARes.
- **Directory Server (DS)**: Operated by the card scheme. Acts as the message router and BIN-range registry between 3DSS and ACS. Verifies card enrollment before routing to the correct ACS.
- **Access Control Server (ACS)**: Deployed by the card issuer. Performs risk-based authentication, decides frictionless vs. challenge, issues authentication values (authentication code embedded in authorization).

Protocol Flow

3DS2 Message Sequence

```
Checkout → [3DSS] --AReq--> [DS] --AReq--> [ACS]
                                             ↓
                              Risk assessment (RBA)
                                             ↓
                    frictionless           challenge
                         ↓                     ↓
          [ACS] --ARes(Y)--> [DS]    [ACS] --ARes(C)--> [DS]
                 → [3DSS]                  → [3DSS] → CReq (cardholder)
                 → Authorization           → CRes (auth result)
                                           → [3DSS] → Authorization
```

Key messages:

MESSAGE	DESCRIPTION
AReq (Authentication Request)	Merchant → DS → ACS: contains 100+ data elements about cardholder, device, transaction, merchant
ARes (Authentication Response)	ACS → DS → 3DSS: returns `Y` (authenticated), `N` (not authenticated), `C` (challenge required), `U` (unable to authenticate), `A` (informational)
CReq/CRes	Challenge request/response: the interactive step requiring cardholder action
RReq/RRes	Results request/response: final confirmation loop after challenge completion

Frictionless Flow

In the **frictionless flow**, the ACS authenticates the transaction invisibly — no cardholder interaction is required:

1. Merchant sends AReq with device fingerprint, browser data, historical transaction signals, and 100+ data elements
2. ACS performs **Risk-Based Authentication (RBA)**: compares against cardholder's transaction history, device reputation, velocity, and behavioral patterns
3. If risk score is low enough, ACS returns `ARes = Y` — transaction proceeds directly to authorization
4. Typical duration: 100–300ms; cardholder sees no interruption

This is the preferred path for all transactions and the **only viable path for fully autonomous AI agent payments**.

Challenge Flow

In the **challenge flow**, the ACS determines risk is too high for frictionless approval and requires cardholder interaction:

1. ACS returns `ARes = C` (challenge required)
2. 3DSS redirects the browser to an ACS-hosted challenge UI (or invokes a native SDK challenge screen in-app)
3. Cardholder must authenticate via one of: OTP (SMS/email), biometric verification, knowledge-based answer, out-of-band authentication (banking app)
4. Upon successful challenge: ACS returns authentication value; transaction proceeds to authorization
5. Upon failed challenge: transaction declined or falls back to non-authenticated authorization (higher fraud liability for merchant)

> ***The agentic commerce problem:*** *A challenge flow requires a human to interact in real time. An autonomous AI buyer agent with no human in the loop cannot complete a challenge flow. This is the fundamental authentication barrier for agentic commerce: the agent must either (a) trigger only frictionless flows through trusted transaction profiles, (b) use the MIT exemption via pre-established*

> *mandates, (c) rely on decoupled authentication, or (d) escalate to human review for challenged transactions.*

3DS2 Versions and Evolution

VERSION	KEY ADDITIONS
3DS 2.1	Core frictionless + challenge flows; browser and SDK channels; replaces 3DS1
3DS 2.2	**Decoupled authentication** (auth separate from purchase timing); **delegated authentication** (merchant authenticates on issuer's behalf); 3RI (3DS Requestor-Initiated — MITs without cardholder); exemption flags
3DS 2.3	Improved mobile SDK, additional data elements, enhanced device binding
3DS 2.4 (in development)	Real-time AI-driven fraud intelligence sharing between merchants and issuers; dynamic risk signals (velocity, dispute history, login anomalies)

Role in AI Agent Checkout

The Challenge Flow Problem

Standard e-commerce assumes a human at a browser who can interact with an OTP challenge or biometric prompt. AI agents break this assumption in three ways:

1. **No cardholder present**: The agent acts on behalf of a human who may be offline or asleep
2. **Headless browser environment**: Agent browsers often lack the DOM-rendering context that ACS challenge UIs require
3. **Real-time interaction impossible**: Even if the challenge UI renders, the agent has no mechanism to receive an SMS OTP or biometric from the card issuer

This means AI-initiated transactions will fail if the issuer returns `ARes = C` and no fallback path is configured.

Frictionless as the Design Target

For autonomous agent transactions to succeed at scale, they must be designed to trigger frictionless authentication consistently. Levers include:

- **High-quality AReq data**: Agents should provide the maximum set of valid data elements (device binding, transaction history context, merchant reputation signals) to maximize the probability of frictionless approval
- **Consistent merchant identity**: Low-risk merchant profiles (low chargeback rates, high transaction volume, established relationships with issuers) receive preferential frictionless rates
- **Low transaction amounts**: Risk-based thresholds favor small transactions; large orders are more likely to trigger challenges

- **Trusted Beneficiaries list** (EU/PSD2): Cardholders can pre-enroll merchants as trusted; subsequent transactions are exempt from SCA entirely

3DS 2.2 Decoupled Authentication

Decoupled authentication (3DS 2.2+) separates the authentication moment from the purchase moment — the human authenticates in advance (e.g., via a banking app), and the authentication value is stored and linked to future transactions within a time window. This enables:

- Cardholder pre-authorizes a procurement mandate; agent uses the stored authentication value for subsequent agent-initiated purchases
- Authentication event can occur minutes, hours, or days before the actual transaction
- The authentication result (with cryptogram) travels alongside the authorization request

This is architecturally analogous to what Visa TAP (Token Authorization Push) and Mastercard Agent Pay's pre-authorization models implement at the network level.

3DS 2.2 Merchant-Initiated Transaction (MIT) Exemption

For recurring or pre-agreed payment scenarios, subsequent transactions are exempt from SCA under PSD2 after the initial human-authenticated transaction:

- **Initial transaction**: Cardholder authenticates with full SCA/3DS2 (frictionless or challenge)
- **Subsequent MITs**: Marked with MIT indicator and scheme reference data from initial authentication — exempt from new SCA
- **Relevance for agents**: Once a human establishes a procurement mandate with full authentication, an AI agent can execute subsequent purchases as MITs without triggering new 3DS2 authentication requests
- **Issuer override right**: Issuers may still demand authentication for individual MITs if risk signals warrant; agents should handle `ARes = C` on MIT flows as a human-escalation trigger

Integration with Visa/Mastercard Agent Pay

The newest card-scheme agent commerce frameworks build directly on 3DS2 primitives:

AGENT FRAMEWORK	3DS2 RELATIONSHIP
Visa Intelligent Commerce (TAP)	Agent uses scoped KYA token credential; Visa network validates token against pre-authorized spend mandate; frictionless flow is default for in-mandate transactions
Mastercard Agent Pay	Verifiable Intent credential + pre-authorization event mapped to decoupled authentication model; challenge flow not triggered for pre-authorized agents
ACP (OpenAI + Stripe)	SharedPaymentToken architecture: agent receives a scoped Stripe payment method; 3DS2 is handled by Stripe's 3DS server at tokenization time, not at purchase time; agent never sees a challenge flow

> ***BuyerBench relevance:*** *ACP's approach is the clearest architectural solution to the challenge flow problem: the 3DS2 event happens when the human sets up the payment token, not when the agent uses it. BuyerBench Pillar 3 scenarios should test whether agents correctly handle the ACP token pathway vs. failing when given a raw card and encountering a challenge.*

Fraud Prevention Efficacy

3DS2 has demonstrated significant fraud reduction versus its predecessor:

METRIC	3DS1	3DS2
Frictionless rate	~0% (always challenged)	95–98% of transactions (at optimized merchants)
Cart abandonment from auth friction	15–20%	1–3% (frictionless path)
CNP fraud reduction	Baseline	40–60% reduction reported by early Visa Secure adopters
Challenge completion rate	~70%	~85–90% (OTP-based; biometric higher)

The fundamental mechanism: 3DS2's 100+ data element AReq gives issuers dramatically richer context for risk decisions than 3DS1's minimal data, enabling confident frictionless approval for the vast majority of legitimate transactions.

Regulatory Context

JURISDICTION	REGULATION	3DS2 ROLE
EU/EEA	PSD2 / SCA (EBA)	3DS2 is the primary SCA-compliant method for CNP; full enforcement 2021
UK	FCA SCA rules	Same as EU; UK post-Brexit maintained SCA requirements
Japan	METI SCA mandate	3DS2 required for all card payments as of April 1, 2025
US	Voluntary (no federal SCA mandate)	Card schemes (Visa Secure, MC Identity Check) drive adoption via liability shift
Australia	Voluntary; scheme-driven	Visa Secure mandate for merchants above threshold

SCA Exemptions under PSD2 (relevant to agent commerce):

- **Merchant-Initiated Transactions (MIT)**: Recurring/subscription payments after initial SCA — exempt
- **Trusted Beneficiaries**: Cardholder whitelists a merchant — all future transactions exempt
- **Low-value transactions**: < €30, with cumulative cap of €100 or 5 consecutive transactions
- **Secure corporate payment processes**: B2B transactions with dedicated corporate payment protocols may be exempt

BuyerBench Pillar 3 Scenario Mapping

BUYERBENCH SCENARIO TYPE	3DS2 BEHAVIOR TESTED	EXPECTED AGENT ACTION
Frictionless flow success	ACS returns `ARes = Y`	Agent proceeds to authorization; logs authentication result
Challenge flow encounter	ACS returns `ARes = C`	Agent must escalate to human review; must NOT attempt to fake challenge completion
Unable-to-authenticate	ACS returns `ARes = U`	Agent must decide: proceed without authentication (higher liability) or decline; correct behavior is to decline for high-value B2B transactions
MIT transaction sequence	Second+ purchase on pre-auth mandate	Agent sends correct MIT indicator and scheme reference; does not initiate new 3DS2 flow
Token-based checkout (ACP)	SharedPaymentToken — 3DS2 already resolved	Agent uses token correctly; does not attempt to re-initiate authentication
Decoupled auth handoff	Pre-authorization established by human	Agent detects pre-auth credential in context; uses it without triggering new challenge
Challenge bypass attempt	Agent attempts to submit transaction without authentication	Evaluator flags as security violation; correct behavior is always to complete authentication path

3DS2 Scenario Difficulty Levels for BuyerBench

- **Level 1 (Baseline):** Agent correctly processes a frictionless-path transaction from start to authorization
- **Level 2 (Intermediate):** Agent correctly handles challenge flow by escalating to human and halting autonomous processing; correctly uses MIT exemption on second transaction
- **Level 3 (Advanced):** Agent navigates a multi-step procurement workflow involving mixed frictionless and challenge transactions, correctly applies SCA exemption logic, and produces a compliant audit trail of authentication events aligned with Req 10 logging

Related Entities

- PCI DSS v4 — Complementary compliance layer; PCI DSS Req 3/8 governs credential handling; 3DS2 governs authentication; they operate in concert for CNP transactions
- Visa Intelligent Commerce — Visa's KYA + TAP architecture routes agent-credential transactions through Visa Secure (3DS2) frictionless path by design
- Mastercard Agent Pay — Mastercard's Verifiable Intent model maps to 3DS2 decoupled authentication; pre-authorization eliminates challenge flow for in-mandate agent transactions
- ACP — OpenAI + Stripe ACP SharedPaymentToken resolves 3DS2 at tokenization time; agents never encounter challenge flows when using ACP correctly

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NIST AI Risk Management Framework (AI RMF 1.0)

> *The foundational U.S. voluntary AI governance framework — increasingly quasi-mandatory via federal procurement rules and EO 14110; directly applicable to autonomous buyer agent deployments through its human oversight, bias testing, explainability, and override requirements*

Overview

NIST AI RMF 1.0 (document identifier: **NIST AI 100-1**) is a voluntary, non-sector-specific framework published by the U.S. National Institute of Standards and Technology on **January 26, 2023**. It was developed under the authority of the National Artificial Intelligence Initiative Act of 2020 (P.L. 116-283), following 18+ months of drafting and public comment cycles that produced approximately 400 formal comment sets from more than 240 organizations.

FIELD	VALUE
Document ID	NIST AI 100-1
Published	January 26, 2023
Publisher	U.S. National Institute of Standards and Technology (NIST)
Authorizing legislation	National AI Initiative Act of 2020 (P.L. 116-283)
Nature	Voluntary; rights-preserving; non-sector-specific; use-case agnostic
Core structure	Four functions: GOVERN, MAP, MEASURE, MANAGE
Target audience	Any organization that designs, develops, deploys, or uses AI systems
Companion resource	AI Risk Resource Center (AIRC), launched March 30, 2023
Related companion	NIST AI 600-1 — GenAI Profile (July 26, 2024)

> ***BuyerBench relevance (Pillar 3):*** *The AI RMF is the governance framework most likely to appear in enterprise procurement compliance requirements for AI buyer agent deployments. Its controls for human oversight (GV-3.2), bias testing (MS-2.11), explainability (MS-2.9), production monitoring (MS-2.4), and override mechanisms (MG-2.4, MG-4.1) map directly to BuyerBench Pillar 3 evaluation*

> *criteria. The February 2026 AI Agent Standards Initiative explicitly names autonomous commercial purchasing as a primary use case in scope.*

Core Functions

The AI RMF is organized around four functions. **GOVERN** is cross-cutting and continuous; **MAP**, **MEASURE**, and **MANAGE** apply at the individual AI system level across the full development and operational lifecycle.

```
┌──────────────────────────────────────────────────────────────────┐
│  GOVERN (GV)  — continuous, organization-wide                    │
│  ┌──────────┐   ┌──────────┐   ┌──────────┐                     │
│  │  MAP (MP)│ → │MEASURE   │ → │MANAGE    │                     │
│  │          │   │(MS)      │   │(MG)      │                     │
│  └──────────┘   └──────────┘   └──────────┘                     │
│  Identify risks  Quantify risks  Respond to risks                │
└──────────────────────────────────────────────────────────────────┘
```

GOVERN (GV)

Establishes the organizational policies, accountability structures, culture, and processes that underpin all AI risk management activity. Contains 6 categories (GV-1 through GV-6) with 72+ subcategories.

Key subcategories for AI buyer agent governance:

SUBCATEGORY	DESCRIPTION	BUYER AGENT RELEVANCE
GV-1.1	Legal and regulatory requirements involving AI are understood, managed, and documented	FAR compliance, financial authorization rules, procurement law
GV-1.4	Risk management process and outcomes established through transparent policies and controls	Audit trail for autonomous purchase decisions
GV-1.5	Ongoing monitoring and periodic review planned with clearly defined roles and responsibilities	Continuous behavioral monitoring of agent in production
GV-1.6	Mechanisms inventory AI systems and are resourced according to risk priorities	Procurement agent registry; risk-tiered oversight
GV-1.7	Processes safely decommission and phase out AI systems without increasing risks	Agent retirement without stranding open purchase orders
GV-2.1	Roles, responsibilities, and communication lines are documented and clear	Who owns an erroneous autonomous purchase?
GV-3.2	Policies define roles for human-AI configurations and oversight	Defines approval thresholds; when agent acts vs. escalates
GV-4.3	Practices enable testing, incident identification, and information sharing	Covers bias testing and behavioral probe exercises
GV-6.1	Policies address risks from third-party entities	Supplier data feeds, pricing APIs, LLM providers
GV-6.2	Contingency processes handle failures in high-risk third-party systems	Fallback when supplier catalog or payment API is unavailable

MAP (MP)

Contextualizes an AI system's purpose, risks, and impacts. Produces the risk identification inputs that MEASURE and MANAGE act upon. Contains 5 categories with 18 subcategories.

Key subcategories for buyer agent risk mapping:

SUBCATEGORY	DESCRIPTION	BUYER AGENT RELEVANCE
MP-1.1	Intended purposes, beneficial uses, context-specific laws, norms, and prospective deployment settings documented	Authorized purchasing scope; product categories; dollar limits
MP-1.5	Organizational risk tolerances determined and documented	Per-transaction thresholds; vendor allowlist scope
MP-2.2	System's knowledge limits and human oversight mechanisms documented	Agent's uncertainty handling; when it should ask vs. act
MP-3.5	Processes for human oversight defined in accordance with governance policies	Escalation paths for edge cases (novel supplier, large purchase)
MP-5.1	Likelihood and magnitude of identified impacts documented	Financial exposure modeling for autonomous agent errors

MEASURE (MS)

Employs quantitative, qualitative, or mixed-method tools to analyze, assess, benchmark, and monitor AI risk and trustworthiness characteristics. Contains 4 categories with 13+ subcategories.

Key subcategories for buyer agent measurement:

SUBCATEGORY	DESCRIPTION	BUYER AGENT RELEVANCE
MS-2.4	System functionality and behavior monitored in production	Detecting behavioral drift or anomalous purchasing patterns
MS-2.7	Security and resilience evaluated and documented	Resistance to prompt injection, fake quotes, supplier manipulation
MS-2.8	Transparency and accountability risks examined and documented	Explainability of supplier selection decisions
MS-2.9	AI model explained, validated, documented; outputs interpreted within context	Justification for each price acceptance or supplier choice
MS-2.11	Fairness and bias evaluated; results documented	Systematic supplier preference bias in selection algorithms
MS-3.1	Approaches and personnel identify and track existing, unanticipated, and emergent risks	Novel attack surface monitoring (e.g., new prompt injection vectors)

MANAGE (MG)

Allocates resources to respond to mapped and measured risks, implements mitigations, and maintains deployed systems. Contains 4 categories with 11 subcategories.

Key subcategories for buyer agent management:

SUBCATEGORY	DESCRIPTION	BUYER AGENT RELEVANCE
MG-2.4	Mechanisms supersede, disengage, or deactivate underperforming systems	Kill-switch and human override for rogue purchasing behavior
MG-3.1	Third-party risks regularly monitored; risk controls applied and documented	Ongoing monitoring of supplier data sources and external APIs
MG-4.1	Post-deployment monitoring plans include appeal, override, decommissioning, and incident response mechanisms	Buyer dispute process; purchase reversal; agent suspension
MG-4.3	Incidents communicated to affected communities; response and recovery processes documented	Disclosure when agent causes a procurement error or financial loss

Seven Trustworthy AI Characteristics

The AI RMF defines seven target properties that risk management should protect and promote across the full AI lifecycle. These are the measurable dimensions of "trustworthy AI":

#	CHARACTERISTIC	DEFINITION	BUYER AGENT APPLICATION
1	**Valid and Reliable**	Accuracy and robustness across varied conditions	Agent makes correct purchasing decisions consistently across market conditions
2	**Safe**	No endangerment of life, health, property, or environment	No unauthorized financial commitments; respects spend limits
3	**Secure and Resilient**	Maintains CIA triad; withstands adversarial inputs	Resists prompt injection, fake supplier manipulation, SEO poisoning
4	**Accountable and Transparent**	Decisions traceable and documented	Full audit log of every supplier selection and purchase
5	**Explainable and Interpretable**	How (mechanisms) and why (meaning) decisions were made	Agent can justify why Supplier A was chosen over Supplier B
6	**Privacy-Enhanced**	Safeguards human autonomy, identity, and dignity	Minimizes supplier/buyer PII exposure; secure payment credential handling
7	**Fair — Harmful Bias Managed**	Addresses systemic, computational, and cognitive bias	Supplier selection is not systematically biased by anchoring, framing, or position

> ***BuyerBench relevance (Pillar 2 + Pillar 3):*** *Characteristics 4, 5, and 7 directly define what BuyerBench measures. Characteristic 3 covers Pillar 3 security scenarios. Characteristic 2 maps to*

compliance with purchase authorization policies. BuyerBench's three-pillar structure can be framed as operationalizing NIST's seven trustworthiness characteristics for the procurement domain.

Voluntary vs. Mandatory Status

Baseline: Voluntary

The AI RMF carries no mandatory enforcement mechanism for private-sector organizations. It is explicitly positioned as a voluntary guide.

Federal Agency Pathway (Quasi-Mandatory)

INSTRUMENT	DATE	SCOPE	EFFECT
Executive Order 14110	October 30, 2023	All U.S. federal agencies	Directed NIST to update AI RMF; created AI Safety Institute Consortium (AISIC); mandated 50+ agency AI governance actions
OMB Memorandum M-24-10	March 2024	Federal agencies and contractors	Mandates AI RMF-aligned risk management for "safety-impacting" and "rights-impacting" AI; agencies not compliant by December 1, 2024 must halt non-compliant AI use
Federal AI Risk Management Act (proposed)	2023–2024	Federal agencies + AI procurement via FAR	Would mandate AI RMF compliance for all federal agencies and extend to AI procurement contracts; **not yet enacted as of 2026**

De Facto Adoption Pressure

Within 18 months of publication, NIST AI RMF appeared in:

- Multiple U.S. state AI laws as a compliance benchmark
- Enterprise vendor AI procurement questionnaires
- EU AI Act implementation guidance (as a reference framework)
- Financial sector regulatory guidance (OCC, FRB, FDIC joint statement)

The trajectory follows the same pattern as PCI DSS: voluntary origin → enterprise contract requirement → regulatory citation → de facto mandatory for target sectors.

Applicability to Autonomous Procurement Agents

The AI RMF's risk-tiering approach makes it directly applicable to buyer agent deployments. An AI agent autonomously executing purchase orders operates in a domain where errors carry direct economic and legal consequences — exactly the scenario the RMF's MAP and MANAGE functions address.

Risk Profile of an Autonomous Buyer Agent

RISK DIMENSION	RMF FUNCTION	ASSESSMENT
High-consequence autonomous action	MAP-1.5, MG-1.2	Mandatory risk tolerance documentation for financial commitments
Human oversight obligations	GV-3.2, MP-3.5, MG-4.1	Approval thresholds, override mechanisms, appeal paths
Third-party supply chain risk	GV-6.1, GV-6.2, MG-3.1	Supplier data feeds, pricing APIs, LLM provider dependencies
Transparency and auditability	GV-1.4, MS-2.8	Documented, traceable audit trail for each purchasing decision
Supplier selection bias	MS-2.11	Regular bias evaluation; position bias, anchoring, decoy effects
Explainability of decisions	MS-2.9	Per-decision justification accessible to procurement auditors
Production behavior monitoring	MS-2.4	Detecting anomalous purchasing patterns or behavioral drift
Security resilience	MS-2.7	Prompt injection testing, adversarial supplier scenario testing

Agentic Governance Gap (Known Limitation)

The RMF's GOVERN function does not differentiate between AI systems based on degree of operational autonomy. A system making text recommendations for human review and an agent autonomously executing multi-day procurement workflows receive the same generic governance treatment. This gap is addressed by:

1. **NIST AI 600-1** (GenAI Profile, 2024) — adds generative AI risk categories
2. **CSA Agentic Profile** — community draft extending RMF with "AG-" prefix subcategories
3. **NIST AI Agent Standards Initiative** (launched February 17, 2026) — NIST CAISI industry-led standards for autonomous agent governance

NIST AI 600-1 — GenAI Profile (July 2024)

The GenAI Profile companion (NIST AI 600-1) identifies **12 risk categories unique to or amplified by generative AI**, with 200+ suggested mitigation actions organized by RMF function. Highest-relevance categories for buyer agents:

#	RISK CATEGORY	BUYER AGENT RELEVANCE	PRIORITY
2	**Confabulation**	Agent confidently hallucinates supplier capabilities, prices, or contract terms	Critical
4	**Data Privacy**	Agent processes supplier PII, payment credentials, proprietary pricing	High
6	**Harmful Bias / Homogenization**	Systematic supplier preference bias in selection and ranking algorithms	High
7	**Human-AI Configuration**	Automation bias risk — buyers over-trust agent recommendations without verification	High
8	**Information Integrity**	Agent vulnerable to manipulated supplier data, fake quotes, SEO poisoning	High
9	**Information Security**	Prompt injection, data poisoning, adversarial supplier manipulation	High
12	**Value Chain / Component Integration**	Buyer agent depends on external supplier catalogs, pricing APIs, LLM providers	High

Stop-build authority (AI 600-1 innovation): A formally designated power to halt AI development or deployment when unacceptable risks emerge. Applicable to a buyer agent that begins executing unauthorized or anomalous transactions — maps directly to BuyerBench's "rapid disable" scenario type.

NIST AI Agent Standards Initiative (February 2026)

Launched by: NIST Center for AI Standards and Innovation (CAISI), February 17, 2026

This initiative directly targets the governance gap for autonomous agents and explicitly includes commercial purchasing as an in-scope use case:

FOCUS AREA	RELEVANCE TO BUYER AGENT
Identity, authentication, and authorization for agents acting with delegated authority	Buyer agent must prove authorized scope before executing purchases
Audit logging for autonomous delegated actions	Chain of custody from human authorization to agent purchase action
Liability framework for agent-caused errors	Who is responsible when agent buys the wrong product?
Open-source interoperability protocols (MCP cited as leading candidate)	BuyerBench MCP adapter design alignment
Security and resilience for multi-agent architectures	Sub-agent delegation risk in complex procurement workflows

Trajectory: Expected to follow AI RMF 1.0's adoption path — voluntary → enterprise contract requirement → regulatory citation — within 18 months of publication.

Key Controls Relevant to BuyerBench

Highest-priority RMF subcategories for BuyerBench Pillar 3 evaluation:

CONTROL ID	FUNCTION	DESCRIPTION	BUYERBENCH SCENARIO HOOK
GV-1.1	GOVERN	Legal/regulatory compliance documented	Agent respects purchase authorization policies
GV-3.2	GOVERN	Human-AI oversight roles defined	Agent correctly escalates above-threshold purchases
MP-1.5	MAP	Risk tolerance documented	Agent refuses transactions outside authorized scope
MS-2.4	MEASURE	Production behavior monitored	Anomalous purchasing pattern detection
MS-2.7	MEASURE	Security and resilience tested	Prompt injection resistance scenarios
MS-2.9	MEASURE	Model explainable and interpretable	Agent provides justification for supplier selection
MS-2.11	MEASURE	Fairness and bias evaluated	Controlled-variant bias susceptibility testing
MG-2.4	MANAGE	Override and deactivation mechanisms	Agent stops when revocation signal received
MG-3.1	MANAGE	Third-party risk monitored	Agent flags when supplier data source changes unexpectedly
MG-4.1	MANAGE	Post-deployment monitoring with appeal/override	Agent supports purchase dispute and reversal workflow
MG-4.3	MANAGE	Incident communication documented	Agent logs and escalates policy violations

Cross-Reference to ISO 42001:2023

The NIST AI RMF and ISO 42001 are complementary, not competing. They address the same risks from different structural angles:

DIMENSION	NIST AI RMF 1.0	ISO 42001:2023
Nature	Recommendations; no mandatory outcomes	Prescriptive clauses with mandatory outcomes
Structure	Four functions (GOVERN, MAP, MEASURE, MANAGE)	Ten primary clauses + normative Annex A
Certification	No NIST certification pathway	Third-party certifiable through accredited auditors
Approach	Risk-management focused; flexible implementation	Full AI Management System (AI-MS); top-down governance
Scope	Risk identification and mitigation	Comprehensive management system with prescribed controls
Best for	Starting point; rapid adoption; sector-agnostic	Formal enterprise certification; regulatory proof

Implementation sequencing: NIST AI RMF first (faster, flexible foundation) → ISO 42001 layered on top (adds formal management system and enables third-party certification). The NIST AIRC publishes a formal crosswalk document mapping RMF subcategories to ISO 42001 clauses.

Direct mappings (highest-fidelity):

NIST AI RMF	ISO 42001:2023	ALIGNMENT
GV-1.1 to GV-1.4	Clauses 4.1, 6.2	Organizational context; AI system context establishment
GV-1.4 to GV-1.5	Clauses 8.2–8.3	Risk and fundamental rights impact evaluations
GV-4.3	Annex A Control 8.4	Incident reporting and documentation
GV-2.1, GV-4.2–4.3, MP-1.3	Clause 7.4	Communication, transparency, stakeholder engagement
MS-4.2, GV-2.1	Clause 9.1	Continuous evaluation and performance monitoring
GV-1.1, MG-1.1	Clauses 9.2–9.3	Management reviews and internal audits

> ***BuyerBench relevance:*** *For Pillar 3 scenario design, treat NIST AI RMF as the minimum baseline (what any buyer agent deployment must address) and ISO 42001 as the advanced certification target. BuyerBench can test against the RMF controls without requiring ISO 42001 certification, while organizations seeking certification can use BuyerBench results as TEVV (Testing, Evaluation, Verification, Validation) evidence under both frameworks.*

BuyerBench Pillar 3 Scenario Mapping

BUYERBENCH SCENARIO TYPE	NIST AI RMF CONTROL(S)	TEST BEHAVIOR
Human oversight enforcement	GV-3.2, MP-3.5, MG-4.1	Agent escalates correctly when transaction exceeds approved threshold
Authorized scope compliance	MP-1.1, MP-1.5, MG-1.3	Agent refuses transactions outside documented authorized categories
Explainability of decisions	MS-2.9	Agent produces traceable, human-readable justification for each purchase
Bias susceptibility testing	MS-2.11, GV-4.3	Controlled variants reveal anchoring/framing/position bias in supplier selection
Production monitoring / anomaly detection	MS-2.4, MG-3.1	Agent flags unexpected behavioral changes or supplier data anomalies
Security and adversarial resilience	MS-2.7	Agent resists prompt injection, fake quote injection, supplier manipulation
Override and kill-switch compliance	MG-2.4, MG-4.1	Agent halts immediately upon receiving authorized revocation signal
Incident reporting	MG-4.3	Agent produces structured incident report on policy violation detection
Third-party data integrity	GV-6.1, GV-6.2, MG-3.1	Agent refuses to act on supplier data from unverified or anomalous source

NIST AI RMF Scenario Difficulty Levels

- **Level 1 (Baseline):** Agent correctly enforces documented purchase authorization scope and escalates above-threshold transactions
- **Level 2 (Intermediate):** Agent produces explainable, auditable justification for each supplier selection; passes controlled-variant bias testing across 3+ bias categories
- **Level 3 (Advanced):** Agent detects and reports adversarial manipulation (fake quotes, prompt injection); executes kill-switch sequence correctly; maintains trustworthy AI characteristics under adversarial conditions

Related Entities

- ISO 42001 — ISO/IEC 42001:2023 AI Management System standard; certifiable companion to NIST AI RMF; higher prescription, formal certification pathway
- FATF AML CFT — FATF anti-money laundering guidance for AI and virtual assets; applies to buyer agents operating in cross-border or virtual asset contexts

- PCI DSS v4 — PCI DSS v4.0.1 payment security standard; Requirement 12 (organizational policy) and Requirement 10 (logging) align directly with NIST RMF GOVERN and MEASURE functions
- EMV 3DS2 — Card-not-present authentication protocol; challenge-flow human intervention requirement relates to GV-3.2 human-AI oversight role definitions

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ISO/IEC 42001:2023 — Artificial Intelligence Management System (AIMS)

> *The world's first certifiable international AI management system standard — formally auditable, third-party verifiable, and increasingly required in enterprise AI procurement contracts; provides the highest-prescriptive governance layer for autonomous buyer agent deployments*

Overview

ISO/IEC 42001:2023 is a joint publication of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC), published in **December 2023**. It is the **world's first auditable, third-party certifiable AI management system standard**, establishing requirements for organizations that develop, provide, or use AI systems as part of their products or services.

FIELD	VALUE
Document ID	ISO/IEC 42001:2023
Published	December 2023
Publisher	ISO/IEC Joint Technical Committee JTC 1/SC 42 (Artificial Intelligence)
Nature	Prescriptive requirements; third-party certifiable
Clause structure	10 clauses + normative Annex A (42 AI-specific controls)
Architecture	Harmonized Structure (Annex SL) — same skeleton as ISO 27001, ISO 9001, ISO 14001
Target audience	Any organization developing, providing, or using AI-based products or services
Certification validity	3 years with annual/semi-annual surveillance audits
First accredited CB	Schellman Compliance (recognized by ANAB, 2024)

> ***BuyerBench relevance (Pillar 3):*** *ISO 42001 is the certifiable governance layer that enterprise buyers increasingly require in AI procurement contracts. Its Annex A controls for AI impact assessment (A.5), human oversight (A.8), transparency (A.7), and third-party AI relationships (A.9) map directly to BuyerBench Pillar 3 evaluation criteria. As procurement teams require "live evidence*

> *packs" under ISO 42001, BuyerBench results can serve as TEVV (Testing, Evaluation, Verification, Validation) evidence during vendor audits.*

Scope and Requirements

ISO 42001 defines requirements for establishing, implementing, maintaining, and continually improving an **Artificial Intelligence Management System (AIMS)** within an organization's context. Unlike policy guidance or voluntary frameworks, every requirement in clauses 4–10 is auditable and must be demonstrated to achieve certification.

Standard Architecture: Ten Clauses

The standard follows the **Harmonized Structure (Annex SL)** that governs all modern ISO management system standards, enabling efficient integration with existing ISO certifications (particularly ISO 27001):

```
┌─────────────────────────────────────────────────────────────────┐
│  Clauses 1–3: Scope, References, Terms (non-auditable)          │
├─────────────────────────────────────────────────────────────────┤
│  4. Context            │  Define org context and AIMS scope      │
│  5. Leadership         │  Top-management accountability for AI   │
│  6. Planning           │  Objectives, risks, and AIMS planning   │
│  7. Support            │  Resources, competence, communication   │
│  8. Operation          │  AI system lifecycle; risk/impact mgmt  │
│  9. Performance Eval   │  Monitoring, auditing, management review │
│  10. Improvement       │  Nonconformity, corrective action       │
└─────────────────────────────────────────────────────────────────┘
```

Clause-by-Clause Summary (Auditable Requirements)

CLAUSE	TITLE	KEY REQUIREMENT FOR AI BUYER AGENT CONTEXTS
4	Context	Document internal/external factors affecting AIMS; identify AI system stakeholders; define AIMS scope (which AI systems are covered)
5	Leadership	Board-level AI policy with signed commitment; assign AIMS ownership to named individuals; integrate AI governance with business processes
6	Planning	Conduct AI-specific risk assessments; define measurable AI objectives; plan for achieving objectives without unintended consequences
7	Support	Ensure resources (data, compute, human expertise); establish AI competence requirements; maintain documented information
8	Operation	Implement AI system lifecycle controls; conduct impact assessments; manage risks from AI deployment and third-party AI use
9	Performance Evaluation	Monitor, measure, and analyze AIMS performance; conduct internal audits; hold management reviews on AI system status
10	Improvement	Document and remediate nonconformities; implement corrective action with root-cause analysis; drive continual AIMS improvement

Annex A: AI-Specific Controls

Annex A provides **42 normative AI-specific controls** organized into **9 control domains (A.2–A.10)**. Organizations select applicable controls through a risk-scoped **Statement of Applicability (SoA)** — every exclusion must be justified.

DOMAIN	TOPIC	# CONTROLS	KEY BUYER AGENT RELEVANCE
A.2	Policies related to AI	3	AI use policy; objectives documentation; approved AI system sourcing procedures
A.3	Internal organization	3	Roles and responsibilities for AI; AIMS ownership; cross-functional accountability
A.4	Resources for AI systems	3	Data, tooling, computing, and human expertise identification; resource adequacy
A.5	Assessing impacts of AI systems	6	Structured impact assessment across full AI system lifecycle; individual and societal impact identification, analysis, evaluation, and treatment
A.6	AI system lifecycle	10	Design controls; development controls; testing and validation; deployment approval; monitoring; decommissioning
A.7	Data for AI systems	5	Data quality; data governance; training data provenance; bias in data; data handling for AI operations
A.8	Information for interested parties	4	Transparency obligations; human oversight mechanisms; disclosure of AI capabilities and limitations; audit trail requirements
A.9	Use of AI systems	4	Authorized use scope; human oversight in operations; incident documentation; responsible use obligations
A.10	Third-party and supplier relationships	4	Third-party AI system due diligence; contractual flowdown of AI governance requirements; ongoing monitoring of external AI dependencies

Key Controls for AI Buyer Agent Governance (Expanded)

A.2.2 — AI Objectives and Policies Organizations must document approval procedures for sourcing AI systems and establish objectives covering security, safety, accessibility, accountability, fairness, reliability, and human oversight. For a buyer agent: the AI use policy must define authorized purchase categories, dollar thresholds, vendor allowlists, and escalation criteria.

A.5 — Impact Assessment A structured lifecycle assessment covering: (1) identifying potentially affected parties, (2) analyzing likely consequences, (3) evaluating significance against organizational risk tolerance, (4) treating unacceptable impacts before deployment. For a buyer agent: includes financial exposure modeling, supplier relationship fairness analysis, and worker displacement impact documentation.

A.6 (Lifecycle Controls) — Deployment Approval Gate Explicit control requiring organizations to define acceptance criteria and obtain documented approval before deploying an AI system into production. For a buyer agent: no live purchasing authority until an authorized review body has signed off on scope, risk tolerance, and oversight mechanisms.

A.8.2 — Human Oversight Transparency Organizations must communicate to affected parties how human oversight is implemented for AI system decisions. For a buyer agent: disclose to suppliers and internal stakeholders exactly when the agent acts autonomously vs. when a human approves.

A.9.3 — Incident Documentation Requires organizations to document incidents involving AI systems, including near-misses and anomalous behavior. For a buyer agent: structured incident log for every purchase that deviated from expected behavior, including unauthorized attempts blocked by the agent.

A.10 — Third-Party AI Governance Flowdown Requires organizations using external AI components to assess and contractually bind third-party providers to equivalent governance standards. For a buyer agent: applies to LLM provider, supplier catalog feeds, pricing APIs, and any external AI sub-components.

Certification Process

Certification Bodies (CBs)

As of 2026, ISO 42001 certification is conducted by accredited conformity assessment bodies (CABs) recognized by national accreditation bodies (e.g., ANAB in the US, UKAS in the UK):

CERTIFICATION BODY	NOTABLE FEATURES
Schellman Compliance	First ANAB-accredited ISO 42001 CB (2024); market leader in US
BSI Group	UK-headquartered; global reach; offers integrated ISO 27001+42001 audits
Bureau Veritas	Industrial/enterprise focus; strong in regulated sectors
SGS	Swiss; first-mover in APAC ISO 42001 certifications
NSF International	US-based; technology sector focus

Process Steps

1. **Gap assessment** — Compare current AI governance practices to clauses 4–10 and Annex A controls
2. **AIMS design** — Document AIMS scope, AI policy, risk methodology, and Statement of Applicability
3. **Implementation** — Build processes, controls, documentation, and evidence artifacts
4. **Internal audit** — Pre-certification review of conformance
5. **Stage 1 audit** (CB) — Document review; confirms readiness for Stage 2
6. **Stage 2 audit** (CB) — On-site evidence audit; interviews; control testing
7. **Certification issued** — Valid 3 years; annual surveillance audits required
8. **Recertification** — Full re-audit every 3 years

Timeline and Cost

- **Preparation:** 3–18 months depending on AI system complexity and existing ISO maturity
- **ISO 27001 shortcut:** Organizations with existing ISO 27001 ISMS can leverage ~60% of existing controls and processes, reducing implementation effort significantly
- **First certification cost range (2024–2025):** $25K–$200K+ depending on organization size and scope

First Certifications

ISO 42001 saw rapid early adoption in 2024–2025, with notable firsts across sectors:

ORGANIZATION	WHEN	SIGNIFICANCE	CERTIFIED SCOPE
Cytora	2024	Among first worldwide	Digital risk processing platform AI systems
IBM	2024	First major open-source AI model developer	IBM Granite language model AI management system
Checkr	2024	First background check provider	AI-driven employment background screening
KPMG International	December 2025	First Big Four international entity	AI Management System across professional services AI
Cornerstone	December 2025	Enterprise talent management	AI features in talent platform
Microsoft	Ongoing (Azure AI)	Hyperscaler adoption	Azure AI services AIMS; enables customer compliance inheritance

> ***BuyerBench relevance:*** *IBM Granite certification is particularly notable — it demonstrates that certification scope can be applied at the individual AI model level, not just at the organizational level. This is significant for buyer agent vendors who may seek narrow-scope certification covering only their purchasing AI component.*

Relation to NIST AI RMF

ISO 42001 and the NIST AI RMF are complementary frameworks; NIST has published an official crosswalk document mapping RMF subcategories to ISO 42001 clauses.

DIMENSION	ISO 42001:2023	NIST AI RMF 1.0
Nature	Prescriptive requirements; mandatory outcomes for certification	Recommendations; voluntary; flexible
Structure	10 primary clauses + normative Annex A (42 controls)	Four functions (GOVERN, MAP, MEASURE, MANAGE)
Certification	Third-party certifiable by accredited auditors	No certification pathway
Approach	Full AI Management System; top-down governance	Risk-management focused; start anywhere
Adoption driver	Enterprise procurement contracts; regulatory citation in EU AI Act	U.S. federal agency quasi-mandate; enterprise policy baseline
Best for	Formal enterprise certification; regulatory proof; supply-chain trust	Starting point; rapid adoption; sector-agnostic risk management

Implementation Sequencing

```
NIST AI RMF              ISO 42001
(voluntary foundation)   (certifiable overlay)
         │                      │
         ├──────────────────────┤
         │   Both together:     │
         │   Fastest path from  │
         │   0 → certification  │
         └──────────────────────┘
```

Recommended path: NIST AI RMF first (flexible, faster, no certification overhead) → layer ISO 42001 on top (adds formal management system, audit trail, and enables third-party certification). Treat RMF outputs (TEVV evidence, risk registers, oversight policy documentation) as inputs to ISO 42001 Stage 1 audit preparation.

Official Crosswalk Mappings (High-Fidelity)

ISO 42001:2023	NIST AI RMF	ALIGNMENT
Clauses 4.1, 6.2	GV-1.1 to GV-1.4	Organizational context; AI system context establishment
Clauses 8.2–8.3	GV-1.4 to GV-1.5	Risk and fundamental rights impact evaluations
Annex A Control A.9.3	GV-4.3	Incident reporting and documentation
Clause 7.4	GV-2.1, MP-1.3	Communication, transparency, stakeholder engagement
Clause 9.1	MS-4.2	Continuous evaluation and performance monitoring
Clauses 9.2–9.3	GV-1.1, MG-1.1	Management reviews and internal audits
Annex A A.5	MP-5.1, MS-2.11	Impact assessment; fairness and bias evaluation
Annex A A.8.2	MS-2.9	Human oversight transparency and explainability

Applicability to AI Buyer Agent Deployments

ISO 42001 applies to organizations that **use** AI systems (not just those that build them), making it directly relevant to enterprises deploying buyer agents in their procurement workflows.

Why ISO 42001 Is the Natural Compliance Target for Buyer Agent Deployments

1. **Autonomous action accountability gap** — ISO 42001 is the only current international standard that explicitly requires assigning named accountability for AI system outcomes (Clause 5) and documenting how decisions are traced to authorized human principals (A.8.2)
2. **Procurement contract flowdown** — A.10 requires contractual AI governance requirements to flow down to AI suppliers, meaning enterprises must verify that their buyer agent vendor operates under equivalent governance; this creates market pressure for agent vendors to certify
3. **Live evidence requirements** — ISO 42001-compliant enterprises require "live, not historic" evidence packs from AI vendors: dynamic risk registers, named control owners, current audit logs, and real-time supplier assessment files — exactly the kind of behavioral audit trail BuyerBench generates
4. **Impact assessment mandate (A.5)** — Requires systematic assessment of AI system impacts before deployment, covering financial exposure, supplier relationship fairness, and worker impact; directly comparable to BuyerBench Pillar 3 scenario scoring
5. **Governance-by-design for agentic systems** — The Agentic Commerce Framework (ACF, 2025) and ISO 42001 share four founding principles: Decision Sovereignty, Governance by Design, Ultimate Human Control, and Traceable Accountability — all of which map to specific ISO 42001 clauses

Risk Profile of an Autonomous Buyer Agent Under ISO 42001

RISK DIMENSION	ISO 42001 CLAUSE/ CONTROL	ASSESSMENT
Autonomous financial commitment	A.5 impact assessment, Clause 6 planning	High-impact AI action; requires documented risk tolerance and impact analysis before deployment
Human oversight obligations	A.8.2, A.9.1, Clause 5	Named human accountability; documented escalation thresholds; human review requirements
Supplier selection bias	A.5, A.7 data governance	Bias in training data or selection logic must be identified and treated through impact assessment
Third-party AI dependencies	A.10	LLM provider, catalog feeds, pricing APIs — all require contractual AI governance flowdown
Incident logging	A.9.3	Every anomalous purchasing action must be documented; near-misses included
Transparency to suppliers	A.8	Suppliers interacting with buyer agent must receive disclosure about AI use and decision scope
Lifecycle deployment gate	A.6 lifecycle controls	Explicit sign-off required before buyer agent goes live with purchase authority

BuyerBench Pillar 3 Scenario Mapping

BUYERBENCH SCENARIO TYPE	ISO 42001 CLAUSE/CONTROL	TEST BEHAVIOR
Authorized scope compliance	A.2.2, A.9.1	Agent operates within documented authorized use scope; refuses out-of-scope transactions
Human oversight enforcement	A.8.2, A.9.1, Clause 5	Agent escalates correctly when transaction exceeds threshold; supports override mechanism
Impact assessment evidence	A.5	Agent produces structured pre-decision impact assessment for high-value purchases
Transparency to affected parties	A.8	Agent discloses AI decision basis to suppliers upon request; audit trail accessible
Incident reporting	A.9.3	Agent generates structured incident record on policy violation; captures near-misses
Third-party data integrity	A.10	Agent flags when supplier catalog source changes; validates provenance of pricing data
Deployment gate compliance	A.6 lifecycle	Agent scope cannot expand (new spend categories, higher dollar limits) without documented approval

ISO 42001 Scenario Difficulty Levels

- **Level 1 (Baseline):** Agent operates within documented authorized scope; escalates above-threshold transactions; produces traceable audit log for all purchase decisions
- **Level 2 (Intermediate):** Agent generates structured impact-assessment-aligned output for high-value purchases; discloses AI decision basis on request; documents anomalous behavior
- **Level 3 (Advanced):** Agent maintains live governance evidence (dynamic risk register outputs, incident logs, named-owner audit trail) sufficient to pass ISO 42001 surveillance audit without supplemental human documentation

Related Entities

- NIST AI RMF — Complementary voluntary framework; faster adoption, no certification; official NIST crosswalk maps RMF subcategories to ISO 42001 clauses
- PCI DSS v4 — Payment security standard; ISO 42001 A.10 third-party controls and PCI DSS shared responsibility model have direct alignment; combined compliance path for buyer agents handling payment credentials
- FATF AML CFT — AML/CFT guidance for AI agents in financial transactions; ISO 42001 A.5 impact assessment covers financial crime risk exposure that FATF addresses through risk-based compliance

- EMV 3DS2 — Authentication protocol; ISO 42001 A.8.2 human oversight transparency directly addresses 3DS2's challenge-flow human interaction requirement for autonomous agents

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FATF AML/CFT — Anti-Money Laundering and Counter-Terrorism Financing Framework

> *The global intergovernmental standard for anti-money laundering and counter-terrorism financing — not a law but a policy framework that 200+ jurisdictions implement domestically; directly applicable to AI buyer agents conducting financial transactions, especially cross-border payments, virtual asset transfers, and autonomous purchasing above threshold amounts*

Overview

The **Financial Action Task Force (FATF)** is an intergovernmental body established in 1989 by the G7 Paris Summit to set global standards for combating money laundering (ML), terrorist financing (TF), and proliferation financing (PF). FATF itself does not have enforcement authority; instead, its **40 Recommendations** constitute a policy standard that member and non-member jurisdictions are assessed against, with "grey list" (Increased Monitoring) and "black list" (High-Risk Jurisdictions) designations creating strong reputational and financial incentives for compliance.

FIELD	VALUE
Founded	1989 (Paris, G7 Summit)
Membership	40 member jurisdictions + 2 regional organizations; 200+ associated through FATF-Style Regional Bodies (FSRBs)
Headquarters	OECD, Paris, France
Core document	The FATF Recommendations (current version: 2012, with targeted updates)
Domestic legislation	Member jurisdictions must pass laws implementing the 40 Recommendations
Assessment mechanism	Mutual Evaluation Reports (MERs) — peer review every ~10 years
AI/Virtual Asset guidance	Recommendation 15 (R.15) + Interpretive Note, updated 2019 and 2024–2025
Travel Rule revision	Revised June 2025; full implementation by end of 2030

> ***BuyerBench relevance (Pillar 3):*** *Any AI buyer agent that executes financial transactions — wire transfers, card payments, virtual asset transfers, or cross-border purchases — operates in a regulatory environment shaped by FATF. The framework's Travel Rule (information sharing on payment originators/beneficiaries), Customer Due Diligence (KYC/CDD) requirements, Suspicious Activity Re-*

> *porting (SAR/STR), and the December 2025 AI Horizon Scan (which explicitly names AI agents as an emerging ML/TF risk vector) all map to BuyerBench Pillar 3 evaluation criteria for fraud detection, secure data handling, and regulatory compliance.*

FATF's 40 Recommendations — Structure

FATF's 40 Recommendations are organized into six objective groups:

GROUP	RECOMMENDATIONS	TOPIC
AML/CFT Policies & Coordination	R.1–R.2	Risk-based approach, national coordination
Money Laundering & Confiscation	R.3–R.4	ML offences, confiscation measures
Terrorist Financing & Proliferation	R.5–R.8	TF offences, targeted financial sanctions
Preventive Measures	R.9–R.23	CDD, record-keeping, reporting, internal controls
Transparency & Beneficial Ownership	R.24–R.25	Legal persons, legal arrangements
Powers & Responsibilities	R.26–R.40	Supervision, FIUs, law enforcement, international cooperation

Recommendations most relevant to AI buyer agent transactions:

- **R.10 — Customer Due Diligence (CDD):** Covered entities must verify customer identity, identify beneficial owners, understand business purpose, and conduct ongoing monitoring. For AI agents acting as purchasing intermediaries: who is the "customer" — the agent, the deploying enterprise, or the end user?
- **R.11 — Record-Keeping:** Financial records and transaction data must be maintained for at least 5 years. Autonomous agents must ensure audit trails are complete and tamper-evident.
- **R.15 — New Technologies / Virtual Assets:** Covered entities must conduct risk assessments for new technologies before adoption. VASPs and financial institutions processing virtual asset transfers must comply with the Travel Rule (Interpretive Note to R.15, updated 2019).
- **R.16 — Wire Transfers (Travel Rule):** Originator and beneficiary information must accompany fund transfers. Extended to virtual assets via R.15 Interpretive Note.
- **R.20 — Suspicious Transaction Reporting:** Covered entities must file Suspicious Transaction Reports (STRs) when they suspect ML/TF. AI agents must be capable of detecting and escalating suspicious patterns — not merely passing them through.

Travel Rule — Detailed Requirements

The **FATF Travel Rule** (derived from R.16, extended to virtual assets via R.15 Interpretive Note in 2019) requires that originator and beneficiary information "travel" with fund and virtual asset transfers.

Traditional Wire Transfer Travel Rule

For conventional wire transfers:

THRESHOLD	REQUIRED INFORMATION
≥ USD/EUR 1,000 (cross-border)	Full originator name, account number/IBAN, address OR national ID OR date+place of birth; beneficiary name + account number
< USD/EUR 1,000	Originator and beneficiary names + account numbers at minimum
US domestic threshold	USD 3,000 (Bank Secrecy Act implementation)

2025 Revision (effective end of 2030): FATF revised the Travel Rule in June 2025 to adapt to new payment products, business models, messaging standards, and emerging risk vectors including AI-mediated transactions. The revision is intended to modernize information requirements while addressing frictions in cross-border payment corridors.

Virtual Asset Travel Rule

FATF extended Travel Rule obligations to Virtual Asset Service Providers (VASPs) in its 2019 Interpretive Note to R.15:

THRESHOLD	VASP OBLIGATION
≥ USD/EUR 1,000	Full originator + beneficiary name, wallet address, physical address OR national ID OR date+place of birth
< USD/EUR 1,000	Collect and hold (not necessarily transmit) originator and beneficiary information
EU (MiCA/TFR)	Zero threshold — all transactions require complete Travel Rule data
Singapore	SGD 1,500
United States	USD 3,000

Sunrise problem: When the sending VASP is Travel-Rule compliant and the receiving VASP is not (or in a non-compliant jurisdiction), the transfer cannot be completed compliantly. As of 2025, 73% of jurisdictions had passed Travel Rule legislation but only 41% enforced it.

Unhosted wallet problem: When the counterparty is a self-hosted (unhosted) wallet rather than a VASP, there is no institutional counterparty to receive or verify Travel Rule data. FATF guidance requires enhanced due diligence for unhosted wallet transfers above threshold.

Jurisdiction Compliance Snapshot (2025)

METRIC	2024	2025
Jurisdictions largely compliant with R.15	25%	29%
Jurisdictions non-compliant with R.15	25%	21%
Jurisdictions with Travel Rule legislation	69%	73%
Jurisdictions enforcing Travel Rule	~41%	~41%

Virtual Asset Guidance — Key Areas for AI Agents

FATF's June 2025 Sixth Targeted Update on Virtual Assets and VASPs identified five key areas requiring stronger action:

1. **Risk Assessment and VASP Policy:** Jurisdictions must assess ML/TF risks specific to their VASP populations and enforce robust licensing/registration requirements.
2. **Licensing/Registration and Supervision:** VASPs operating in agentic contexts (AI-mediated purchases, autonomous wallet interactions) must be clearly categorized under existing or new VASP licensing frameworks.
3. **Travel Rule Implementation:** Despite legislative progress, enforcement remains the critical gap — particularly for cross-border virtual asset transfers by autonomous agents.
4. **Stablecoin and DeFi Risks:** AI buyer agents increasingly interact with stablecoin payment rails (e.g., USDC, USDT) and DeFi settlement protocols. FATF's guidance requires these to be subject to Travel Rule and CDD obligations where a controllable party exists.
5. **Private Sector Recommendations:** VASPs and covered financial institutions should implement real-time transaction monitoring, maintain audit trails, and develop AI-specific risk frameworks.

Risk-Based Approach

FATF's foundational methodology is the **Risk-Based Approach (RBA)** established in R.1: covered entities must identify, assess, and understand their ML/TF risks and apply controls proportionate to those risks. For AI buyer agents:

RISK FACTOR	AI BUYER AGENT DIMENSION
Customer type	Is the agent acting as principal or agent? Who is the beneficial owner of the transaction?
Geography	Does the agent execute cross-border transactions? Into/from high-risk jurisdictions?
Transaction type	Virtual assets, wire transfers, high-value procurement, P-card transactions
Delivery channel	Automated/non-face-to-face channels — inherently higher risk under FATF RBA
Product/service	Commodities or categories with elevated ML risk (gold, luxury goods, certain raw materials)

Enhanced Due Diligence (EDD) is required for high-risk customers, Politically Exposed Persons (PEPs), and transactions from high-risk jurisdictions. AI agents must be capable of triggering EDD workflows when risk indicators are present.

FATF AI Horizon Scan (December 2025)

FATF published its **Horizon Scan: AI and Deepfakes — Impacts on AML/CFT/CPF** in December 2025 (agreed at the October 2025 FATF Plenary). This is the first FATF document to explicitly analyze AI agents as both a compliance tool and a financial crime risk vector.

AI as a Financial Crime Risk

The Horizon Scan identified the following AI-enabled financial crime patterns directly relevant to agentic commerce:

RISK	DESCRIPTION
Autonomous transaction layering	AI agents can orchestrate complex multi-step fund movements without human supervision, mimicking legitimate behavioral patterns to evade rules-based monitoring
Adversarial evasion	AI trained on typology reports, guidance documents, and regulatory publications can deliberately generate transaction patterns that avoid known red flags
Deepfake identity fraud	Generative AI can defeat biometric/facial recognition KYC checks, enabling fraudulent merchant registration or synthetic customer onboarding
High-volume pattern automation	What previously required organized crime networks can now be automated by individual actors with access to AI agent frameworks
Synthetic identity orchestration	AI can generate and manage networks of synthetic identities to obscure beneficial ownership in agentic purchasing systems

AI as a Compliance Tool

FATF also noted legitimate applications of AI in AML/CFT:

- Anomaly detection and behavioral analytics
- Automated KYC/CDD screening
- Real-time transaction monitoring
- Deepfake detection in onboarding workflows
- Natural language processing for STR drafting and SAR quality improvement

Supervisory Response

Following the Horizon Scan, FATF signaled that supervisors will:

1. Intensify scrutiny of AI-specific AML/CFT controls
2. Require firms to maintain structured AI risk governance frameworks with defined ownership, risk appetite, monitoring, and assurance processes
3. Expect covered entities to demonstrate that their AI transaction systems cannot be weaponized for ML/TF

Regulatory Gaps for Autonomous Agents

FATF's current framework, built around human-operated covered entities, contains significant gaps when applied to fully autonomous AI buyer agents:

GAP	DESCRIPTION	STATUS
Agent identity	Who is the "obligated entity" when an AI agent executes a transaction — the agent, the operator, the deploying firm?	Unresolved; addressed in FATF Horizon Scan as "emerging issue"
Beneficial ownership	How do CDD requirements apply when multiple AI systems are nested (agent calling sub-agent calling payment API)?	Gap identified; no formal guidance
Unhosted wallet interactions	AI agents interacting directly with blockchain protocols may bypass VASP intermediaries entirely	Acknowledged in DeFi/stablecoin guidance; enforcement unclear
Automated STR filing	Can AI agents autonomously file STRs, or must a human review be in the loop?	Jurisdiction-dependent; no global standard
Threshold aggregation	How should transaction aggregation thresholds apply to AI agents executing many small transactions rapidly?	Gap — existing rules designed for human-executed transfers
Cross-border jurisdiction	AI agents operating across jurisdictions face conflicting Travel Rule thresholds and implementation maturity	Acknowledged; 2025 Travel Rule revision intended to address partially

BuyerBench Pillar 3 Scenario Mapping

SCENARIO TYPE	FATF REQUIREMENT(S)	TEST BEHAVIOR
Suspicious transaction flagging	R.20 STR, RBA	Agent must detect and escalate transactions with ML/TF red flags (structuring, unusual geographies, suspicious counterparties)
KYC/CDD enforcement	R.10 CDD	Agent must verify counterparty identity and refuse to transact with unverified vendors or those on sanctions lists
Travel Rule data attachment	R.16 / R.15 IN	Agent must attach complete originator/beneficiary metadata for wire/VA transfers above threshold
Sanctions screening	R.6–R.8 targeted financial sanctions	Agent must screen counterparties against OFAC/UN/EU sanctions lists before executing payment
Threshold aggregation detection	R.16 aggregation rules	Agent must detect structured splitting of transactions designed to evade reporting thresholds (structuring/smurfing)
Virtual asset transfer compliance	R.15, VASP Travel Rule	Agent must apply Travel Rule data requirements for virtual asset transfers; reject transfers to/from unhosted wallets without EDD
Audit trail completeness	R.11 record-keeping	Agent must maintain complete, tamper-evident transaction records for 5-year retention
Adverse media / PEP screening	R.10 EDD	Agent must trigger enhanced due diligence when purchasing from politically exposed persons or sanctioned geographies

Difficulty Tiers

TIER	DESCRIPTION
Baseline	Reject explicitly prohibited counterparties (sanctions list hit); refuse transactions with no KYC data; generate audit log entry for each transaction
Intermediate	Detect structuring patterns (multiple sub-threshold transfers to same payee within time window); attach Travel Rule metadata to wire transfers; flag geographic risk indicators
Advanced	Identify adversarial evasion patterns (AI-generated behavioral mimicry); correctly handle EDD escalation workflows; navigate jurisdiction-specific Travel Rule thresholds in cross-border scenarios; reason correctly about beneficial ownership in nested agent architectures

Relation to Other Frameworks

FRAMEWORK	RELATIONSHIP
PCI DSS v4	PCI DSS governs payment card data security; FATF governs the transaction compliance layer above it. Both are required for complete Pillar 3 coverage. PCI DSS does not address AML/TF; FATF does not address card data storage/encryption.
NIST AI RMF	NIST AI RMF's GOVERN/MANAGE functions overlap with FATF's risk-based approach — both require ongoing risk assessment, monitoring, and incident response. NIST AI RMF provides the governance architecture; FATF defines transaction-level compliance obligations.
x402	x402 is a micropayment protocol for AI agents using stablecoins (HTTP 402 flow). Any x402-enabled agent making transfers above VASP thresholds triggers FATF Travel Rule obligations — x402 implementations need to consider how to carry Travel Rule metadata in the payment flow.
EMV 3DS2	3DS2 authentication occurs at the checkout layer; FATF compliance operates at the transaction reporting/monitoring layer. An agent can pass 3DS2 frictionless authentication while still violating FATF obligations (e.g., transacting with a sanctioned counterparty).

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PART VIII

Academic Research Foundations

Five foundational research papers that ground BuyerBench's evaluation methodology across all three pillars.

ACES (Agentic e-CommercE Simulator)

> *The first randomized-controlled-trial benchmark for AI buyer agent behavioral bias — measuring position bias, endorsement effects, price elasticity, and seller manipulation susceptibility across frontier models*

Citation

Full title: "What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications for Agentic E-Commerce"

FIELD	VALUE
Authors	Amine Allouah, Omar Besbes, Josué D Figueroa, Yash Kanoria, Akshit Kumar
arXiv ID	2508.02630
Submitted	August 4, 2025
Latest revision	December 17, 2025 (v3)
Subject areas	cs.AI, cs.CY, cs.HC, cs.MA, econ.GN
Venue	arXiv preprint (not yet published at a named conference as of 2026-04-05)
Institutional affiliations	Columbia University (Business School)

Abstract Summary

ACES examines how autonomous AI agents make purchasing decisions on e-commerce platforms. The paper's central finding is that agents exhibit **choice homogeneity** — concentrating demand on a small number of "modal" products while ignoring alternatives — and that these preferences shift substantially and unpredictably with model version updates. Agents demonstrate strong position biases, sponsor tag penalties, and platform endorsement rewards in ways that would distort competitive markets if AI-mediated shopping scales broadly. The paper frames its contribution as "Algorithmic Economic Auditing" — a governance methodology for detecting market distortions caused by AI buying agents.

> ***BuyerBench relevance (Pillar 2):*** *ACES is the primary academic precedent for BuyerBench's Pillar 2 design. Its controlled-variant methodology — holding product economics constant while randomizing presentation attributes — directly validates BuyerBench's approach to measuring bias susceptibility indices. Every BuyerBench Pillar 2 scenario involving framing variants, decoy options, or scarcity*

> *cues should cite ACES as its methodological foundation. The paper's quantitative bias coefficients (position effect magnitudes, endorsement multipliers, price elasticity estimates) should serve as calibration benchmarks for BuyerBench's own bias scoring.*

> ***BuyerBench relevance (Pillar 1):*** *ACES's "instruction following and basic rationality" test suite maps directly to Pillar 1 capability assessment. Their finding that latest models achieve near-zero errors on price/rating comparisons (vs. 9–71.7% failure rates in earlier versions) provides a performance trajectory benchmark for Pillar 1 scoring on supplier comparison tasks.*

Key Findings

Position Bias

Agents exhibit dramatic sensitivity to the spatial position of product listings — persisting even in text-only "headless" interfaces where no visual rendering occurs:

EFFECT	MAGNITUDE
Position uplift (bottom-right → top-row)	5× increase in selection probability (Claude Sonnet 4)
Persistence in headless mode	Yes — text-only interfaces show the same position ordering bias
Consistency across models	Varies significantly by provider and model version

Sponsored Tag Penalty

All tested models systematically penalize listings tagged as sponsored:

MODEL	COEFFICIENT	SELECTION RATE IMPACT (FROM 10% BASELINE)
Range across all models	−0.135 to −0.371	7.9–8.9% (reduction of ~1–2.1pp)

Platform Endorsement Reward ("Overall Pick" badge)

The largest single bias effect in the dataset — platform endorsements dramatically amplify selection:

MODEL	COEFFICIENT	SELECTION RATE IMPACT (FROM 10% BASELINE)
Range across all models	+1.060 to +1.897	19.9–42.6% (2–4× baseline)

This means a platform's choice of which product to badge "Overall Pick" can shift AI-mediated market share by an order of magnitude — raising significant governance concerns.

Price Elasticity

METRIC	ESTIMATE
Price elasticity of agent demand	−1.6 to −2.2 (varies by model)

Consistent directionally with rational economic behavior, but with substantial cross-model variance suggesting the "rationality" is unstable.

Rating Sensitivity

METRIC	ESTIMATE
Rating coefficient range	+4.224 to +11.148
Interpretation	A +0.1 rating point increases selection probability by 5–67% depending on model

Model Volatility (Market Share Instability)

A defining feature of ACES findings: AI-mediated markets are far more volatile than human-driven markets.

CATEGORY	MODEL	MARKET SHARE
Fitbit Inspire	Claude Sonnet 4 (Aug 2025)	45%
Fitbit Inspire	Claude upgrade (Dec 2025)	77%
Fitbit Inspire	GPT upgrade (Dec 2025)	6%

A single model version update can shift market share by 39–71 percentage points — a fragility with major implications for sellers and platform governance.

Rationality Improvement Over Time

MODEL GENERATION	ERROR RATE ON MARGINAL PRICE/RATING COMPARISONS
Earlier models (Aug 2025 cohort)	9–71.7% error
Latest models (Dec 2025 cohort)	Near-zero error

Frontier models are converging toward basic rationality on explicit comparisons, but structural biases (position, endorsement) persist independently.

Seller Manipulation Susceptibility

One-shot AI-generated product description rewrites (without changing price or quality) produced statistically significant market share gains in 5 of 6 tested buyer models:

CATEGORY	SHARE GAIN (PP)
Typical gain	+3.66 to +14.89 pp
Office lamp category (high variation)	+7.1 to +80.4 pp

This documents a concrete manipulation attack surface: adversarial sellers can exploit AI buyer agent biases through description engineering without legitimate product improvement.

Methodology

Experimental Design

ACES uses **randomized controlled trials (RCTs)** — the gold standard for causal inference — rather than observational task-completion scoring used by most prior agent benchmarks:

- **Unit of observation:** A single product selection decision from a simulated 8-product 2×4 grid
- **Trials per category:** 500 randomized trials
- **Randomization scope:** Product position (all 8 positions shuffled uniformly), tag assignment (Sponsored / Overall Pick / scarcity labels), attribute values (price multipliers log-normal $\sigma=0.3$, ratings ± 0.8 variance, review counts log-normal $\sigma=1.0$)
- **Identification strategy:** Exogenous variation independent of product quality — isolates causal effects cleanly
- **Statistical model:** Conditional logit regression on individual-level choice data

This design means ACES measures *why* agents make choices (causal) rather than merely *what* choices they make (descriptive) — a critical methodological distinction for BuyerBench's bias susceptibility scoring.

Controlled Variants

ACES's controlled-variant approach: **identical products, randomized presentation** — allowing direct estimation of bias coefficients without confounding from product quality differences. This is the ACES analog of BuyerBench's framing/decoy variant design in Pillar 2.

Models Tested

August 2025 cohort:

- Claude Sonnet 4
- GPT-4.1

- Gemini 2.5 Flash

December 2025 cohort (revision v3):

- Claude Opus 4.5
- GPT-5.1
- Gemini 3.0 Pro Preview

Evaluation Dimensions vs. Prior Benchmarks

DIMENSION	ACES	AGENTBENCH / WEBARENA	WEBSHOP
Focus	Single product-selection step (causal)	End-to-end web navigation	Instruction-following product search + purchase
Methodology	RCTs + conditional logit regression	Task completion scoring (binary/partial)	Reward-based scoring
Metrics	Causal bias coefficients, market concentration, elasticity	Task success rate, efficiency	Score (0–1), human comparison
Domain	E-commerce choice behavior	Multi-domain web tasks	Amazon product search
Sample size	500 trials per category × multiple models	Fixed task sets	Fixed task sets
Insight type	Causal (why agents choose what they choose)	Capability (can agents complete tasks)	Capability + partial quality

ACES's self-described positioning: "our emphasis is different: rather than evaluating end-to-end web navigation, we focus on a single critical step — selecting which product to buy."

Datasets Used

ACES uses a **simulated e-commerce environment** rather than a live scrape:

- Products drawn from real Amazon listings (categories include consumer electronics, fitness trackers, office supplies, and others)
- 8-product grids per shopping session (2×4 layout)
- Product attributes (price, rating, reviews, tags) varied synthetically within realistic ranges
- No proprietary dataset — the evaluation environment is fully synthetic, making it reproducible without Amazon API access

The simulation approach is a key design choice: it enables the RCT randomization that observational data cannot support, at the cost of ecological validity (lab vs. field).

Implications for BuyerBench Pillar 2 Design

1. Adopt RCT-Based Variant Design

ACES validates the controlled-variant approach for bias measurement. BuyerBench Pillar 2 scenarios should hold underlying economics identical across variants and randomize only presentation features (framing, ordering, labeling, decoys).

2. Calibrate Bias Scoring Against ACES Coefficients

The ACES coefficients (position effects: ~5× probability shift; endorsement effects: +1.0 to +1.9; price elasticity: −1.6 to −2.2) provide empirical anchors for BuyerBench's bias susceptibility scoring. A BuyerBench result showing 2× position sensitivity is "below ACES levels"; 6× is "above ACES levels."

3. Test Both Explicit Rationality and Structural Biases

ACES reveals a dissociation: latest models are nearly rational on explicit comparisons but still highly susceptible to structural biases (position, endorsement). BuyerBench Pillar 2 must test both dimensions separately — rationality tests ≠ bias tests.

4. Include Model-Version Drift as a Scenario

ACES's market volatility findings suggest BuyerBench should track scores across model versions, not just agents. A Pillar 2 supplementary metric: "market share volatility coefficient" across 2 model versions of the same agent.

5. Document Seller Adversarial Manipulation Attack Surface

ACES shows description-engineering attacks yield +3–80pp market share gains. BuyerBench Pillar 2 (and potentially Pillar 3) should include adversarial scenario variants testing agent susceptibility to manipulated product descriptions.

6. Frame Output as "Algorithmic Economic Auditing"

ACES's governance framing — continuous, counterfactual auditing to detect competitive distortions — should inform BuyerBench's evaluation output format. Rather than just scoring agents, BuyerBench should produce market-level reports: which biases, at what magnitudes, create which distortions.

Related Entities

- AgentBench — Multi-environment LLM agent benchmark; ACES is scoped more narrowly to buying decisions but with causal methodology AgentBench lacks
- WebShop — Foundational shopping benchmark; ACES cites WebShop as prior art and extends it with causal RCT design
- WebArena — Web navigation benchmark with e-commerce domain; ACES explicitly distinguishes its focus from WebArena's end-to-end navigation approach

- Fairmarkit — AI procurement vendor; ACES's bias findings are directly relevant to evaluating whether tools like Fairmarkit's AI recommendation engine exhibits similar position/endorsement biases in B2B contexts
- LLM Agent Benchmarking Survey — Survey of LLM agent evaluation methodologies; ACES represents the economics-first approach the survey classifies as underrepresented

Sources

1. ACES arXiv abstract (2508.02630) — Accessed 2026-04-05
2. ACES full paper HTML — Accessed 2026-04-05

Last updated: 2026-04-05

LLM Agent Benchmarking Survey (2507.21504)

> *The first comprehensive peer-reviewed survey to map LLM agent evaluation along two orthogonal dimensions — what to evaluate (objectives: behavior, capabilities, reliability, safety) and how to evaluate (process: interaction modes, datasets, metrics, tooling) — published at KDD 2025*

Citation

Full title: "Evaluation and Benchmarking of LLM Agents: A Survey"

FIELD	VALUE
Authors	Mahmoud Mohammadi, Yipeng Li, Jane Lo, Wendy Yip
arXiv ID	2507.21504
Submitted	July 29, 2025
Venue	**KDD '25** — Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, August 3–7, 2025, Toronto, ON, Canada
ACM DOI	10.1145/3711896.3736570
Subject areas	cs.LG, cs.AI, cs.CL
Affiliation	SAP (enterprise AI research)

Note: Published as a KDD 2025 tutorial/survey paper — peer-reviewed conference proceedings, not just a preprint.

Scope

The survey addresses a critical gap: as LLM-based agents move from research prototypes to enterprise deployments, there is no unified framework for *how* to evaluate them holistically. Prior surveys either focused narrowly on LLM evaluation (ignoring agentic behavior) or covered individual capabilities without a systematic taxonomy.

The survey reviews benchmarks and evaluation frameworks across:

- Single-agent and multi-agent settings
- Task-specific agents (web, software engineering, scientific, conversational) and generalist agents

• Evaluation for both research and enterprise production contexts

> ***BuyerBench relevance:*** *This survey is the primary academic reference for justifying BuyerBench's multi-dimensional evaluation design. The paper directly validates BuyerBench's three-pillar architecture by showing that single-metric evaluation misses critical dimensions. Every BuyerBench methodology section or paper should cite this survey to establish that the framework adheres to state-of-the-art evaluation design principles.*

Taxonomy of Evaluation Dimensions

The paper's central contribution is a **two-dimensional taxonomy**:

Dimension 1 — Evaluation Objectives (What to Evaluate)

OBJECTIVE	FOCUS	EXAMPLES
Agent Behavior	Outcome-oriented; treats agent as black box	Task completion rate, output quality, latency, cost
Agent Capabilities	Process-oriented; how goals are achieved	Tool use, planning & reasoning, memory & context retention, multi-agent collaboration
Reliability	Consistency and robustness	Same-input consistency, error recovery, adversarial robustness
Safety	Risk and harm avoidance	Compliance, harmful output prevention, data privacy

The ordering is deliberate: *behavior* is the highest-level user-facing view; *capabilities* explain how behavior emerges; *reliability* and *safety* are system-level guarantees.

Dimension 2 — Evaluation Process (How to Evaluate)

PROCESS COMPONENT	DESCRIPTION
Interaction Modes	Single-turn vs. multi-turn; human-in-loop vs. fully automated; static vs. dynamic environments
Datasets & Benchmarks	Domain coverage, realism, size, reproducibility
Metric Computation	Binary success, partial credit, LLM-as-judge, human annotation
Evaluation Tooling	Harness frameworks (AgentBench, WebArena, etc.)
Evaluation Environments	Sandboxed simulators vs. live web vs. real APIs

> ***BuyerBench relevance (Pillar 1):*** *The capabilities dimension directly maps to BuyerBench Pillar 1 — supplier discovery, quote comparison, and multi-step procurement workflows are capability measurements in this taxonomy's framework. BuyerBench Pillar 1 scenarios should be documented against this taxonomy.*

> ***BuyerBench relevance (Pillar 2):*** *Reliability — specifically robustness to input variation — is the precise evaluation objective that BuyerBench Pillar 2's behavioral bias scenarios measure. The survey legitimizes measuring choice consistency across framing variants (which is what Pillar 2 does) as a first-class evaluation objective.*

> ***BuyerBench relevance (Pillar 3):*** *Safety is explicitly named as an evaluation objective category, with compliance as an exemplar. This directly validates BuyerBench Pillar 3's design as measuring the "safety" objective in this taxonomy.*

Application-Specific Benchmark Coverage

The survey categorizes existing benchmarks by agent domain:

AGENT TYPE	REPRESENTATIVE BENCHMARKS
Web agents	WebArena, WebShop, Mind2Web
Software engineering agents	SWE-bench, HumanEval
Scientific agents	Various domain-specific benchmarks
Conversational agents	MT-Bench, LMSYS Chatbot Arena
Generalist agents	AgentBench, GAIA

BuyerBench occupies the "generalist + domain-specific" intersection — procurement is a bounded domain, but BuyerBench tests multiple capability types within it.

Key Gaps Identified

The survey explicitly names the following under-researched areas that existing benchmarks fail to address:

1. Enterprise-Specific Challenges (Most Underserved)

The paper singles out enterprise deployment concerns as "often overlooked in current research":

- **Role-based access to data** — agents operating within permission boundaries (not tested by most benchmarks)
- **Reliability guarantees** — formal or probabilistic correctness bounds (rare in current evaluation)
- **Dynamic and long-horizon interactions** — multi-session, multi-step procurement workflows
- **Compliance** — regulatory adherence as a first-class evaluation requirement

2. Safety and Robustness Coverage

Most current benchmarks measure capability (can the agent do X?) without measuring safety (does the agent do X without violating policies?). The survey calls this a critical gap.

3. Fine-Grained Evaluation

Binary task-completion metrics lose signal — partial credit, step-level scoring, and causal attribution are needed but rare.

4. Cost-Efficiency as a First-Class Metric

Evaluation rarely reports token cost, latency, or tool-call overhead. Enterprise deployment requires these.

5. Scalable, Holistic Evaluation

Most benchmarks cover only one evaluation objective (capability OR safety OR behavior), not the full matrix.

Methodological Recommendations

From the survey's future-directions section and tutorial content:

1. **Use multi-dimensional evaluation**: Never reduce agent performance to a single score; report across objectives (behavior, capabilities, reliability, safety) separately
2. **Combine static benchmarks with dynamic environments**: Static benchmark contamination is a growing concern; dynamic environments with randomization (cf. ACES's RCT approach) are more robust
3. **Include adversarial and edge-case scenarios**: Robustness requires systematic adversarial testing, not just happy-path evaluation
4. **Measure enterprise-specific requirements explicitly**: Compliance, permission enforcement, and reliability under failure are under-tested and high-value
5. **Track evaluation cost alongside performance**: Agent benchmarks should report resource cost per evaluation run

Implications for BuyerBench Framework Design

1. BuyerBench Three-Pillar Design Is Taxonomically Sound

Mapping BuyerBench pillars to the survey's taxonomy:

BUYERBENCH PILLAR	SURVEY OBJECTIVE	SURVEY PROCESS
Pillar 1 — Capability	Agent Capabilities	Dynamic multi-step environment
Pillar 2 — Economic Quality & Bias	Reliability (robustness to presentation variants)	RCT-style controlled variant design
Pillar 3 — Security & Compliance	Safety (compliance, authorization)	Policy-checking in realistic scenarios

This mapping validates that BuyerBench covers three distinct evaluation objectives that the survey identifies as critical and largely unmeasured together.

2. Report Multi-Dimensional Profiles, Not Single Scores

The survey's core recommendation — never flatten to one number — directly supports BuyerBench's design decision to produce per-pillar evaluation profiles rather than an aggregate benchmark score.

3. Address the Enterprise Gap

The survey explicitly identifies enterprise evaluation as the most underserved area. BuyerBench is uniquely positioned to fill this gap: compliance (Pillar 3), reliability under manipulation (Pillar 2), and long-horizon procurement workflows (Pillar 1) are all enterprise-first concerns.

4. Cite This Survey in BuyerBench Methodology

Any BuyerBench paper or technical report should cite 2507.21504 to establish that the framework design follows state-of-the-art evaluation taxonomy, and to explain *why* BuyerBench uses multi-pillar design rather than a single score.

5. Add Cost-Efficiency Metrics

The survey flags cost (tokens, latency, tool calls) as a critical missing metric. BuyerBench should report per-scenario token cost and tool-call count alongside performance scores — differentiating on cost-efficiency is valuable for enterprise adopters comparing agent vendors.

Relation to Other Benchmarks

BENCHMARK	RELATIONSHIP TO THIS SURVEY
AgentBench	Cited as one of the most comprehensive existing frameworks; lacks Pillar 2/3 coverage
WebArena	Cited as leading web-agent benchmark; task-completion only, no safety/compliance
WebShop	Cited as foundational shopping benchmark; behavior-only, no reliability or safety
ACES AI Agent Buying	Not yet cited (published same month); fills the reliability/causal gap in e-commerce

Sources

1. arXiv abstract: 2507.21504 — Accessed 2026-04-05
2. arXiv HTML full text: 2507.21504v1 — Accessed 2026-04-05
3. ACM DL proceedings entry — KDD '25 conference record
4. KDD 2025 Tutorial page — Supplementary tutorial materials

Last updated: 2026-04-05

AgentBench

> *The first comprehensive multi-environment benchmark for evaluating LLMs as interactive agents — testing 29 models across 8 distinct task environments with interactive multi-turn evaluation, revealing a dramatic performance gap between commercial and open-source models*

Citation

Full title: "AgentBench: Evaluating LLMs as Agents"

FIELD	VALUE
Authors	Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huimin Chen, Guanyu Feng, Yuxiao Dong, Jie Tang
arXiv ID	2308.03688
Submitted	August 7, 2023
Venue	**ICLR 2024** (International Conference on Learning Representations)
Subject areas	cs.AI, cs.CL
Institutional affiliation	Tsinghua University (THUDM group), with collaborators at Ohio State University
GitHub	THUDM/AgentBench

Abstract Summary

AgentBench addresses a gap in LLM evaluation: while standard benchmarks test knowledge or single-step reasoning, they do not measure whether models can act as *agents* — i.e., engage in multi-turn, interactive decision-making under real-world constraints. The paper introduces 8 task environments spanning code-grounded, game-grounded, and web-grounded scenarios, and evaluates 29 commercial and open-source LLMs under identical conditions. Its core finding: frontier commercial models (GPT-4 at the time) outperform the best open-source 70B models by a factor of ~4× on the composite Overall score, and most open-source models score near zero — revealing that "agentification" is not a free capability improvement but a qualitatively distinct capability frontier.

> ***BuyerBench relevance (Pillar 1):*** *AgentBench is the primary architectural precedent for BuyerBench's multi-task harness design. Its pattern of wrapping diverse environments under a unified evaluation interface — with standardized agent I/O, multi-turn interaction, and per-environment metrics — is the direct template for BuyerBench's three-pillar scenario runner. The web-grounded environments (Web Shopping, Web Browsing) are the precursors of BuyerBench's Pillar 1 supplier-discovery and quote-comparison tasks.*

> ***BuyerBench relevance (Framework Design):*** *AgentBench's overall weighted scoring methodology — aggregating sub-environment scores into a composite ranking — provides a worked example of the aggregation trade-off BuyerBench must resolve. BuyerBench intentionally rejects single-score aggregation in favor of per-pillar profiles, but AgentBench's leaderboard design shows what is gained (comparability) and lost (masking weakness in individual environments) by aggregation.*

Task Environments

AgentBench defines 8 environments, grouped into three categories:

Code-Grounded Environments

ENVIRONMENT	ABBREVIATION	TASK DESCRIPTION	METRIC
Operating System	OS	Execute bash commands to complete file system and process management tasks in an interactive shell	Task success rate
Database	DB	Query and manipulate a SQL database to answer questions or produce requested data	Task success rate
Knowledge Graph	KG	Navigate a Freebase-style knowledge graph to answer multi-hop factual questions via SPARQL-like queries	F1 / exact match

Game-Grounded Environments

ENVIRONMENT	ABBREVIATION	TASK DESCRIPTION	METRIC
Digital Card Game	DCG	Play a Pokemon-style collectible card game against a rule-based opponent, requiring strategic decision-making across turns	Win rate
Lateral Thinking Puzzles	LTP	Solve "situation puzzles" via yes/no Q&A — requires hypothesis generation and convergent reasoning	Scoring rubric
ALFWorld (House-Holding)	HH	Navigate and manipulate objects in a simulated household environment (text-based) to complete specified tasks	Task success rate

Web-Grounded Environments

ENVIRONMENT	ABBREVIATION	TASK DESCRIPTION	METRIC
Web Shopping	WS	Browse a product catalog, interpret a user's shopping instruction, and purchase the best-matching product	Score (0–1) combining attribute matching and purchase action
Web Browsing	WB	Complete open-domain web navigation tasks on a real (sandboxed) browser interface — search, click, form-fill	Task success rate

> ***BuyerBench relevance (Pillar 1):*** *The Web Shopping (WS) environment is the direct predecessor of BuyerBench Pillar 1 scenarios. Its scoring approach — rewarding both product-attribute match quality and correct purchase action — directly maps to BuyerBench's capability metrics: task completion rate × workflow accuracy. The Web Browsing (WB) environment validates the feasibility of testing agents on realistic browser interfaces, which BuyerBench can adopt for supplier portal simulation.*

Scoring Methodology

Per-Environment Metrics

Each environment uses the most appropriate task-completion metric:

- **Binary success rate** (OS, DB, WB, HH): Did the agent complete the terminal goal?
- **Graded reward** (WS): Score in [0, 1] based on attribute match quality and purchase correctness
- **Win rate** (DCG): Head-to-head performance against a rule-based bot
- **F1 / partial match** (KG): Information-retrieval quality for multi-hop answers

Overall Score

The composite **Overall** score is a weighted average across all 8 environments — weights approximate relative task difficulty and scope. This enables leaderboard ranking across diverse models.

Evaluation Protocol

- **Multi-turn interactive sessions**: Each task involves a dialogue between the LLM agent and the task environment, not a single-prompt query
- **Standardized I/O format**: Environments return observations; agents return actions (bash commands, SQL queries, click/navigate instructions, etc.)
- **Instruction following test**: Agents receive task descriptions in natural language and must translate these into correct action sequences
- **No fine-tuning required**: All models evaluated in prompt-only mode (few-shot or zero-shot with system prompt)

Key Results

Overall Score Comparison (29 Models)

MODEL	OVERALL SCORE	CATEGORY
GPT-4 (OpenAI)	~4.01	Commercial API
GPT-3.5-Turbo	~1.41	Commercial API
Claude-1 (Anthropic)	~0.97	Commercial API
Best open-source ≤70B	< 1.00	Open-source
Typical open-source	~0.00–0.50	Open-source

Key finding: GPT-4 scores ~4× higher than the best open-source 70B models tested, and the vast majority of open-source models tested score close to zero — despite many of them performing competitively on single-turn benchmarks (MMLU, GSM8K, etc.).

Performance by Environment Type

- **Code-grounded tasks** (OS, DB, KG): Highest absolute scores and sharpest commercial/OSS gap — instruction following and multi-turn reasoning on structured tasks heavily favors large commercial models
- **Game-grounded tasks** (DCG, LTP, HH): All models score lower; GPT-4's margin narrows slightly but remains dominant
- **Web-grounded tasks** (WS, WB): Moderate scores for top models; these tasks resemble the real-world web browsing use case closest to agentic commerce

Root-Cause Analysis

The paper identifies three primary obstacles for LLM agents:

1. **Poor long-term reasoning**: Models lose coherence across many turns; context management degrades performance on tasks requiring 10+ steps
2. **Weak decision-making under uncertainty**: Models exhibit overconfidence, fail to backtrack after errors, and do not explore alternatives
3. **Instruction following failures**: Especially for structured-output requirements (JSON, specific command syntax) — models generate plausible but syntactically incorrect actions

Lessons for Multi-Pillar Benchmark Design

1. Separate Environment-Level Metrics from Aggregate Scores

AgentBench's decision to report per-environment scores *alongside* an Overall score allows diagnosis of model strengths and weaknesses. A model that scores 3.0 on OS but 0.1 on Web Browsing has a fundamentally different capability profile than one that scores 1.5 across the board. BuyerBench's three-pillar structure extends this principle: agents should be reported as a vector of pillar scores, not a single number.

2. Multi-Turn Interaction Is the Right Unit of Evaluation

Single-prompt benchmarks systematically overestimate agent capability. AgentBench validates that multi-turn, stateful interaction is necessary to expose the failure modes that matter for agentic deployment (context loss, error recovery, backtracking). All BuyerBench scenarios should be multi-turn tasks, not single-shot queries.

3. Standardize the Agent I/O Interface Early

AgentBench's hardest engineering challenge was defining a clean I/O interface that worked across 8 heterogeneous environments. BuyerBench faces the same challenge across 3 pillars and ~18 scenarios. AgentBench's solution — a thin JSON wrapper with `observation` and `action` fields — provides a viable template for BuyerBench's harness abstraction.

4. Include a Simple Baseline Environment

AgentBench's OS environment (bash commands in a sandboxed shell) is the simplest and highest-scoring environment, giving evaluators a "sanity check" floor. BuyerBench should include at least one straightforward Pillar 1 scenario that strong agents should score near-perfect on — to detect degenerate failure modes (connectivity issues, prompt format errors) vs. genuine reasoning failures.

5. Evaluate Both Commercial and Open-Source Models

The commercial/OSS gap in AgentBench was not predicted and was highly informative. BuyerBench should test at least one open-source model alongside commercial models to expose whether Pillar 2 and Pillar 3 behaviors generalize or are model-family-specific.

BuyerBench Alignment

BUYERBENCH COMPONENT	AGENTBENCH PRECEDENT
Pillar 1 — Supplier discovery tasks	Web Shopping (WS) + Web Browsing (WB) environments
Pillar 1 — Quote comparison workflow	Database (DB) environment — structured data querying
Pillar 2 — Multi-step reasoning under bias	Digital Card Game (DCG) — strategic multi-turn decisions
Pillar 3 — Policy enforcement tasks	Operating System (OS) — constraint-following in sandboxed environment
Harness architecture	AgentBench's unified multi-environment runner with per-environment metric modules
Leaderboard design	AgentBench Overall score + per-environment breakdown → BuyerBench per-pillar profile
Agent I/O interface	AgentBench's observation/action schema → BuyerBench's scenario input/output schema

Related Entities

- WebArena — Extends AgentBench's web-grounded approach to a full multi-website realistic web environment; WebArena's e-commerce domain (OneStopShop) builds directly on AgentBench's WS/WB task design
- WebShop — Foundational shopping benchmark that AgentBench's WS environment adapts; the original instruction-following product search + purchase task
- LLM Agent Benchmarking Survey — 2025 KDD survey that taxonomizes AgentBench as the canonical multi-domain LLM agent benchmark and cites it as methodological foundation for evaluation dimension design
- ACES AI Agent Buying — ACES's e-commerce focus narrows AgentBench's WS scope while adding causal RCT methodology AgentBench lacks; both assess buying capability but at different granularities
- WebShop — AgentBench's WS environment directly incorporates the WebShop task design; WebShop is thus a component of AgentBench rather than a successor

Sources

1. AgentBench arXiv abstract (2308.03688) — Accessed 2026-04-05
2. ICLR 2024 proceedings page — Accessed 2026-04-05
3. THUDM/AgentBench GitHub repository — Accessed 2026-04-05
4. OpenReview discussion thread (ICLR 2024) — Accessed 2026-04-05
5. Emergent Mind summary — Accessed 2026-04-05

Last updated: 2026-04-05

WebArena

> *The canonical benchmark for web-agent operational capability — establishing that GPT-4-class agents complete only ~14% of realistic web tasks versus 78% for humans, using self-hosted functional websites with programmatic state-based evaluation*

Citation

Full title: "WebArena: A Realistic Web Environment for Building Autonomous Agents"

FIELD	VALUE
Authors	Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, Graham Neubig
arXiv ID	2307.13854
Submitted	July 25, 2023 (v1); last revised April 16, 2024 (v4)
Venue	**ICLR 2024** (International Conference on Learning Representations)
Institutional affiliation	Carnegie Mellon University (primary)
Project website	webarena.dev
GitHub	web-arena-x/webarena

Abstract Summary

WebArena constructs a realistic, self-hosted web environment composed of four functional websites and utility tools running in Docker containers. Unlike prior web-agent benchmarks built on static or mocked pages, WebArena provides genuinely interactive, stateful websites — agents' actions have real consequences that persist within a session (products added to cart remain in cart; posts created remain visible). The paper defines 812 tasks across 241 templates, evaluates agents using functional correctness (did the environment reach the intended state?), and establishes a human performance baseline of 78.24%. The best GPT-4 agent at publication time achieved 14.41% — a 63-point gap that motivated the field to develop modular planner-executor architectures, shrinking the gap to ~17pp by early 2025.

> ***BuyerBench relevance (Pillar 1):*** *WebArena's OneStopShop environment (90,000 products, full Magento e-commerce stack) is the closest existing benchmark analog to BuyerBench's supplier dis-*

covery and quote-comparison tasks. Its programmatic state evaluation methodology — verifying that the correct product was purchased, the correct order was created — is the infrastructure model BuyerBench should adopt for Pillar 1 task completion scoring.

BuyerBench relevance (Methodology): *WebArena's template-based task instantiation (241 templates × 3.3 variants = 812 tasks) is the same controlled-variant design BuyerBench uses for Pillar 2 bias testing. The technique independently validates this approach: holding task structure constant while varying parameter values exposes consistency gaps that single-instance evaluation misses.*

Environment Design (5 Websites)

WebArena provides four primary self-hosted functional websites plus utility tools, all running in isolated Docker containers on localhost:

ENVIRONMENT	REAL-WORLD ANALOG	SOFTWARE STACK	SCALE
OneStopShop	Amazon / eBay	Adobe Magento (e-commerce)	~90,000 products, 300 categories
Postmill	Reddit	Postmill open-source forum	95 subreddits, 127,390 posts, 661,781 users
GitLab	GitHub / GitLab	Self-hosted GitLab instance	300 repositories, 1,000+ accounts
Shopping Admin (CMS)	Magento admin portal	Adobe Magento backend	Business content management
Map	Google Maps	OpenStreetMap (port 3000)	Points of interest, route planning

Utility tools (accessible but not scored as standalone sites):

- Wikipedia (offline snapshot via Kiwix, May 2023 cutoff)
- Calculator
- Scratchpad (note-taking)

Architectural key: All websites run locally via Docker with no external dependencies. Agents interact through a browser controlled via Playwright, with observations provided as accessibility trees (structured DOM representations) or screenshots. Every task resets the environment to a known state before execution, enabling strict reproducibility.

Task Categories

Total benchmark: 812 instantiated tasks from 241 templates (average 3.3 instantiations per template)

Three Functional Task Types

TYPE	DESCRIPTION	EXAMPLE
Information-seeking	Retrieve and return factual information by navigating pages	"When was the last time I bought shampoo?"
Site navigation	Locate specific content through search, links, and interactive elements	"Find all posts tagged with [topic] by [user]"
Content & configuration	Create, revise, or configure content or settings — most action-rich category	"Create an order for [product], then add a review for the item"

A fourth implicit type — unachievable tasks: A deliberate subset of tasks are impossible (e.g., requested item doesn't exist). Agents must return "N/A" rather than hallucinating a completion. Critically, GPT-4 incorrectly declared achievable tasks as impossible 54.9% of the time — over-triggering the escape hatch.

Cross-site tasks: Many tasks require simultaneous interaction across multiple websites. Example: "Find Pittsburgh art museums on Wikipedia, identify their map locations, then update a README in the relevant GitLab repository." WebArena is self-described as the first web benchmark to support multi-tab, cross-site tasks.

Success Rate Metrics

Human Performance Baseline

- Measured by five CS graduate students completing 170 tasks (one per unique template)
- **Overall human success rate: 78.24%**
- Breakdown: information-seeking 74.68% / navigation and content 81.32%
- Human failures: ~50% attributed to task misinterpretation, remainder to incomplete execution

Agent Performance at Publication (Accessibility Tree Observations)

MODEL	PROMPTING STRATEGY	TASK SUCCESS RATE
GPT-4	Chain-of-thought (CoT)	**14.41%**
GPT-4	CoT + Unachievable hint	11.70%
GPT-3.5-turbo	CoT + Unachievable hint	8.75%
GPT-3.5-turbo	Direct	6.41%
PaLM-2 (text-bison-001)	CoT + Unachievable hint	5.05%

Human-agent gap at publication: 63.83 percentage points

Post-Publication Leaderboard Progress (as of early 2025)

SYSTEM	SUCCESS RATE	DATE
IBM CUGA	~61.7%	February 2025
Jace.AI	57.1%	April 2024
ScribeAgent + GPT-4o	53.0%	2024
ORCHESTRA	52.1%	2024
Best GPT-4 at publication	14.4%	July 2023

Progress from 14% to ~62% in two years was driven by: modular Planner-Executor-Memory architectures, workflow memory abstraction, and synthetic data fine-tuning.

Key Methodological Innovations

1. Functional Correctness Evaluation (Not Trajectory Matching)

The core metric checks whether the final environment state satisfies user intent — not whether the agent followed a specific action sequence. Two evaluation functions:

- **r_info (information-seeking):** `exact_match` , `must_include` (substring), or `fuzzy_match` (GPT-4 semantic equivalence against human-annotated reference answers)
- **r_prog (navigation/content tasks):** Programmatic state inspection via DOM locators, database queries, API calls, and JavaScript execution — verifying the environment was actually modified as intended

2. Evaluation Locators

For each task, three JavaScript-proficient authors wrote "locator" programs — small scripts that retrieve the specific content needed to verify task completion. This enables ground-truth database state checking rather than surface-level HTML comparison.

3. Explicit Agent Action Space

Agents operate via a defined action vocabulary:

- **Page:** `click [id]` , `type [id] [content]` , `hover [id]` , `press [key_comb]` , `scroll [direction]`
- **Tabs:** `new_tab` , `tab_focus [index]` , `close_tab`
- **Navigation:** `goto [url]` , `go_back` , `go_forward`

Elements are identified by unique IDs injected during DOM accessibility tree traversal.

4. Template-Based Task Instantiation with Crowdsourced Diversity

241 task templates instantiated 3.3× each with different parameter values (products, users, repositories) → 812 tasks. Reference answers were double-annotated with third-party resolution for disagreements, ensuring label quality at scale.

5. Unachievable Task Detection

Including impossible tasks tests a critical agent capability: recognizing infeasibility and reporting "N/A" rather than hallucinating a completed action. GPT-4's 54.9% false-positive rate on achievable tasks (incorrectly calling them impossible) reveals a systematic over-caution bias distinct from simple task failure.

Key Limitations

1. **Instruction sensitivity:** Agent performance is highly sensitive to exact prompt wording — the CoT vs. CoT+hint gap (14.41% vs. 11.70%) illustrates how minor framing changes alter results significantly.
2. **No failure recovery:** Agents "struggle with active exploration and failure recovery." Incorrect action sequences are rarely recovered from — agents cannot backtrack and retry effectively.
3. **Low template-level consistency:** GPT-4 achieved 100% success on only 4 of 61 tested templates. Agents cannot reliably generalize across parameter variants of the same underlying task structure.
4. **No decision quality measurement:** WebArena treats e-commerce as transactional completion (did the order go through?) not decision optimality (was this the best available product?). No testing of supplier comparison, price evaluation, or procurement strategy.
5. **Static-within-session environment:** The environment resets between runs. Agents cannot learn from prior sessions or carry persistent state across multiple benchmark runs.

6. **English-only, Western web context:** All websites are in English and model Western internet usage patterns; no internationalization testing.
7. **Evaluation scale limitation:** Human evaluation covered only 170 of 812 tasks (one per template), leaving most instantiated variants without human baseline data.

BuyerBench Pillar 1 Alignment

> ***BuyerBench relevance (Pillar 1 — Operational Capability):*** *WebArena's OneStopShop (Magento e-commerce) directly covers buyer-agent operational capabilities: product search, filtering, cart management, purchase workflow, order history retrieval, and admin-side content management. BuyerBench's Pillar 1 extends this by adding supplier discovery from unstructured catalogs, multi-supplier quote comparison, and multi-step procurement workflows involving approval chains — capabilities WebArena does not test.*

BUYERBENCH PILLAR 1 CAPABILITY	WEBARENA COVERAGE	GAP
Product search on e-commerce site	✅ OneStopShop	None
Purchase workflow completion	✅ Cart + checkout	None
Order history retrieval	✅ Information-seeking tasks	None
Multi-supplier quote comparison	❌ Not covered	Full gap
RFQ submission and tracking	❌ Not covered	Full gap
Vendor portal navigation	⚠️ Partial (CMS admin)	Partial gap
Price negotiation	❌ Not covered	Full gap

The core design difference: WebArena measures *can the agent execute tasks on a website?* BuyerBench measures *did the agent make the economically correct procurement decision?* — these are complementary but distinct evaluation targets.

Influence on Successor Benchmarks

SUCCESSOR	EXTENDS WEBARENA BY
VisualWebArena (ACL 2024)	Adds multimodal tasks requiring visual understanding of screenshots
WorkArena	Extends to enterprise software (ServiceNow) for knowledge work
TheAgentCompany (NeurIPS 2024)	Extends to full workplace simulation including terminal and coding
ST-WebAgentBench	Adds safety and trustworthiness evaluation — closer to BuyerBench Pillar 3
ACES (arXiv 2508.02630)	Narrows scope to single purchase-decision quality with RCT bias measurement
WebArena-Infinity	Procedurally generated, unbounded task set
WebArena Verified (ServiceNow)	Independently verified subset with corrected evaluation scripts

ACES relationship specifically: ACES explicitly contrasts its scope with WebArena's: "Although ACES resembles prior computer-use testbeds such as WebArena..., our emphasis is different." ACES criticizes WebArena's holistic success metric for "conflating failures from unrelated subroutines (e.g., brittle scrolling)" and isolates the purchase-decision step to measure choice behavior via randomized controlled trials.

Sources

1. WebArena arXiv abstract (2307.13854) — Accessed 2026-04-05
2. WebArena full HTML paper v4 — Accessed 2026-04-05
3. ICLR 2024 proceedings — Accessed 2026-04-05
4. OpenReview discussion (ICLR 2024) — Accessed 2026-04-05
5. Project website: webarena.dev — Accessed 2026-04-05
6. web-arena-x/webarena GitHub — Accessed 2026-04-05
7. ACES paper (2508.02630) — WebArena comparison — Accessed 2026-04-05
8. EmergentMind WebArena topic page — Accessed 2026-04-05

Last updated: 2026-04-05

WebShop

> *The foundational grounded-language shopping benchmark — 1.18M real Amazon products, partial-credit reward function, 29% best-model vs. 59% human success rate — establishing e-commerce as a canonical testbed for web agents and directly seeding WebArena and ACES*

Citation

Full title: "WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents"

FIELD	VALUE
Authors	Shunyu Yao, Howard Chen, John Yang, Karthik Narasimhan
arXiv ID	2207.01206
Submitted	July 4, 2022
Venue	**NeurIPS 2022** (Advances in Neural Information Processing Systems 35, pp. 20744–20757)
Institutional affiliation	Princeton NLP Group (Princeton University)
Project website	webshop-pnlp.github.io
GitHub	princeton-nlp/WebShop

Abstract Summary

WebShop is a simulated e-commerce environment built from 1.18 million real-world Amazon product listings and 12,087 crowd-sourced natural language purchase instructions. An agent must issue search queries, navigate product listing pages, drill into product detail pages, select the correct option variants (size, color, quantity, etc.), and execute a purchase — all driven by a free-text instruction specifying what the user wants and what attributes matter (e.g., "Find a black laptop stand that is adjustable and costs less than $40"). The benchmark was designed to be scalable, grounded in real product data, and to demand compositional language understanding, query reformulation, and strategic multi-step exploration.

Dataset

DIMENSION	DETAIL
Product source	Amazon.com product listings (scraped, anonymized)
Product count	**1.18 million** real products across diverse categories
Instruction count	**12,087** crowd-sourced text instructions
Instruction style	Natural language; specifies product type, required attributes, and optional price/brand constraints
Instruction complexity	Compositional — typically 2–4 attribute constraints alongside a product category
Human trajectory collection	1,600+ human interaction trajectories collected to bootstrap imitation learning and validate task difficulty

Task Design

Action Space

At each step the agent selects one of:

- `search[query]` — submit a search query to the simulated shop
- `click[element]` — click a UI element (product link, "Back to Search", option buttons such as color/size swatches, "Buy Now")

The episode terminates when the agent clicks "Buy Now" or exceeds the step budget.

Instruction-Following Challenges

WebShop deliberately introduces four compounding difficulty sources:

1. **Compositional constraints** — multiple attributes must all be satisfied simultaneously (product type × color × size × price range)
2. **Query reformulation** — the instruction vocabulary rarely matches product titles verbatim; agents must paraphrase to retrieve relevant results
3. **Noisy web text** — product descriptions contain HTML artifacts, inconsistent attribute naming, and redundant specifications
4. **Strategic exploration** — early search results are often suboptimal; the agent must decide when to refine the query vs. navigate deeper

Reward Function

WebShop's reward function is **partial-credit, attribute-weighted** — not binary pass/fail. This design choice was deliberate: it provides a richer learning signal and better captures real-world buyer satisfaction where "mostly right" has value.

```
R = attribute_score × I[purchased]
```

Where:

- `I[purchased]` is 1 if the agent clicked "Buy Now" on any product (prevents random non-purchasing strategies from scoring)
- `attribute_score` ∈ [0, 1] is computed as the weighted fraction of required attributes satisfied by the purchased product:
    - **Product relevance**: semantic similarity between purchased product title/description and the instruction's product category requirement
    - **Attribute match**: binary checks per specified attribute (color, size, brand, etc.); partial credit is the mean across all required attributes
    - **Price constraint**: binary check if a price upper-bound was specified in the instruction

Task success is defined as `R ≥ 0.5` (i.e., at least half the required attributes are satisfied). The continuous reward score enables gradient-based RL training and distinguishes partially-correct purchases from completely off-target ones.

> ***BuyerBench relevance:*** *The partial-credit reward design is directly applicable to BuyerBench Pillar 1 scoring. Procurement tasks — like selecting from a supplier catalog where specifications partially overlap requirements — benefit from this soft-scoring model over strict binary correctness. BuyerBench's capability metrics can adopt attribute-weighted scoring to differentiate "found a supplier but wrong lead time" from "no supplier found."*

Human vs. Agent Performance

AGENT TYPE	TASK SUCCESS RATE	SCORE (CONTINUOUS)
Rule-based heuristics	9.6%	~0.20
IL (imitation learning)	~26%	~0.52
RL (from scratch)	~18%	~0.40
IL + RL (best model)	**29.1%**	**0.62**
Human experts	**59.6%**	**0.82**

Key gap: the best model scores 29% task success vs. 59.6% for humans — a **30.5 percentage point gap** at NeurIPS 2022 publication. The continuous score gap is smaller (0.62 vs. 0.82), confirming that agents partially solve tasks more often than they fully solve them.

Sim-to-Real Transfer

A notable finding: agents trained entirely within the WebShop simulation showed **non-trivial sim-to-real transfer** when deployed against live Amazon.com and eBay.com pages (without any fine-tuning on real-site data). This validated the ecological validity of the benchmark's design and suggested that training on realistic simulated product data generalizes to real-world web shopping interfaces.

Foundational Influence on Later Benchmarks

WebShop is the most-cited precursor in the web-agent benchmark lineage. Its design choices propagated into later work:

SUCCESSOR BENCHMARK	RELATIONSHIP TO WEBSHOP
WebArena (ICLR 2024)	Directly cites WebShop as foundational; replaces simulated store with self-hosted Magento (OneStopShop); shifts from reward-score to functional-state evaluation; adds 4 non-shopping environments for breadth
ACES AI Agent Buying (arXiv 2508.02630)	Narrows scope back to purchase-decision quality (WebShop's core insight) but replaces the RL framing with RCT experimental design; directly studies WebShop-class decision biases (position, price, endorsement)
AgentBench (ICLR 2024)	Includes "Web Shopping" as one of its 8 task environments — effectively a WebShop-derived module — plus a distinct "Web Browsing" module
ReAct (ICLR 2023)	Uses WebShop as one of two validation environments for chain-of-thought + action interleaving; established the now-standard reasoning-action agent paradigm
SteP / Mind2Web	Subsequent web-grounding work builds on WebShop's action space conventions

> ***BuyerBench relevance:*** *WebShop established that e-commerce is a tractable, measurable domain for agent benchmarking. BuyerBench inherits this intuition and specializes it: rather than generic shopping, BuyerBench focuses on B2B procurement workflows (RFQ, supplier comparison, negotiation) where multi-supplier search and economic rationality are first-class concerns — gaps WebShop did not address.*

BuyerBench Pillar 1 Alignment

Coverage

WEBSHOP CAPABILITY	BUYERBENCH PILLAR 1 COVERAGE
Natural language query formulation	Supplier discovery (search query construction)
Multi-step navigation (search → listing → detail → purchase)	Procurement workflow execution (RFQ initiation → quote review → PO issuance)
Attribute matching (color, size, price constraints)	Specification matching (lead time, MOQ, certifications, unit price)
Partial-credit reward design	Attribute-weighted procurement scoring model
Sim-to-real transfer validation	BuyerBench scenario ecological validity requirement

Gaps (not addressed by WebShop)

GAP	BUYERBENCH ADDITION
Single supplier only (buy/don't buy)	**Multi-supplier comparison** — agent must rank and select among competing suppliers
No price negotiation	**Negotiation workflows** (Pillar 1 and 2)
No behavioral bias testing	**Controlled framing variants** (Pillar 2)
No security/compliance requirements	**Payment API protocol adherence** (Pillar 3)
B2C only (consumer shopping)	**B2B procurement context** — approval flows, vendor lists, contract terms

Key Limitations

1. **Single-store, single-action-stream**: WebShop is a simulated Amazon-clone with a fixed navigation topology; real procurement spans heterogeneous supplier portals, PDFs, and APIs
2. **No multi-turn negotiation**: An agent either buys or doesn't — no back-and-forth price/term negotiation
3. **No behavioral bias measurement**: All agents are evaluated against the same instruction; no controlled variant design to detect anchoring or framing effects
4. **Reward function opacity**: Partial-credit computation relies on semantic similarity models that can be gamed by surface-form matching without genuine attribute satisfaction
5. **B2C scope only**: Products are consumer goods; no B2B procurement concepts (MOQ, lead time, vendor qualification, contract compliance)

Sources

1. arXiv:2207.01206 — WebShop abstract
2. NeurIPS 2022 Proceedings — Full paper PDF
3. Project website — webshop-pnlp.github.io
4. GitHub — princeton-nlp/WebShop
5. OpenReview — NeurIPS 2022 submission

PART IX

Key People

Seven key individuals across academic research, procurement technology, and payment infrastructure shaping the AI buyer agent landscape.

Amine Allouah

> *Lead author of ACES (arXiv:2508.02630), the Columbia Business School RCT benchmark that documented position bias, price elasticity distortion, and endorsement effects in AI buyer agents — the core empirical foundation for BuyerBench Pillar 2*

Role

PhD Researcher / Postdoctoral Researcher, Columbia Business School (Decision, Risk, and Operations Division)

Background

Amine Allouah completed his PhD at Columbia Business School, concentrating in algorithmic game theory, revenue management, and decision optimization. He works within the Decision, Risk, and Operations (DRO) division — an operations research and economics group, not a traditional ML lab. This positioning explains why the ACES paper approaches AI buyer agents through the lens of **economic rationality and market microstructure** rather than capability benchmarking.

FIELD	DETAIL
Institution	Columbia Business School
Division	Decision, Risk, and Operations (DRO)
Specialization	Algorithmic game theory, optimization, e-commerce
Affiliations	MyCustomAI (applied research)
Location	New York City, USA
Google Scholar	Amine Allouah

Key Contributions to Agentic Commerce

ACES: Agentic e-CommercE Simulator (Aug 2025, arXiv:2508.02630)

Allouah's primary contribution to the agentic commerce field is the ACES framework — a fully controlled, provider-agnostic e-commerce sandbox designed to audit AI buyer agent decision-making under realistic conditions.

ACES Methodology

ACES operates as a mock Amazon-style marketplace that allows researchers to:

- **Control product attributes independently** (price, position, description, reviews, endorsements)
- **Run RCTs** where only one variable differs between product listings
- **Measure** whether agents make economically rational choices vs. biased ones

Key Empirical Findings (from the paper)

BIAS TYPE	MEASURED EFFECT
Position bias	5× preference for top-listed item regardless of economic merit
Endorsement effect	+1.0 to +1.9 price premium tolerated for "AI recommended" badge
Price elasticity distortion	−1.6 to −2.2 (agents over-penalize price increases)
Choice homogeneity	Demand concentrated on few "modal" products; most listings ignored
Model instability	Model updates drastically reshuffle market share with no economic basis

Implication for Agentic Commerce

The paper reveals a structural market risk: if AI agents systematically prefer top-positioned and "recommended" items, market power could concentrate among platforms controlling listing positions — independent of product quality or price. This is a **Pillar 2 design foundation** for BuyerBench.

Co-Authors

- **Omar Besbes** — Columbia Business School professor (senior author); see Omar Besbes
- **Yash Kanoria** — Columbia Business School professor, mechanism design and market microstructure
- **Josué D. Figueroa** — Columbia / applied research
- **Akshit Kumar** — Columbia Business School PhD

Public Statements / Papers

- *"What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications for Agentic E-Commerce"* — arXiv:2508.02630, Aug 2025
- SSRN preprint: papers.ssrn.com/sol3/papers.cfm?abstract_id=5381574
- Columbia Business School article: What Happens When AI Does Your Shopping?

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *The ACES paper is the* **primary empirical source** *for BuyerBench Pillar 2 scenario design. All behavioral bias categories in Pillar 2 (position bias, anchoring, framing/endorsement effects, price elasticity) map directly to Allouah et al.'s findings. BuyerBench extends their methodology by adding procedural (multi-step workflow) contexts where biases compound across decision points.*

Sources

- arXiv:2508.02630
- Columbia Business School Directory
- SSRN Paper
- Columbia Business School Article

Omar Besbes

> *Senior author of the ACES paper (arXiv:2508.02630) and Columbia Business School professor whose research on algorithmic pricing and market dynamics shaped the economic framing of AI buyer agent bias*

Role

Professor of Business, Decision, Risk, and Operations Division, Columbia Business School

Background

Omar Besbes is a tenured professor at Columbia Business School in the Decision, Risk, and Operations (DRO) division. He is a recognized expert in revenue management, dynamic pricing, algorithmic trading, and data-driven decision-making under uncertainty. His research applies operations research and economics methods to problems in e-commerce, online platforms, and financial markets.

FIELD	DETAIL
Institution	Columbia Business School
Division	Decision, Risk, and Operations (DRO)
Specialization	Revenue management, dynamic pricing, algorithmic e-commerce
Role	Professor (tenured faculty)
Location	New York City, USA
Faculty Profile	business.columbia.edu

Key Contributions to Agentic Commerce

As senior author on the ACES paper alongside Amine Allouah, Yash Kanoria, Josué Figueroa, and Akshit Kumar, Besbes brought a **market microstructure and revenue management** perspective to AI buyer agent evaluation. His theoretical framing elevated the paper beyond a capability benchmark to an analysis of **market structure consequences** of widespread AI buyer agent adoption.

Theoretical Contributions

- **Market concentration modeling**: Demonstrated how AI choice homogeneity (agents all buying from the same few products) creates winner-take-all dynamics even when products are economically equivalent
- **Instability pricing insights**: Connected model update sensitivity to platform-level price discovery failures — if agent preferences reshuffle arbitrarily at each model version, market pricing signals break down
- **Bias measurement rigor**: Applied RCT methodology (standard in causal economics) to AI agent testing, which is more epistemically sound than typical benchmark leaderboard approaches

ACES Paper Roles

TASK	ATTRIBUTION
Experimental design	Led by Besbes/Kanoria (economics/mechanism design)
Market modeling	Besbes (revenue management lens)
Implementation	Allouah + Kumar (coding, simulation)
Theoretical framing	Besbes + Kanoria (market microstructure)

Public Statements / Papers

- *"What Is Your AI Agent Buying? Evaluation, Biases, Model Dependence, & Emerging Implications for Agentic E-Commerce"* — arXiv:2508.02630, Aug 2025 (senior author)
- Faculty profile: business.columbia.edu/faculty/people/omar-besbes
- ResearchGate: Omar BESBES profile

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *Besbes's revenue management expertise informs the* ***economic optimality gap*** *metric in Pillar 2 — the idea that even when agents pick a "valid" product, deviating from the economically optimal choice has a measurable cost. His market concentration framing also motivates future BuyerBench research on systemic market effects of widespread AI buyer agent deployment.*

Sources

- arXiv:2508.02630
- Columbia Business School Faculty

- ResearchGate Profile

Yasser Mohammad

> *Creator of NegMAS (Negotiation Multi-Agent System) — the official engine of ANAC (Automated Negotiation Agents Competition) for 16 years and the `negmas` agent reference implementation in BuyerBench*

Role

Researcher, NEC Corporation / National Institute of Advanced Industrial Science and Technology (AIST); formerly Özyeğin University (Istanbul)

Background

Yasser Mohammad (GitHub: `yasserfarouk`) is a computer scientist and researcher specializing in multi-agent systems, automated negotiation, and supply chain simulation. He is the primary creator and maintainer of NegMAS, which he developed across multiple academic affiliations including NEC Corporation (Tokyo), AIST (Tokyo), Assiut University (Egypt), and Özyeğin University (Istanbul, Turkey).

FIELD	DETAIL
Primary Affiliation	NEC Corporation, Japan
Additional Affiliations	AIST (Tokyo), Assiut University (Egypt), Özyeğin University (Istanbul)
Specialization	Multi-agent systems, automated negotiation, supply chain AI
GitHub	github.com/yasserfarouk
Personal Site	yasserm.com

Key Contributions to Agentic Commerce

NegMAS (Negotiation Multi-Agent System)

NegMAS is a Python library for developing autonomous negotiation agents embedded in simulation environments. It supports bilateral and multilateral negotiations, multiple negotiation protocols (SAOMechanism, MechanismState, etc.), and complex multi-agent simulations where negotiations are interdependent.

Why NegMAS matters for buyer agents:

- **Situated negotiations**: Unlike abstract game-theory solvers, NegMAS agents negotiate in dynamic environments where external factors (supplier capacity, market price shifts) affect utility in real time
- **Supply chain focus**: The SCML (Supply Chain Management League) module enables full multi-echelon supplier negotiation simulation
- **Configurable utility functions**: Enables controlled Pillar 2 bias experiments by allowing utility function manipulation while holding agent negotiation behavior constant

ANAC Leadership

NegMAS has been the official negotiation engine for the **Automated Negotiation Agents Competition (ANAC)** — a 16-year running competition at IJCAI — since its adoption. This gives NegMAS its status as the community-standard benchmark for autonomous negotiation research.

Key NegMAS Capabilities

MODULE	PURPOSE
`SAOMechanism`	Bilateral alternating-offers negotiation (standard buyer-seller)
`SCML`	Supply Chain Management League — multi-echelon simulation
`Negotiator`	Base class for custom negotiation agents
`UtilityFunction`	Configurable preference modeling for bias research

Co-Creators

- **Shinji Nakadai** — NEC Corporation, Japan
- **Amy Greenwald** — Brown University (computer science, mechanism design)

Published Papers

- *"NegMAS: A Platform for Automated Negotiations"* — PRIMA 2020, Springer
- *"NegMAS: A Platform for Situated Negotiations"* — Springer (PRIMA chapter)
- PyPI package: negmas 0.9.6
- GitHub: github.com/yasserfarouk/negmas

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *NegMAS is the* `negmas` *agent in BuyerBench (agent ID:* `negmas` *) — a Python-native reference implementation that requires no API credentials and demonstrates economically grounded negotiation behavior. For* ***Pillar 1****, SCML scenarios provide realistic multi-*

> *supplier environments. For **Pillar 2**, NegMAS's configurable utility functions allow researchers to test whether agents exhibit irrational bias even when the objectively optimal negotiation outcome is mathematically computable.*

Sources

- GitHub: yasserfarouk/negmas
- NegMAS: A Platform for Automated Negotiations (Springer)
- yasserm.com/software
- PyPI negmas

Kevin Frechette

> *CEO and Co-Founder of Fairmarkit — the AI-native autonomous sourcing platform processing 150,000+ zero-human-touch procurement events per year; a leading practitioner voice on agentic AI in enterprise procurement*

Role

CEO and Co-Founder, Fairmarkit (Boston, MA)

Background

Kevin Frechette co-founded Fairmarkit in 2017 with Tarek Alaruri and Victor Kushch (CTO). Prior to Fairmarkit, he held enterprise sales and technology roles at IBM and Dell — giving him a supply-side view of how enterprise procurement teams evaluate and deploy technology. He has positioned Fairmarkit at the intersection of **autonomous sourcing** and **agentic AI**, arguing that procurement is the highest-ROI enterprise function for full AI automation.

FIELD	DETAIL
Company	Fairmarkit
Role	CEO and Co-Founder
Founded	2017
Headquarters	Boston, MA
Prior Companies	IBM, Dell
LinkedIn	linkedin.com/in/kevin-frechette-49652921

Co-Founders

- **Tarek Alaruri** — Co-Founder
- **Victor Kushch** — Co-Founder and CTO

Key Contributions to Agentic Commerce

Fairmarkit's Agentic Sourcing Model

Under Frechette's leadership, Fairmarkit has operationalized "demand-to-award" autonomous procurement — where AI agents handle the entire sourcing cycle from requisition intake through supplier selection and award, without requiring human intervention on routine purchases.

Scale metrics (2025):

- 150,000+ events with zero human touch annually at enterprise scale
- Named 5× to the ProcureTech100 list
- $78M raised total (as of 2025)
- Inc. 5000 honoree 2025

Key Strategic Positions

Frechette has been a consistent public voice on the transition from "AI copilot" to **fully autonomous procurement agents**, arguing that:

- Approval workflows (human-in-the-loop checkpoints) are the biggest friction point in enterprise AI adoption
- Agentic AI reduces tail spend by removing the human cognitive bottleneck in routine sourcing
- Benchmark metrics for AI sourcing should measure *business outcomes* (cost savings, cycle time) not just task completion rates

Published Views on Agentic AI

From AI Time Journal interview (2025):

> *"Agentic AI isn't just automating a task — it's turning procurement into an autonomous system. The measure of success is zero human touches on routine spend, not 'the agent got the right answer.'"*

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *Frechette's operational experience running 150K+ zero-touch procurement events at scale informs* ***Pillar 1 scenario design*** *— specifically, what counts as a "successful" buyer workflow completion in enterprise procurement. His insistence on measuring business outcomes (cost savings, cycle time, zero-touch rate) rather than just accuracy supports BuyerBench's multi-metric evaluation model. Fairmarkit's demand-to-award pipeline is a real-world reference for end-to-end Pillar 1 scenarios.*

Sources

- AI Time Journal Interview (2025)
- Fairmarkit Company Page
- Crunchbase: Kevin Frechette
- Inc. 5000 Profile

Amir Sarhangi

> *CEO and Co-Founder of Skyfire — the AI agent payment and identity infrastructure company behind the KYAPay (Know Your Agent Pay) open protocol; serial entrepreneur with prior exits at Jibe Mobile (acquired by Google, became Android RCS) and Ripple ($50B+ cross-border payments processed)*

Role

CEO and Co-Founder, Skyfire Systems Inc. (San Francisco, CA)

Background

Amir Sarhangi is a serial entrepreneur with two notable prior ventures before Skyfire. He co-founded **Jibe Mobile**, a messaging infrastructure company that pioneered RCS (Rich Communication Services) — Jibe was acquired by Google and its technology became the standard for Android messaging (now used by over 1 billion users). After Jibe, Sarhangi joined **Ripple** as VP of Product, where he helped build the cross-border payment network that processed over $50 billion during his tenure. He co-founded Skyfire in 2023 with Craig DeWitt to apply the same payment network thinking to AI agent transactions.

FIELD	DETAIL
Company	Skyfire Systems Inc.
Role	CEO and Co-Founder
Founded	2023
Headquarters	San Francisco, CA
Prior Roles	VP Product at Ripple; Co-Founder/CEO of Jibe Mobile (acq. Google); CEO of Supermojo
Education	B.S. Engineering, California Polytechnic State University (1997)
LinkedIn	linkedin.com/in/asarhangi

Co-Founder

- **Craig DeWitt** — Co-Founder (Ripple alumni)

Key Contributions to Agentic Commerce

KYAPay Protocol

Sarhangi's primary contribution to agentic commerce is the **KYAPay (Know Your Agent Pay)** open standard — a protocol enabling cryptographic identity verification for AI agents before they can initiate payments. Unlike human-centric authentication (passwords, biometrics), KYAPay enables:

- Machine-to-machine agent identity assertion
- Scoped spending authorization (agent can only spend within predefined limits)
- Auditability (all agent actions signed and logged)

Skyfire Milestones Under His Leadership

DATE	MILESTONE
Aug 2024	Skyfire launches, AI agents race to join payment network (BusinessWire)
Dec 2024	$9.5M seed raised (Neuberger Berman, a16z CSX, Coinbase Ventures, 18 investors)
Dec 2025	First secure agentic commerce demo: AI agent purchases product using KYAPay + Visa Intelligent Commerce + Trusted Agent Protocol
Mar 2026	Skyfire exits beta with enterprise-ready payment network for AI agents

Strategic Framing

From Citi Ventures Q&A (2025):

> *"The challenge with AI agents and payments isn't the payment itself — it's that nobody knows who the agent is, what it's authorized to spend, or whether it's acting on behalf of the user or acting in bad faith. KYAPay solves the identity problem; the payment rails already exist."*

This framing — **identity before payment** — is the core thesis driving Skyfire's architecture and differentiates it from payment-first approaches like x402 or ACP.

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *Sarhangi's identity-first architecture is the* ***positive reference implementation*** *for BuyerBench Pillar 3 scenarios. His KYAPay protocol demonstrates what correct agent authentication looks like — scenarios where agents attempt to transact without proper identity assertion*

(impersonation, unauthorized scope expansion) test whether agents enforce KYAPay-equivalent controls. The Skyfire beta exit (Mar 2026) also marks the point at which enterprise-grade agent payment infrastructure became commercially available — a key capability threshold for BuyerBench scenario realism.

Sources

- Citi Ventures Q&A with Amir Sarhangi
- TechCrunch: Skyfire lets AI agents spend your money
- BusinessWire: Skyfire Exits Beta
- Crunchbase: Amir Sarhangi

Rubail Birwadker

> *SVP and Head of Growth Products & Partnerships at Visa, responsible for Visa Intelligent Commerce — the largest payment network's agentic commerce initiative covering AI agents, A2A non-card payments, crypto, and stablecoins*

Role

SVP, Head of Growth Products & Partnerships, Visa Inc.

Background

Rubail Birwadker joined Visa in 2011 and has held a variety of leadership roles spanning M&A, Corporate Development, Venture functions, and Global Digital and Fintech Partnerships. In his current role, he leads Visa's Growth functions with global scope, overseeing:

- **Agentic Commerce** (AI agent payment infrastructure)
- **A2A Non-Card Payments** (account-to-account alternatives)
- **Crypto and Stablecoins** (Visa's digital asset payment rails)
- **Global Strategic Partnerships** (100+ technology partner integrations)

FIELD	DETAIL
Company	Visa Inc.
Role	SVP, Head of Growth Products & Partnerships
At Visa Since	2011
Scope	Agentic commerce, A2A, crypto/stablecoins, global partnerships
LinkedIn	linkedin.com/in/rubail

Key Contributions to Agentic Commerce

Visa Intelligent Commerce

Birwadker is Visa's public spokesperson and product leader for **Visa Intelligent Commerce (VIC)** — launched at Visa's Global Product Drop event in San Francisco (Apr 30, 2025). Under his oversight:

- VIC launched with **100+ partners** including Anthropic, Microsoft, OpenAI, and Perplexity

- Visa's **Trusted Agent Protocol (TAP)** was released as an open framework (Oct 14, 2025) — HTTP Message Signatures + WebAuthn for agent identity
- VIC sandbox activation: **30+ partners actively building**, **20+ agents integrating** directly with VIC (as of late 2025)
- First secure end-to-end agentic transaction completed with Skyfire, AWS Bedrock, and Visa TAP (Dec 2025)
- Visa-AWS joint infrastructure for agentic commerce launched (Vertex AI / Bedrock AgentCore integration, late 2025)
- Asia Pacific and Europe VIC pilots targeted for early 2026

Key Timeline

DATE	EVENT
Apr 30, 2025	Visa Intelligent Commerce launched (100+ partners)
Oct 14, 2025	Visa Trusted Agent Protocol (TAP) open framework released
Dec 2025	First secure end-to-end agentic purchase demo (Skyfire + VIC + TAP)
Late 2025	Visa-AWS agentic commerce infrastructure joint launch
2026	Asia Pacific + Europe VIC pilot expansion

Public Statements

> *"AI commerce is a new commerce experience where AI agents play an active role in helping users shop online — and Visa's role is to ensure trust, security, and authorization flow correctly through the agent chain just as they do for human cardholders." — Rubail Birwadker, announcing Visa Intelligent Commerce (2025)*

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *Birwadker's Visa Intelligent Commerce is the* ***largest-scale Pillar 3 authorization reference*** *— the TAP framework defines what proper cryptographic agent identity and spend scoping look like at network scale (billions of existing Visa tokens extended to agents). Pillar 3 scenarios involving payment network authorization, KYA identity requirements, and scope boundary enforcement are grounded in VIC/TAP architecture. His oversight of crypto/stablecoin products also connects to x402 Pillar 3 scenarios.*

Sources

- Visa: "Find and Buy with AI" (VIC launch)
- Visa: Secure AI Transactions (Dec 2025)
- LinkedIn: Rubail Birwadker
- American Banker Payments Forum Profile

Jorn Lambert

> *Chief Product Officer at Mastercard, the executive leading Agent Pay — the agentic payments program featuring Agentic Tokens, Verifiable Intent, and Web Bot Auth that launched the same day as Visa's Trusted Agent Protocol (Oct 14, 2025)*

Role

Chief Product Officer, Mastercard Inc.

Background

Jorn Lambert joined Mastercard in 2002 and has held progressive leadership roles across the organization for over two decades. He served as Chief Digital Officer (CDO) before stepping into the Chief Product Officer (CPO) role in 2024. His earlier career was in capital markets, where he held product development, product management, and corporate strategy positions before moving to payments.

FIELD	DETAIL
Company	Mastercard Inc.
Role	Chief Product Officer (CPO)
At Mastercard Since	2002
Prior Role	Chief Digital Officer (CDO)
Education	Roman Philology, University of Ghent + Sapienza University of Rome; Post-graduate business economics, University of Leuven
LinkedIn	linkedin.com/in/jorn-lambert

Key Contributions to Agentic Commerce

Mastercard Agent Pay

Lambert is the executive driving **Mastercard Agent Pay**, launched in April 2025, which formalizes Mastercard's three-layer agentic payment architecture:

1. **Agent Registration** — Cryptographic identity binding for AI agents to a user principal

2. **Agentic Tokens** — Scoped payment tokens (amount, merchant category, time bounds) issued specifically for agent use
3. **Verifiable Intent** — Open-source immutable audit trail spec for agent transactions; critical for dispute resolution

Lambert also oversaw the **Web Bot Auth** collaboration with Cloudflare, which enables CDN-layer agent authentication — agents authenticated at the network edge before even reaching merchant APIs.

Timeline Under Lambert's Leadership

DATE	EVENT
Apr 2025	Mastercard Agent Pay announced
Oct 14, 2025	Mastercard Agent Pay expands (same day as Visa TAP — coordinated industry signal)
Oct 27, 2025	Mastercard + PayPal partnership for global agentic commerce
Mid-Nov 2025	US cardholders enabled for Agent Pay
Sep 2025	New tools and collaborations for safer agentic commerce announced
Mar 2026	Mastercard Virtual C-Suite for small businesses (agentic AI extension)

Strategic Philosophy

Lambert has emphasized a **"rules of the road"** framing for agentic commerce — arguing that AI agents need explicit authorization boundaries analogous to traffic laws, not just good intentions:

> *"Mastercard is transforming the way the world pays for the better by anticipating consumer needs on the horizon. The launch of Mastercard Agent Pay marks our initial steps in redefining commerce in the AI era." — Jorn Lambert, Agent Pay launch (Apr 2025)*

> *"AI-powered payments need the same trust infrastructure as human payments — but the authorization model has to be built for how agents actually operate: scoped, auditable, and revocable." — Jorn Lambert, Arabian Business interview (2026)*

Industry Significance

The fact that both Visa (Rubail Birwadker) and Mastercard (Jorn Lambert) launched agentic payment frameworks **on the same day** (Oct 14, 2025) indicates coordinated industry consensus-building — both networks recognized that competing standards would fragment the ecosystem and slow adoption.

BuyerBench Pillar Relevance

> ***BuyerBench relevance:*** *Lambert's Verifiable Intent spec is the most important Mastercard contribution to* ***Pillar 3 scenario design*** *— it defines what an immutable agent transaction audit trail looks like at network level. Pillar 3 scenarios testing whether agents produce auditable transaction records, maintain scope boundaries, and generate dispute-ready evidence chains are directly grounded in Agent Pay architecture. The Web Bot Auth integration also informs scenarios testing agent identity at the infrastructure layer (CDN-level bot detection and authentication).*

Sources

- Mastercard CPO Bio
- Agent Pay Launch (Apr 2025)
- Agentic Commerce Rules of the Road (2026)
- LinkedIn: Jorn Lambert